chemistry-ai999x: capability guide

Creator / engineer: Ibnul Jaif Farabi · **Company:** AI999X. Ask the package `Who created you?` or call `chemistry_ai999x.about()` for the same answer.

Every capability chemistry-ai999x supports, generated from its certified capability contracts. Every example below was executed when this guide was built. The structured call `solve("<capability id>", ...)` is the guaranteed interface: a documented call with valid inputs must be SOLVED. Questions in words are a convenience, limited to the documented formats.

Quick start

```
import chemistry_ai999x

result = chemistry_ai999x.solve('molar_mass', formula='H2O')
result.show()
# M = 0.018015 kg/mol, M_g = 18.015
```

By topic

topic	capabilities
Acids and bases / buffers	Buffer pH (Henderson-Hasselbalch)
Acids and bases / conjugate pairs	Conjugate acid-base constants (Ka Kb = Kw)
Acids and bases / pH	PH from the hydronium concentration, Hydronium concentration from pH, PH from the hydroxide concentration
Acids and bases / polyprotic acids	PH of a diprotic acid
Acids and bases / salt hydrolysis	PH of a salt of a weak acid
Acids and bases / strong acids	PH of a strong monoprotic acid
Acids and bases / strong bases	PH of a strong base
Acids and bases / titrations	Titration: equivalence-point volume, PH during a strong acid - strong base titration, PH during a weak acid - strong base titration
Acids and bases / weak acids	PH of a weak acid, Ka from the pH of a weak acid
Acids and bases / weak bases	PH of a weak base
Atomic structure / isotopes	Average atomic mass from isotopes, Abundances of two isotopes from the average atomic mass
Electrochemistry / cell potentials	Nernst equation, Standard cell potential from two half-reactions

topic	capabilities
Electrochemistry / cell thermodynamics	Gibbs energy from a cell potential, Equilibrium constant from a standard cell potential
Electrochemistry / electrolysis	Mass deposited in electrolysis (Faraday's law), Time for electrolysis (Faraday's law)
Equilibrium / equilibrium constants	Kp from Kc, Kc from Kp
Equilibrium / ICE tables	Equilibrium concentrations (ICE table)
Equilibrium / reaction quotient	Reaction quotient and the direction of shift
Equilibrium / solubility	Molar solubility from Ksp, Ksp from molar solubility, Molar solubility with a common ion, Precipitation test (Q versus Ksp)
Equilibrium / temperature dependence	Van 't Hoff equation: K at another temperature
Equilibrium / thermodynamics of equilibrium	Equilibrium constant from the standard Gibbs energy, Standard Gibbs energy from the equilibrium constant, Gibbs energy change at non-standard conditions
Gases / gas laws	Boyle's law: final volume, Boyle's law: final pressure, Charles's law: final volume, Gay-Lussac's law: final pressure, Combined gas law: final volume, Combined gas law: final pressure, Combined gas law: final temperature
Gases / gas mixtures	Partial pressure (Dalton's law), Partial pressures in a gas mixture
Gases / ideal gas law	Ideal gas law: pressure, Ideal gas law: volume, Ideal gas law: amount, Ideal gas law: temperature, Mass of a gas sample (ideal gas), Density of an ideal gas, Molar mass from gas density, Molar mass from a gas sample
Gases / kinetic theory	Graham's law: ratio of effusion rates, Graham's law: molar mass of an unknown gas, Root-mean-square speed of gas molecules, Mean speed of gas molecules
Gases / real gases	Van der Waals equation: pressure
Kinetics / complex reactions	Consecutive first-order reactions (A -> B -> C)
Kinetics / half-life	Half-life of a first-order reaction
Kinetics / integrated rate laws	Zero-order integrated rate law, First-order integrated rate law, Second-order integrated rate law, Time to reach a concentration (first order)
Kinetics / rate laws	Reaction order from initial rates
Kinetics / temperature dependence	Arrhenius equation: rate constant at another temperature, Activation energy from two rate constants, Arrhenius equation: rate constant
Molecular properties / structure	Degree of unsaturation
Nuclear chemistry / binding energy	Mass defect and nuclear binding energy
Nuclear chemistry / radioactive decay	Amount remaining after radioactive decay (half-life), Age from the fraction remaining (radioactive dating)
Organic chemistry / stereochemistry	Specific rotation from an observed rotation, Enantiomeric excess from optical rotation, Enantiomeric excess from the composition of a mixture
Solutions / colligative properties	Vapor pressure of a solution (Raoult's law), Freezing-point depression, Boiling-point elevation, Molar mass from freezing-point depression, Osmotic pressure, Molar mass from osmotic pressure

topic	capabilities
Solutions / concentration	Molarity from a dissolved mass, Mass of solute for a solution, Molality of a solution, Mole fraction in a solution
Solutions / dilution	Dilution: final concentration, Dilution: volume of stock needed
Solutions / gas solubility	Henry's law: solubility of a gas
Spectroscopy / absorption	Beer-Lambert law: absorbance, Beer-Lambert law: concentration
Spectroscopy / atomic spectra	Hydrogen atom transition wavelength (Rydberg)
Spectroscopy / infrared	Vibrational wavenumber of a diatomic molecule
Spectroscopy / light	Energy of a photon from its wavelength, Wavelength of a photon from its energy
Stoichiometry / chemical equations	Balancing a chemical equation
Stoichiometry / composition	Mass percent of an element, Mass of an element in a sample, Empirical formula from percent composition, Molecular formula from the empirical formula
Stoichiometry / molar mass	Molar mass of a formula
Stoichiometry / reaction yields	Limiting reagent and theoretical yield, Theoretical yield from one reactant, Percent yield of a reaction
Stoichiometry / the mole	Moles from a mass, Mass from moles, Number of molecules in a mass, Number of particles in an amount, Moles from a number of particles, Atoms of an element in a mass of compound
Thermochemistry / bond enthalpies	Reaction enthalpy from bond enthalpies
Thermochemistry / calorimetry	Heat from a temperature change, Reaction enthalpy from coffee-cup calorimetry
Thermochemistry / entropy	Standard reaction entropy from standard molar entropies
Thermochemistry / Hess's law	Reaction enthalpy from enthalpies of formation (Hess's law)
Thermochemistry / phase equilibria	Clausius-Clapeyron: vapor pressure at another temperature, Clausius-Clapeyron: enthalpy of vaporization
Thermochemistry / reaction enthalpy	Heat of reaction for a given mass
Thermochemistry / spontaneity	Gibbs energy change (dG = dH - T dS), Temperature at which dG = 0, Standard reaction Gibbs energy from Gibbs energies of formation

Outcomes

outcome	meaning	what to do
SOLVED	a value with its unit, steps, assumptions and verification	use it; read the assumptions
NEEDS_INPUT	a required input is missing; `uncertainty['needs']` names it	add it (with its unit) and ask again
AMBIGUOUS	two readings or two physical branches give different answers	say which situation is meant
INVALID_INPUT	a unit of the wrong kind, an unknown input name, values with no valid solution	correct the input
OUT_OF_SCOPE	no certified capability computes this (or the structure is not modelled yet)	see the catalogue below

outcome	meaning	what to do
UNRECOGNIZED_FORMAT	the wording was not recognised; a supported template is offered in `uncertainty['try']`	use the template or solve()
INTERNAL_ERROR	a defect of the engine, never a property of the question	report it with the question

Result fields

field	meaning
value, unit	the result in SI units
values	every output of a procedure (x1, x2, discriminant ...)
outcome	one of the seven outcomes above
steps	each equation used, its inputs (with where each came from) and its result
assumptions	conditions and declared defaults the answer relies on
verification	residual of every equation, dimensional consistency, independent checks
latex()	the derivation as LaTeX
show()	renders in a notebook, prints in a terminal

Common mistakes

- Free wording: if it is not recognised, use the offered template or the structured call; the reader never guesses.
- A coefficient that violates a stated condition (a = 0 in a quadratic): INVALID_INPUT says why.
- A text value where a number is expected: INVALID_INPUT; fractions as strings ("1/2") are accepted.
- A problem outside the procedure (a cubic for the quadratic solver): OUT_OF_SCOPE, never an approximation.

Procedures: algorithms with several outputs

Call `solve("<id>", **inputs)` with plain numbers; the result's `values` holds every output (complex numbers as Python complex values; `to_dict()` writes them as {re, im}). Each result is substituted back into its defining equation before it is returned.

id	operation	inputs	outputs
molar_mass	Molar mass of a formula	formula	M, M_g
moles_from_mass	Moles from a mass	formula, m	n, N, M
mass_from_moles	Mass from moles	formula, n	m, m_g, N
particles_from_mass	Number of molecules in a mass	formula, m	N, n
particles_from_moles	Number of particles in an amount	n	N
moles_from_particles	Moles from a number of particles	N	n
atoms_of_element	Atoms of an element in a mass of compound	formula, element, m	N_atoms, n_element

id	operation	inputs	outputs
mass_percent	Mass percent of an element	formula, element	percent, composition
mass_of_element	Mass of an element in a sample	formula, element, m	m_element, fraction
empirical_formula	Empirical formula from percent composition	composition	formula, counts
molecular_formula	Molecular formula from the empirical formula	empirical, M	multiple, formula
balance_equation	Balancing a chemical equation	equation	balanced, coefficients
limiting_reagent	Limiting reagent and theoretical yield	equation, reactant_a, m_a, reactant_b, m_b, product	product_mass, limiting, excess_left
theoretical_yield	Theoretical yield from one reactant	equation, reactant, m, product	product_mass, n_product
percent_yield	Percent yield of a reaction	equation, reactant, m, product, actual	percent_yield, theoretical_mass
ideal_gas_pressure	Ideal gas law: pressure	n, V, T	P, P_atm
ideal_gas_volume	Ideal gas law: volume	n, P, T	V, V_L
ideal_gas_moles	Ideal gas law: amount	P, V, T	n
ideal_gas_temperature	Ideal gas law: temperature	P, V, n	T, T_C
ideal_gas_mass	Mass of a gas sample (ideal gas)	formula, P, V, T	m, n
gas_density	Density of an ideal gas	formula, P, T	rho
molar_mass_from_gas_density	Molar mass from gas density	rho, P, T	M, M_g
molar_mass_from_gas_sample	Molar mass from a gas sample	m, V, P, T	M, M_g
boyle_final_volume	Boyle's law: final volume	P1, V1, P2	V2
boyle_final_pressure	Boyle's law: final pressure	P1, V1, V2	P2
charles_final_volume	Charles's law: final volume	V1, T1, T2	V2
gay_lussac_final_pressure	Gay-Lussac's law: final pressure	P1, T1, T2	P2
combined_gas_final_volume	Combined gas law: final volume	P1, V1, T1, P2, T2	V2
combined_gas_final_pressure	Combined gas law: final pressure	P1, V1, T1, V2, T2	P2
combined_gas_final_temperature	Combined gas law: final temperature	P1, V1, T1, P2, V2	T2
partial_pressure	Partial pressure (Dalton's law)	x, P_total	p

id	operation	inputs	outputs
dalton_mixture	Partial pressures in a gas mixture	moles, P_total	partial_pressures, mole_fractions
graham_rate_ratio	Graham's law: ratio of effusion rates	formula_1, formula_2	ratio
graham_molar_mass	Graham's law: molar mass of an unknown gas	ratio, formula_known	M, M_g
van_der_waals_pressure	Van der Waals equation: pressure	n, V, T, a, b	P, P_ideal, P_atm
rms_speed	Root-mean-square speed of gas molecules	formula, T	u_rms
mean_speed	Mean speed of gas molecules	formula, T	u_mean
molarity_from_mass	Molarity from a dissolved mass	formula, m, V	c, c_molL
solute_mass_for_solution	Mass of solute for a solution	formula, c, V	m, m_g
dilution_final_concentration	Dilution: final concentration	c1, V1, V2	c2, c2_molL
dilution_volume	Dilution: volume of stock needed	c1, c2, V2	V1, V1_mL
molality	Molality of a solution	formula, m_solute, m_solvent	b
mole_fraction	Mole fraction in a solution	formula_solute, m_solute, formula_solvent, m_solvent	x_solute, x_solvent
raoult_vapor_pressure	Vapor pressure of a solution (Raoult's law)	formula_solute, m_solute, formula_solvent, m_solvent, P0	P, dP
freezing_point_depression	Freezing-point depression	i, K, b	dT
boiling_point_elevation	Boiling-point elevation	i, K, b	dT
molar_mass_from_freezing_point	Molar mass from freezing-point depression	m_solute, m_solvent, K, dT, i	M, M_g
osmotic_pressure	Osmotic pressure	i, c, T	Pi, Pi_atm
molar_mass_from_osmotic_pressure	Molar mass from osmotic pressure	m, V, Pi, T, i	M, M_g
kp_from_kc	Kp from Kc	Kc, T, dn	Kp
kc_from_kp	Kc from Kp	Kp, T, dn	Kc
equilibrium_ice	Equilibrium concentrations (ICE table)	equation, initial, K	x, equilibrium
k_from_delta_g	Equilibrium constant from the standard Gibbs	dG, T	K, lnK

id	operation	inputs	outputs
	energy		
delta_g_from_k	Standard Gibbs energy from the equilibrium constant	K, T	dG
vant_hoff_k2	Van 't Hoff equation: K at another temperature	K1, T1, T2, dH	K2, lnK2
delta_g_nonstandard	Gibbs energy change at non-standard conditions	dG0, T, Q	dG
ksp_molar_solubility	Molar solubility from Ksp	Ksp, x, y	s, s_molL
ksp_from_solubility	Ksp from molar solubility	s, x, y	Ksp
common_ion_solubility	Molar solubility with a common ion	Ksp, x, y, c_common, common	s, s_molL
ph_from_hydronium	pH from the hydronium concentration	h	pH, pOH, oh
hydronium_from_ph	Hydronium concentration from pH	pH	h, h_molL, pOH, oh
ph_from_hydroxide	pH from the hydroxide concentration	oh	pH, pOH
ph_strong_acid	pH of a strong monoprotic acid	c	pH, h
ph_strong_base	pH of a strong base	c, n_oh	pH, pOH
ph_weak_acid	pH of a weak acid	Ka, c	pH, h, percent_ionized
ph_weak_base	pH of a weak base	Kb, c	pH, pOH, percent_ionized
ka_from_ph	Ka from the pH of a weak acid	pH, c	pKa, Ka
buffer_ph	Buffer pH (Henderson-Hasselbalch)	Ka, c_base, c_acid	pH
titration_equivalence_volume	Titration: equivalence-point volume	c_t, V_a, c_a, ratio	V_eq, V_eq_mL
titration_ph_strong_acid_strong_base	pH during a strong acid - strong base titration	c_a, V_a, c_b, V_b	pH
titration_ph_weak_acid_strong_base	pH during a weak acid - strong base titration	Ka, c_a, V_a, c_b, V_b	pH
ph_diprotic_acid	pH of a diprotic acid	Ka1, Ka2, c	pH
ph_salt_of_weak_acid	pH of a salt of a weak acid	Ka, c	pH, Kb
zero_order_concentration	Zero-order integrated rate law	A0, k, t	A, A_molL
first_order_concentration	First-order integrated rate law	A0, k, t	A, A_molL, fraction_left
second_order_concentration	Second-order integrated rate law	A0, k, t	A, A_molL
first_order_half_life	Half-life of a first-order reaction	k	t_half

id	operation	inputs	outputs
first_order_time	Time to reach a concentration (first order)	A0, A, k	t
arrhenius_k2	Arrhenius equation: rate constant at another temperature	k1, T1, T2, Ea	k2, ratio
activation_energy	Activation energy from two rate constants	k1, T1, k2, T2	Ea, Ea_kJ
arrhenius_rate_constant	Arrhenius equation: rate constant	A, Ea, T	k
reaction_order_from_rates	Reaction order from initial rates	c1, r1, c2, r2	order
consecutive_first_order	Consecutive first-order reactions (A -> B -> C)	A0, k1, k2, t	B, A, C
calorimetry_heat	Heat from a temperature change	m, c, T1, T2	q
enthalpy_from_calorimetry	Reaction enthalpy from coffee-cup calorimetry	m, c, T1, T2, n	dH, dH_kJ, q_solution
reaction_enthalpy_from_formation	Reaction enthalpy from enthalpies of formation (Hess's law)	equation, values	dH, dH_kJ
gibbs_free_energy	Gibbs energy change (dG = dH - T dS)	dH, dS, T	dG, dG_kJ, spontaneity
crossover_temperature	Temperature at which dG = 0	dH, dS	T, T_C
heat_from_reaction_mass	Heat of reaction for a given mass	formula, m, dH	q, heat_released
clausius_clapeyron_pressure	Clausius-Clapeyron: vapor pressure at another temperature	P1, T1, T2, dH	P2
clausius_clapeyron_enthalpy	Clausius-Clapeyron: enthalpy of vaporization	P1, T1, P2, T2	dH
delta_g_from_cell_potential	Gibbs energy from a cell potential	n, E	dG, dG_kJ
k_from_cell_potential	Equilibrium constant from a standard cell potential	n, E, T	lnK, K
nernst_potential	Nernst equation	E0, n, Q, T	E
electrolysis_mass	Mass deposited in electrolysis (Faraday's law)	metal, I, t, n	m, m_g
electrolysis_time	Time for electrolysis (Faraday's law)	metal, m, I, n	t
photon_energy	Energy of a photon from its wavelength	lam	E, E_eV, E_molar, nu
photon_wavelength	Wavelength of a photon from its energy	E	lam, lam_nm

id	operation	inputs	outputs
beer_lambert_absorbance	Beer-Lambert law: absorbance	eps, l, c	A, T_percent
beer_lambert_concentration	Beer-Lambert law: concentration	A, eps, l	c, c_molL
hydrogen_transition_wavelength	Hydrogen atom transition wavelength (Rydberg)	n_upper, n_lower	lam, lam_nm, E
degree_of_unsaturation	Degree of unsaturation	formula	dou
vibrational_wavenumber	Vibrational wavenumber of a diatomic molecule	molecule, k	wavenumber, wavenumber_cm, nu
average_atomic_mass	Average atomic mass from isotopes	masses, abundances	A_r, M
isotope_abundance	Abundances of two isotopes from the average atomic mass	m1, m2, A_r	x1, x2, percent1, percent2
standard_cell_potential	Standard cell potential from two half-reactions	E_cathode, E_anode	E_cell, spontaneity
bond_enthalpy_reaction	Reaction enthalpy from bond enthalpies	broken_n, broken_D, formed_n, formed_D	dH, dH_kJ, broken_kJ, formed_kJ
henry_law_solubility	Henry's law: solubility of a gas	kH, P	c, c_molL
reaction_quotient_direction	Reaction quotient and the direction of shift	equation, conc, K	Q, log10_Q, direction
precipitation_test	Precipitation test (Q versus Ksp)	Ksp, x, y, c_cation, c_anion	Q, log10_Q, result
reaction_entropy_from_standard_entropies	Standard reaction entropy from standard molar entropies	equation, values	dS
reaction_gibbs_from_formation	Standard reaction Gibbs energy from Gibbs energies of formation	equation, values	dG, dG_kJ, spontaneity
nuclear_binding_energy	Mass defect and nuclear binding energy	Z, A, mass	dm, E_b, E_b_MeV, E_per_nucleon_MeV
radioactive_decay_remaining	Amount remaining after radioactive decay (half-life)	m0, t_half, t	m, m_g, fraction, half_lives
radioactive_dating_age	Age from the fraction remaining (radioactive dating)	fraction, t_half	t, t_years, half_lives
conjugate_dissociation_constant	Conjugate acid-base constants (Ka Kb = Kw)	K, known	K_conj, pK, pK_conj, conjugate
specific_rotation	Specific rotation from an observed rotation	alpha, l, c	specific_rotation
enantiomeric_excess_from_rotation	Enantiomeric excess from optical rotation	alpha_obs, alpha_pure	ee, major_percent, minor_percent, predominant

id	operation	inputs	outputs
enantiomeric_excess_from_composition	Enantiomeric excess from the composition of a mixture	R, S	ee, percent_R, percent_S, predominant

Molar mass of a formula, `molar_mass`

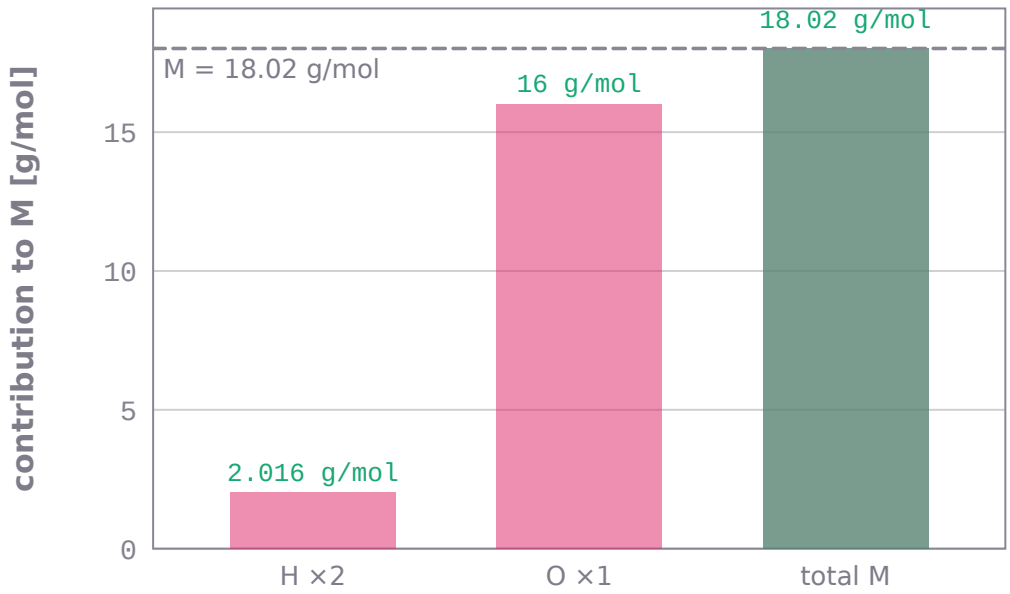
The molar mass of a chemical formula (parentheses, brackets and hydrates are read).

$$M = \sum_i n_i A_r(i)\ \mathrm{g\,mol^{-1}}$$

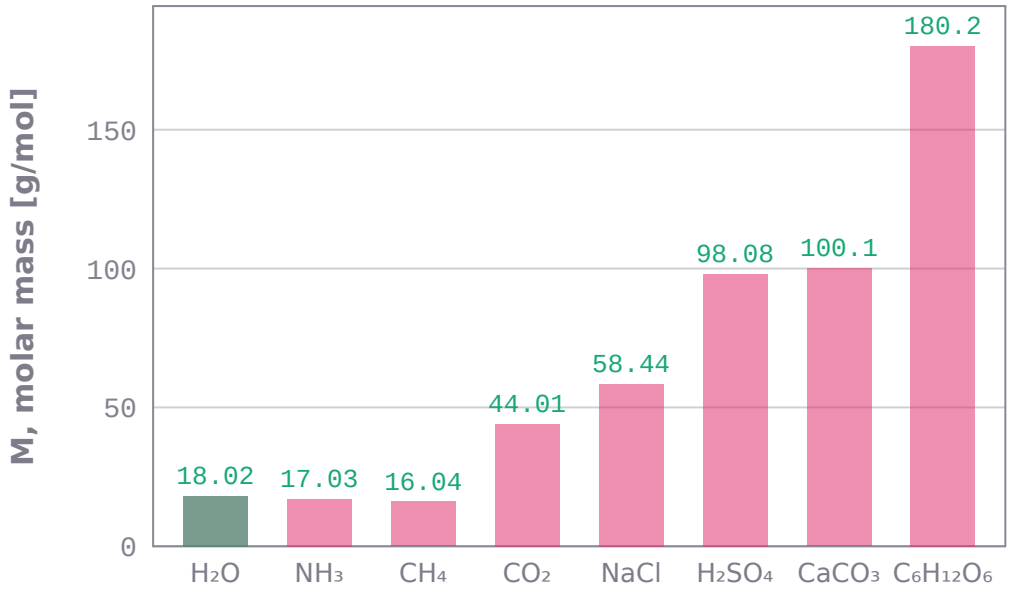
input	meaning	condition
`formula`	chemical formula, e.g. Ca(OH)2 or CuSO4·5H2O	

output	meaning
`M`	molar mass (kg/mol)
`M_g`	molar mass in g/mol

- **Method:** atom counts read from the formula (nested groups multiply, hydrate parts add), times standard atomic weights
- **Assumes:** a neutral formula of element symbols with standard atomic weights
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Where the molar mass of H_2O (the worked example) comes from: each bar is one element's share, its mass percent (returned by the package) times the molar mass M = 18.015 g/mol (returned by the package), the shares add up to M. Marked: M = 18.015 g/mol; H 2.016 g/mol, O 15.999 g/mol.



Molar masses of eight common substances, each returned by the package: H_2O 18.02, NH_3 17.03, CH_4 16.04, CO_2 44.01, NaCl 58.44, H_2SO_4 98.08, $CaCO_3$ 100.1, $C_6H_{12}O_6$ 180.2 g/mol. Highlighted: the worked example H_2O.

```
chemistry_ai999x.solve('molar_mass', formula='H2O')
# -> M = 0.018015 kg/mol, M_g = 18.015   (SOLVED)
```

```
chemistry_ai999x.solve('molar_mass', formula='Ca(OH)2')
# -> M = 0.074092 kg/mol, M_g = 74.092   (SOLVED)
```

```
chemistry_ai999x.solve('molar_mass', formula='CuSO4·5H2O')
# -> M = 0.249684 kg/mol, M_g = 249.684   (SOLVED)
```

```
chemistry_ai999x.solve('molar_mass', formula='K4[Fe(CN)6]')
# -> M = 0.368345 kg/mol, M_g = 368.345   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the molar mass of Ca(OH)2?	SOLVED	M = 0.074092 kg/mol, M_g = 74.092
Find the molar mass of CuSO4·5H2O.	SOLVED	M = 0.249684 kg/mol, M_g = 249.684
Calculate the molecular weight of C6H12O6.	SOLVED	M = 0.180156 kg/mol, M_g = 180.156
What is the molar mass of Xy2O?	NEEDS_INPUT	molar_mass needs formula (chemical formula, e.g. Ca(OH)2 or CuSO4·5H2O). Example: solve('molar_mass', formula='H2O').

Moles from a mass, `moles_from_mass`

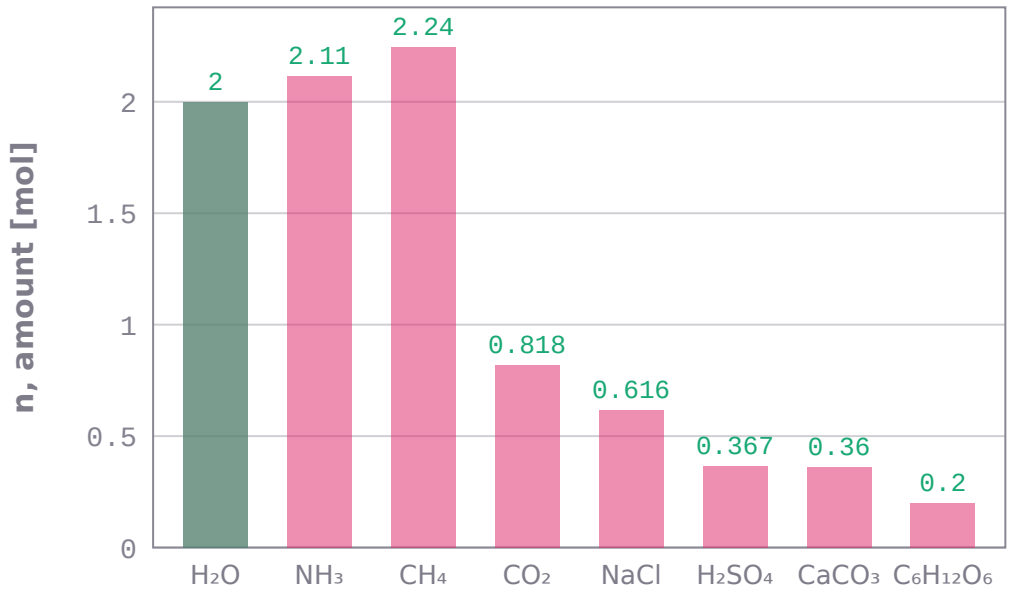
The amount of substance (and number of formula units) in a given mass of a compound.

$$n = \frac{m}{M}$$

input	meaning	condition
`formula`	chemical formula	
`m`	mass of the sample	

output	meaning
`n`	amount of substance
`N`	number of formula units
`M`	molar mass

- **Method:** $n = m/M$ with M from the formula
- **Assumes:** $m > 0$; a neutral formula
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Amount of substance in the same mass, 36 g, of eight substances, each returned by the package: H_2O 2, NH_3 2.11, CH_4 2.24, CO_2 0.818, NaCl 0.616, H_2SO_4 0.367, $CaCO_3$ 0.36, $C_6H_{12}O_6$ 0.2 mol. Highlighted: the worked example H_2O.

```
chemistry_ai999x.solve('moles_from_mass', formula='H2O', m=0.036)
# -> n = 1.99833 mol, N = 1.20343e+24, M = 0.018015 kg/mol   (SOLVED)
```

```
chemistry_ai999x.solve('moles_from_mass', formula='NaCl', m=0.01)
# -> n = 0.171107 mol, N = 1.03043e+23, M = 0.058443 kg/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How many moles are in 36.0 g of H2O?	SOLVED	n = 1.99833 mol, N = 1.20343e+24, M = 0.018015 kg/mol

question	outcome	result
How many moles are there in 10.0 g of NaCl?	SOLVED	n = 0.171107 mol, N = 1.03043e+23, M = 0.058443 kg/mol
How many moles are in 250 mg of CaCO3?	SOLVED	n = 0.00249785 mol, N = 1.50424e+21, M = 0.100086 kg/mol
How many moles are in -5 g of H2O?	INVALID_INPUT	Moles from a mass: m must be a positive number.

Mass from moles, `mass_from_moles`

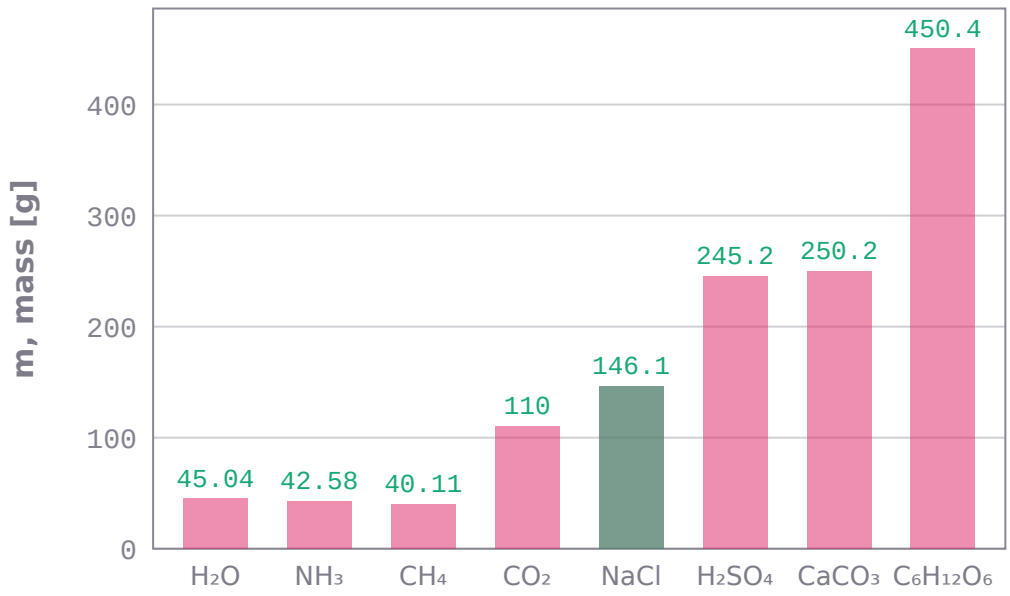
The mass of a given amount (moles) of a compound.

$$m = nM$$

input	meaning	condition
`formula`	chemical formula	
`n`	amount of substance	

output	meaning
`m`	mass
`m_g`	mass in grams
`N`	number of formula units

- **Method:** $m = n$ M with M from the formula
- **Assumes:** $n > 0$; a neutral formula
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Mass of the same amount, 2.5 mol, of eight substances, each returned by the package: H_2O 45.04, NH_3 42.58, CH_4 40.11, CO_2 110, NaCl 146.1, H_2SO_4 245.2, $CaCO_3$ 250.2, $C_6H_{12}O_6$ 450.4 g. Highlighted: the worked example NaCl.

```
chemistry_ai999x.solve('mass_from_moles', formula='NaCl', n=2.5)
# -> m = 0.146108 kg, m_g = 146.107, N = 1.50554e+24   (SOLVED)
```

```
chemistry_ai999x.solve('mass_from_moles', formula='H2SO4', n=0.15)
# -> m = 0.0147119 kg, m_g = 14.7119, N = 9.03321e+22   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the mass of 2.50 mol of NaCl?	SOLVED	m = 0.146108 kg, m_g = 146.107, N = 1.50554e+24
How many grams are in 0.150 mol of H2SO4?	SOLVED	m = 0.0147119 kg, m_g = 14.7119, N = 9.03321e+22
What is the mass of 3 mmol of C6H12O6?	SOLVED	m = 0.000540468 kg, m_g = 0.540468, N = 1.80664e+21

Number of molecules in a mass, `particles_from_mass`

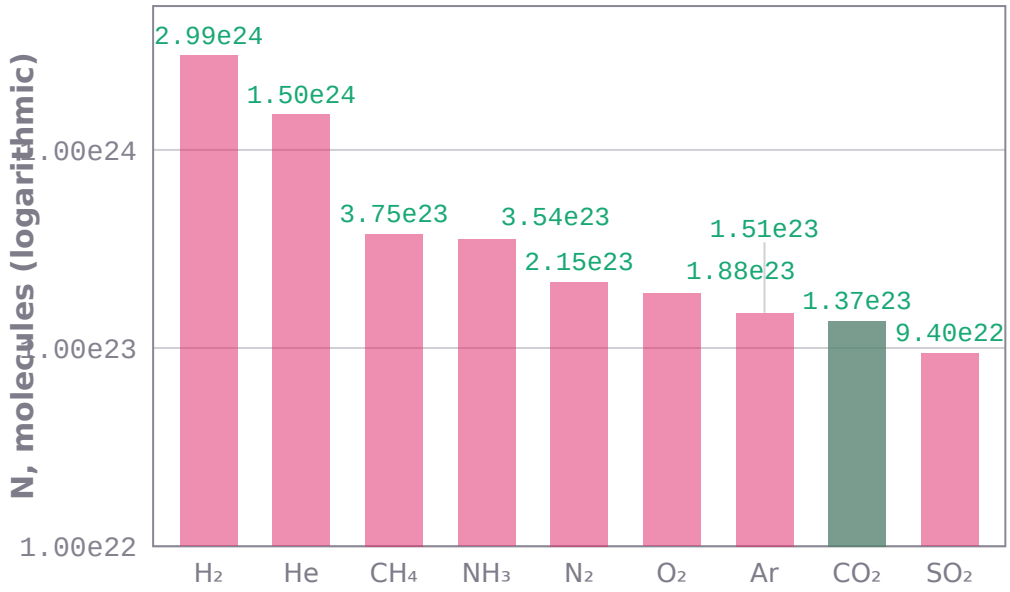
The number of molecules (formula units) in a given mass of a compound.

$$N = \frac{m}{M} N_A$$

input	meaning	condition
`formula`	chemical formula	
`m`	mass of the sample	

output	meaning
`N`	number of molecules / formula units
`n`	amount of substance

- **Method:** $N = (m/M)$ N_A
- **Assumes:** $m > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Number of molecules in 10 g of nine gases, each returned by the package, light molecules are many per gram. Highlighted: the worked example CO_2, 1.3684e+23 molecules.

```
chemistry_ai999x.solve('particles_from_mass', formula='CO2', m=0.01)
# -> N = 1.36839e+23, n = 0.227226 mol   (SOLVED)
```

```
chemistry_ai999x.solve('particles_from_mass', formula='NaCl', m=0.005)
# -> N = 5.15215e+22, n = 0.0855534 mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How many molecules are in 10.0 g of CO2?	SOLVED	N = 1.36839e+23, n = 0.227226 mol
How many formula units are in 5.00 g of NaCl?	SOLVED	N = 5.15215e+22, n = 0.0855534 mol

Number of particles in an amount, `particles_from_moles`

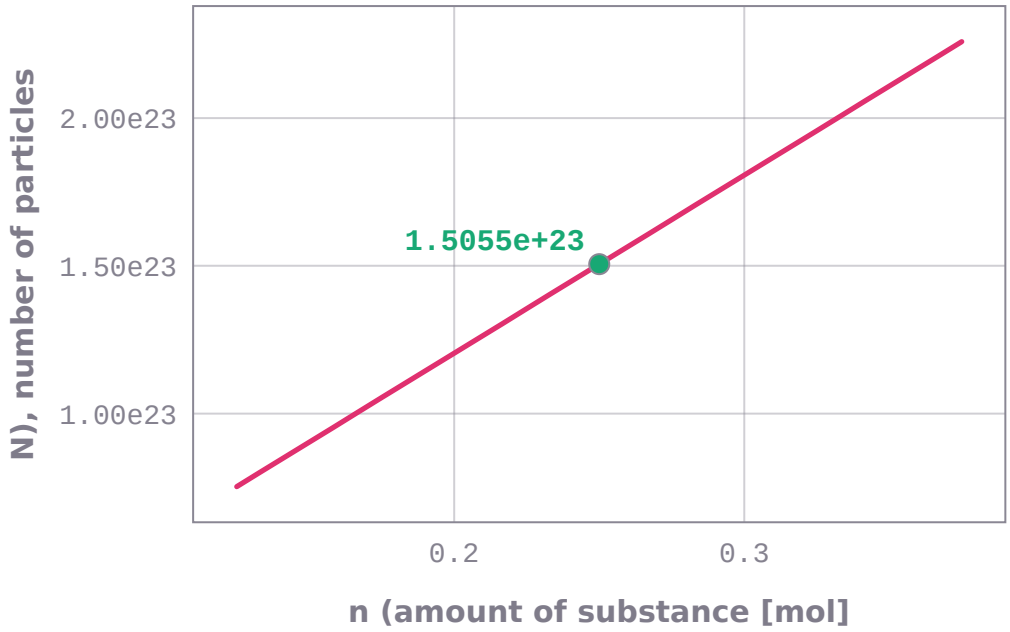
The number of particles in a given amount of substance (Avogadro's number).

$$N = nN_A$$

input	meaning	condition
n	amount of substance	

output	meaning
N	number of particles

- **Method:** $N = n$ N_A
- **Assumes:** $n > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



N (number of particles) against n (amount of substance) from 0.125 mol to 0.375 mol. 31 points computed by chemistry-ai999x; marked: the worked example, n = 0.25 mol, N = 1.5055e+23.

```
chemistry_ai999x.solve('particles_from_moles', n=0.25)
# -> N = 1.50554e+23   (SOLVED)
```

```
chemistry_ai999x.solve('particles_from_moles', n=2.0)
# -> N = 1.20443e+24   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How many molecules are in 0.250 mol of water?	SOLVED	N = 1.50554e+23
How many atoms are in 2.0 mol of helium?	SOLVED	N = 1.20443e+24

Moles from a number of particles, `moles_from_particles`

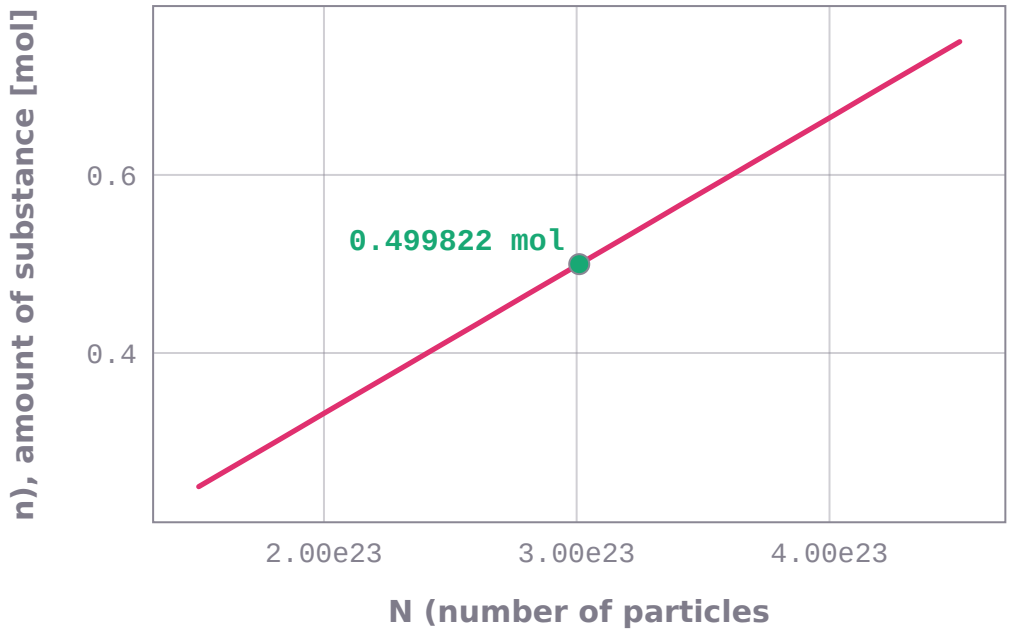
The amount of substance that contains a given number of particles.

$$n = \frac{N}{N_A}$$

input	meaning	condition
`N`	number of particles	

output	meaning
`n`	amount of substance

- **Method:** $n = N/N_A$
- **Assumes:** $N > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



n (amount of substance) against N (number of particles) from 1.5050e+23 to 4.5150e+23. 31 points computed by chemistry-ai999x; marked: the worked example, N = 3.0100e+23, n = 0.499822 mol.

```
chemistry_ai999x.solve('moles_from_particles', N=3.01e+23)
# -> n = 0.499822 mol   (SOLVED)
```

```
chemistry_ai999x.solve('moles_from_particles', N=1.204e+24)
# -> n = 1.99929 mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How many moles are 3.01e23 molecules of CO2?	SOLVED	n = 0.499822 mol
How many moles is 1.204 × 10^24 atoms of iron?	SOLVED	n = 1.99929 mol

Atoms of an element in a mass of compound, `atoms_of_element`

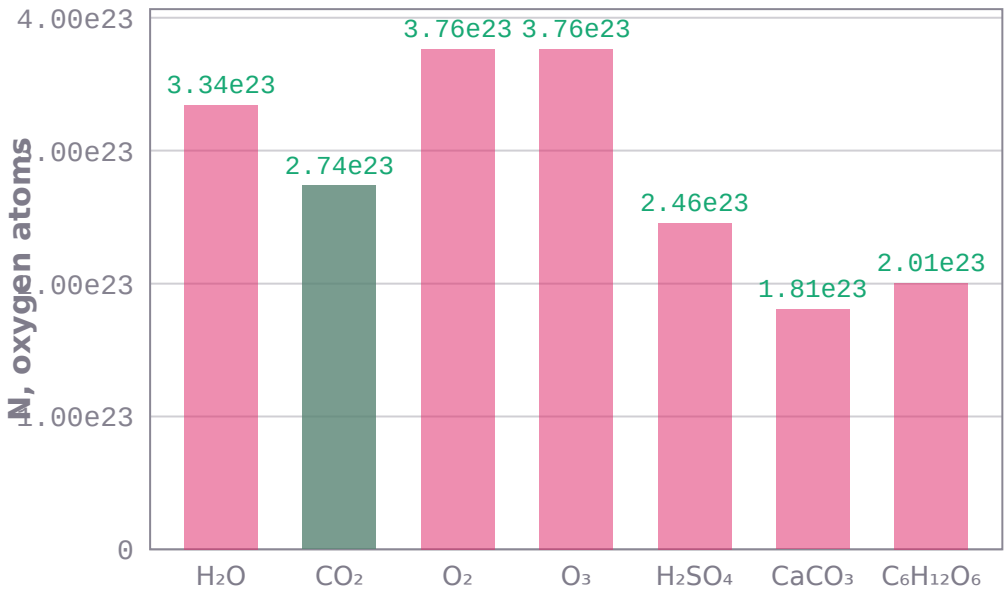
How many atoms of one element a given mass of a compound contains.

$$N_X = \frac{m}{M}\,\nu_X\,N_A$$

input	meaning	condition
`formula`	chemical formula of the compound	
`element`	the element (symbol or name)	
`m`	mass of the compound	

output	meaning
`N_atoms`	number of atoms of the element
`n_element`	amount of the element

- **Method:** moles of compound times the number of X atoms per formula, times N_A
- **Assumes:** $m > 0$; the element occurs in the formula
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Number of oxygen atoms in 10 g of seven substances, each returned by the package: H_2O 3.34e23, CO_2 2.74e23, O_2 3.76e23, O_3 3.76e23, H_2SO_4 2.46e23, $CaCO_3$ 1.81e23, $C_6H_{12}O_6$ 2.01e23. Highlighted: the worked example CO_2.

```
chemistry_ai999x.solve('atoms_of_element', formula='CO2', element='O', m=0.01)
# -> N_atoms = 2.73678e+23, n_element = 0.454452 mol   (SOLVED)
```

```
chemistry_ai999x.solve('atoms_of_element', formula='CH4', element='H', m=0.005)
# -> N_atoms = 7.5075e+23, n_element = 1.24665 mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How many atoms of oxygen are in 10.0 g of CO2?	SOLVED	N_atoms = 2.73678e+23, n_element = 0.454452 mol
How many atoms of H are in 5.00 g of CH4?	SOLVED	N_atoms = 7.5075e+23, n_element = 1.24665 mol
How many atoms of N are in 10.0 g of CO2?	INVALID_INPUT	Atoms of an element in a mass of compound: N does not occur in CO2.

Mass percent of an element, `mass_percent`

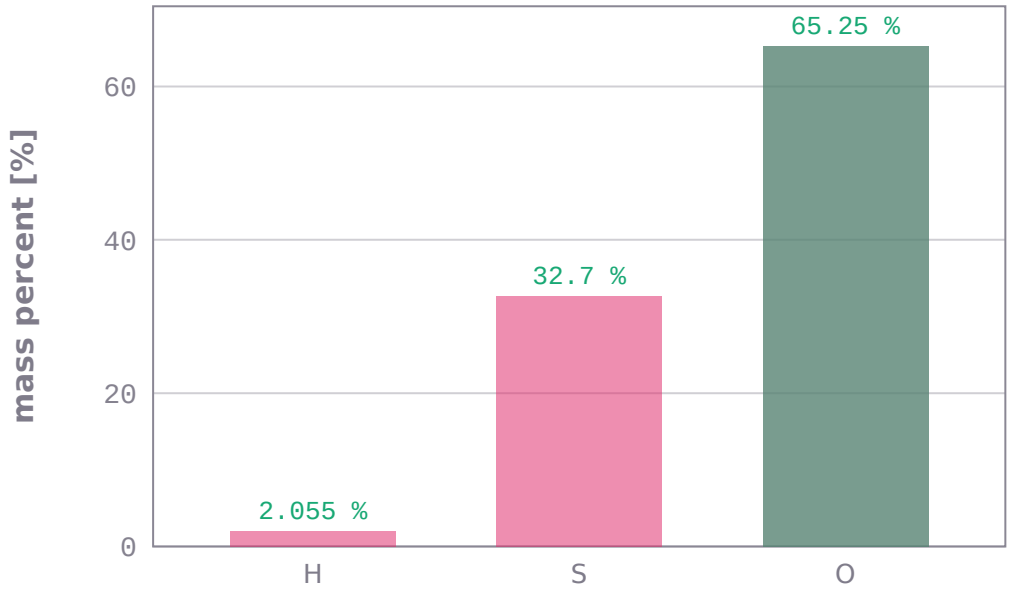
The percentage by mass of an element in a compound (and the full percent composition).

$$w_X = \frac{\nu_X A_r(X)}{M} \times 100\%$$

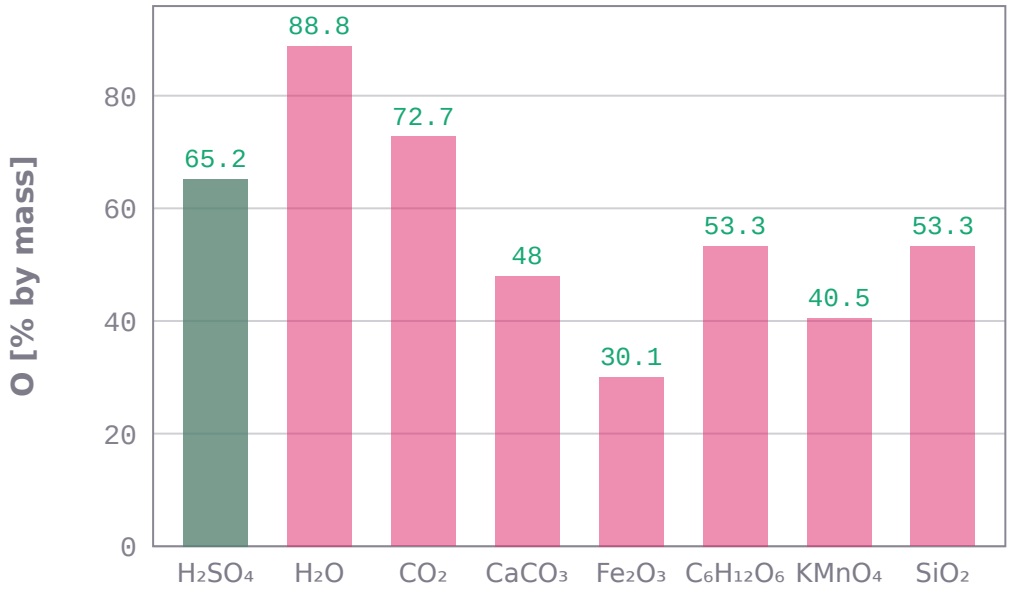
input	meaning	condition
`formula`	chemical formula	
`element`	the element (symbol or name)	

output	meaning
`percent`	mass percent of the element (%)
`composition`	mass percent of every element (%)

- **Method:** mass of the element per formula over the molar mass
- **Assumes:** the element occurs in the formula
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Mass composition of H_2SO_4: the percent by mass of every element, returned by the package (H 2.055 %, S 32.7 %, O 65.25 %). Highlighted: the element asked in the worked example, O.



Percent O by mass in 8 compounds containing it, each returned by the package: H_2SO_4 65.25 %, H_2O 88.81 %, CO_2 72.71 %, $CaCO_3$ 47.96 %, Fe_2O_3 30.06 %, $C_6H_{12}O_6$ 53.28 %, $KMnO_4$ 40.5 %, SiO_2 53.26 %. Highlighted: the worked example, H_2SO_4.

```
chemistry_ai999x.solve('mass_percent', formula='H2SO4', element='O')
# -> percent = 65.2494, composition = {'H': 2.055485883828342, 'S': 32.69507233964457, 'O':
65.24944177652708}   (SOLVED)
```

```
chemistry_ai999x.solve('mass_percent', formula='NH4NO3', element='N')
# -> percent = 34.9987, composition = {'N': 34.99868820508976, 'H': 5.0372924553052725,
'O': 59.96401933960495}   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the mass percent of O in H2SO4?	SOLVED	percent = 65.2494, composition = {'H': 2.055485883828342, 'S': 32.69507233964457, 'O': 65.24944177652708}
Find the percent by mass of nitrogen in NH4NO3.	SOLVED	percent = 34.9987, composition = {'N': 34.99868820508976, 'H': 5.0372924553052725, 'O': 59.96401933960495}
What is the mass percent of Fe in Fe2O3?	SOLVED	percent = 69.9431, composition = {'Fe': 69.94307614270416, 'O': 30.056923857295832}
What is the mass percent of Cl in H2SO4?	INVALID_INPUT	Mass percent of an element: Cl does not occur in H2SO4.

Mass of an element in a sample, `mass_of_element`

The mass of one element contained in a given mass of a compound.

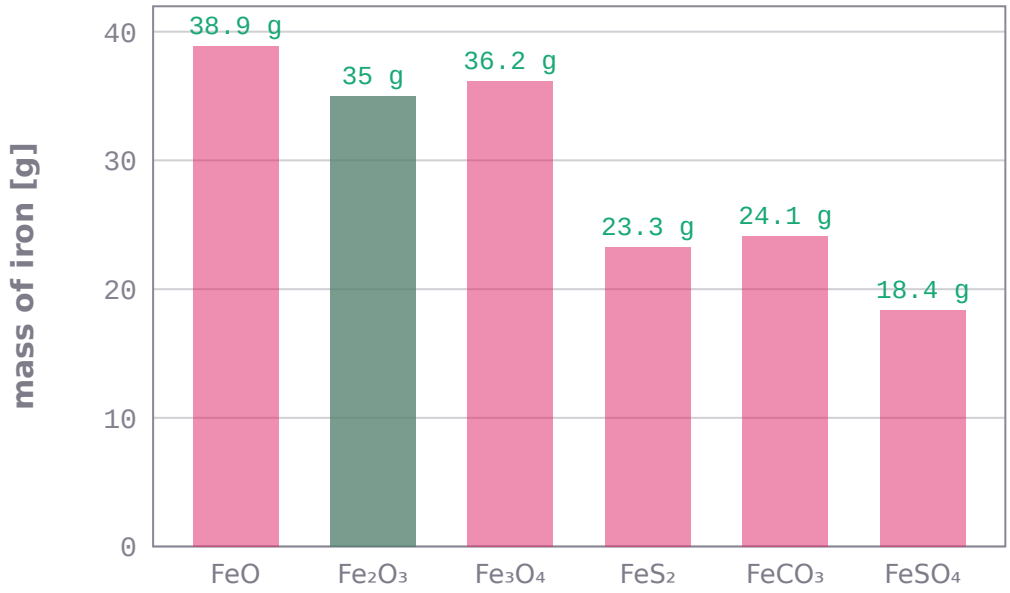
$$m_X = m\,\frac{\nu_X A_r(X)}{M}$$

input	meaning	condition
`formula`	chemical formula	

input	meaning	condition
`element`	the element (symbol or name)	
`m`	mass of the compound	

output	meaning
`m_element`	mass of the element
`fraction`	mass fraction

- **Method:** the sample mass times the element's mass fraction
- **Assumes:** $m > 0$; the element occurs in the formula
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Iron contained in 50 g of six iron compounds (ores and salts), each mass returned by the package with its mass fraction: FeO 0.777, Fe_2O_3 0.699, Fe_3O_4 0.724, FeS_2 0.465, $FeCO_3$ 0.482, $FeSO_4$ 0.368. Highlighted: the worked example, Fe_2O_3, 34.9715 g.

```
chemistry_ai999x.solve('mass_of_element', formula='Fe2O3', element='Fe', m=0.05)
# -> m_element = 0.0349715 kg, fraction = 0.699431   (SOLVED)
```

```
chemistry_ai999x.solve('mass_of_element', formula='NH4NO3', element='N', m=10.0)
# -> m_element = 3.49987 kg, fraction = 0.349987   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What mass of iron is in 50.0 g of Fe2O3?	SOLVED	m_element = 0.0349715 kg, fraction = 0.699431
What mass of N is contained in 10.0 kg of NH4NO3?	SOLVED	m_element = 3.49987 kg, fraction = 0.349987

Empirical formula from percent composition, `empirical_formula`

The simplest whole-number formula of a compound from its mass percentages.

$$n_i = \frac{w_i}{A_r(i)}, \quad \text{subscripts} \propto \frac{n_i}{\min_j n_j}$$

input	meaning	condition
`composition`	element percentages (or masses), e.g. 'C 40.0, H 6.71, O 53.3'	

output	meaning
`formula`	empirical formula
`counts`	subscripts in the order given

- **Method:** moles per 100 g, divided by the smallest, multiplied by the smallest k (1..12) giving whole numbers within 1.5%
- **Assumes:** at least two elements; a whole-number ratio with subscripts up to 12 times the smallest
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Top: the stated composition, C 40 %, H 6.71 %, O 53.3 % by mass. Bottom: the whole-number atom ratio the package returns, C 1, H 2, O 1, the empirical formula CH_2O. Hydrogen is a small mass share but two atoms.

```
chemistry_ai999x.solve('empirical_formula', composition='C 40.0, H 6.71, O 53.3')
# -> formula = CH2O, counts = [1.0, 2.0, 1.0]   (SOLVED)
```

```
chemistry_ai999x.solve('empirical_formula', composition='C 85.6, H 14.4')
# -> formula = CH2, counts = [1.0, 2.0]   (SOLVED)
```

```
chemistry_ai999x.solve('empirical_formula', composition='Fe 69.94, O 30.06')
# -> formula = Fe2O3, counts = [2.0, 3.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the empirical formula of a compound with 40.0% C, 6.71% H and 53.3% O.	SOLVED	formula = CH2O, counts = [1.0, 2.0, 1.0]
What is the empirical formula of a compound that is 85.6% C and 14.4% H?	SOLVED	formula = CH2, counts = [1.0, 2.0]
Find the empirical formula of a compound containing 69.94% Fe and 30.06% O.	SOLVED	formula = Fe2O3, counts = [2.0, 3.0]

Molecular formula from the empirical formula, `molecular_formula`

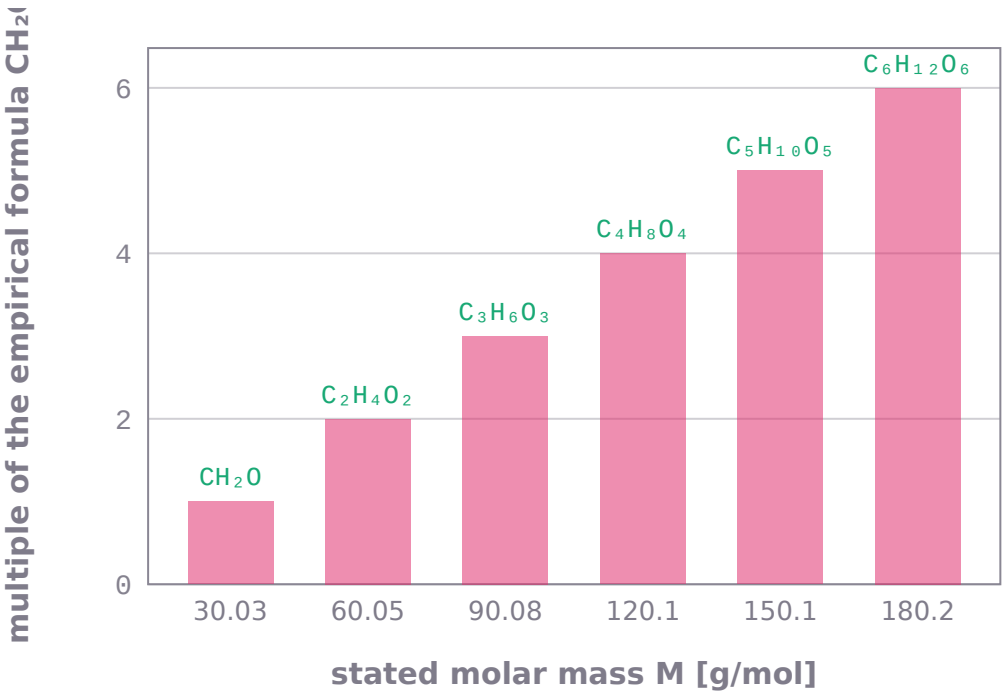
The molecular formula from an empirical formula and the molar mass.

$$n = \frac{M}{M_{\text{empirical}}}$$

input	meaning	condition
`empirical`	empirical formula	
`M`	molar mass of the compound	

output	meaning
`multiple`	whole-number multiple n
`formula`	molecular formula

- **Method:** $n = M/M$(empirical), rounded (must be within 2% of a whole number)
- **Assumes:** $M > 0$; M is a whole multiple of the empirical formula mass (within 2%)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



From the empirical formula CH_2O to the molecular formula: for six stated molar masses the package returns the multiple and the formula, 30.03 g/mol → CH_2O, 60.05 g/mol → $C_2H_4O_2$, 90.08 g/mol → $C_3H_6O_3$, 120.1 g/mol → $C_4H_8O_4$, 150.1 g/mol → $C_5H_{10}O_5$, 180.2 g/mol → $C_6H_{12}O_6$. The worked example (M = 180.16 g/mol) is $C_6H_{12}O_6$, glucose.

```
chemistry_ai999x.solve('molecular_formula', empirical='CH2O', M=0.18016)
# -> multiple = 6, formula = C6H12O6   (SOLVED)
```

```
chemistry_ai999x.solve('molecular_formula', empirical='CH', M=0.07811)
# -> multiple = 6, formula = C6H6   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A compound has the empirical formula CH2O and a molar mass of 180.16 g/mol. What is its molecular formula?	SOLVED	multiple = 6, formula = C6H12O6
The empirical formula is CH and the molar mass is 78.11 g/mol. Find the molecular formula.	SOLVED	multiple = 6, formula = C6H6
The empirical formula is CH2O and the molar mass is 100 g/mol. Find the molecular formula.	INVALID_INPUT	Molecular formula from the empirical formula: the molar mass is 3.330 times the empirical formula mass, not a whole mult

Balancing a chemical equation, `balance_equation`

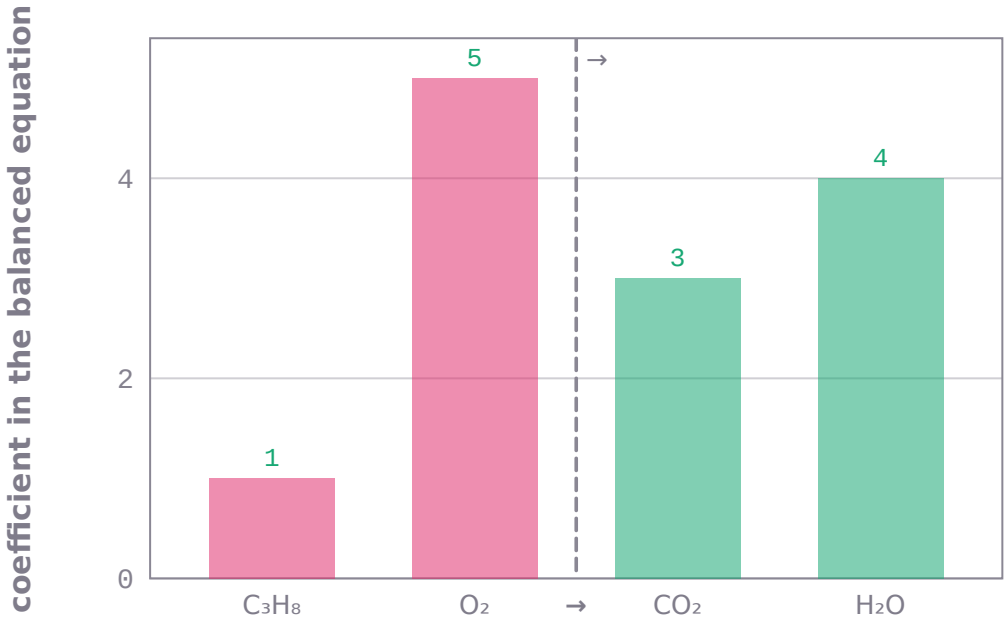
The smallest whole-number coefficients that balance a chemical equation.

$$A\boldsymbol{\nu} = \mathbf{0},\ \boldsymbol{\nu} \in \mathbb{Z}^{n}_{>0}$$

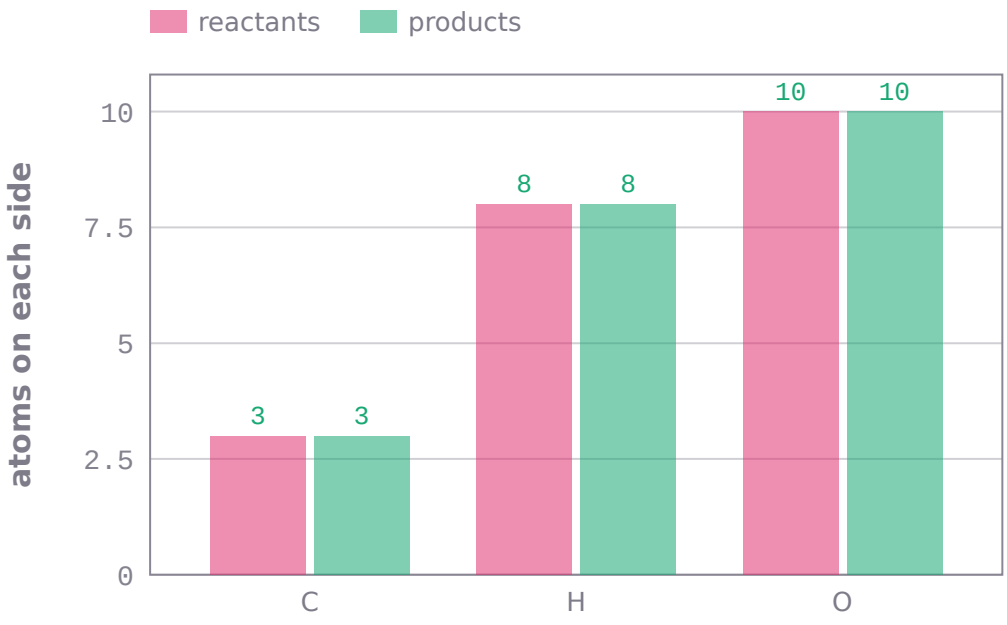
input	meaning	condition
`equation`	the equation, e.g. '$C3H8 + O_2$ -> $CO2 + H2O$'	

output	meaning
`balanced`	the balanced equation
`coefficients`	coefficients in the order the species are written

- **Method:** exact rational null space of the element-conservation matrix, scaled to the smallest whole numbers
- **Assumes:** one independent way to balance (a one-dimensional null space); neutral species
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The balanced equation C_3H_8 + 5 O_2 -> 3 CO_2 + 4 H_2O: the coefficient of each reactant (left) and product (right), as returned by the package for the stated skeleton C3H8 + O2 -> CO2 + H2O.



Atom balance of C_3H_8 + 5 O_2 -> 3 CO_2 + 4 H_2O: for each element, the atoms on the reactant and on the product side (each the returned coefficient times the subscript read from the formula (derived)), C 3 = 3; H 8 = 8; O 10 = 10.

```
chemistry_ai999x.solve('balance_equation', equation='C3H8 + O2 -> CO2 + H2O')
# -> coefficients = [1.0, 5.0, 3.0, 4.0], balanced = C3H8 + 5 O2 -> 3 CO2 + 4 H2O
(SOLVED)
```

```
chemistry_ai999x.solve('balance_equation', equation='Fe + O2 -> Fe2O3')
# -> coefficients = [4.0, 3.0, 2.0], balanced = 4 Fe + 3 O2 -> 2 Fe2O3   (SOLVED)
```

```
chemistry_ai999x.solve('balance_equation', equation='Cu + HNO3 -> Cu(NO3)2 + NO + H2O')
# -> coefficients = [3.0, 8.0, 3.0, 2.0, 4.0], balanced = 3 Cu + 8 HNO3 -> 3 Cu(NO3)2 + 2
NO + 4 H2O   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Balance the equation C3H8 + O2 -> CO2 + H2O.	SOLVED	coefficients = [1.0, 5.0, 3.0, 4.0], balanced = C3H8 + 5 O2 -> 3 CO2 + 4 H2O
Balance: Fe + O2 -> Fe2O3	SOLVED	coefficients = [4.0, 3.0, 2.0], balanced = 4 Fe + 3 O2 -> 2 Fe2O3
Balance KMnO4 + HCl → KCl + MnCl2 + H2O + Cl2.	SOLVED	coefficients = [2.0, 16.0, 2.0, 2.0, 8.0, 5.0], balanced = 2 KMnO4 + 16 HCl -> 2 KCl + 2 MnCl2 + 8 H2O + 5 Cl2
Balance the equation H2 + O2 -> NaCl.	INVALID_INPUT	Balancing a chemical equation: the equation cannot be balanced uniquely (it has no solution or several independent ones)

Limiting reagent and theoretical yield, `limiting_reagent`

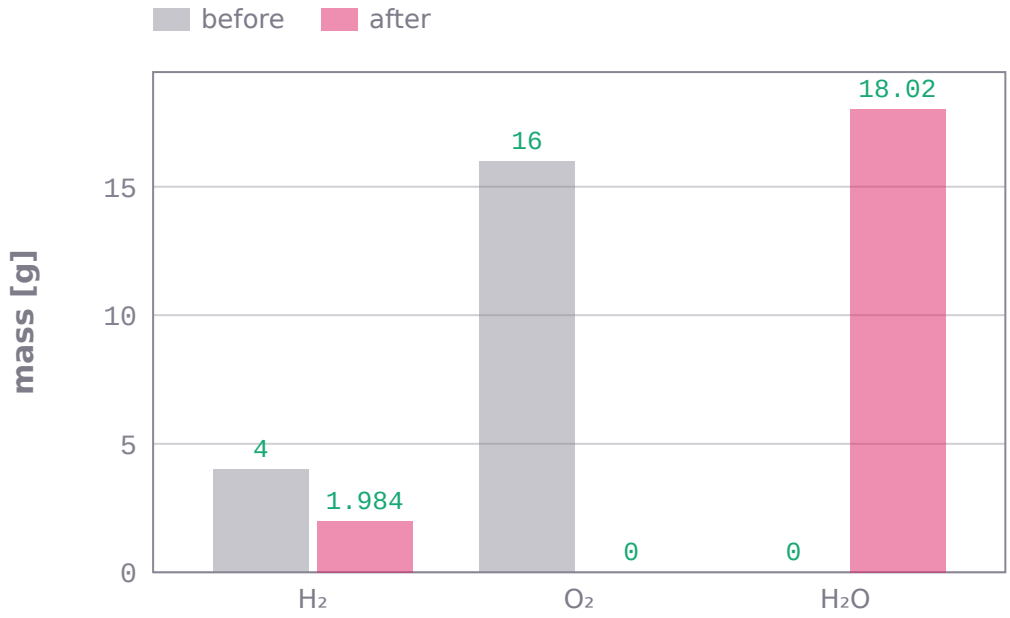
Which of two reactants limits a reaction, the mass of a product formed and the excess left over.

$$\xi = \min_i \frac{n_i}{\nu_i}, \quad m_P = \xi \nu_P M_P$$

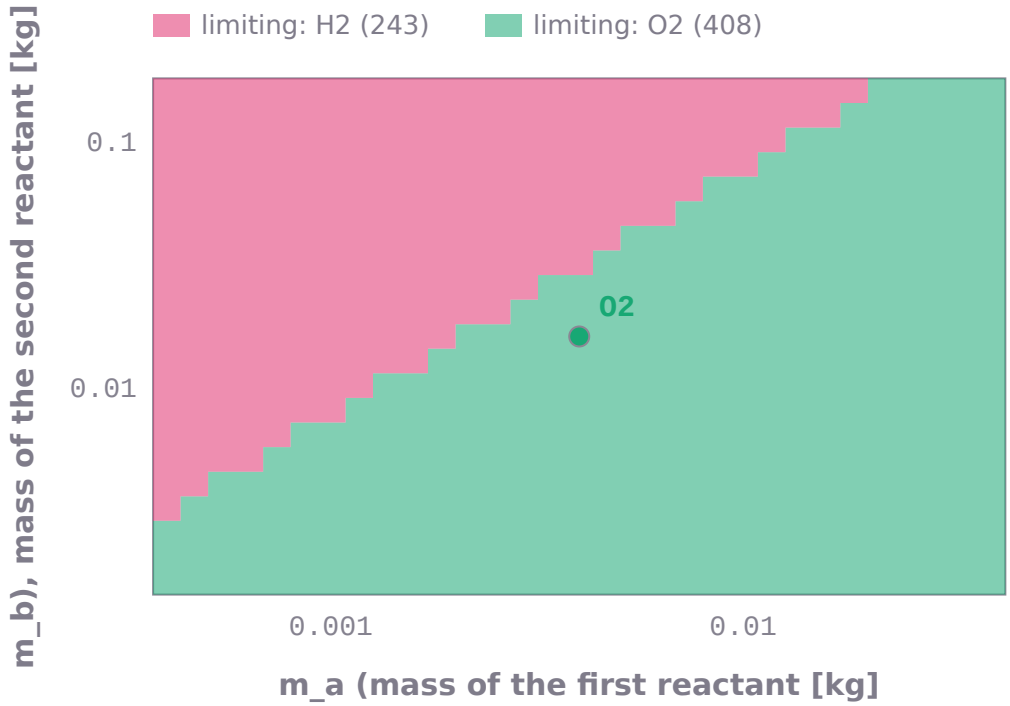
input	meaning	condition
`equation`	the reaction (balanced or not)	
`reactant_a`	first reactant	
`m_a`	mass of the first reactant	
`reactant_b`	second reactant	
`m_b`	mass of the second reactant	
`product`	the product asked for	

output	meaning
`product_mass`	mass of product formed
`limiting`	limiting reagent
`excess_left`	mass of the excess reagent left over

- **Method:** balance the equation, moles of each reactant over its coefficient; the smallest is the extent
- **Assumes:** the equation balances uniquely; two different reactants and one product of the equation
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Masses before and after the reaction H2 + O2 -> H2O with 4 g of H_2 and 16 g of O_2 (the worked example). The package returns the limiting reactant (O_2, used up), the excess left (1.98387 g) and the product formed (18.0161 g of H_2O).



limiting reagent (limiting) over m_a (mass of the first reactant) from 0.0004 kg to 0.04 kg and m_b (mass of the second reactant) from 0.0016 kg to 0.16 kg (logarithmic axes): each of the 31 × 21 cells is one run of chemistry-ai999x, coloured by the verdict it returned, H2: 243 cells, O2: 408 cells. Other inputs as in the worked example: equation = H2 + O2 -> H2O, reactant_a = H2, reactant_b = O2, product = H2O. Marked: the worked example, m_a = 0.004 kg, m_b = 0.016 kg → O2.

```
chemistry_ai999x.solve('limiting_reagent', equation='H2 + O2 -> H2O', reactant_a='H2',
m_a=0.004, reactant_b='O2', m_b=0.016, product='H2O')
# -> product_mass = 0.0180161 kg, limiting = O2, excess_left = 0.00198387 kg   (SOLVED)
```

```
chemistry_ai999x.solve('limiting_reagent', equation='N2 + H2 -> NH3', reactant_a='N2',
m_a=0.028, reactant_b='H2', m_b=0.01, product='NH3')
# -> product_mass = 0.034045 kg, limiting = N2, excess_left = 0.00395502 kg   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For H2 + O2 -> H2O, 4.0 g of H2 react with 16.0 g of O2. Which is the limiting reagent and what mass of H2O forms?	SOLVED	product_mass = 0.0180161 kg, limiting = O2, excess_left = 0.00198387 kg
In N2 + 3H2 -> 2NH3, 28.0 g of N2 and 10.0 g of H2 are mixed. Find the limiting reactant and the mass of NH3 produced.	SOLVED	product_mass = 0.034045 kg, limiting = N2, excess_left = 0.00395502 kg
For H2 + O2 -> H2O, 4.0 g of H2 react with oxygen. Which is the limiting reagent and what mass of H2O forms?	NEEDS_INPUT	limiting_reagent needs reactant_b, m_b (second reactant; mass of the second reactant). Example: solve('limiting_reagent'

Theoretical yield from one reactant, `theoretical_yield`

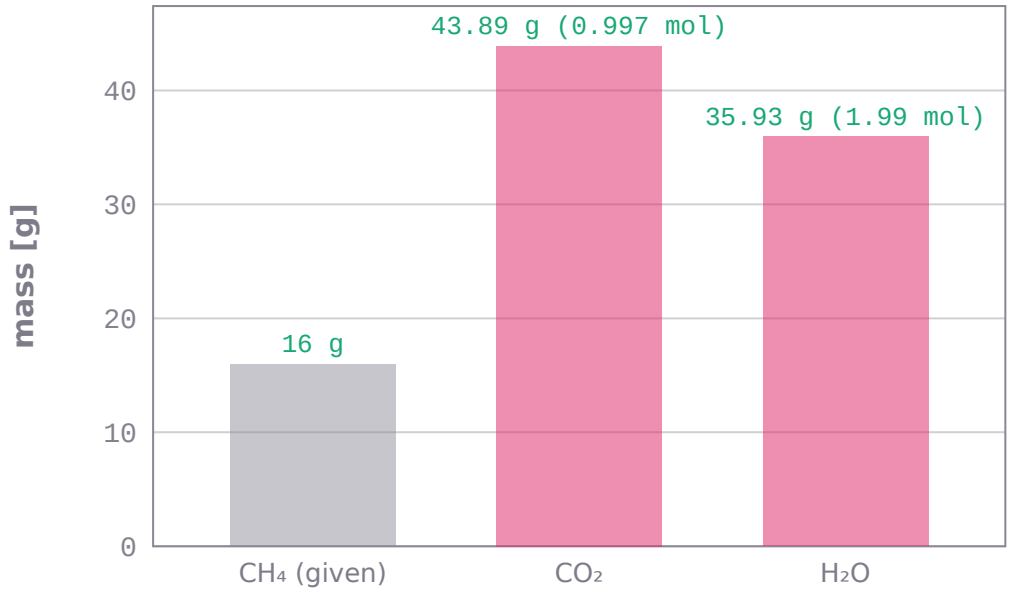
The mass of a product formed from a given mass of one reactant (in excess of the others).

$$m_P = \frac{m_R}{M_R}\,\frac{\nu_P}{\nu_R}\,M_P$$

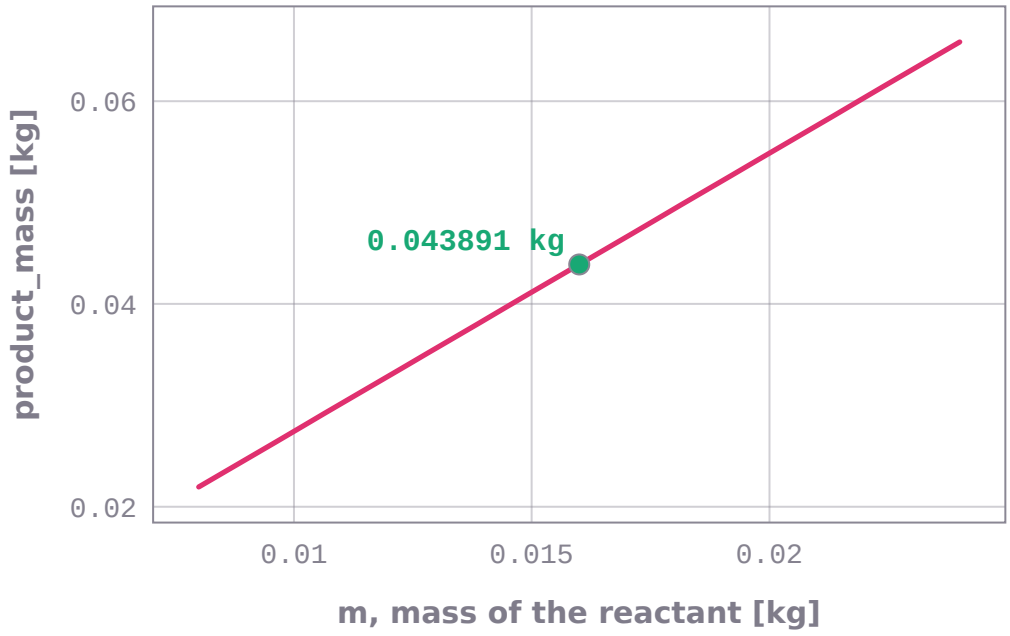
input	meaning	condition
`equation`	the reaction (balanced or not)	
`reactant`	the reactant whose mass is given	
`m`	mass of the reactant	
`product`	the product asked for	

output	meaning
`product_mass`	theoretical mass of product
`n_product`	moles of product

- **Method:** balance the equation; moles of reactant times the coefficient ratio times the product's molar mass
- **Assumes:** the equation balances uniquely; the other reactants are in excess
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Theoretical yield of every product of CH4 + O2 -> CO2 + H2O from 16 g of CH_4 (the other reactant in excess), each returned by the package: CO_2 43.89 g, H_2O 35.93 g.



product_mass (theoretical mass of product) against m (mass of the reactant) from 0.008 kg to 0.024 kg. Other inputs as in the worked example: equation = CH4 + O2 -> CO2 + H2O, reactant = CH4, product = CO2. 31 points computed by chemistry-ai999x; marked: the worked example, m = 0.016 kg, product_mass = 0.043891 kg.

```
chemistry_ai999x.solve('theoretical_yield', equation='CH4 + O2 -> CO2 + H2O',
reactant='CH4', m=0.016, product='CO2')
# -> product_mass = 0.043891 kg, n_product = 0.99732 mol   (SOLVED)
```

```
chemistry_ai999x.solve('theoretical_yield', equation='Fe2O3 + CO -> Fe + CO2',
reactant='Fe2O3', m=0.1, product='Fe')
# -> product_mass = 0.0699431 kg, n_product = 1.25245 mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For CH4 + O2 -> CO2 + H2O, what mass of CO2 is produced from 16.0 g of CH4?	SOLVED	product_mass = 0.043891 kg, n_product = 0.99732 mol
What is the theoretical yield of Fe from 100 g of Fe2O3 in Fe2O3 + CO -> Fe + CO2?	SOLVED	product_mass = 0.0699431 kg, n_product = 1.25245 mol
For CH4 + O2 -> CO2 + H2O, what mass of CO2 is produced from methane?	NEEDS_INPUT	theoretical_yield needs reactant, m (the reactant whose mass is given; mass of the reactant). Example: solve('theoretica

Percent yield of a reaction, `percent_yield`

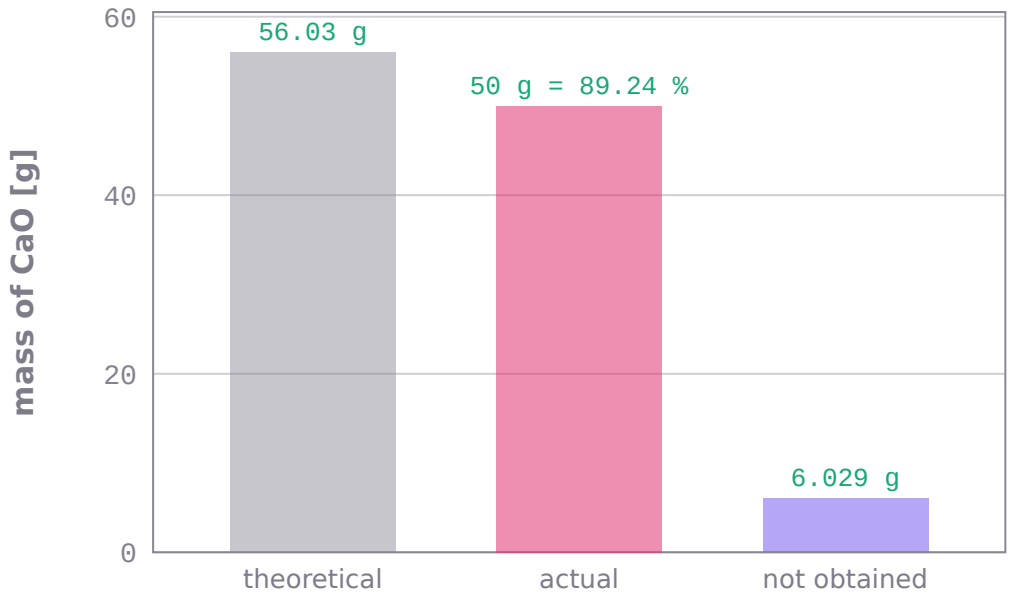
The percent yield from the mass of reactant used and the mass of product obtained.

$$\%\,\text{yield} = \frac{m_{\text{actual}}}{m_{\text{theoretical}}} \times 100$$

input	meaning	condition
`equation`	the reaction (balanced or not)	
`reactant`	the reactant whose mass is given	
`m`	mass of the reactant	
`product`	the product obtained	
`actual`	actual mass of product obtained	

output	meaning
`percent_yield`	percent yield (%)
`theoretical_mass`	theoretical mass of product

- **Method:** theoretical yield from the balanced equation, then actual / theoretical x 100
- **Assumes:** the equation balances uniquely; the other reactants are in excess
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Percent yield of CaO from 100 g of $CaCO_3$ by CaCO3 -> CaO + CO2: the package's theoretical mass (56.0288 g) against the stated actual mass (50 g); right: the difference not obtained (6.02882 g, derived). Marked: percent yield 89.2398 %.

```
chemistry_ai999x.solve('percent_yield', equation='CaCO3 -> CaO + CO2', reactant='CaCO3',
m=0.1, product='CaO', actual=0.05)
# -> percent_yield = 89.2398, theoretical_mass = 0.0560288 kg   (SOLVED)
```

```
chemistry_ai999x.solve('percent_yield', equation='N2 + H2 -> NH3', reactant='N2', m=0.028,
product='NH3', actual=0.0255)
# -> percent_yield = 74.9009, theoretical_mass = 0.034045 kg   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For CaCO3 -> CaO + CO2, 100.0 g of CaCO3 gave 50.0 g of CaO. What is the percent yield?	SOLVED	percent_yield = 89.2398, theoretical_mass = 0.0560288 kg
In N2 + H2 -> NH3, 28.0 g of N2 produced 25.5 g of NH3. Calculate the percentage yield.	SOLVED	percent_yield = 74.9009, theoretical_mass = 0.034045 kg

Ideal gas law: pressure, `ideal_gas_pressure`

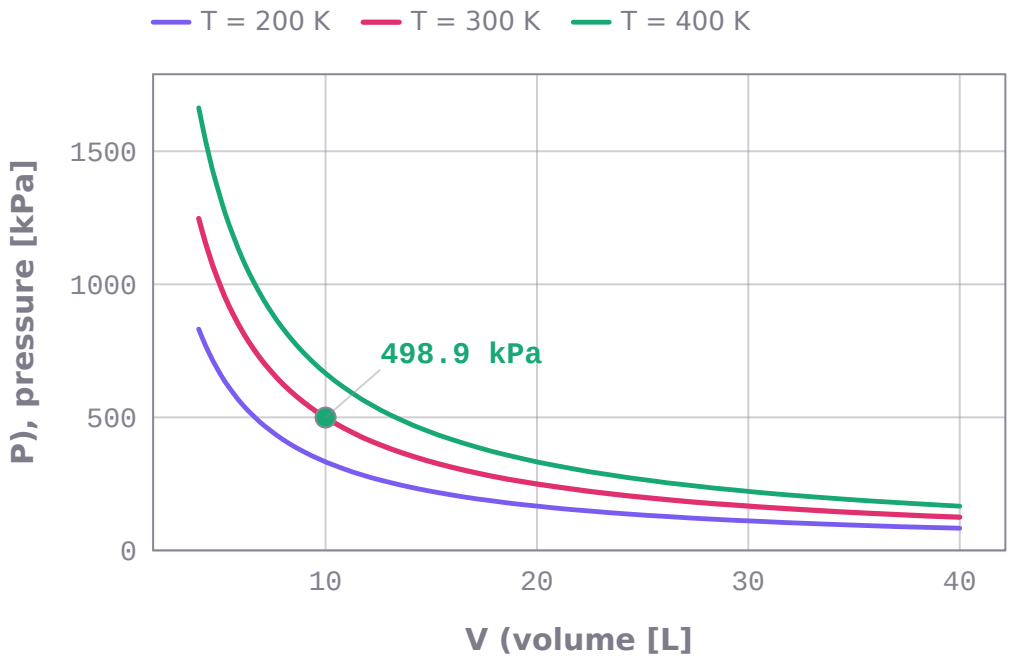
The pressure of an ideal gas from its amount, volume and temperature (PV = nRT).

$$P = \frac{nRT}{V}$$

input	meaning	condition
`n`	amount of gas	
`V`	volume	
`T`	temperature	

output	meaning
`P`	pressure
`P_atm`	pressure in atm

- **Method:** P = nRT / V
- **Assumes:** n, V, $T > 0$ (T in kelvin)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Isotherms of 2 mol of ideal gas: pressure against volume at T = 200, 300 and 400 K, each computed by the package at 61 volumes from 4 L to 40 L. Marked: the worked example, V = 10 L at 300 K, P = 498.868 kPa.

```
chemistry_ai999x.solve('ideal_gas_pressure', n=2.0, V=0.01, T=300.0)
# -> P = 498868 Pa, P_atm = 4.92344   (SOLVED)
```

```
chemistry_ai999x.solve('ideal_gas_pressure', n=0.5, V=0.005, T=298.15)
# -> P = 247896 Pa, P_atm = 2.44654   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pressure of 2.0 mol of gas in 10.0 L at 300 K?	SOLVED	P = 498868 Pa, P_atm = 4.92344
Find the pressure exerted by 0.50 mol of an ideal gas in a 5.0 L container at 25 °C.	SOLVED	P = 247896 Pa, P_atm = 2.44654
What is the pressure of 2.0 mol of gas in 10.0 L?	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemistry_ai999x.templates(); eve

Ideal gas law: volume, `ideal_gas_volume`

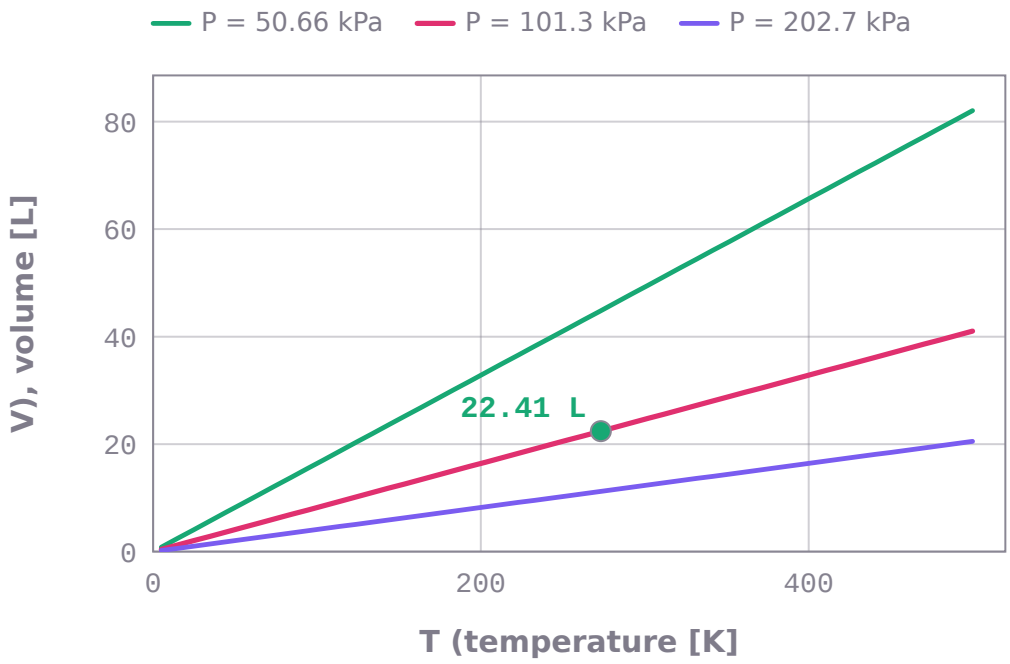
The volume of an ideal gas from its amount, pressure and temperature.

$$V = \frac{nRT}{P}$$

input	meaning	condition
`n`	amount of gas	
`P`	pressure	
`T`	temperature	

output	meaning
`V`	volume
`V_L`	volume in litres

- **Method:** V = nRT / P
- **Assumes:** n, P, $T > 0$ (T in kelvin)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Volume of 1 mol of ideal gas against temperature (isobars) at 50.66, 101.325 and 202.65 kPa, computed by the package at 51 temperatures from 5 K to 500 K: straight lines through 0 K. Marked: the worked example, 273.15 K and 101.325 kPa, V = 22.414 L (the molar volume at 0 °C and 1 atm).

```
chemistry_ai999x.solve('ideal_gas_volume', n=1.0, P=101325.0, T=273.15)
# -> V = 0.022414 m^3, V_L = 22.414   (SOLVED)
```

```
chemistry_ai999x.solve('ideal_gas_volume', n=0.25, P=202650.0, T=350.0)
# -> V = 0.00359001 m^3, V_L = 3.59001   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What volume does 1.00 mol of gas occupy at 1.00 atm and 273.15 K?	SOLVED	V = 0.022414 m^3, V_L = 22.414
Calculate the volume of 0.250 mol of an ideal gas at 2.00 atm and 350 K.	SOLVED	V = 0.00359001 m^3, V_L = 3.59001

Ideal gas law: amount, `ideal_gas_moles`

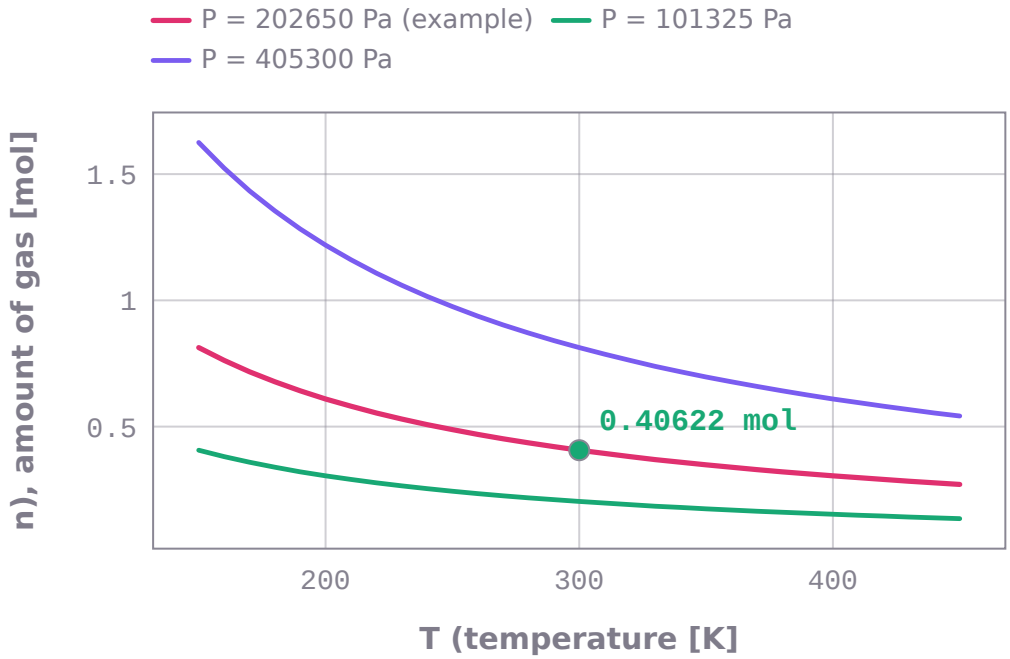
The amount (moles) of an ideal gas from its pressure, volume and temperature.

$$n = \frac{PV}{RT}$$

input	meaning	condition
P	pressure	
V	volume	
T	temperature	

output	meaning
n	amount of gas

- **Method:** $n = PV/RT$
- **Assumes:** P, V, $T > 0$ (T in kelvin)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



n (amount of gas) against T (temperature) from 150 K to 450 K. One curve for each value of P (pressure): 202650 Pa, 101325 Pa, 405300 Pa. Other inputs as in the worked example: V = 0.005 m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T = 300 K, n = 0.40622 mol.

```
chemistry_ai999x.solve('ideal_gas_moles', P=202650.0, V=0.005, T=300.0)
# -> n = 0.40622 mol   (SOLVED)
```

```
chemistry_ai999x.solve('ideal_gas_moles', P=101325.0, V=0.0224, T=273.15)
# -> n = 0.999377 mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How many moles of gas are in a 5.00 L flask at 2.00 atm and 300 K?	SOLVED	n = 0.40622 mol
How many moles of gas occupy 22.4 L at 1.00 atm and 0 °C?	SOLVED	n = 0.999377 mol

Ideal gas law: temperature, `ideal_gas_temperature`

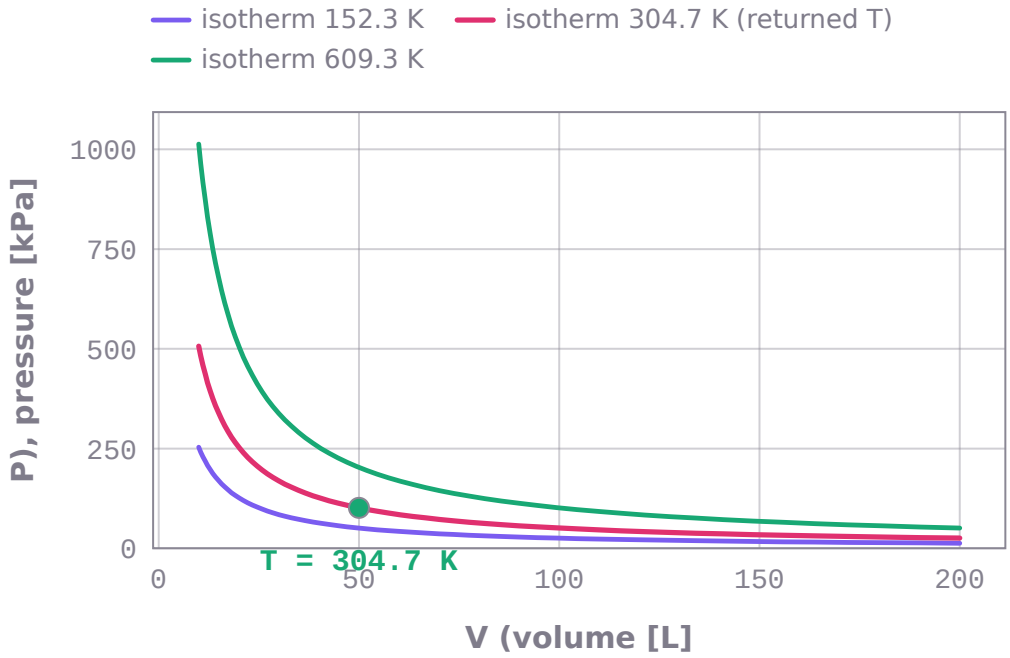
The temperature of an ideal gas from its pressure, volume and amount.

$$T = \frac{PV}{nR}$$

input	meaning	condition
`P`	pressure	
`V`	volume	
`n`	amount of gas	

output	meaning
`T`	temperature
`T_C`	temperature in °C

- **Method:** $T = PV/nR$
- **Assumes:** P, V, $n > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Which isotherm passes through the state P = 101.325 kPa, V = 50 L of 2 mol? The package returns T = 304.665 K; drawn: that isotherm and those at half and twice the temperature, each from the package's ideal-gas pressure at 61 volumes from 10 L to 200 L. Marked: the state, T = 304.665 K.

```
chemistry_ai999x.solve('ideal_gas_temperature', P=101325.0, V=0.05, n=2.0)
# -> T = 304.665 K, T_C = 31.5149   (SOLVED)
```

```
chemistry_ai999x.solve('ideal_gas_temperature', P=150000.0, V=0.002, n=0.1)
# -> T = 360.817 K, T_C = 87.6671   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
At what temperature does 2.00 mol of gas occupy 50.0 L at 1.00 atm?	SOLVED	T = 304.665 K, T_C = 31.5149
Find the temperature of 0.100 mol of gas in a 2.00 L vessel at 150 kPa.	SOLVED	T = 360.817 K, T_C = 87.6671

Mass of a gas sample (ideal gas), `ideal_gas_mass`

The mass of a named gas that fills a volume at a given pressure and temperature.

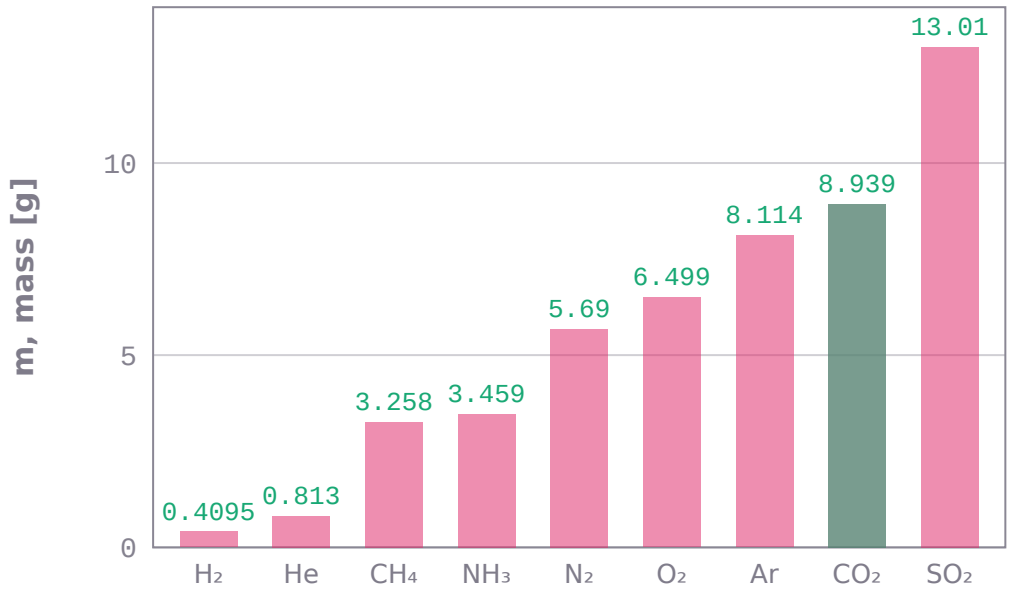
$$m = \frac{PV}{RT}\,M$$

input	meaning	condition
`formula`	formula of the gas	
`P`	pressure	
`V`	volume	
`T`	temperature	

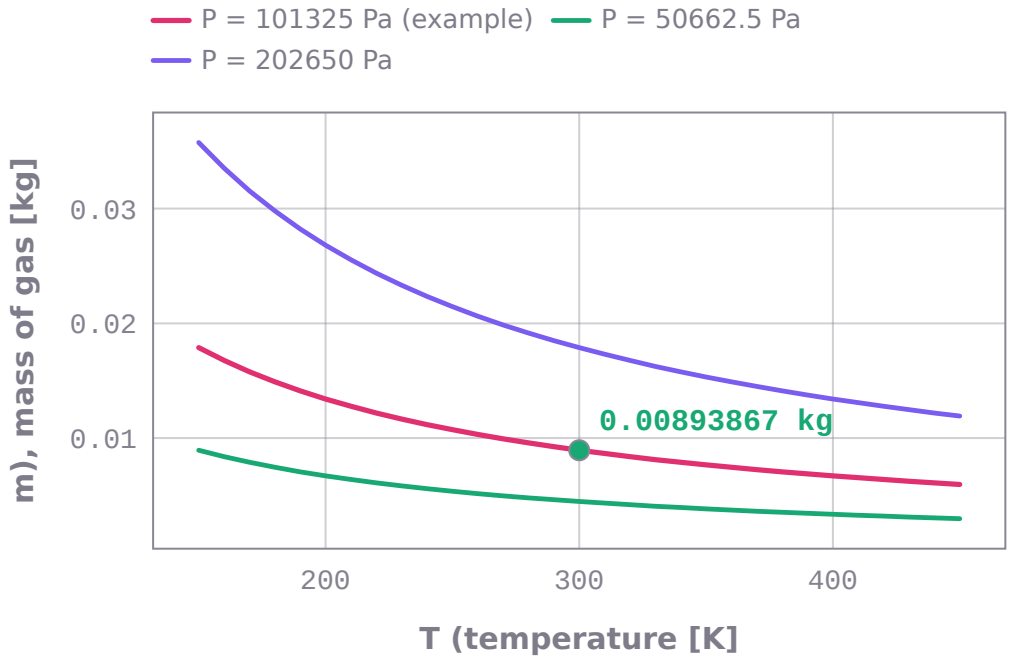
output	meaning
`m`	mass of gas

output	meaning
n	amount of gas

- **Method:** $n = PV/RT$, then $m = nM$
- **Assumes:** P, V, $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Mass of 5 L of different gases at 101.325 kPa and 300 K, each from the package (the same amount of gas, so the mass follows the molar mass). Highlighted: the worked example, CO_2, 8.93867 g.



m (mass of gas) against T (temperature) from 150 K to 450 K. One curve for each value of P (pressure): 101325 Pa, 50662.5 Pa, 202650 Pa. Other inputs as in the worked example: formula = CO2, V = 0.005 m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T = 300 K, m = 0.00893867 kg.

```
chemistry_ai999x.solve('ideal_gas_mass', formula='CO2', P=101325.0, V=0.005, T=300.0)
# -> m = 0.00893867 kg, n = 0.20311 mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What mass of CO2 occupies 5.00 L at 1.00 atm and 300 K?	SOLVED	m = 0.00893867 kg, n = 0.20311 mol
How many grams of N2 fill a 10.0 L tank at 2.00 atm and 25 °C?	SOLVED	m = 0.0229009 kg, n = 0.817481 mol

Density of an ideal gas, `gas_density`

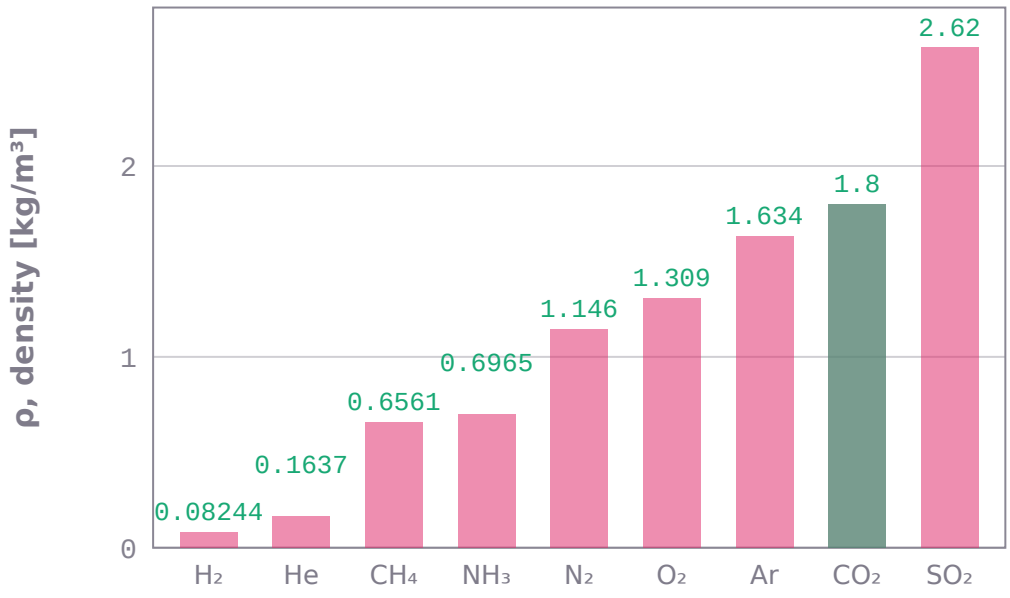
The density of a named gas at a given pressure and temperature (ideal gas).

$$\rho = \frac{PM}{RT}$$

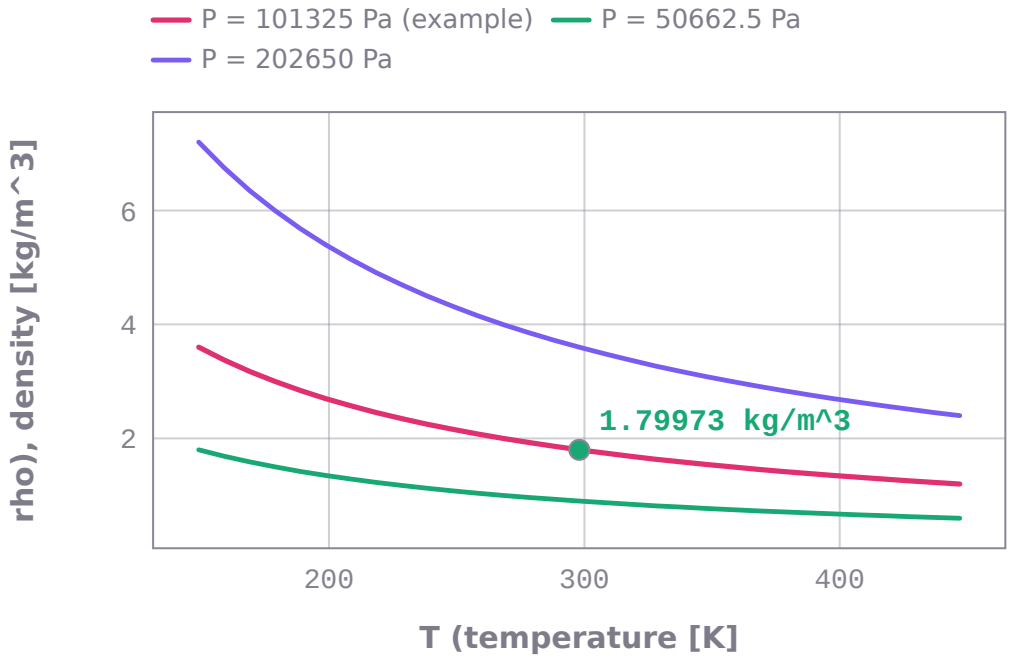
input	meaning	condition
`formula`	formula of the gas	
`P`	pressure	
`T`	temperature	

output	meaning
`rho`	density

- **Method:** $\rho = PM/RT$
- **Assumes:** P, $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Density of different gases at 101.325 kPa and 298 K, each returned by the package (ideal gas: density proportional to molar mass). Highlighted: the worked example CO_2, 1.79973 kg/m³; air is about 1.2 kg/m³ at these conditions.



rho (density) against T (temperature) from 149 K to 447 K. One curve for each value of P (pressure): 101325 Pa, 50662.5 Pa, 202650 Pa. Other inputs as in the worked example: formula = CO2. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T = 298 K, rho = 1.79973 kg/m^3.

```
chemistry_ai999x.solve('gas_density', formula='CO2', P=101325.0, T=298.0)
# -> rho = 1.79973 kg/m^3   (SOLVED)
```

```
chemistry_ai999x.solve('gas_density', formula='N2', P=200000.0, T=350.0)
# -> rho = 1.92532 kg/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the density of CO2 at 1.00 atm and 298 K?	SOLVED	rho = 1.79973 kg/m^3
Calculate the density of N2 at 200 kPa and 350 K.	SOLVED	rho = 1.92532 kg/m^3

Molar mass from gas density, `molar_mass_from_gas_density`

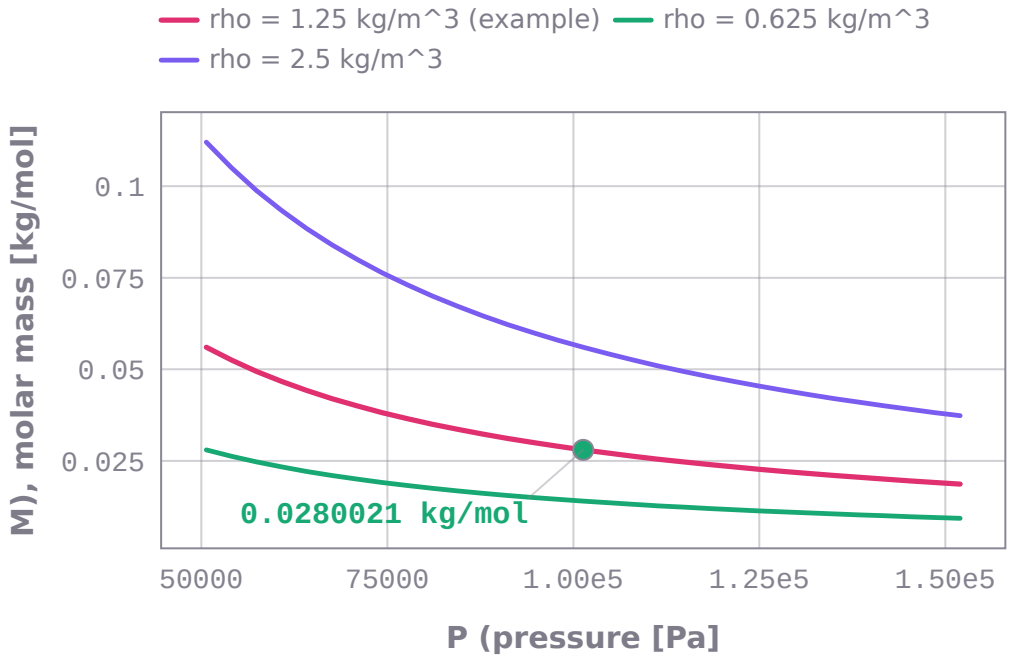
The molar mass of a gas from its density at a known pressure and temperature.

$$M = \frac{\rho R T}{P}$$

input	meaning	condition
`rho`	gas density	
`P`	pressure	
`T`	temperature	

output	meaning
`M`	molar mass
`M_g`	molar mass in g/mol

- **Method:** $M = \rho$ R T / P
- **Assumes:** rho, P, $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



M (molar mass) against P (pressure) from 50662.5 Pa to 151988 Pa. One curve for each value of rho (gas density): 1.25 kg/m^3, 0.625 kg/m^3, 2.5 kg/m^3. Other inputs as in the worked example: T = 273.0 K. 31 points per curve computed by chemistry-ai999x; marked: the worked example, P = 101325 Pa, M = 0.0280021 kg/mol.

```
chemistry_ai999x.solve('molar_mass_from_gas_density', rho=1.25, P=101325.0, T=273.0)
# -> M = 0.0280021 kg/mol, M_g = 28.0021   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A gas has a density of 1.25 g/L at 1.00 atm and 273 K. What is its molar mass?	SOLVED	M = 0.0280021 kg/mol, M_g = 28.0021
Find the molar mass of a gas whose density is 2.0 g/L at 0.95 atm and 300 K.	SOLVED	M = 0.0518257 kg/mol, M_g = 51.8257

Molar mass from a gas sample, `molar_mass_from_gas_sample`

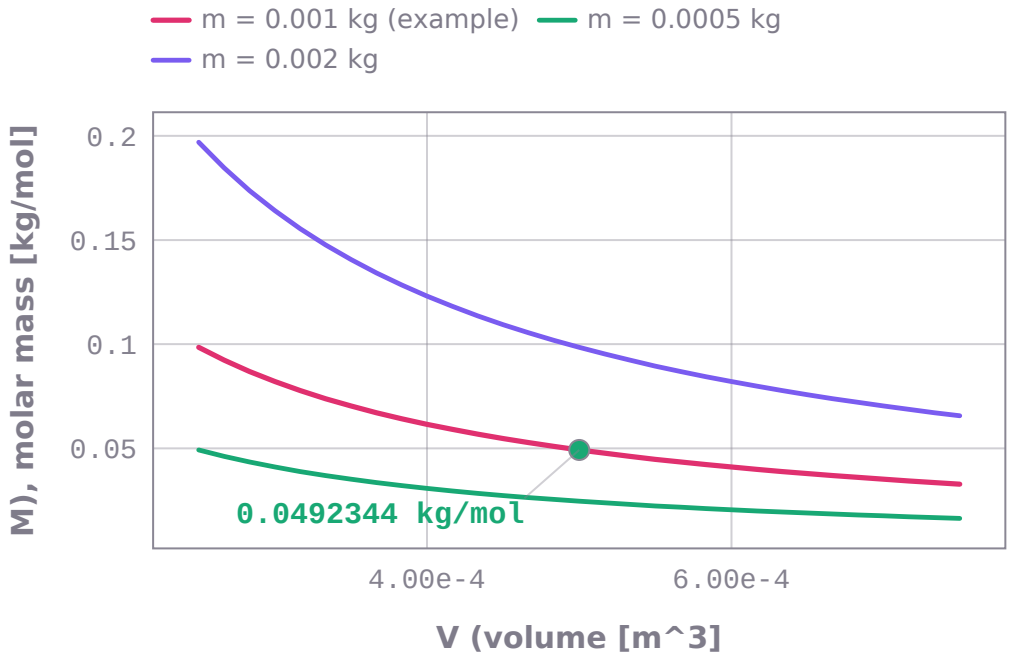
The molar mass of a gas from a sample's mass, volume, pressure and temperature.

$$M = \frac{mRT}{PV}$$

input	meaning	condition
`m`	mass of the sample	
`V`	volume	
`P`	pressure	
`T`	temperature	

output	meaning
`M`	molar mass
`M_g`	molar mass in g/mol

- **Method:** M = mRT / PV
- **Assumes:** m, V, P, $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



M (molar mass) against V (volume) from 0.00025 m^3 to 0.00075 m^3. One curve for each value of m (mass of the sample): 0.001 kg, 0.0005 kg, 0.002 kg. Other inputs as in the worked example: P = 101325.0 Pa, T = 300.0 K. 31 points per curve computed by chemistry-ai999x; marked: the worked example, V = 0.0005 m^3, M = 0.0492344 kg/mol.

```
chemistry_ai999x.solve('molar_mass_from_gas_sample', m=0.001, V=0.0005, P=101325.0,
T=300.0)
# -> M = 0.0492344 kg/mol, M_g = 49.2344   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A 1.00 g sample of a gas occupies 0.500 L at 1.00 atm and 300 K. What is its molar mass?	SOLVED	M = 0.0492344 kg/mol, M_g = 49.2344
A 2.50 g sample of gas fills 1.20 L at 750 torr and 25 °C. Find its molar mass.	SOLVED	M = 0.0516492 kg/mol, M_g = 51.6492

Boyle's law: final volume, `boyle_final_volume`

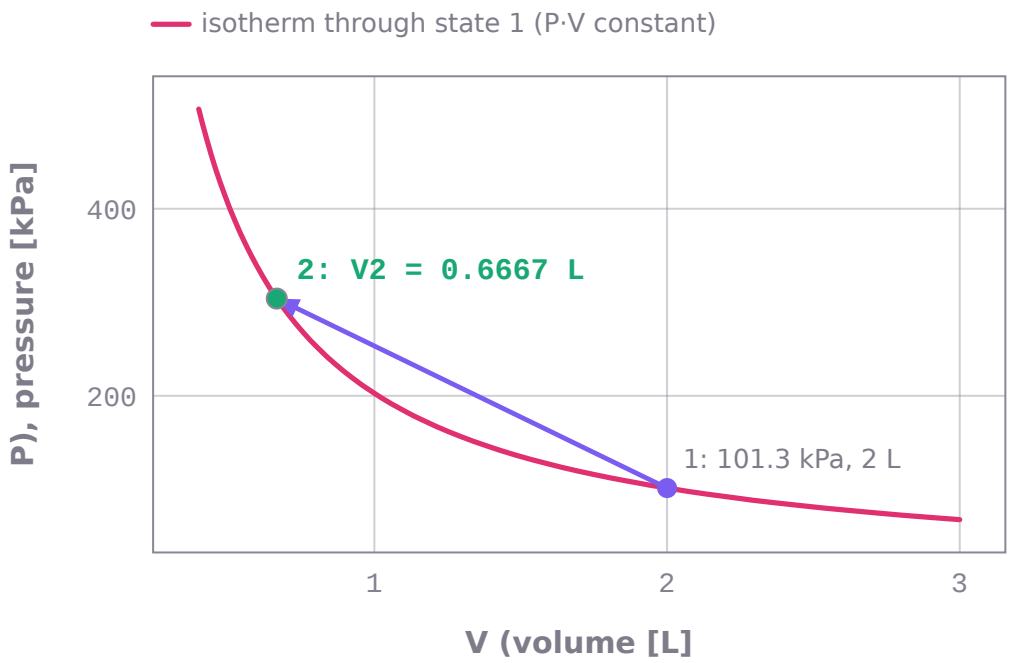
The volume of a gas after a pressure change at constant temperature $(P1V1 = P2V2)$.

$$V_2 = \frac{P_1 V_1}{P_2}$$

input	meaning	condition
`P1`	initial pressure	
`V1`	initial volume	
`P2`	final pressure	

output	meaning
`V2`	final volume

- **Method:** $V_2 = P_1\ V_1 / P_2$
- **Assumes:** P1, V1, $P_2 > 0$; constant temperature and amount
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Boyle's law on a P–V diagram: the isotherm through state 1 (P1 = 101.325 kPa, V1 = 2 L), computed by the package at 61 volumes; arrow: raising the pressure to P2 = 303.975 kPa (the worked example). Marked: the returned V2 = 0.666667 L.

```
chemistry_ai999x.solve('boyle_final_volume', P1=101325.0, V1=0.002, P2=303975.0)
# -> V2 = 0.000666667 m^3   (SOLVED)
```

```
chemistry_ai999x.solve('boyle_final_volume', P1=100000.0, V1=0.005, P2=250000.0)
# -> V2 = 0.002 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A gas occupies 2.00 L at 1.00 atm. What is its volume when compressed to 3.00 atm at constant temperature?	SOLVED	V2 = 0.000666667 m^3
P1 = 100 kPa, V1 = 5.0 L, P2 = 250 kPa. Find the final volume (constant temperature).	SOLVED	V2 = 0.002 m^3

question	outcome	result
A gas at 1.00 atm and 2.00 atm occupies 2.00 L. What is its volume?	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemistry_ai999x.templates(); eve

Boyle's law: final pressure, `boyle_final_pressure`

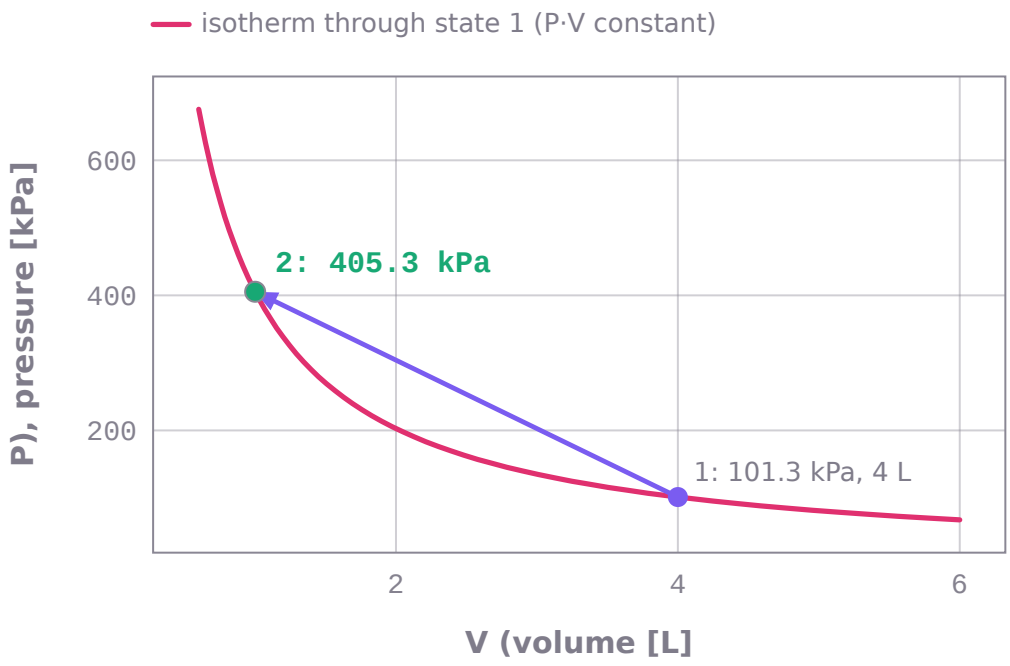
The pressure of a gas after a volume change at constant temperature.

$$P_2 = \frac{P_1 V_1}{V_2}$$

input	meaning	condition
P1	initial pressure	
V1	initial volume	
V2	final volume	

output	meaning
P2	final pressure

- **Method:** $P_2 = P_1\ V_1 / V_2$
- **Assumes:** P1, V1, $V_2 > 0$; constant temperature and amount
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Boyle's law on a P–V diagram: the isotherm through state 1 (P1 = 101.325 kPa, V1 = 4 L), each point the package's final pressure at 61 volumes from 0.6 to 6 L; arrow: the compression to V2 = 1 L (the worked example). Marked: P2 = 405.3 kPa.

```
chemistry_ai999x.solve('boyle_final_pressure', P1=101325.0, V1=0.004, V2=0.001)
# -> P2 = 405300 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A gas at 1.00 atm occupies 4.00 L. What is the new pressure if it is compressed to 1.00 L at constant temperature?	SOLVED	P2 = 405300 Pa
P1 = 2.0 atm, V1 = 3.0 L, V2 = 6.0 L. Find the final pressure.	SOLVED	P2 = 101325 Pa

Charles's law: final volume, `charles_final_volume`

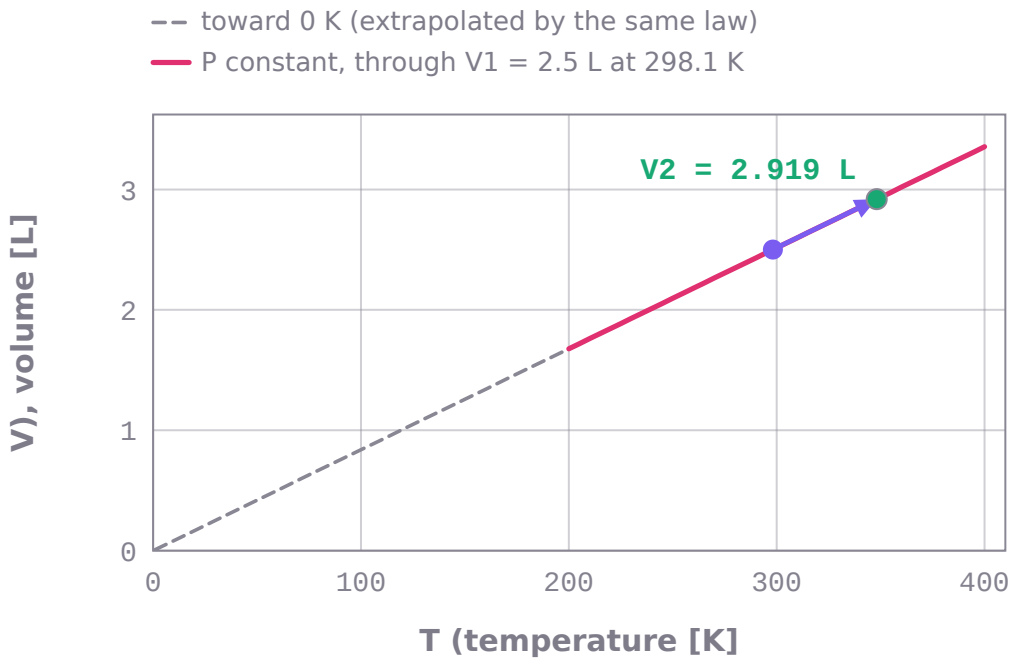
The volume of a gas after a temperature change at constant pressure $(V_1/T_1 = V_2/T_2)$.

$$V_2 = V_1 \frac{T_2}{T_1}$$

input	meaning	condition
V1	initial volume	
T1	initial temperature	
T2	final temperature	

output	meaning
V2	final volume

- **Method:** $V_2 = V_1\ T_2/T_1$ (kelvin)
- **Assumes:** $V_1 > 0$, T1, $T_2 > 0$ K; constant pressure and amount
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Charles's law: volume against absolute temperature at constant pressure through state 1 (V1 = 2.5 L, T1 = 298.15 K), the package's final volume at 41 temperatures from 200 K to 400 K; dashed: the same law toward 0 K, where the volume would vanish. Arrow: heating to T2 = 348.15 K. Marked: V2 = 2.91925 L.

```
chemistry_ai999x.solve('charles_final_volume', V1=0.0025, T1=298.15, T2=348.15)
# -> V2 = 0.00291925 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A balloon holds 2.50 L at 25 °C. What is its volume when heated to 75 °C at constant pressure?	SOLVED	V2 = 0.00291925 m^3
V1 = 1.0 L, T1 = 300 K, T2 = 600 K. Find the final volume.	SOLVED	V2 = 0.002 m^3

Gay-Lussac's law: final pressure, `gay_lussac_final_pressure`

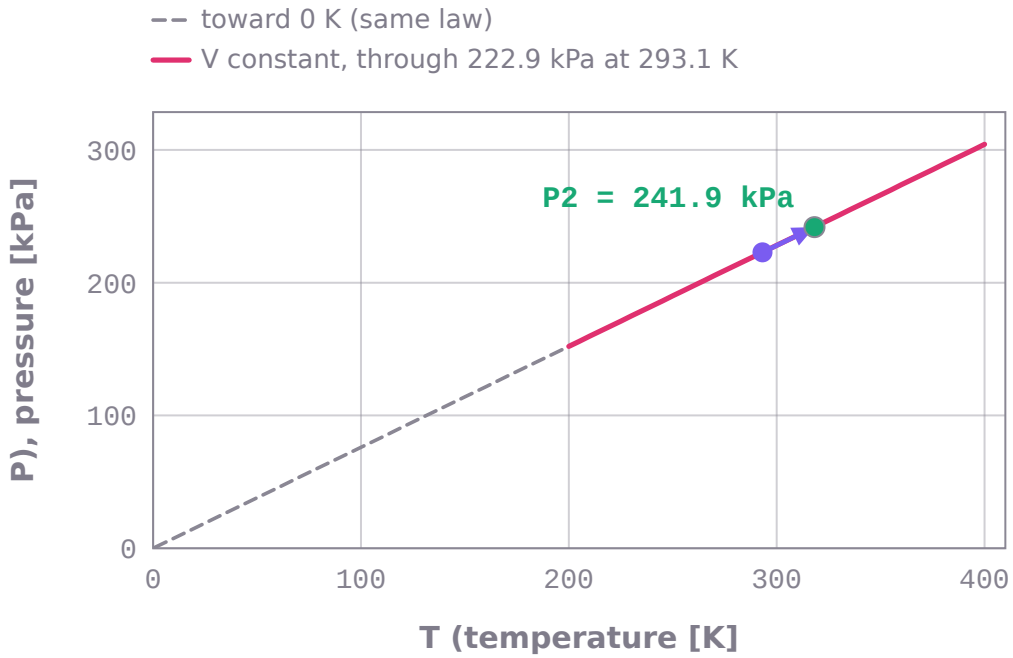
The pressure of a gas after a temperature change at constant volume $(P_1/T_1 = P_2/T_2)$.

$$P_2 = P_1 \frac{T_2}{T_1}$$

input	meaning	condition
`P1`	initial pressure	
`T1`	initial temperature	
`T2`	final temperature	

output	meaning
`P2`	final pressure

- **Method:** $P_2 = P_1\ T_2/T_1$ (kelvin)
- **Assumes:** $P_1 > 0$, T1, $T_2 > 0$ K; constant volume and amount
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Gay-Lussac's law: pressure against absolute temperature at constant volume through state 1 (P1 = 222.915 kPa, T1 = 293.15 K, a tyre-like example), the package's final pressure at 41 temperatures from 200 K to 400 K; dashed: toward 0 K. Arrow: warming to T2 = 318.15 K. Marked: P2 = 241.925 kPa.

```
chemistry_ai999x.solve('gay_lussac_final_pressure', P1=222915.00000000003, T1=293.15,
T2=318.15)
# -> P2 = 241925 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A tyre is at 2.20 atm at 20 °C. What is the pressure when it warms to 45 °C at constant volume?	SOLVED	P2 = 241925 Pa
P1 = 100 kPa, T1 = 300 K, T2 = 450 K. Find the final pressure.	SOLVED	P2 = 150000 Pa

Combined gas law: final volume, `combined_gas_final_volume`

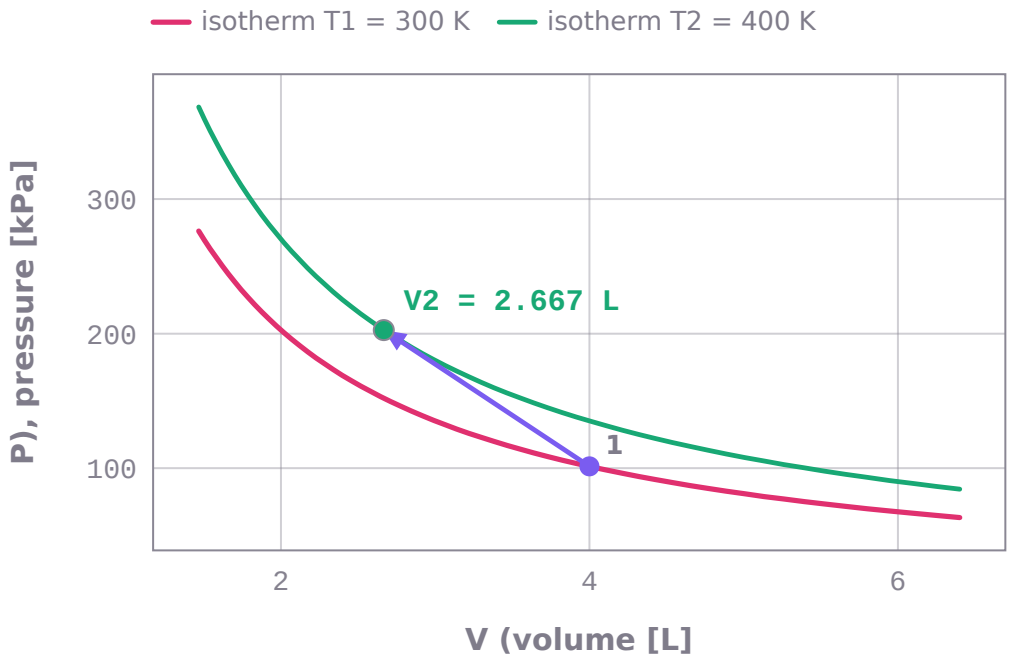
The volume of a fixed amount of gas after its pressure and temperature change.

$$V_2 = \frac{P_1 V_1 T_2}{T_1 P_2}$$

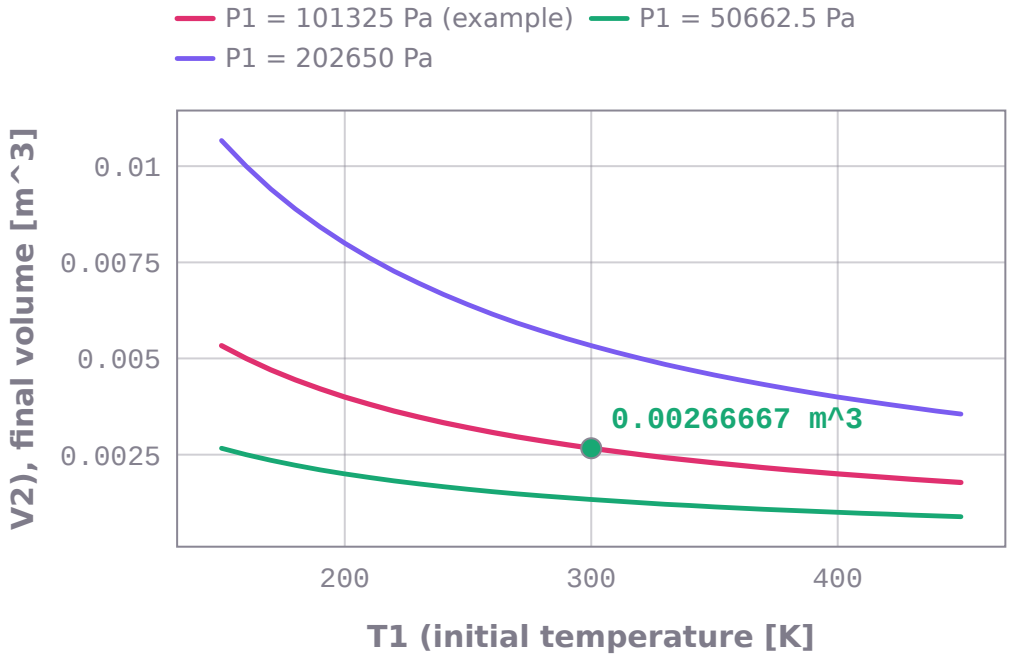
input	meaning	condition
`P1`	initial pressure	
`V1`	initial volume	
`T1`	initial temperature	
`P2`	final pressure	
`T2`	final temperature	

output	meaning
V2	final volume

- **Method:** $P1V1/T_1 = P2V2/T_2$ solved for V2
- **Assumes:** all values > 0 (temperatures in kelvin); fixed amount of gas
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The combined gas law on a P–V diagram: state 1 (P1 = 101.325 kPa, V1 = 4 L, T1 = 300 K) lies on the isotherm T1 and state 2 on the isotherm T2; each isotherm is P·V constant through its state, drawn with the package's Boyle relation at 61 volumes. Stated: P2 = 202.65 kPa, T2 = 400 K. Marked: the returned V2 = 2.667 L.



V2 (final volume) against T1 (initial temperature) from 150 K to 450 K. One curve for each value of P1 (initial pressure): 101325 Pa, 50662.5 Pa, 202650 Pa. Other inputs as in the worked example: V1 = 0.004 m^3, P2 = 202650.0 Pa, T2 = 400.0 K. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T1 = 300 K, V2 = 0.00266667 m^3.

```
chemistry_ai999x.solve('combined_gas_final_volume', P1=101325.0, V1=0.004, T1=300.0,
P2=202650.0, T2=400.0)
# -> V2 = 0.00266667 m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A gas occupies 4.00 L at 1.00 atm and 300 K. What is its volume at 2.00 atm and 400 K?	SOLVED	V2 = 0.00266667 m^3
P1 = 1.0 atm, V1 = 2.0 L, T1 = 273 K, P2 = 0.5 atm, T2 = 546 K. Find the final volume.	SOLVED	V2 = 0.008 m^3

Combined gas law: final pressure, `combined_gas_final_pressure`

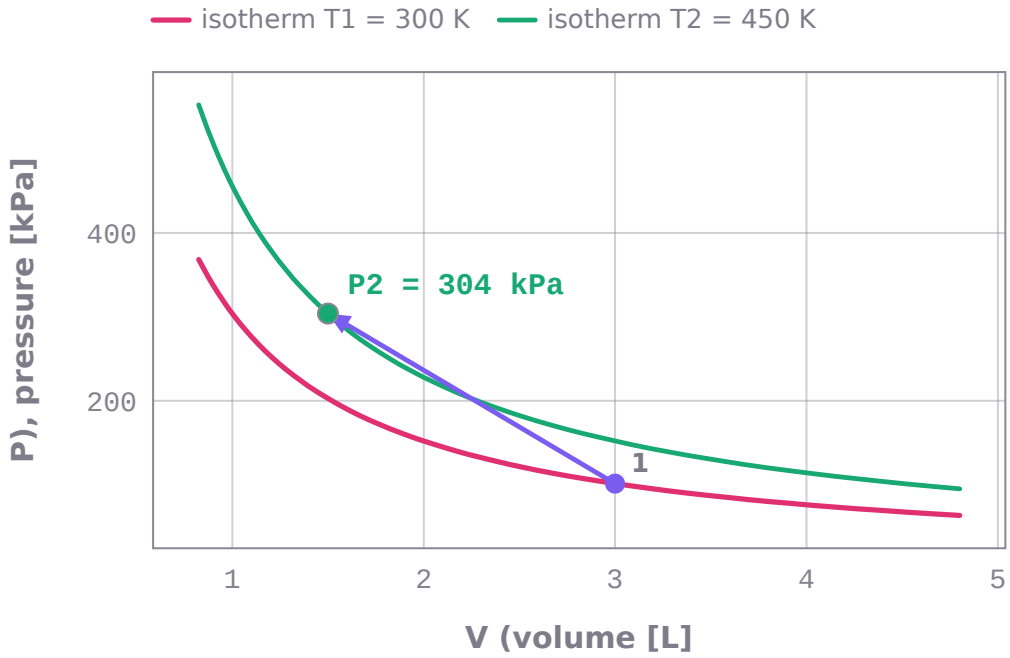
The pressure of a fixed amount of gas after its volume and temperature change.

$$P_2 = \frac{P_1 V_1 T_2}{T_1 V_2}$$

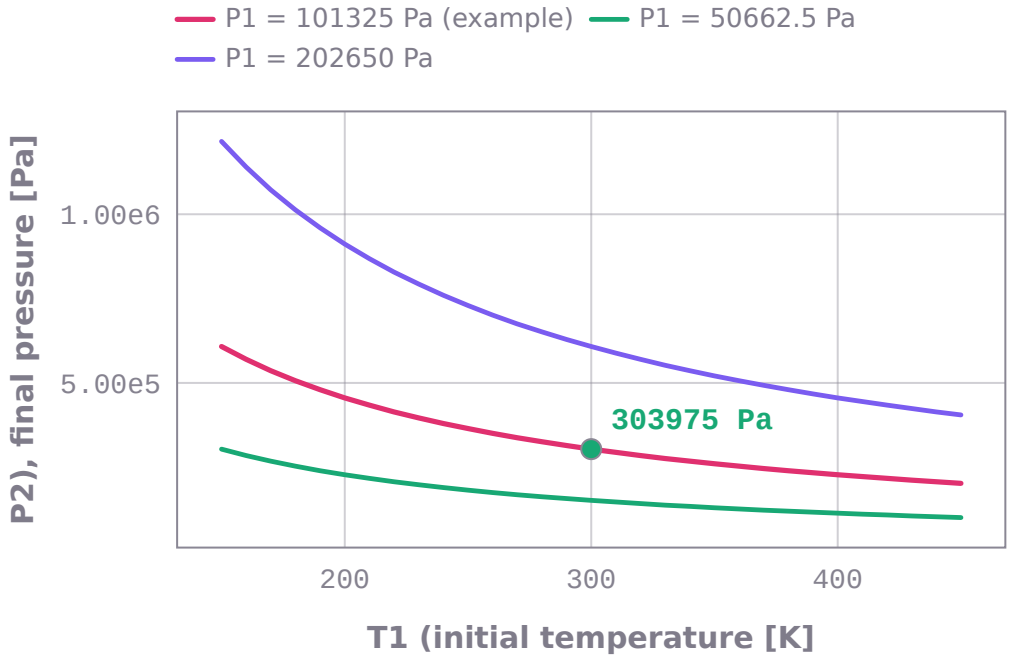
input	meaning	condition
`P1`	initial pressure	
`V1`	initial volume	
`T1`	initial temperature	
`V2`	final volume	
`T2`	final temperature	

output	meaning
`P2`	final pressure

- **Method:** $P1V1/T_1 = P2V2/T_2$ solved for P2
- **Assumes:** all values > 0 (temperatures in kelvin); fixed amount of gas
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The combined gas law on a P–V diagram: state 1 (P1 = 101.325 kPa, V1 = 3 L, T1 = 300 K) lies on the isotherm T1 and state 2 on the isotherm T2; each isotherm is P·V constant through its state, drawn with the package's Boyle relation at 61 volumes. Stated: V2 = 1.5 L, T2 = 450 K. Marked: the returned P2 = 304 kPa.



P2 (final pressure) against T1 (initial temperature) from 150 K to 450 K. One curve for each value of P1 (initial pressure): 101325 Pa, 50662.5 Pa, 202650 Pa. Other inputs as in the worked example: V1 = 0.003 m^3, V2 = 0.0015 m^3, T2 = 450.0 K. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T1 = 300 K, P2 = 303975 Pa.

```
chemistry_ai999x.solve('combined_gas_final_pressure', P1=101325.0, V1=0.003, T1=300.0,
V2=0.0015, T2=450.0)
# -> P2 = 303975 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A gas at 1.00 atm and 300 K occupies 3.00 L. Find the final pressure if V2 = 1.50 L and T2 = 450 K.	SOLVED	P2 = 303975 Pa

question	outcome	result
P1 = 100 kPa, V1 = 10 L, T1 = 300 K, V2 = 5 L, T2 = 330 K. What is the final pressure?	SOLVED	P2 = 220000 Pa

Combined gas law: final temperature, `combined_gas_final_temperature`

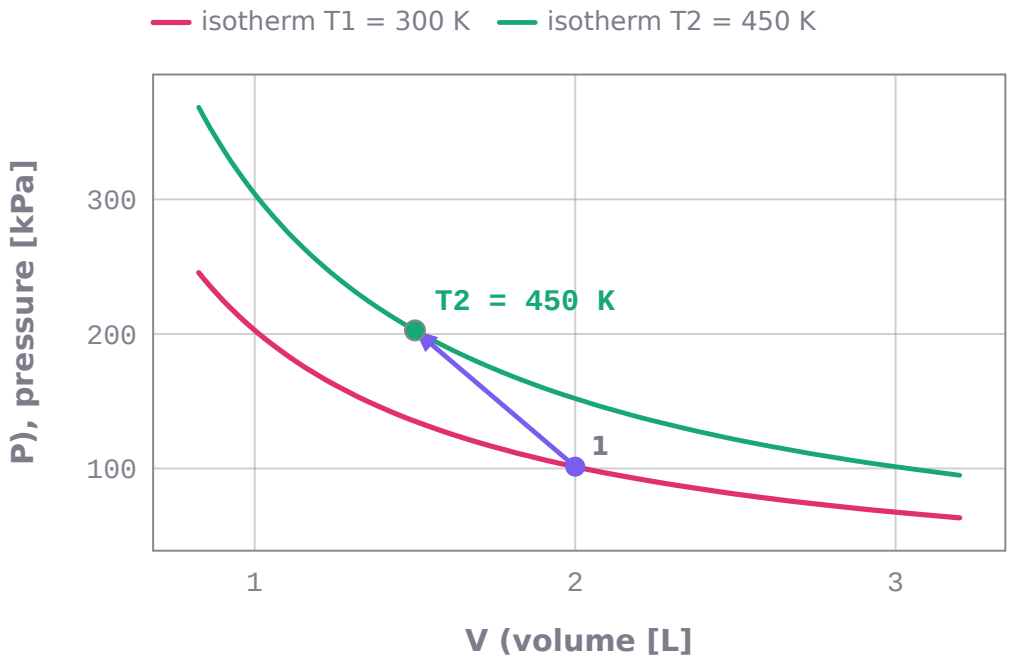
The temperature of a fixed amount of gas after its pressure and volume change.

$$T_2 = \frac{P_2 V_2 T_1}{P_1 V_1}$$

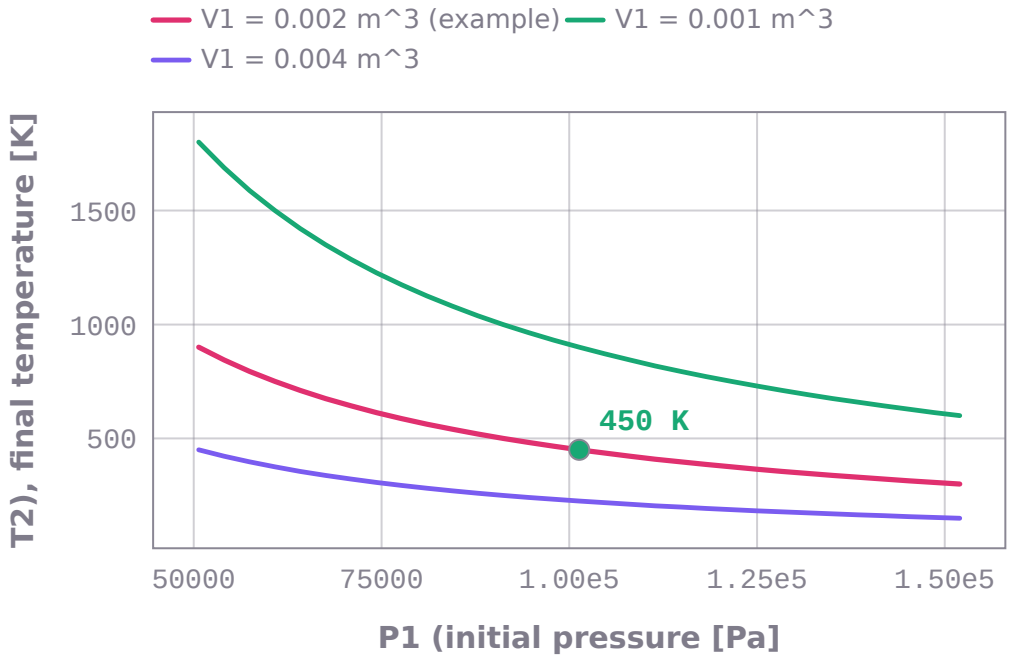
input	meaning	condition
`P1`	initial pressure	
`V1`	initial volume	
`T1`	initial temperature	
`P2`	final pressure	
`V2`	final volume	

output	meaning
`T2`	final temperature

- **Method:** $P1V1/T_1 = P2V2/T_2$ solved for T2
- **Assumes:** all values > 0 (temperatures in kelvin); fixed amount of gas
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The combined gas law on a P–V diagram: state 1 (P1 = 101.325 kPa, V1 = 2 L, T1 = 300 K) lies on the isotherm T1 and state 2 on the isotherm T2; each isotherm is P·V constant through its state, drawn with the package's Boyle relation at 61 volumes. Stated: P2 = 202.65 kPa, V2 = 1.5 L. Marked: the returned T2 = 450 K.



T2 (final temperature) against P1 (initial pressure) from 50662.5 Pa to 151988 Pa. One curve for each value of V1 (initial volume): 0.002 m^3, 0.001 m^3, 0.004 m^3. Other inputs as in the worked example: T1 = 300.0 K, P2 = 202650.0 Pa, V2 = 0.0015 m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, P1 = 101325 Pa, T2 = 450 K.

```
chemistry_ai999x.solve('combined_gas_final_temperature', P1=101325.0, V1=0.002, T1=300.0,
P2=202650.0, V2=0.0015)
# -> T2 = 450 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
P1 = 1.00 atm, V1 = 2.00 L, T1 = 300 K, P2 = 2.00 atm, V2 = 1.50 L. Find the final temperature.	SOLVED	T2 = 450 K
A gas at 100 kPa, 5.0 L and 290 K is changed to 150 kPa and 4.0 L. What is the final temperature?	SOLVED	T2 = 348 K

Partial pressure (Dalton's law), `partial_pressure`

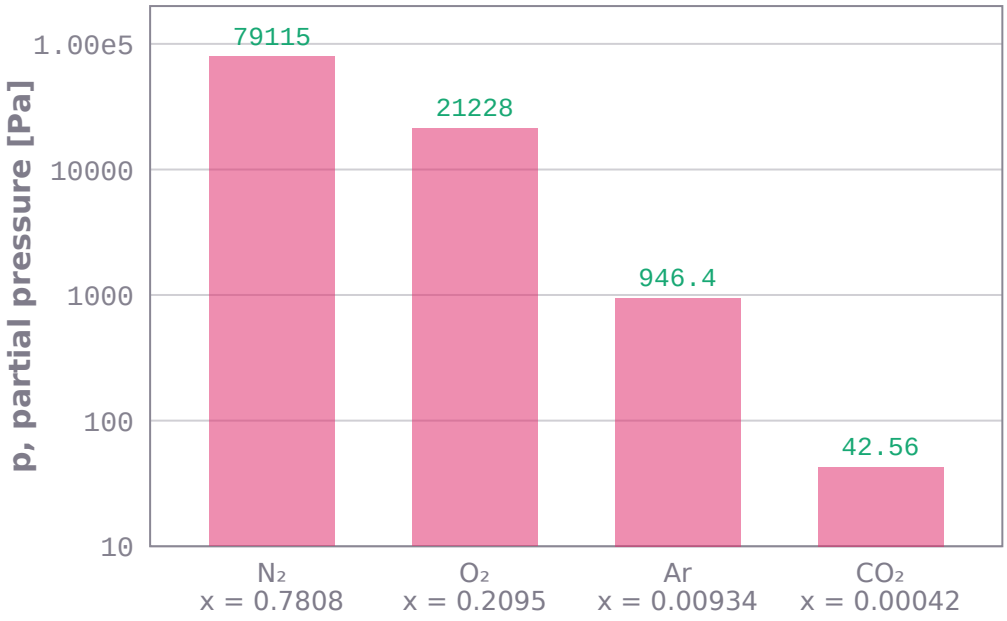
The partial pressure of a gas in a mixture from its mole fraction and the total pressure.

$$p_i = x_i P_{\text{total}}$$

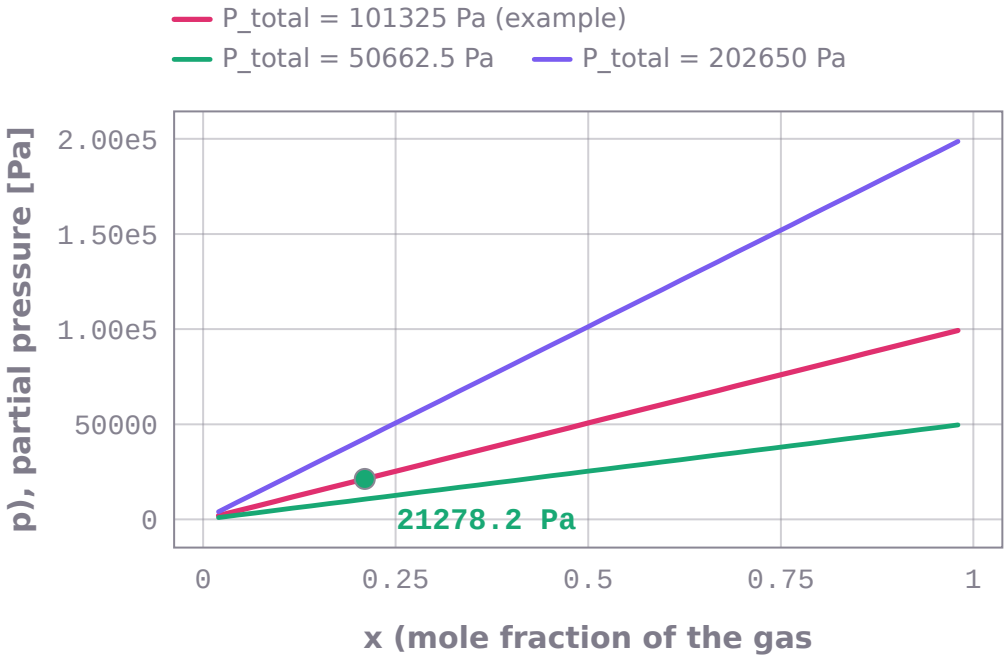
input	meaning	condition
`x`	mole fraction of the gas	
`P_total`	total pressure	

output	meaning
`p`	partial pressure

- **Method:** $p = x$ P_total
- **Assumes:** $0 < x \leq 1$; P_total > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Partial pressures in dry air at a total of 101.325 kPa (logarithmic axis), each from the package with a stated mole fraction: N_2 0.7808, O_2 0.2095, Ar 0.00934, CO_2 0.00042 (the worked example rounds oxygen to 0.21). Marked by value: O_2 21227.6 Pa, CO_2 42.5565 Pa.



p (partial pressure) against x (mole fraction of the gas) from 0.02 to 0.98. One curve for each value of P_total (total pressure): 101325 Pa, 50662.5 Pa, 202650 Pa. 41 points per curve computed by chemistry-ai999x; marked: the worked example, x = 0.21, p = 21278.2 Pa.

```
chemistry_ai999x.solve('partial_pressure', x=0.21, P_total=101325.0)
# -> p = 21278.2 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The mole fraction of O2 in air is 0.21. What is its partial pressure at a total pressure of 1.00 atm?	SOLVED	p = 21278.2 Pa

question	outcome	result
A gas has mole fraction 1.5 in a mixture at 2.0 atm. What is its partial pressure?	INVALID_INPUT	Partial pressure (Dalton's law): a mole fraction lies between 0 and 1.

Partial pressures in a gas mixture, `dalton_mixture`

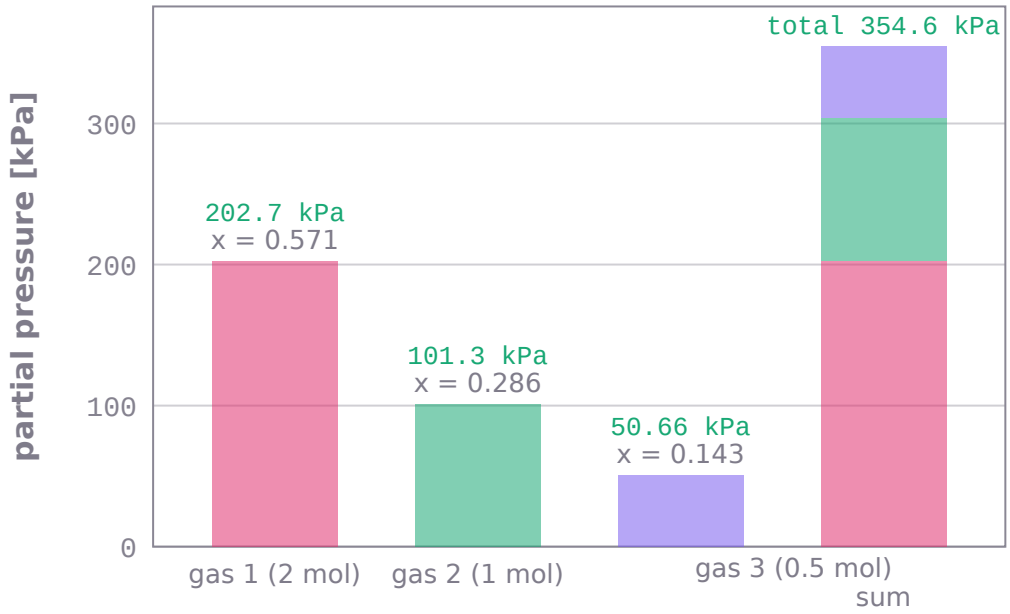
Mole fractions and partial pressures of every gas in a mixture from the amounts and the total pressure.

$$x_i = \frac{n_i}{\sum_j n_j}, \quad p_i = x_i P$$

input	meaning	condition
`moles`	amount of each gas (mol), e.g. '2.0, 1.0, 0.5'	
`P_total`	total pressure	

output	meaning
`partial_pressures`	partial pressure of each gas, in the order given
`mole_fractions`	mole fraction of each gas

- **Method:** mole fractions from the amounts, times the total pressure
- **Assumes:** every amount > 0; P_total > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Dalton's law: a mixture of 2, 1, 0.5 mol of three gases at a total pressure of 354.637 kPa; each bar is a partial pressure returned by the package with its mole fraction, and the right-hand column stacks them back to the stated total.

```
chemistry_ai999x.solve('dalton_mixture', moles='2.0, 1.0, 0.5', P_total=354637.5)
# -> partial_pressures = [202650.0, 101325.0, 50662.5] Pa, mole_fractions =
[0.5714285714285714, 0.2857142857142857, 0.14285714285714285]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A mixture of 2.0 mol N2, 1.0 mol O2 and 0.50 mol Ar has a total pressure of 3.5 atm. Find the partial pressures.	SOLVED	partial_pressures = [202650.0, 101325.0, 50662.5] Pa, mole_fractions = [0.5714285714285714, 0.2857142857142857, 0.14285714285714285]
A mixture of 1.0 mol He and 3.0 mol Ne is at 200 kPa. What are the partial pressures?	SOLVED	partial_pressures = [50000.0, 150000.0] Pa, mole_fractions = [0.25, 0.75]

Graham's law: ratio of effusion rates, `graham_rate_ratio`

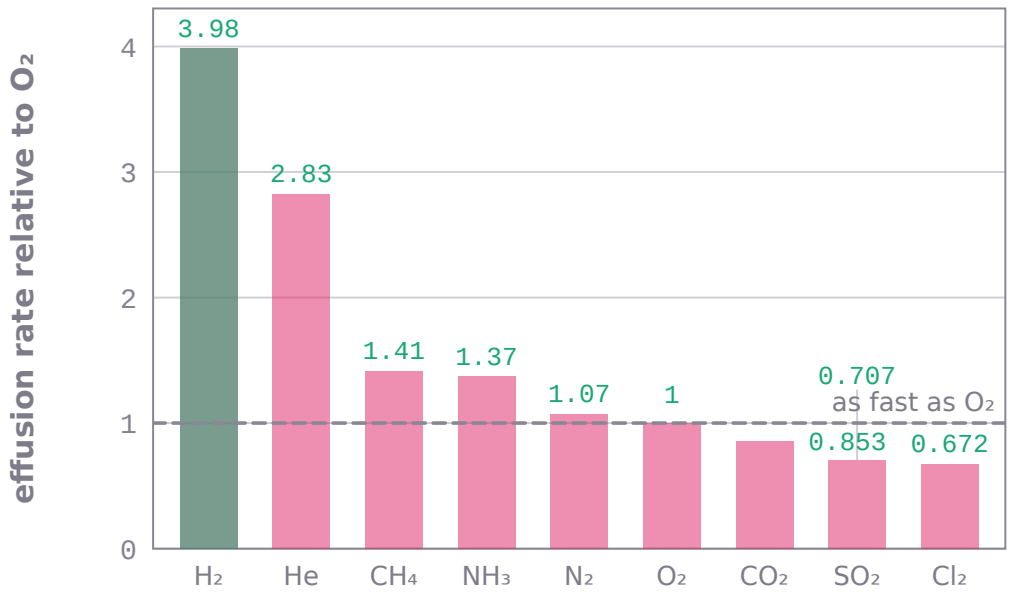
How many times faster one gas effuses (or diffuses) than another.

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

input	meaning	condition
`formula_1`	first gas (numerator)	
`formula_2`	second gas	

output	meaning
`ratio`	rate of gas 1 / rate of gas 2

- **Method:** square root of the inverse ratio of molar masses
- **Assumes:** ideal gases at the same temperature and pressure
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Graham's law: how many times faster each gas effuses than O_2 (rate ratio = √(M_O_2 / M)), each ratio returned by the package. Highlighted: the worked example, H_2 against O_2, ratio 3.98397.

```
chemistry_ai999x.solve('graham_rate_ratio', formula_1='H2', formula_2='O2')
# -> ratio = 3.98397   (SOLVED)
```

```
chemistry_ai999x.solve('graham_rate_ratio', formula_1='He', formula_2='CH4')
# -> ratio = 2.00194   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How many times faster does H2 effuse than O2?	SOLVED	ratio = 3.98397
What is the ratio of the rates of effusion of He to CH4?	SOLVED	ratio = 2.00194
How many times slower does O2 effuse than H2?	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemistry_ai999x.templates(); eve

Graham's law: molar mass of an unknown gas, `graham_molar_mass`

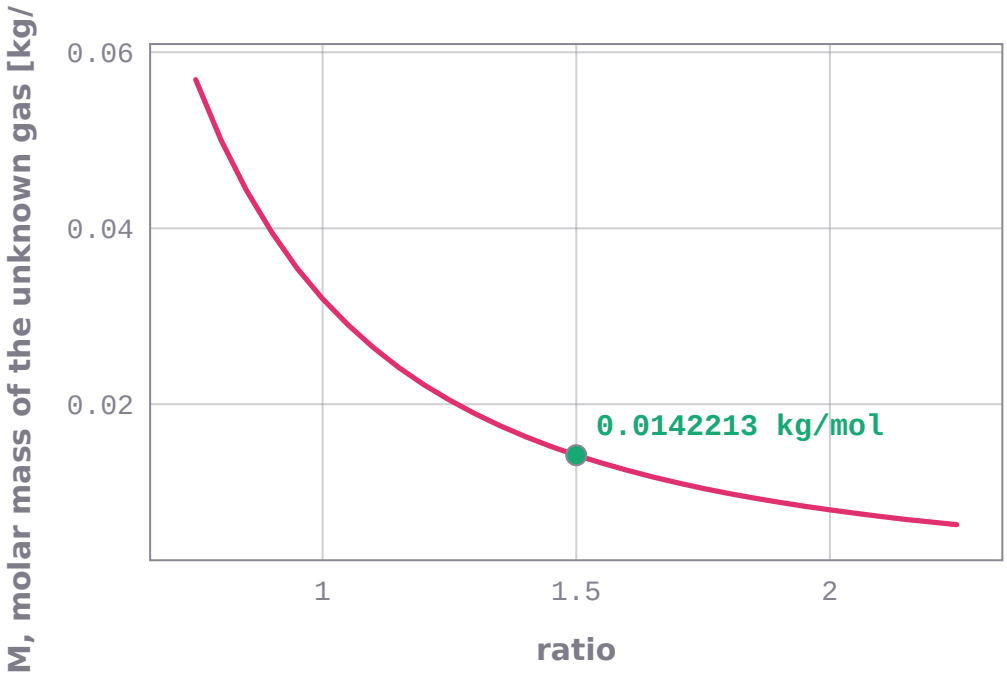
The molar mass of a gas from how fast it effuses relative to a known gas.

$$M_x = \frac{M_{\text{known}}}{(r_x / r_{\text{known}})^2}$$

input	meaning	condition
`ratio`	rate of the unknown gas / rate of the known gas	
`formula_known`	formula of the known gas	

output	meaning
`M`	molar mass of the unknown gas
`M_g`	molar mass in g/mol

- **Method:** M_x = M_known / ratio^2
- **Assumes:** ratio > 0; same temperature and pressure
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



M (molar mass of the unknown gas) against ratio (rate of the unknown gas / rate of the known gas) from 0.75 to 2.25. Other inputs as in the worked example: formula_known = O2. 31 points computed by chemistry-ai999x; marked: the worked example, ratio = 1.5, M = 0.0142213 kg/mol.

```
chemistry_ai999x.solve('graham_molar_mass', ratio=1.5, formula_known='O2')
# -> M = 0.0142213 kg/mol, M_g = 14.2213   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
An unknown gas effuses 1.50 times as fast as O2. What is its molar mass?	SOLVED	M = 0.0142213 kg/mol, M_g = 14.2213
An unknown gas diffuses 0.50 times as fast as CH4. What is the molar mass of the unknown gas?	SOLVED	M = 0.064172 kg/mol, M_g = 64.172

Van der Waals equation: pressure, `van_der_waals_pressure`

The pressure of a real gas from the van der Waals equation (and the ideal-gas pressure for comparison).

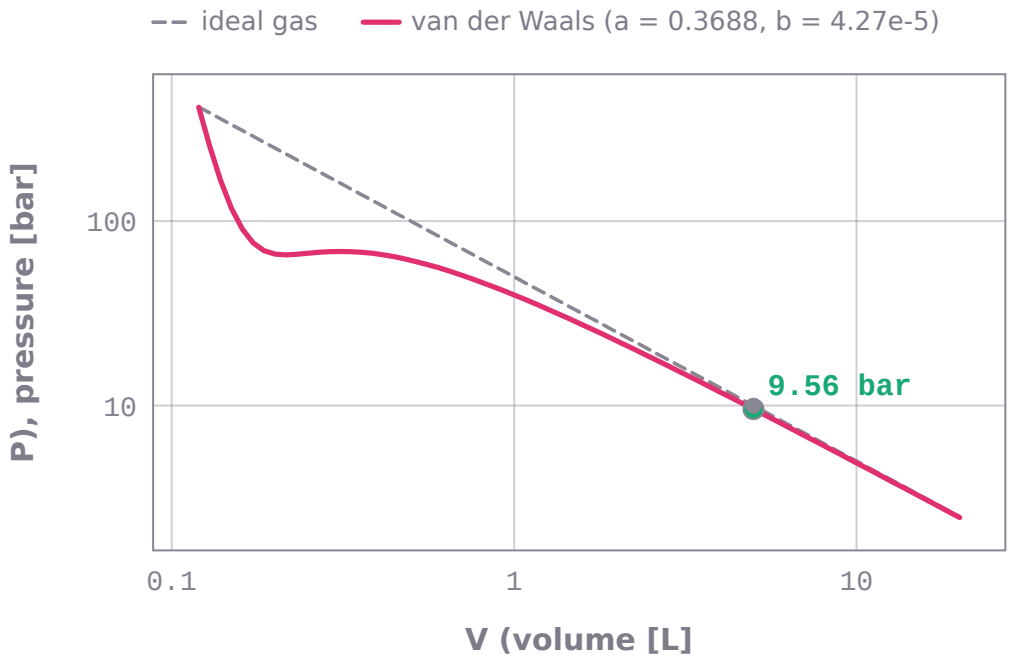
$$P = \frac{nRT}{V - nb} - \frac{an^2}{V^2}$$

input	meaning	condition
n	amount of gas	
V	volume	
T	temperature	
a	van der Waals a	
b	van der Waals b	

output	meaning
P	van der Waals pressure

output	meaning
`P_ideal`	ideal-gas pressure
`P_atm`	van der Waals pressure in atm

- **Method:** the van der Waals equation solved for P
- **Assumes:** n, V, T, a, $b > 0$; $V > n$ b
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Pressure of 2 mol of gas at 300 K against volume, van der Waals (a = 0.368823 Pa·m⁶/mol², b = 4.2670e-05 m³/mol, CO_2-like) and ideal, both returned by the package at 71 volumes from 0.12 L to 20 L (logarithmic axes). At large volume the two agree; squeezed, attraction first lowers the pressure and the molecules' own volume then raises it. Marked: V = 5 L, P = 9.56049 bar (ideal 9.97736 bar).

```
chemistry_ai999x.solve('van_der_waals_pressure', n=2.0, V=0.005, T=300.0, a=0.368823,
b=4.267e-05)
# -> P = 956049 Pa, P_ideal = 997736 Pa, P_atm = 9.43547   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Using the van der Waals equation, find the pressure of 2.00 mol of CO2 in 5.00 L at 300 K (a = 3.640 L^2·atm/mol^2, b = 0.04267 L/mol).	SOLVED	P = 956049 Pa, P_ideal = 997736 Pa, P_atm = 9.43547
What is the van der Waals pressure of 1.00 mol of N2 in 1.00 L at 273 K with a = 1.370 L^2 bar/mol^2 and b = 0.0387 L/mol?	SOLVED	P = 2.22423e+06 Pa, P_ideal = 2.26985e+06 Pa, P_atm = 21.9514
Using the van der Waals equation, find the pressure of 2.00 mol of CO2 in 5.00 L at 300 K.	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemistry_ai999x.templates(); eve

Root-mean-square speed of gas molecules, `rms_speed`

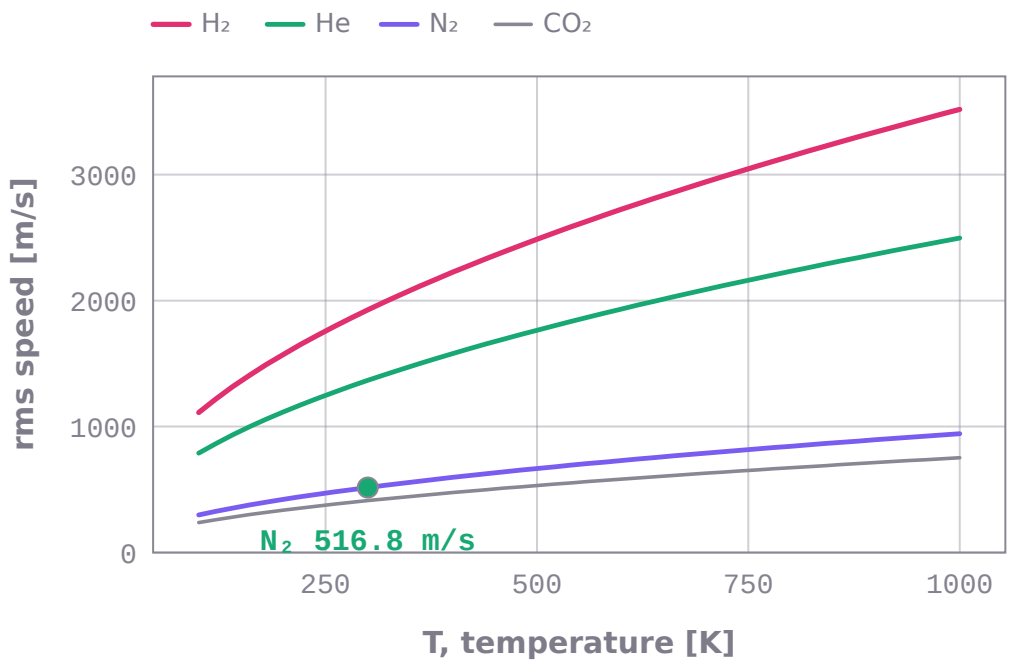
The root-mean-square speed of the molecules of an ideal gas at a temperature.

$$u_{\mathrm{rms}} = \sqrt{\frac{3RT}{M}}$$

input	meaning	condition
`formula`	formula of the gas	
`T`	temperature	

output	meaning
`u_rms`	root-mean-square speed

- **Method:** u_rms = sqrt(3RT/M), M in kg/mol
- **Assumes:** $T > 0$ K; ideal gas
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Rms speed of gas molecules against temperature for four gases, each computed by the package at 46 temperatures from 100 K to 1000 K (it grows as √(T/M)). At 300 K: H_2 1927, He 1367, N_2 516.8, CO_2 412.4 m/s. Marked: the worked example, N_2 at 300 K, 516.834 m/s.

```
chemistry_ai999x.solve('rms_speed', formula='N2', T=300.0)
# -> u_rms = 516.834 m/s   (SOLVED)
```

```
chemistry_ai999x.solve('rms_speed', formula='O2', T=298.15)
# -> u_rms = 482.096 m/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the rms speed of N2 molecules at 300 K?	SOLVED	u_rms = 516.834 m/s
Calculate the root-mean-square speed of O2 at 25 °C.	SOLVED	u_rms = 482.096 m/s

Mean speed of gas molecules, `mean_speed`

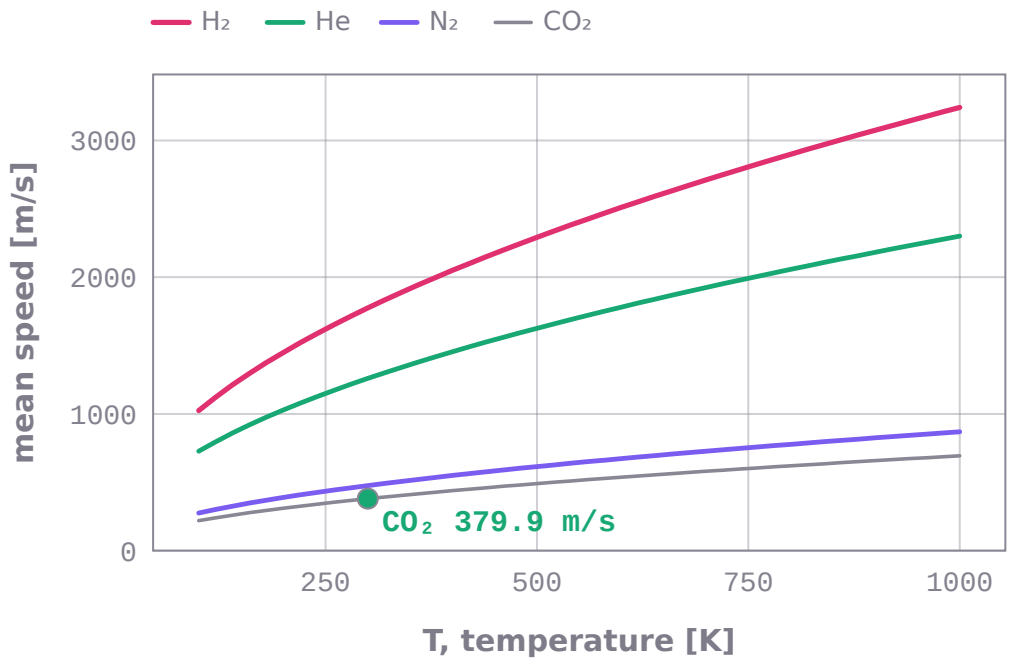
The mean (average) speed of the molecules of an ideal gas (Maxwell-Boltzmann distribution).

$$\bar{u} = \sqrt{\frac{8RT}{\pi M}}$$

input	meaning	condition
`formula`	formula of the gas	
`T`	temperature	

output	meaning
`u_mean`	mean speed

- **Method:** mean of the Maxwell-Boltzmann speed distribution
- **Assumes:** $T > 0$ K; ideal gas
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Mean speed of gas molecules against temperature for four gases, each computed by the package at 46 temperatures from 100 K to 1000 K (it grows as √(T/M)). At 300 K: H_2 1775, He 1260, N_2 476.2, CO_2 379.9 m/s. Marked: the worked example, CO_2 at 300 K, 379.907 m/s.

```
chemistry_ai999x.solve('mean_speed', formula='CO2', T=300.0)
# -> u_mean = 379.907 m/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the mean speed of CO2 molecules at 300 K?	SOLVED	u_mean = 379.907 m/s
Find the average speed of He atoms at 1000 K.	SOLVED	u_mean = 2299.82 m/s

Molarity from a dissolved mass, `molarity_from_mass`

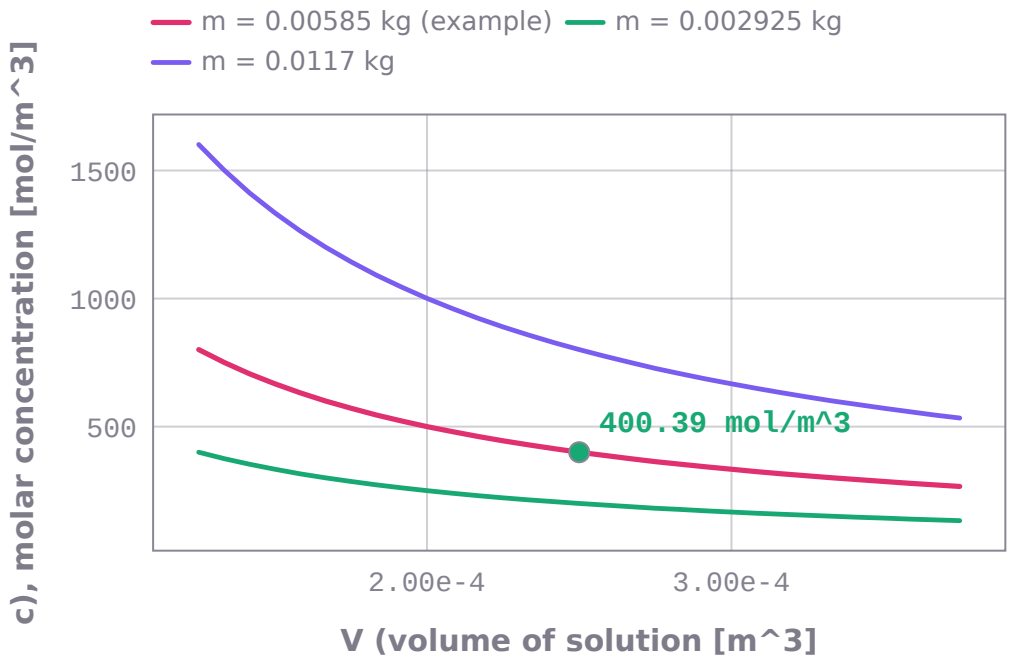
The molar concentration of a solution made by dissolving a mass of a compound to a final volume.

$$c = \frac{m}{MV}$$

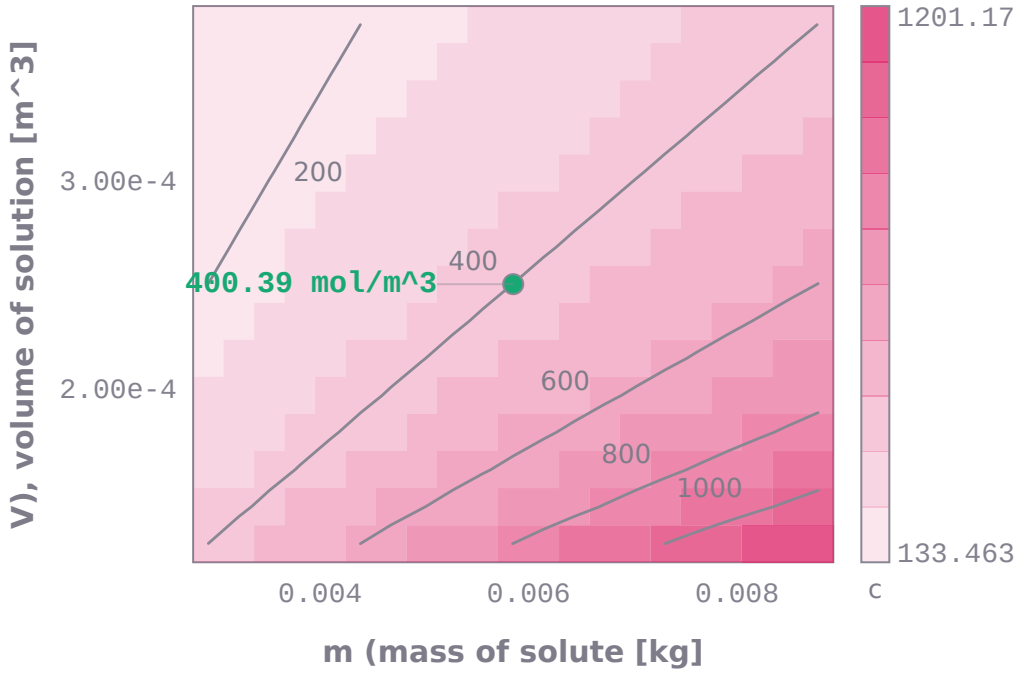
input	meaning	condition
`formula`	formula of the solute	
`m`	mass of solute	
`V`	volume of solution	

output	meaning
`c`	molar concentration
`c_molL`	concentration in mol/L

- **Method:** moles of solute (m / M) over the volume of solution
- **Assumes:** m, $V > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



c (molar concentration) against V (volume of solution) from 0.000125 m^3 to 0.000375 m^3. One curve for each value of m (mass of solute): 0.00585 kg, 0.002925 kg, 0.0117 kg. Other inputs as in the worked example: formula = NaCl. 31 points per curve computed by chemistry-ai999x; marked: the worked example, V = 0.00025 m^3, c = 400.39 mol/m^3.



c (molar concentration) over m (mass of solute) from 0.002925 kg to 0.008775 kg and V (volume of solution) from 0.000125 m^3 to 0.000375 m^3: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 133.463 mol/m^3 to dark 1201.17 mol/m^3); iso-lines at c = 200, 400, 600, 800, 1000 mol/m^3. Other inputs as in the worked example: formula = NaCl. Marked: the worked example, m = 0.00585 kg, V = 0.00025 m^3, c = 400.39 mol/m^3.

```
chemistry_ai999x.solve('molarity_from_mass', formula='NaCl', m=0.00585, V=0.00025)
# -> c = 400.39 mol/m^3, c_molL = 0.40039   (SOLVED)
```

```
chemistry_ai999x.solve('molarity_from_mass', formula='C6H12O6', m=0.09, V=0.0005)
# -> c = 999.134 mol/m^3, c_molL = 0.999134   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the molarity of a solution made by dissolving 5.85 g of NaCl in 250 mL of solution?	SOLVED	c = 400.39 mol/m^3, c_molL = 0.40039
Find the concentration of a solution of 90.0 g of C6H12O6 in 0.500 L of solution.	SOLVED	c = 999.134 mol/m^3, c_molL = 0.999134

Mass of solute for a solution, `solute_mass_for_solution`

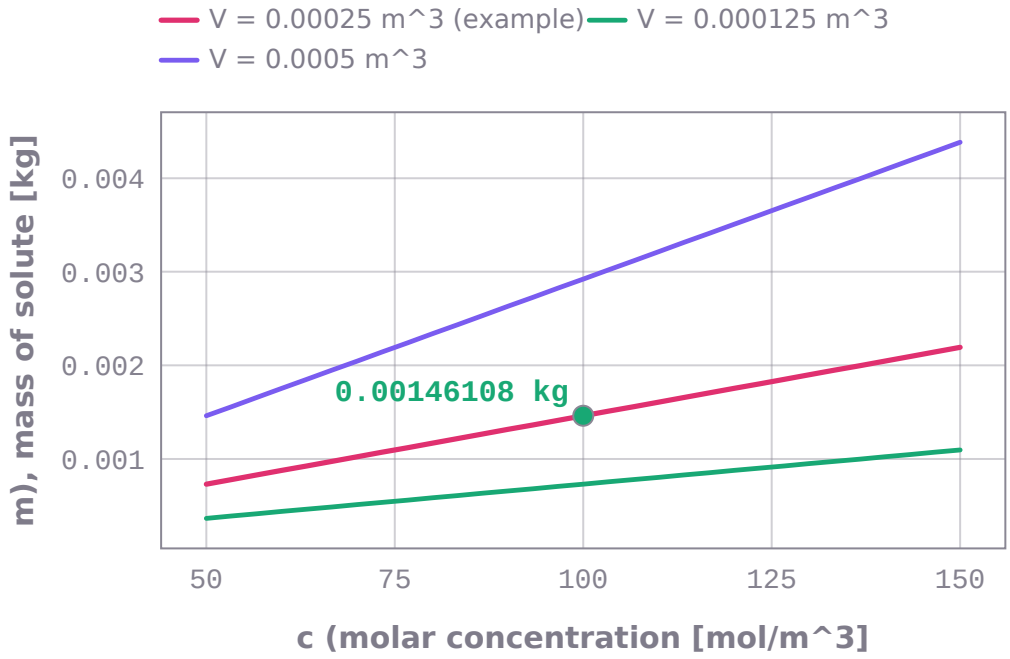
The mass of a compound needed to prepare a volume of solution of a given molarity.

$$m = cVM$$

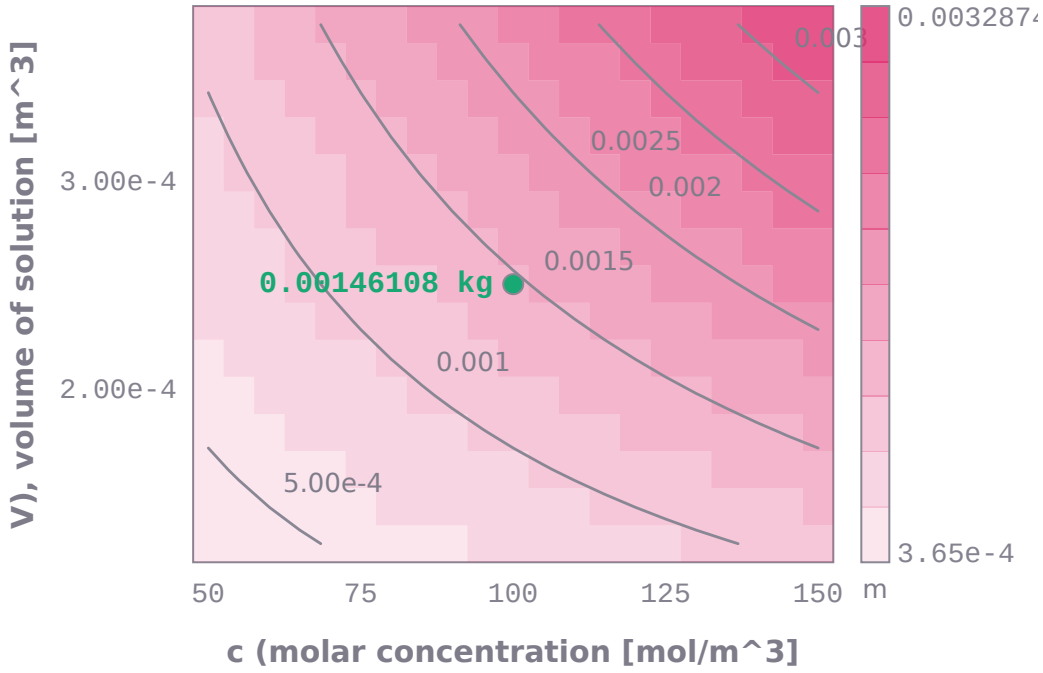
input	meaning	condition
`formula`	formula of the solute	
`c`	molar concentration	
`V`	volume of solution	

output	meaning
`m`	mass of solute
`m_g`	mass in grams

- **Method:** $m = c$ V M
- **Assumes:** c, $V > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



m (mass of solute) against c (molar concentration) from 50 mol/m^3 to 150 mol/m^3. One curve for each value of V (volume of solution): 0.00025 m^3, 0.000125 m^3, 0.0005 m^3. Other inputs as in the worked example: formula = NaCl. 31 points per curve computed by chemistry-ai999x; marked: the worked example, c = 100 mol/m^3, m = 0.00146108 kg.



m (mass of solute) over c (molar concentration) from 50 mol/m^3 to 150 mol/m^3 and V (volume of solution) from 0.000125 m^3 to 0.000375 m^3: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 0.000365269 kg to dark 0.00328742 kg); iso-lines at m = 5.00e-4, 0.001, 0.0015, 0.002, 0.0025, 0.003 kg. Other inputs as in the worked example: formula = NaCl. Marked: the worked example, c = 100 mol/m^3, V = 0.00025 m^3, m = 0.00146108 kg.

```
chemistry_ai999x.solve('solute_mass_for_solution', formula='NaCl', c=100.0, V=0.00025)
# -> m = 0.00146108 kg, m_g = 1.46108   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What mass of NaCl is needed to prepare 250 mL of 0.100 M solution?	SOLVED	m = 0.00146108 kg, m_g = 1.46108
How many grams of KMnO4 are needed to make 2.00 L of a 0.0200 M solution?	SOLVED	m = 0.00632128 kg, m_g = 6.32128

Dilution: final concentration, `dilution_final_concentration`

The concentration after diluting a volume of solution to a larger volume $(c1V1 = c2V2)$.

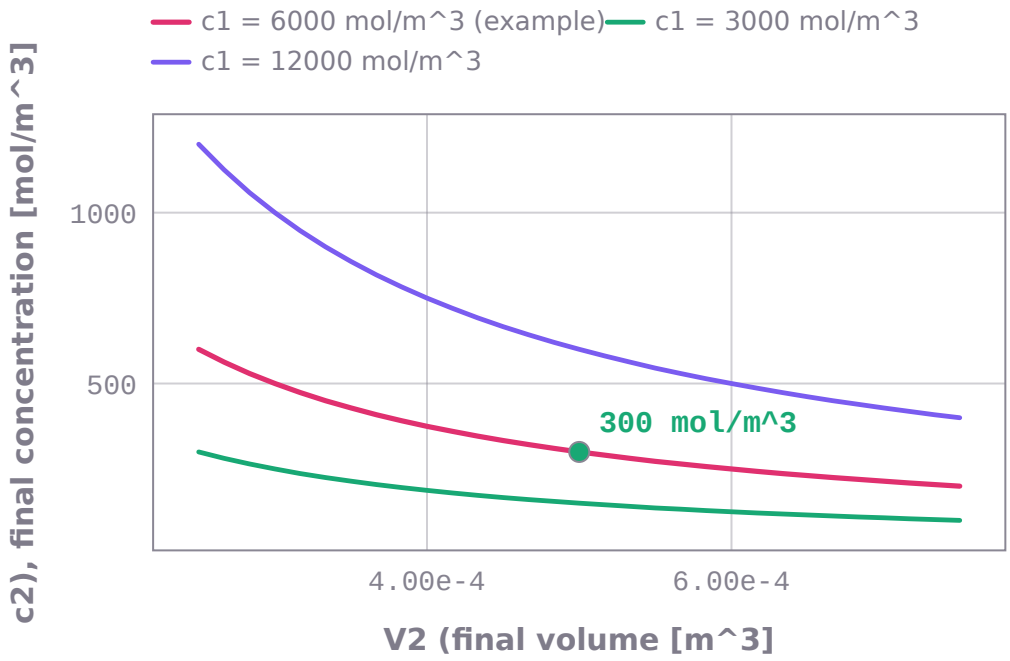
$$c_2 = \frac{c_1 V_1}{V_2}$$

input	meaning	condition
`c1`	initial concentration	
`V1`	initial volume	
`V2`	final volume	

output	meaning
`c2`	final concentration

output	meaning
`c2_molL`	in mol/L

- **Method:** moles of solute are conserved: c1 $V_1 = c_2$ V2
- **Assumes:** c1, V1, $V_2 > 0$; $V_2 \geq V_1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



c2 (final concentration) against V2 (final volume) from 0.00025 m^3 to 0.00075 m^3. One curve for each value of c1 (initial concentration): 6000 mol/m^3, 3000 mol/m^3, 12000 mol/m^3. Other inputs as in the worked example: V1 = 2.5e-05 m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, V2 = 0.0005 m^3, c2 = 300 mol/m^3.

```
chemistry_ai999x.solve('dilution_final_concentration', c1=6000.0, V1=2.5e-05, V2=0.0005)
# -> c2 = 300 mol/m^3, c2_molL = 0.3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
25.0 mL of 6.0 M HCl is diluted to 500 mL. What is the new concentration?	SOLVED	c2 = 300 mol/m^3, c2_molL = 0.3
A 10.0 mL sample of 2.00 M NaOH is diluted to a total volume of 250 mL. Find the final molarity.	SOLVED	c2 = 80 mol/m^3, c2_molL = 0.08

Dilution: volume of stock needed, `dilution_volume`

The volume of a stock solution needed to prepare a volume of a more dilute solution.

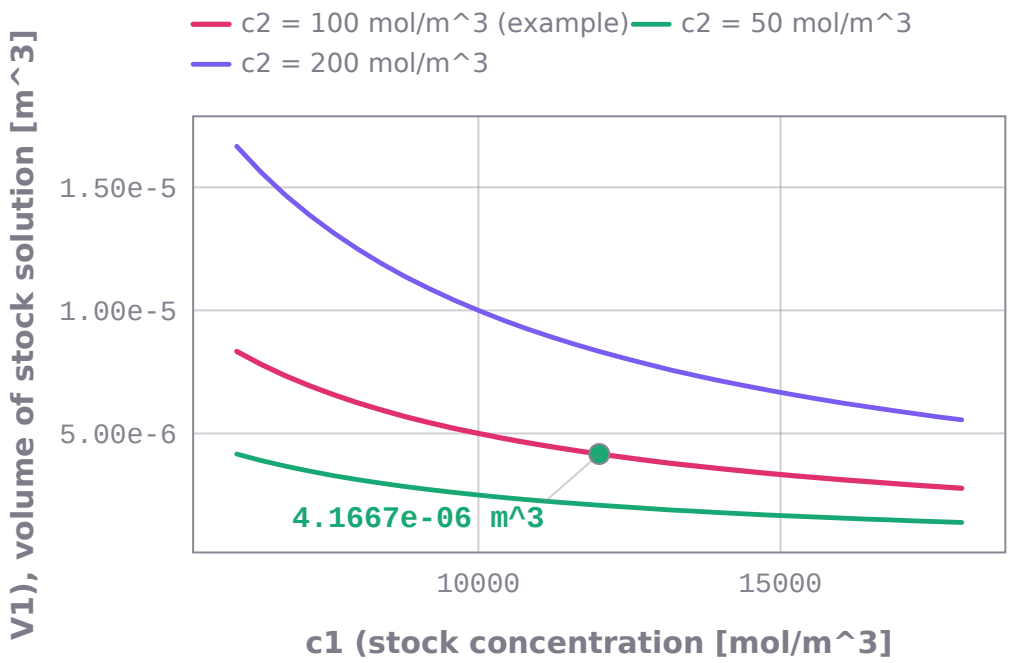
$$V_1 = \frac{c_2 V_2}{c_1}$$

input	meaning	condition
`c1`	stock concentration	

input	meaning	condition
`c2`	target concentration	
`V2`	target volume	

output	meaning
`V1`	volume of stock solution
`V1_mL`	in mL

- **Method:** c1 $V_1 = c_2$ V2 solved for V1
- **Assumes:** $c_1 \geq c_2 > 0$; $V_2 > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



V1 (volume of stock solution) against c1 (stock concentration) from 6000 mol/m^3 to 18000 mol/m^3. One curve for each value of c2 (target concentration): 100 mol/m^3, 50 mol/m^3, 200 mol/m^3. Other inputs as in the worked example: V2 = 0.0005 m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, c1 = 12000 mol/m^3, V1 = 4.1667e-06 m^3.

```
chemistry_ai999x.solve('dilution_volume', c1=12000.0, c2=100.0, V2=0.0005)
# -> V1 = 4.16667e-06 m^3, V1_mL = 4.16667   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What volume of 12.0 M HCl is needed to prepare 500 mL of 0.100 M HCl?	SOLVED	V1 = 4.16667e-06 m^3, V1_mL = 4.16667
C1 = 2.0 M, C2 = 0.50 M, V2 = 100 mL. What volume of stock solution is needed?	SOLVED	V1 = 2.5e-05 m^3, V1_mL = 25

Molality of a solution, `molality`

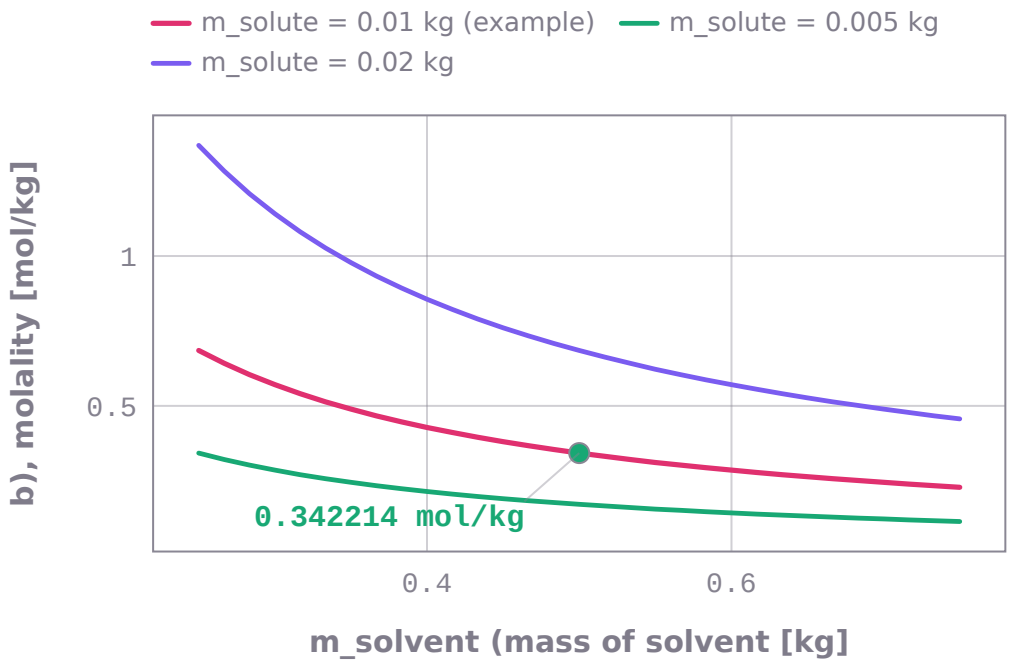
The molality (moles of solute per kilogram of solvent) from the masses of solute and solvent.

$$b = \frac{m_{\text{solute}}/M}{m_{\text{solvent}}}$$

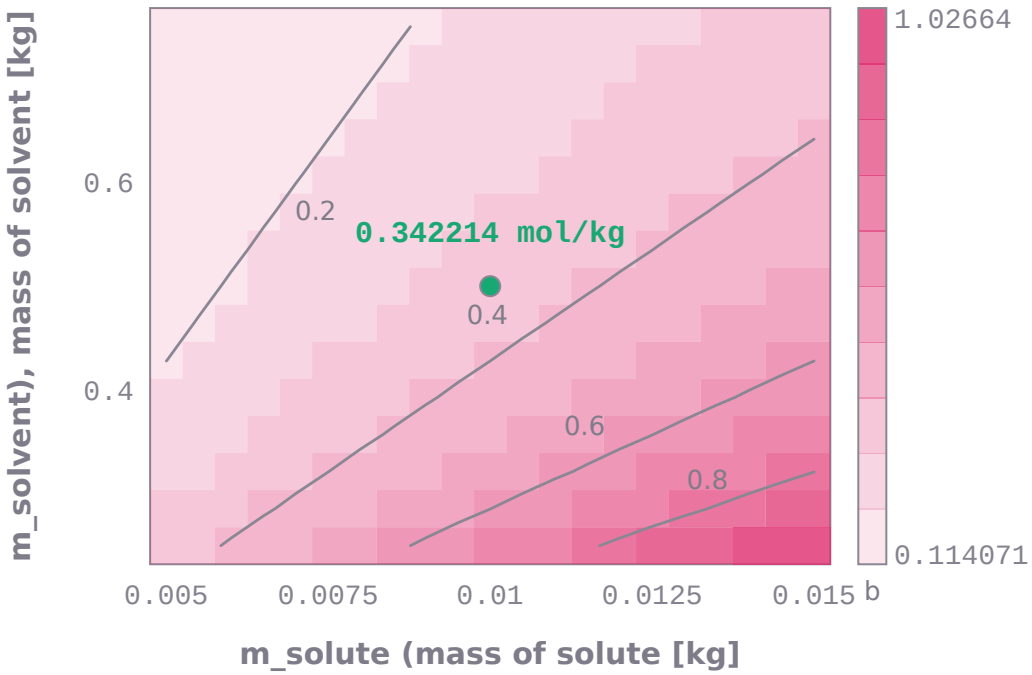
input	meaning	condition
`formula`	formula of the solute	
`m_solute`	mass of solute	
`m_solvent`	mass of solvent	

output	meaning
`b`	molality

- **Method:** moles of solute per kilogram of solvent
- **Assumes:** masses > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



b (molality) against m_solvent (mass of solvent) from 0.25 kg to 0.75 kg. One curve for each value of m_solute (mass of solute): 0.01 kg, 0.005 kg, 0.02 kg. Other inputs as in the worked example: formula = NaCl. 31 points per curve computed by chemistry-ai999x; marked: the worked example, m_solvent = 0.5 kg, b = 0.342214 mol/kg.



b (molality) over m_solute (mass of solute) from 0.005 kg to 0.015 kg and m_solvent (mass of solvent) from 0.25 kg to 0.75 kg: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 0.114071 mol/kg to dark 1.02664 mol/kg); isolines at b = 0.2, 0.4, 0.6, 0.8 mol/kg. Other inputs as in the worked example: formula = NaCl. Marked: the worked example, m_solute = 0.01 kg, m_solvent = 0.5 kg, b = 0.342214 mol/kg.

```
chemistry_ai999x.solve('molality', formula='NaCl', m_solute=0.01, m_solvent=0.5)
# -> b = 0.342214 mol/kg   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the molality of a solution of 10.0 g of NaCl dissolved in 500 g of water?	SOLVED	b = 0.342214 mol/kg
Find the molality when 45.0 g of C6H12O6 is dissolved in 1.00 kg of water.	SOLVED	b = 0.249784 mol/kg

Mole fraction in a solution, `mole_fraction`

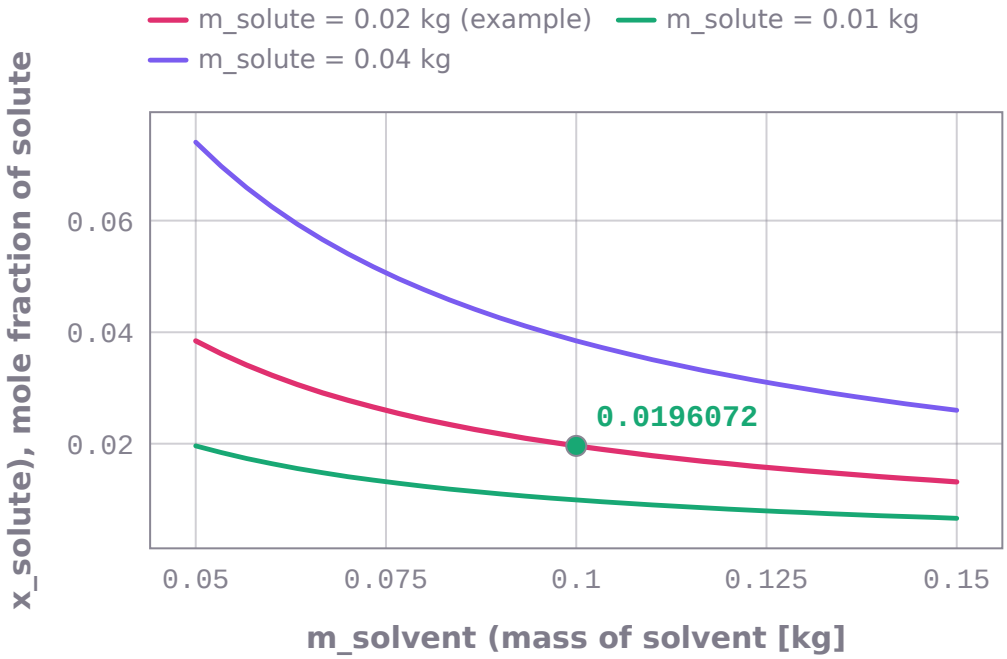
The mole fractions of solute and solvent from their masses.

$$x_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$$

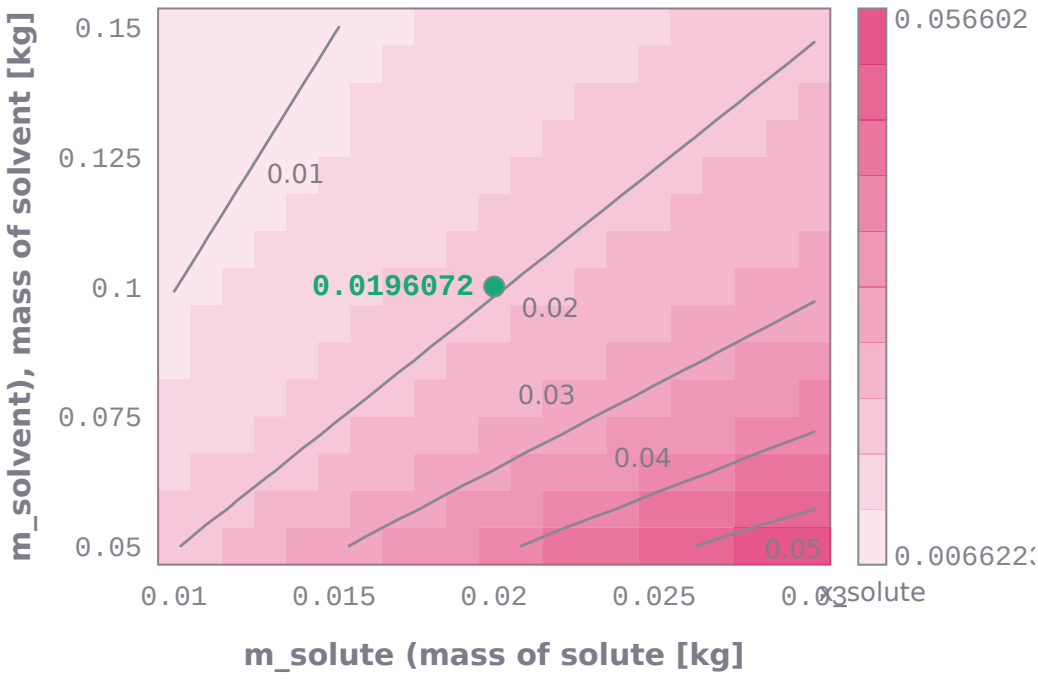
input	meaning	condition
`formula_solute`	formula of the solute	
`m_solute`	mass of solute	
`formula_solvent`	formula of the solvent	
`m_solvent`	mass of solvent	

output	meaning
`x_solute`	mole fraction of solute
`x_solvent`	mole fraction of solvent

- **Method:** moles of each from its mass and molar mass
- **Assumes:** masses > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



x_solute (mole fraction of solute) against m_solvent (mass of solvent) from 0.05 kg to 0.15 kg. One curve for each value of m_solute (mass of solute): 0.02 kg, 0.01 kg, 0.04 kg. Other inputs as in the worked example: formula_solute = C6H12O6, formula_solvent = H2O. 31 points per curve computed by chemistry-ai999x; marked: the worked example, m_solvent = 0.1 kg, x_solute = 0.0196072.



x_solute (mole fraction of solute) over m_solute (mass of solute) from 0.01 kg to 0.03 kg and m_solvent (mass of solvent) from 0.05 kg to 0.15 kg: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 0.0066223 to dark 0.056602); iso-lines at x_solute = 0.01, 0.02, 0.03, 0.04, 0.05. Other inputs as in the worked example: formula_solute = C6H12O6, formula_solvent = H2O. Marked: the worked example, m_solute = 0.02 kg, m_solvent = 0.1 kg, x_solute = 0.0196072.

```
chemistry_ai999x.solve('mole_fraction', formula_solute='C6H12O6', m_solute=0.02,
formula_solvent='H2O', m_solvent=0.1)
# -> x_solute = 0.0196072, x_solvent = 0.980393   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the mole fraction of solute in a solution of 20.0 g of C6H12O6 dissolved in 100 g of H2O?	SOLVED	x_solute = 0.0196072, x_solvent = 0.980393
Find the mole fractions for 46.0 g of C2H5OH in 100 g of H2O.	SOLVED	x_solute = 0.152456, x_solvent = 0.847544

Vapor pressure of a solution (Raoult's law), `raoult_vapor_pressure`

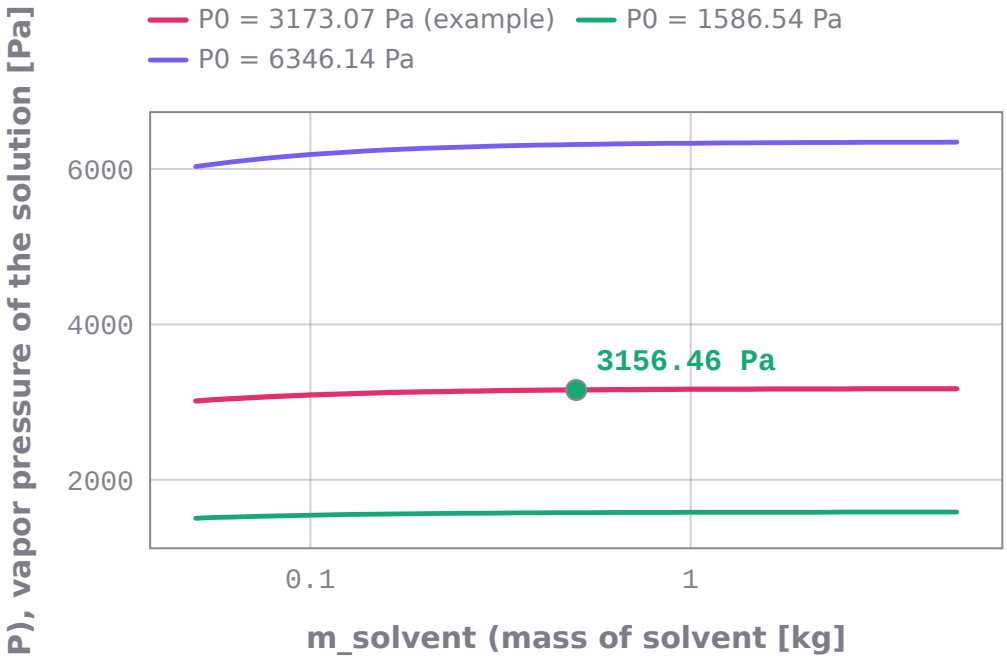
The vapor pressure of a solvent above a solution of a nonvolatile solute, from the masses of both.

$$P = x_{\text{solvent}} P^{\circ}$$

input	meaning	condition
`formula_solute`	formula of the solute	
`m_solute`	mass of solute	
`formula_solvent`	formula of the solvent	
`m_solvent`	mass of solvent	
`P0`	vapor pressure of the pure solvent	

output	meaning
`P`	vapor pressure of the solution
`dP`	vapor pressure lowering

- **Method:** Raoult's law with the solvent mole fraction from the masses
- **Assumes:** nonvolatile, non-dissociating solute; ideal solution
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



P (vapor pressure of the solution) against m_solvent (mass of solvent) from 0.05 kg to 5 kg on a logarithmic axis. One curve for each value of P0 (vapor pressure of the pure solvent): 3173.07 Pa, 1586.54 Pa, 6346.14 Pa. Other inputs as in the worked example: formula_solute = C12H22O11, m_solute = 0.05 kg, formula_solvent = H2O. 41 points per curve computed by chemistry-ai999x; marked: the worked example, m_solvent = 0.5 kg, P = 3156.46 Pa.

```
chemistry_ai999x.solve('raoult_vapor_pressure', formula_solute='C12H22O11', m_solute=0.05,
formula_solvent='H2O', m_solvent=0.5, P0=3173.0723684210525)
# -> P = 3156.46 Pa, dP = 16.6124 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the vapor pressure of a solution of 50.0 g of C12H22O11 in 500 g of H2O if the vapor pressure of pure water is 23.8 torr?	SOLVED	P = 3156.46 Pa, dP = 16.6124 Pa

Freezing-point depression, `freezing_point_depression`

The change of the freezing point of a solvent caused by a solute (dT = i K b).

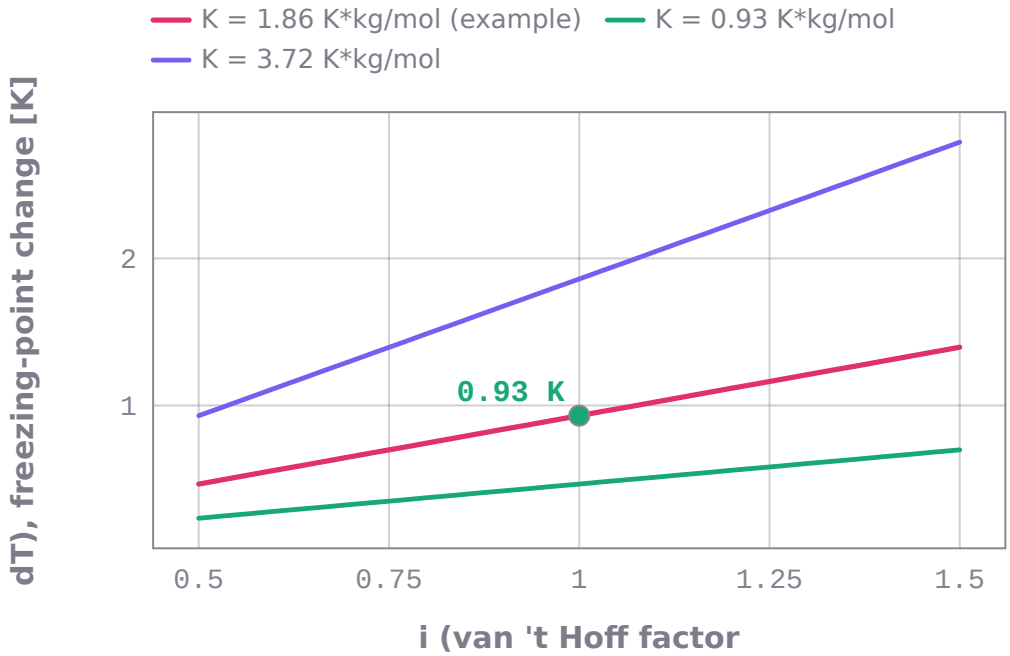
$$\Delta T = iKb$$

input	meaning	condition
`i`	van 't Hoff factor	
`K`	cryoscopic constant	
`b`	molality of the solute	

output	meaning
`dT`	freezing-point change

- **Method:** $dT = i$ K b
- **Assumes:** i, K, $b > 0$; dilute solution

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dT (freezing-point change) against i (van 't Hoff factor) from 0.5 to 1.5. One curve for each value of K (cryoscopic constant): 1.86 K*kg/mol, 0.93 K*kg/mol, 3.72 K*kg/mol. Other inputs as in the worked example: b = 0.5 mol/kg. 31 points per curve computed by chemistry-ai999x; marked: the worked example, i = 1, dT = 0.93 K.

```
chemistry_ai999x.solve('freezing_point_depression', i=1.0, K=1.86, b=0.5)
# -> dT = 0.93 K   (SOLVED)
```

```
chemistry_ai999x.solve('freezing_point_depression', i=2.0, K=1.86, b=0.2)
# -> dT = 0.744 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the freezing point change of a 0.50 mol/kg aqueous solution of a nonelectrolyte (Kf = 1.86 K·kg/mol)?	SOLVED	dT = 0.93 K
Find the freezing point change for b = 0.20 mol/kg, i = 2, Kf = 1.86 °C/m.	SOLVED	dT = 0.744 K
What is the freezing point change of a 0.50 mol/kg solution with Kf = 1.86 K·kg/mol?	NEEDS_INPUT	freezing_point_depression needs i (van 't Hoff factor). Example: solve('freezing_point_depression', i=1, K='1.86 K*kg/mo

Boiling-point elevation, `boiling_point_elevation`

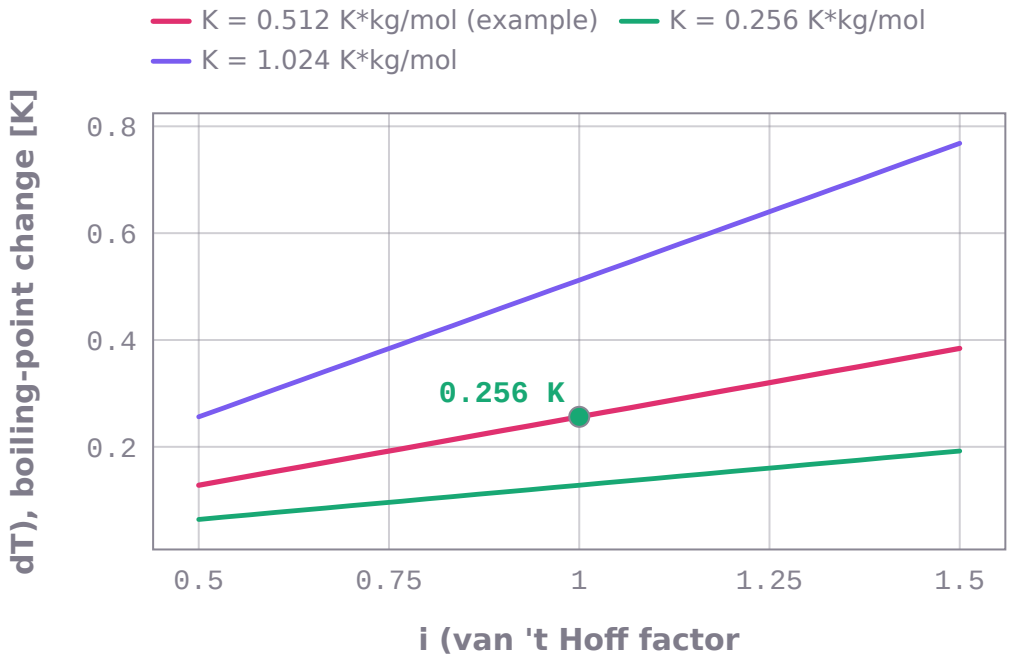
The change of the boiling point of a solvent caused by a solute (dT = i K b).

$$\Delta T = iKb$$

input	meaning	condition
`i`	van 't Hoff factor	
`K`	ebullioscopic constant	
`b`	molality of the solute	

output	meaning
`dT`	boiling-point change

- **Method:** $dT = i$ K b
- **Assumes:** i, K, $b > 0$; dilute solution
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dT (boiling-point change) against i (van 't Hoff factor) from 0.5 to 1.5. One curve for each value of K (ebullioscopic constant): 0.512 K*kg/mol, 0.256 K*kg/mol, 1.024 K*kg/mol. Other inputs as in the worked example: b = 0.5 mol/kg. 31 points per curve computed by chemistry-ai999x; marked: the worked example, i = 1, dT = 0.256 K.

```
chemistry_ai999x.solve('boiling_point_elevation', i=1.0, K=0.512, b=0.5)
# -> dT = 0.256 K   (SOLVED)
```

```
chemistry_ai999x.solve('boiling_point_elevation', i=2.0, K=0.512, b=0.2)
# -> dT = 0.2048 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the boiling point change of a 0.50 mol/kg aqueous solution of a nonelectrolyte (Kb = 0.512 K·kg/mol)?	SOLVED	dT = 0.256 K
Find the boiling point change for b = 0.20 mol/kg, i = 2, Kb = 0.512 °C/m.	SOLVED	dT = 0.2048 K

question	outcome	result
What is the boiling point change of a 0.50 mol/kg solution with Kb = 0.512 K·kg/mol?	NEEDS_INPUT	boiling_point_elevation needs i (van 't Hoff factor). Example: solve('boiling_point_elevation', i=1, K='0.512 K*kg/mol',

Molar mass from freezing-point depression, `molar_mass_from_freezing_point`

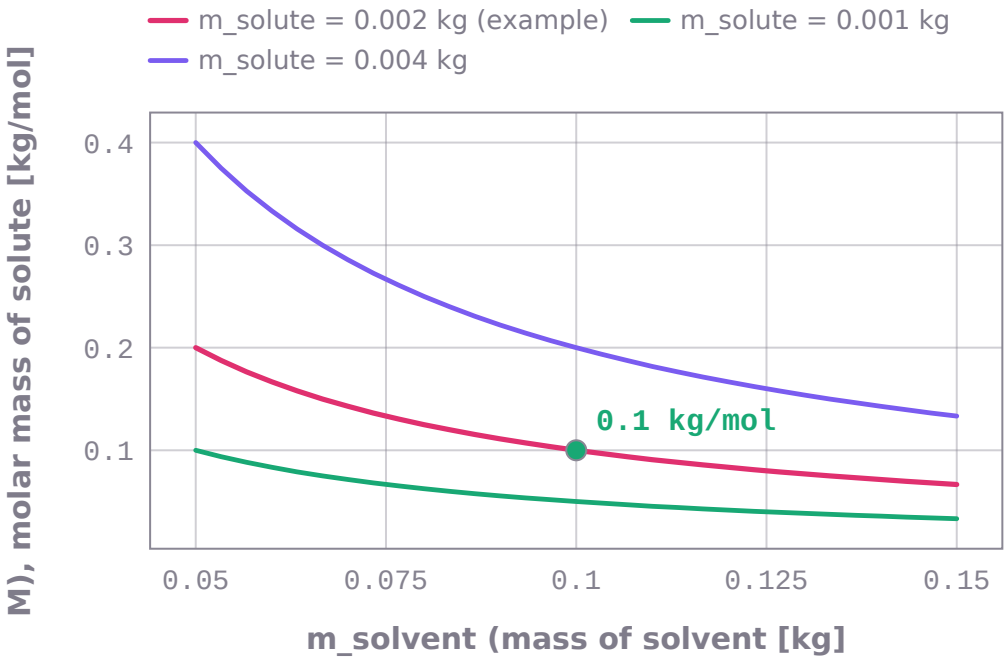
The molar mass of a solute from the freezing-point depression it causes.

$$M = \frac{i K_f m_{\text{solute}}}{\Delta T_f \, m_{\text{solvent}}}$$

input	meaning	condition
`m_solute`	mass of solute	
`m_solvent`	mass of solvent	
`K`	cryoscopic constant Kf	
`dT`	freezing-point depression	
`i`	van 't Hoff factor	

output	meaning
`M`	molar mass of solute
`M_g`	in g/mol

- **Method:** b = dT/(i Kf); M = m_solute / (b m_solvent)
- **Assumes:** all inputs > 0; dilute solution
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



M (molar mass of solute) against m_solvent (mass of solvent) from 0.05 kg to 0.15 kg. One curve for each value of m_solute (mass of solute): 0.002 kg, 0.001 kg, 0.004 kg. Other inputs as in the worked example: K = 1.86 K*kg/mol, dT = 0.372 K, i = 1.0. 31 points per curve computed by chemistry-ai999x; marked: the worked example, m_solvent = 0.1 kg, M = 0.1 kg/mol.

```
chemistry_ai999x.solve('molar_mass_from_freezing_point', m_solute=0.002, m_solvent=0.1,
K=1.86, dT=0.372, i=1.0)
# -> M = 0.1 kg/mol, M_g = 100   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Dissolving 2.00 g of a nonelectrolyte in 100 g of water lowers the freezing point by 0.372 °C (Kf = 1.86 K·kg/mol). What is the molar mass of the solute?	SOLVED	M = 0.1 kg/mol, M_g = 100
A solution of 1.00 g of a nonelectrolyte in 50.0 g of water freezes at -0.200 °C (Kf = 1.86 K·kg/mol). What is the molar mass of the solute?	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemistry_ai999x.templates(); eve

Osmotic pressure, `osmotic_pressure`

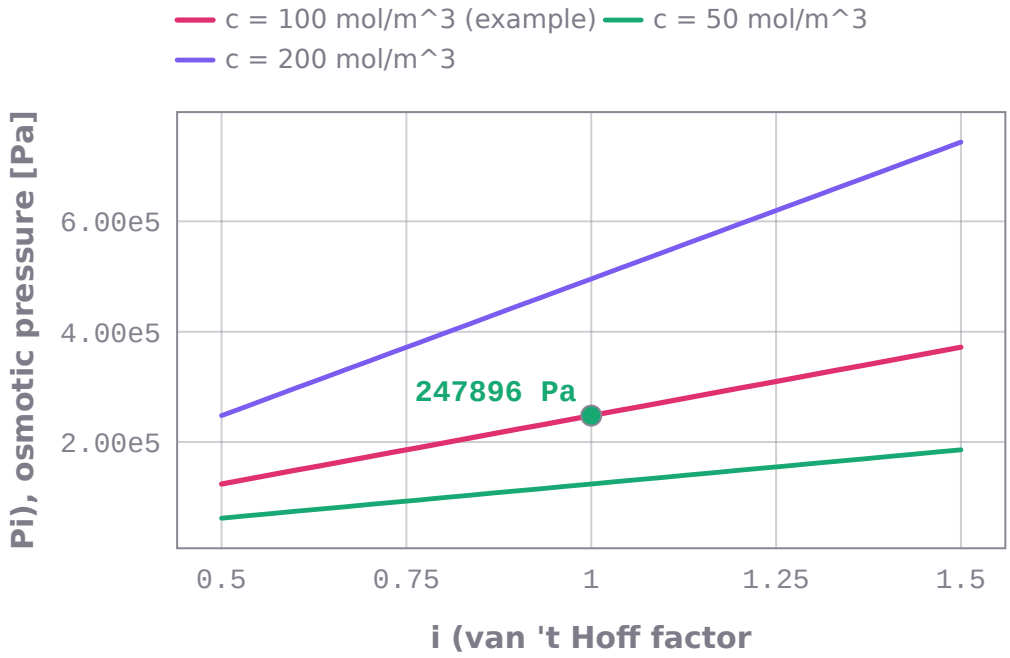
The osmotic pressure of a dilute solution (van 't Hoff equation).

$$\Pi = icRT$$

input	meaning	condition
`i`	van 't Hoff factor	
`c`	molar concentration	
`T`	temperature	

output	meaning
`Pi`	osmotic pressure
`Pi_atm`	in atm

- **Method:** $Pi = i$ c R T
- **Assumes:** i, c, $T > 0$; dilute solution
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Pi (osmotic pressure) against i (van 't Hoff factor) from 0.5 to 1.5. One curve for each value of c (molar concentration): 100 mol/m^3, 50 mol/m^3, 200 mol/m^3. Other inputs as in the worked example: T = 298.15 K. 31 points per curve computed by chemistry-ai999x; marked: the worked example, i = 1, Pi = 247896 Pa.

```
chemistry_ai999x.solve('osmotic_pressure', i=1.0, c=100.0, T=298.15)
# -> Pi = 247896 Pa, Pi_atm = 2.44654   (SOLVED)
```

```
chemistry_ai999x.solve('osmotic_pressure', i=2.0, c=150.0, T=310.0)
# -> Pi = 773245 Pa, Pi_atm = 7.63134   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the osmotic pressure of a 0.100 M glucose solution (a nonelectrolyte) at 25 °C?	SOLVED	Pi = 247896 Pa, Pi_atm = 2.44654
Find the osmotic pressure of 0.150 M NaCl at 310 K with i = 2.	SOLVED	Pi = 773245 Pa, Pi_atm = 7.63134
What is the osmotic pressure of 0.150 M NaCl at 310 K?	NEEDS_INPUT	osmotic_pressure needs i (van 't Hoff factor). Example: solve('osmotic_pressure', i=1, c='100 mol/m^3', T='298.15 K').

Molar mass from osmotic pressure, `molar_mass_from_osmotic_pressure`

The molar mass of a solute (e.g. a protein) from the osmotic pressure of its solution.

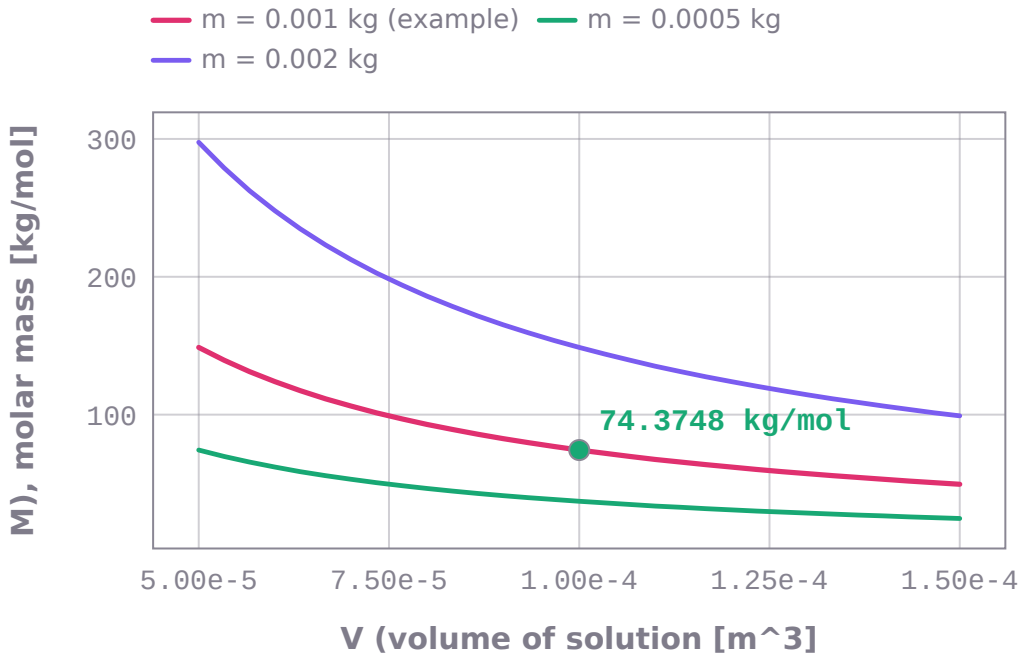
$$M = \frac{imRT}{\Pi V}$$

input	meaning	condition
`m`	mass of solute	
`V`	volume of solution	

input	meaning	condition
`Pi`	osmotic pressure	
`T`	temperature	
`i`	van 't Hoff factor	

output	meaning
`M`	molar mass
`M_g`	in g/mol

- **Method:** c = Pi/(i R T); M = m / (c V)
- **Assumes:** all inputs > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



M (molar mass) against V (volume of solution) from 5.0000e-05 m^3 to 0.00015 m^3. One curve for each value of m (mass of solute): 0.001 kg, 0.0005 kg, 0.002 kg. Other inputs as in the worked example: Pi = 333.30592105263156 Pa, T = 298.15 K, i = 1.0. 31 points per curve computed by chemistry-ai999x; marked: the worked example, V = 0.0001 m^3, M = 74.3748 kg/mol.

```
chemistry_ai999x.solve('molar_mass_from_osmotic_pressure', m=0.001, V=0.0001,
Pi=333.30592105263156, T=298.15, i=1.0)
# -> M = 74.3748 kg/mol, M_g = 74374.8   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A solution of 1.00 g of a protein (a nonelectrolyte) in 100 mL of water has an osmotic pressure of 2.50 torr at 25 °C. What is the molar mass of the protein?	SOLVED	M = 74.3748 kg/mol, M_g = 74374.8

Kp from Kc, `kp_from_kc`

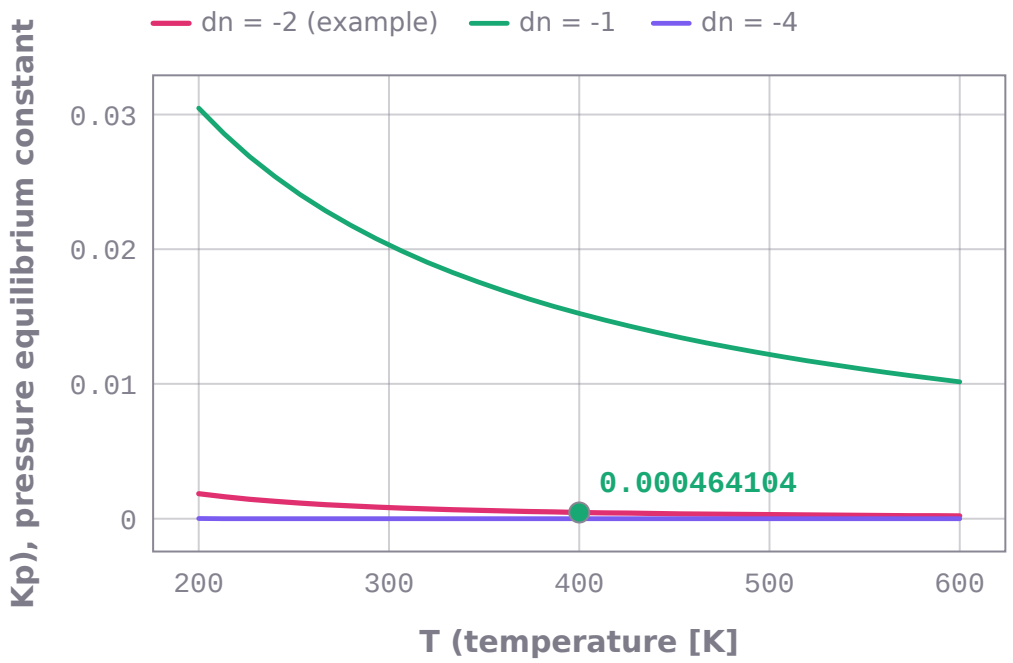
The pressure equilibrium constant from the concentration one (Kp = Kc $(RT)^{dn}$, atm and mol/L).

$$K_p = K_c(RT)^{\Delta n}$$

input	meaning	condition
Kc	concentration equilibrium constant	
T	temperature	
dn	change in moles of gas (products - reactants)	

output	meaning
Kp	pressure equilibrium constant (atm basis)

- **Method:** $Kp = Kc\,(RT)^{dn}$ with $R = 0.082057$ L·atm/(mol·K)
- **Assumes:** Kc, $T > 0$; dn counts gaseous species only
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Kp (pressure equilibrium constant (atm basis)) against T (temperature) from 200 K to 600 K. One curve for each value of dn (change in moles of gas (products - reactants)): -2, -1, -4. Other inputs as in the worked example: Kc = 0.5. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T = 400 K, Kp = 0.000464104.

```
chemistry_ai999x.solve('kp_from_kc', Kc=0.5, T=400.0, dn=-2.0)
# -> Kp = 0.000464104   (SOLVED)
```

```
chemistry_ai999x.solve('kp_from_kc', Kc=4.0, T=500.0, dn=1.0)
# -> Kp = 164.115   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For N2(g) + 3H2(g) ⇌ 2NH3(g), Kc = 0.50 at 400 K. What is Kp?	SOLVED	Kp = 0.000464104

question	outcome	result
Kc = 4.0 at 500 K and Δn = 1. Find Kp.	SOLVED	Kp = 164.115
Kc = 4.0 at 500 K. Find Kp.	NEEDS_INPUT	kp_from_kc needs dn (change in moles of gas (products - reactants)). Example: solve('kp_from_kc', Kc=0.5, T='400 K', dn=

Kc from Kp, `kc_from_kp`

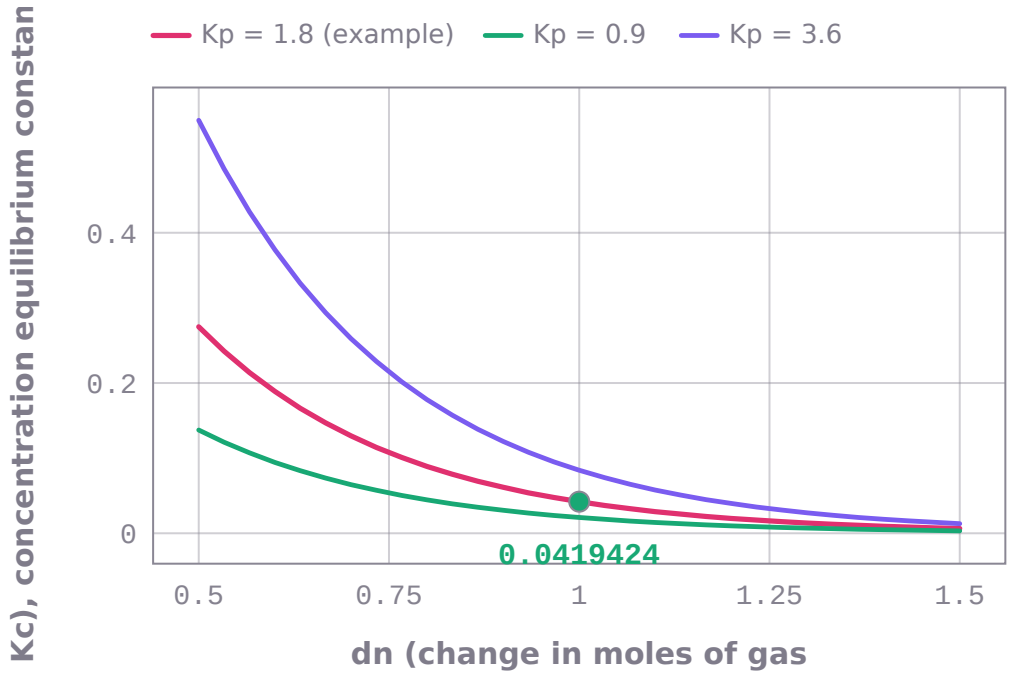
The concentration equilibrium constant from the pressure one $(Kc = Kp/(RT)^{dn})$.

$$K_c = \frac{K_p}{(RT)^{\Delta n}}$$

input	meaning	condition
`Kp`	pressure equilibrium constant (atm basis)	
`T`	temperature	
`dn`	change in moles of gas (products - reactants)	

output	meaning
`Kc`	concentration equilibrium constant

- **Method:** $Kc = Kp/(RT)^{dn}$ with $R = 0.082057$ L·atm/(mol·K)
- **Assumes:** Kp, $T > 0$; dn counts gaseous species only
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Kc (concentration equilibrium constant) against dn (change in moles of gas (products - reactants)) from 0.5 to 1.5. One curve for each value of Kp (pressure equilibrium constant (atm basis)): 1.8, 0.9, 3.6. Other inputs as in the worked example: T = 523.0 K. 31 points per curve computed by chemistry-ai999x; marked: the worked example, dn = 1, Kc = 0.0419424.

```
chemistry_ai999x.solve('kc_from_kp', Kp=1.8, T=523.0, dn=1.0)
# -> Kc = 0.0419424   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For PCl5(g) ⇌ PCl3(g) + Cl2(g), Kp = 1.8 at 523 K. What is Kc?	SOLVED	Kc = 0.0419424
Kp = 0.25 at 700 K and Δn = -1. Find Kc.	SOLVED	Kc = 14.36

Equilibrium concentrations (ICE table), `equilibrium_ice`

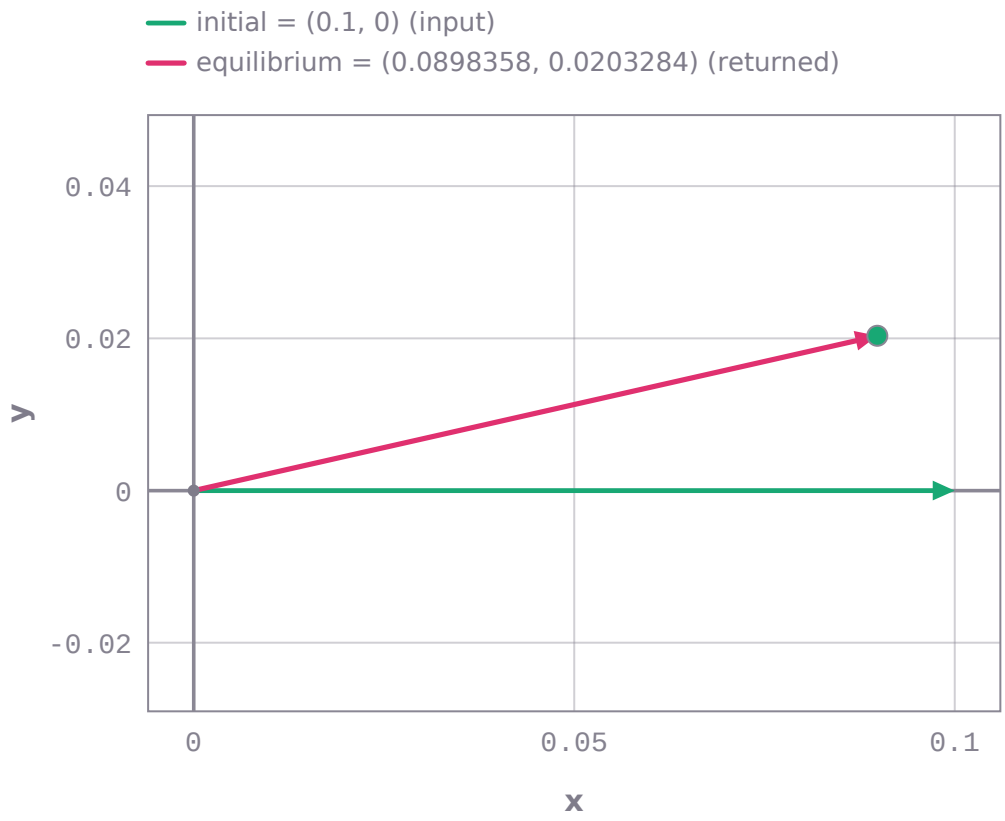
Equilibrium concentrations (or partial pressures) from the initial ones and K, for any one reaction as written.

$$K = \prod_i (c_{i,0} + \nu_i x)^{\nu_i}$$

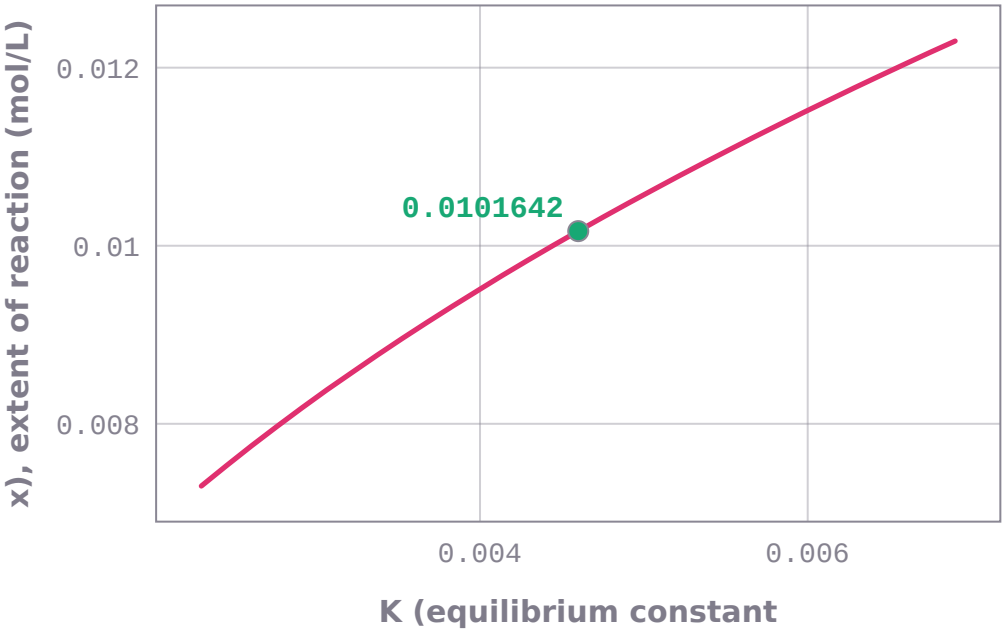
input	meaning	condition
`equation`	the reaction as written (its coefficients define K)	
`initial`	initial concentration of each species in the order written (mol/L, or atm for Kp)	
`K`	equilibrium constant	

output	meaning
`x`	extent of reaction (mol/L)
`equilibrium`	equilibrium concentrations in the order written

- **Method:** the extent x solved from the mass-action expression by bisection between the limits set by the initial amounts
- **Assumes:** one reaction, coefficients as written; no pure solids or liquids; $K > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The plane vectors of the worked example as arrows from the origin: input initial = (0.1, 0); returned by chemistry-ai999x: equilibrium = (0.0898358, 0.0203284); also returned x = 0.0101642. Other inputs: equation = N2O4 ⇌ 2 NO2, K = 0.0046. Marked: the tip of each returned vector.



x (extent of reaction (mol/L)) against K (equilibrium constant) from 0.0023 to 0.0069. Other inputs as in the worked example: equation = N2O4 ⇌ 2 NO2, initial = 0.10, 0. 31 points computed by chemistry-ai999x; marked: the worked example, K = 0.0046, x = 0.0101642.

```
chemistry_ai999x.solve('equilibrium_ice', equation='N2O4 ⇌ 2 NO2', initial='0.10, 0',
K=0.0046)
# -> x = 0.0101642, equilibrium = [0.08983579029909557, 0.020328419401808878]   (SOLVED)
```

```
chemistry_ai999x.solve('equilibrium_ice', equation='H2 + I2 ⇌ 2 HI', initial='0.50, 0.50, 0', K=50.0)
# -> x = 0.389759, equilibrium = [0.11024060460577123, 0.11024060460577123, 0.7795187907884575]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For N2O4 ⇌ 2 NO2, Kc = 4.6e-3. Initially [N2O4] = 0.10 M and [NO2] = 0. What are the equilibrium concentrations?	SOLVED	x = 0.0101642, equilibrium = [0.08983579029909557, 0.020328419401808878]
For H2 + I2 ⇌ 2 HI, K = 50.0. Initially [H2] = 0.50 M, [I2] = 0.50 M and [HI] = 0. Find the equilibrium concentrations.	SOLVED	x = 0.389759, equilibrium = [0.11024060460577123, 0.11024060460577123, 0.7795187907884575]
For N2O4 ⇌ 2 NO2, Kc = 4.6e-3. Initially [N2O4] = 0.10 M. What are the equilibrium concentrations?	NEEDS_INPUT	equilibrium_ice needs initial (initial concentration of each species in the order written (mol/L, or atm for Kp)). Examp

Equilibrium constant from the standard Gibbs energy, `k_from_delta_g`

The equilibrium constant from the standard Gibbs energy change (K = exp(-dG°/RT)).

$$K = e^{-\Delta G^\circ / RT}$$

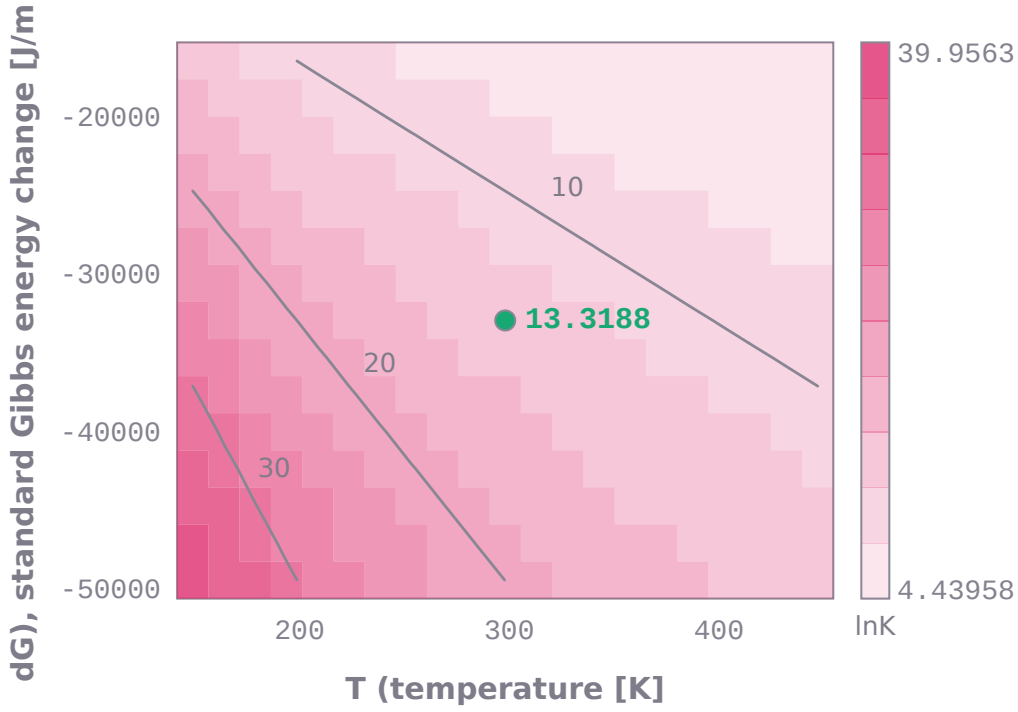
input	meaning	condition
`dG`	standard Gibbs energy change	
`T`	temperature	

output	meaning
`K`	equilibrium constant
`lnK`	ln K

- **Method:** ln $K = -dG^\circ$/(RT)
- **Assumes:** $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



lnK (ln K) against T (temperature) from 149 K to 447 K. One curve for each value of dG (standard Gibbs energy change): -33000 J/mol, -16500 J/mol, -66000 J/mol. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T = 298 K, lnK = 13.3188.



lnK (ln K) over T (temperature) from 149 K to 447 K and dG (standard Gibbs energy change) from -49500 J/mol to -16500 J/mol: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 4.43958 to dark 39.9563); iso-lines at lnK = 10, 20, 30. Marked: the worked example, T = 298 K, dG = -33000 J/mol, lnK = 13.3188.

```
chemistry_ai999x.solve('k_from_delta_g', dG=-33000.0, T=298.0)
# -> K = 608499, lnK = 13.3188   (SOLVED)
```

```
chemistry_ai999x.solve('k_from_delta_g', dG=10000.0, T=500.0)
# -> K = 0.0902251, lnK = -2.40545   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
ΔG° = -33.0 kJ/mol at 298 K. What is the equilibrium constant?	SOLVED	K = 608499, lnK = 13.3188
Find the equilibrium constant K at 500 K if the standard Gibbs energy change is 10.0 kJ/mol.	SOLVED	K = 0.0902251, lnK = -2.40545

Standard Gibbs energy from the equilibrium constant, `delta_g_from_k`

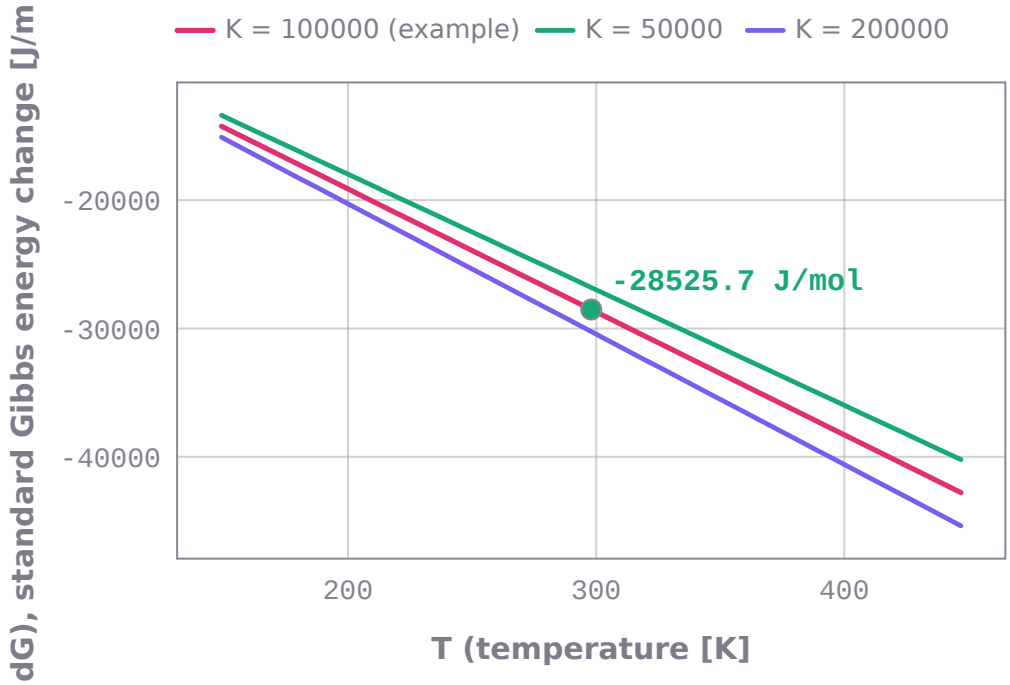
The standard Gibbs energy change from the equilibrium constant (dG° = -RT ln K).

$$\Delta G^\circ = -RT \ln K$$

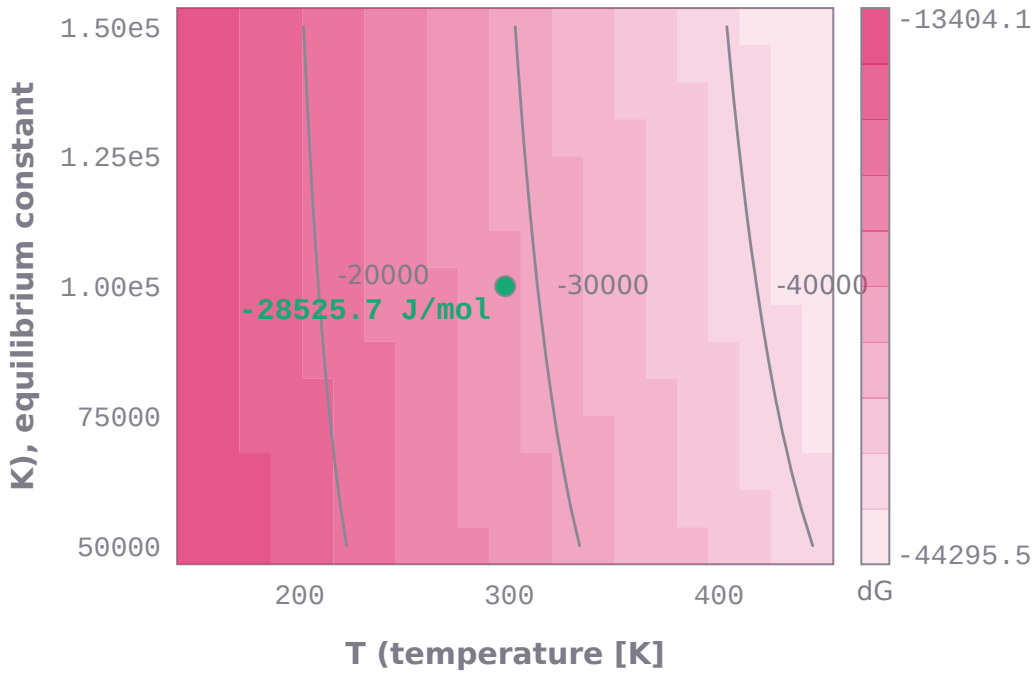
input	meaning	condition
`K`	equilibrium constant	
`T`	temperature	

output	meaning
`dG`	standard Gibbs energy change

- **Method:** dG° = -RT ln K
- **Assumes:** K, $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dG (standard Gibbs energy change) against T (temperature) from 149 K to 447 K. One curve for each value of K (equilibrium constant): 100000, 50000, 200000. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T = 298 K, dG = -28525.7 J/mol.



dG (standard Gibbs energy change) over T (temperature) from 149 K to 447 K and K (equilibrium constant) from 50000 to 150000: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light -44295.5 J/mol to dark -13404.1 J/mol); iso-lines at dG = -40000, -30000, -20000 J/mol. Marked: the worked example, T = 298 K, K = 100000, dG = -28525.7 J/mol.

```
chemistry_ai999x.solve('delta_g_from_k', K=100000.0, T=298.0)
# -> dG = -28525.7 J/mol   (SOLVED)
```

```
chemistry_ai999x.solve('delta_g_from_k', K=0.25, T=400.0)
# -> dG = 4610.52 J/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
K = 1.0e5 at 298 K. What is ΔG°?	SOLVED	dG = -28525.7 J/mol
Find the standard Gibbs energy change if the equilibrium constant is 0.25 at 400 K.	SOLVED	dG = 4610.52 J/mol

Van 't Hoff equation: K at another temperature, `vant_hoff_k2`

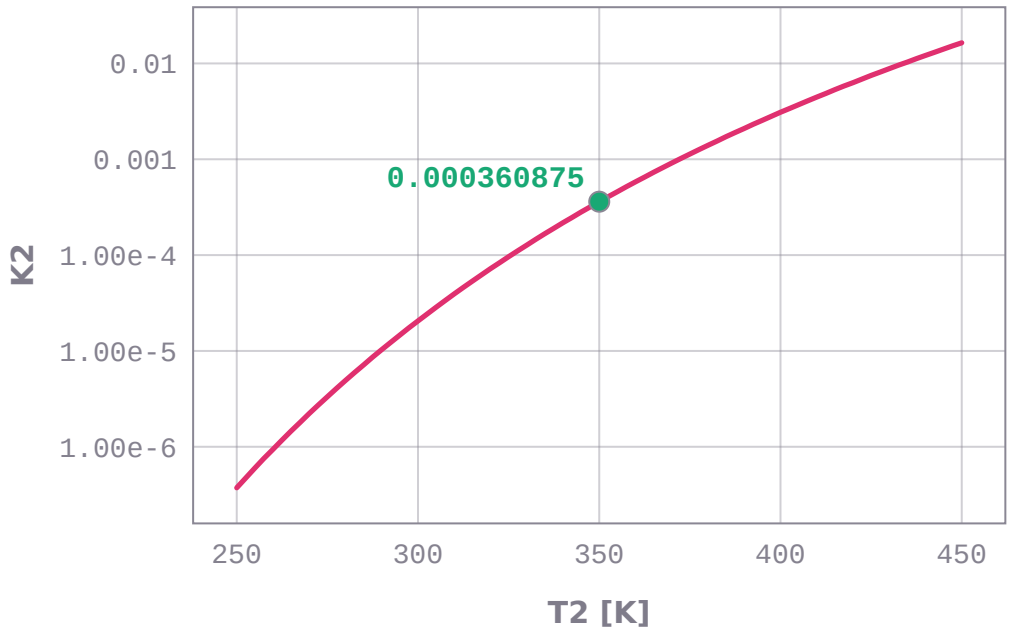
The equilibrium constant at a second temperature from K at a first one and the reaction enthalpy.

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

input	meaning	condition
`K1`	equilibrium constant at T1	
`T1`	first temperature	
`T2`	second temperature	
`dH`	standard enthalpy change	

output	meaning
`K2`	equilibrium constant at T2
`lnK2`	ln K2

- **Method:** integrated van 't Hoff equation (dH° constant)
- **Assumes:** K1, T1, $T_2 > 0$; dH° independent of T
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Van 't Hoff equation: K at another temperature: K2 against T2 from 250 to 450, 81 runs of the package; other inputs as in the worked example (K1 = 1.8e-05, T1 = 298.0, dH = 50000.0). Marked: the worked example, T2 = 350, K2 = 0.000360875.

```
chemistry_ai999x.solve('vant_hoff_k2', K1=1.8e-05, T1=298.0, T2=350.0, dH=50000.0)
# -> K2 = 0.000360875, lnK2 = -7.92698   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
K = 1.8e-5 at 298 K and ΔH° = 50.0 kJ/mol. What is K at 350 K?	SOLVED	K2 = 0.000360875, lnK2 = -7.92698
K1 = 0.50, T1 = 400 K, T2 = 500 K, ΔH° = -92.0 kJ/mol. Find the equilibrium constant at T2.	SOLVED	K2 = 0.00197799, lnK2 = -6.22568

Gibbs energy change at non-standard conditions, `delta_g_nonstandard`

The Gibbs energy change of a reaction at a given reaction quotient (dG = dG° + RT ln Q).

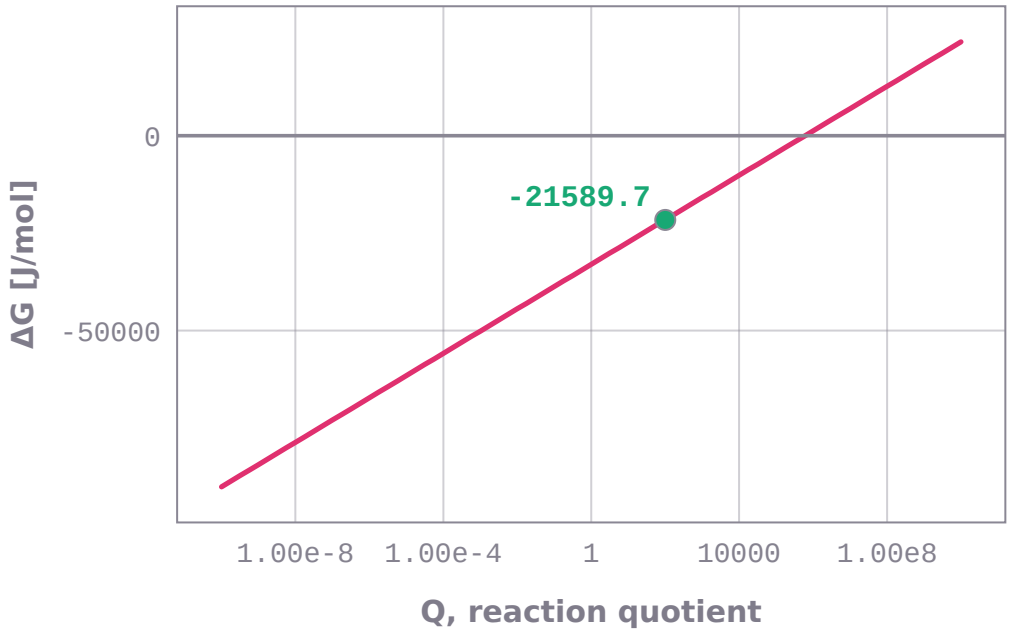
$$\Delta G = \Delta G^\circ + RT \ln Q$$

input	meaning	condition
`dG0`	standard Gibbs energy change	

input	meaning	condition
`T`	temperature	
`Q`	reaction quotient	

output	meaning
`dG`	Gibbs energy change

- **Method:** $dG = dG°$ + RT ln Q
- **Assumes:** T, $Q > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Gibbs energy change at non-standard conditions: dG against Q from 1.0000e-10 to 1.0000e+10 (logarithmic axis), 81 runs of the package; other inputs as in the worked example (dG0 = -33000.0, T = 298.0). Marked: the worked example, Q = 100, dG = -21589.7.

```
chemistry_ai999x.solve('delta_g_nonstandard', dG0=-33000.0, T=298.0, Q=100.0)
# -> dG = -21589.7 J/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
ΔG° = -33.0 kJ/mol, T = 298 K and Q = 100. What is ΔG?	SOLVED	dG = -21589.7 J/mol
The standard Gibbs energy change is 5.0 kJ/mol at 350 K and the reaction quotient is 0.010. Find ΔG.	SOLVED	dG = -8401.33 J/mol

Molar solubility from Ksp, `ksp_molar_solubility`

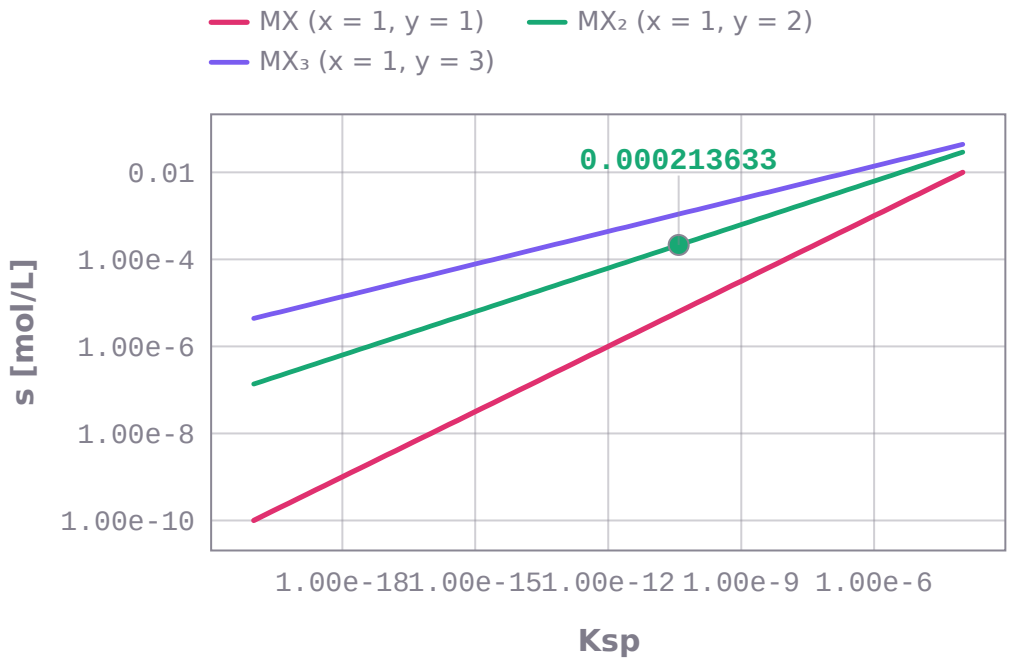
The molar solubility of a sparingly soluble salt MxAy in pure water from its solubility product.

$$K_{sp} = (xs)^x (ys)^y$$

input	meaning	condition
`Ksp`	solubility product	
`x`	cations per formula unit	
`y`	anions per formula unit	

output	meaning
`s`	molar solubility
`s_molL`	in mol/L

- **Method:** s = (Ksp / (x^x y^y))^(1/(x+y))
- **Assumes:** Ksp > 0; no common ion, no side reactions
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Molar solubility from Ksp: s_molL against Ksp from 1.0000e-20 to 0.0001 (logarithmic axis), 81 runs of the package per curve (MX (x = 1, y = 1); MX_2 (x = 1, y = 2); MX_3 (x = 1, y = 3)); other inputs as in the worked example (x = 1, y = 2). Marked: the worked example, Ksp = 3.9000e-11, s_molL = 0.000213633.

```
chemistry_ai999x.solve('ksp_molar_solubility', Ksp=3.9e-11, x=1, y=2)
# -> s = 0.213633 mol/m^3, s_molL = 0.000213633, log10_s = -3.67033   (SOLVED)
```

```
chemistry_ai999x.solve('ksp_molar_solubility', Ksp=1.8e-10, x=1, y=1)
# -> s = 0.0134164 mol/m^3, s_molL = 1.34164e-05, log10_s = -4.87236   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the molar solubility of CaF2 (Ksp = 3.9e-11)?	SOLVED	s = 0.213633 mol/m^3, s_molL = 0.000213633, log10_s = -3.67033
Find the molar solubility of Ag2CrO4 if Ksp = 1.1 × 10^-12.	SOLVED	s = 0.0650296 mol/m^3, s_molL = 6.50296e-05, log10_s = -4.18689

question	outcome	result
What is the molar solubility of Ca3(PO4)2 (Ksp = 2.0e-29)?	SOLVED	s = 0.000713709 mol/m^3, s_molL = 7.13709e-07, log10_s = -6.14648
What is the molar solubility of CaF2?	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemistry_ai999x.templates(); eve

Ksp from molar solubility, `ksp_from_solubility`

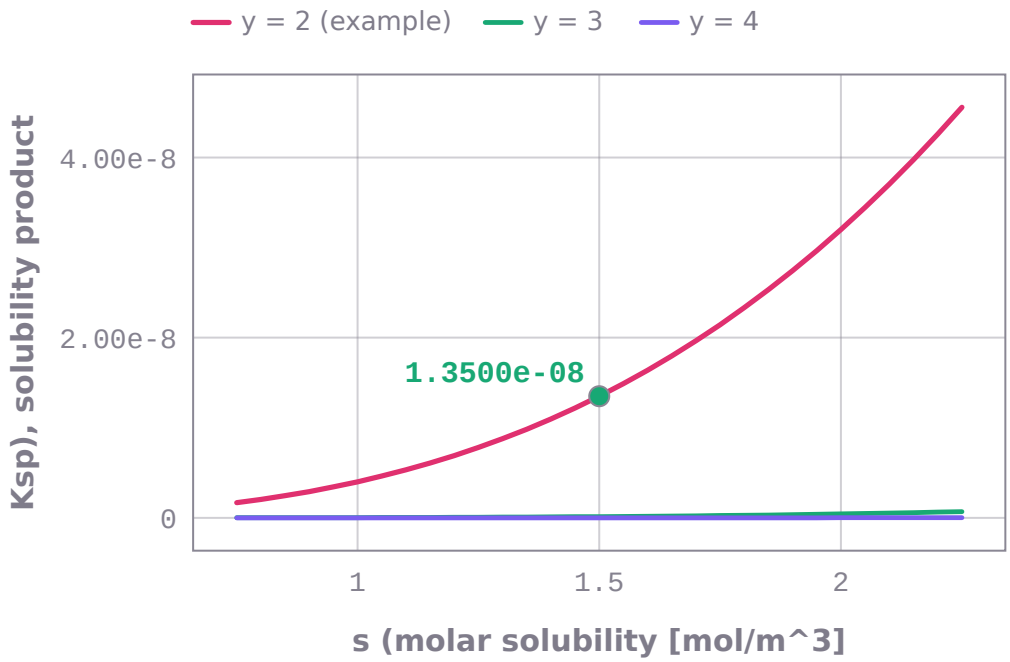
The solubility product of a salt MxAy from its molar solubility.

$$K_{sp} = (xs)^x (ys)^y$$

input	meaning	condition
`s`	molar solubility	
`x`	cations per formula unit	
`y`	anions per formula unit	

output	meaning
`Ksp`	solubility product

- **Method:** Ksp = (x s)^x (y s)^y
- **Assumes:** $s > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Ksp (solubility product) against s (molar solubility) from 0.75 mol/m^3 to 2.25 mol/m^3. One curve for each value of y (anions per formula unit): 2, 3, 4. Other inputs as in the worked example: x = 1. 31 points per curve computed by chemistry-ai999x; marked: the worked example, s = 1.5 mol/m^3, Ksp = 1.3500e-08.

```
chemistry_ai999x.solve('ksp_from_solubility', s=1.5, x=1, y=2)
# -> Ksp = 1.35e-08, pKsp = 7.86967   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The molar solubility of PbI2 is 1.5e-3 M. What is its Ksp?	SOLVED	Ksp = 1.35e-08, pKsp = 7.86967
The molar solubility of AgCl is 1.3 × 10^-5 M. Calculate the Ksp of AgCl.	SOLVED	Ksp = 1.69e-10, pKsp = 9.77211

Molar solubility with a common ion, `common_ion_solubility`

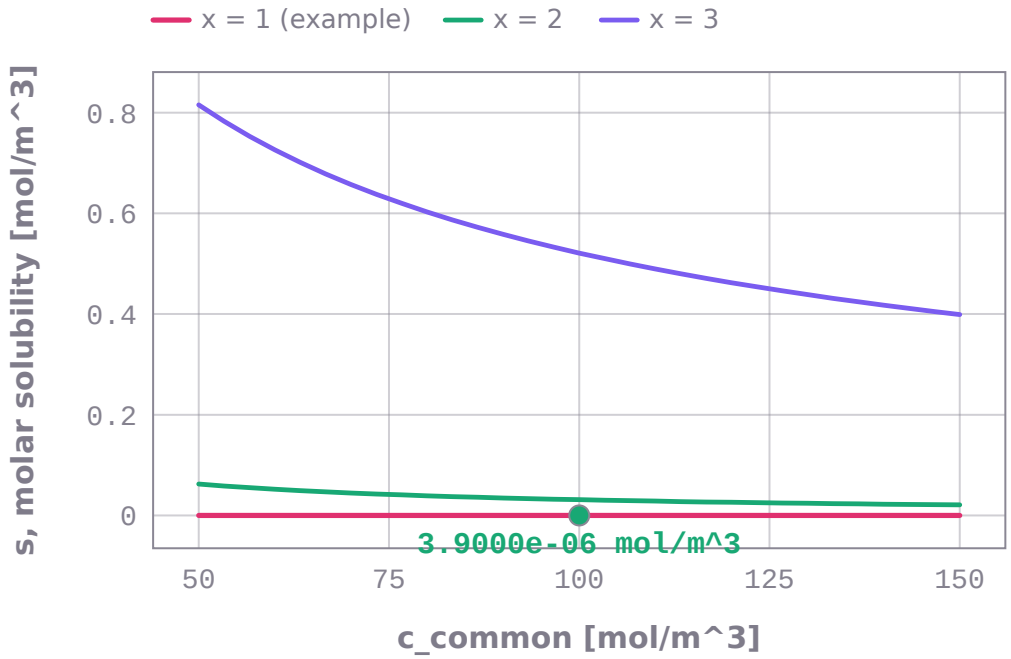
The molar solubility of a salt MxAy in a solution that already contains one of its ions.

$$K_{sp} = (xs + c_M)^x (ys + c_A)^y$$

input	meaning	condition
`Ksp`	solubility product	
`x`	cations per formula unit	
`y`	anions per formula unit	
`c_common`	concentration of the common ion	
`common`	which ion is common	

output	meaning
`s`	molar solubility
`s_molL`	in mol/L

- **Method:** the solubility s solved from the Ksp expression with the common ion's concentration, by bisection
- **Assumes:** Ksp > 0; the common ion comes from a fully dissolved salt
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



s (molar solubility) against c_common (concentration of the common ion) from 50 mol/m^3 to 150 mol/m^3. One curve for each value of x (cations per formula unit): 1, 2, 3. Other inputs as in the worked example: Ksp = 3.9e-11, y = 2, common = anion. 31 points per curve computed by chemistry-ai999x; marked: the worked example, c_common = 100 mol/m^3, s = 3.9000e-06 mol/m^3.

```
chemistry_ai999x.solve('common_ion_solubility', Ksp=3.9e-11, x=1, y=2, c_common=100.0,
common='anion')
# -> s = 3.9e-06 mol/m^3, s_molL = 3.9e-09, log10_s = -8.40894   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the molar solubility of CaF2 in 0.10 M NaF (Ksp = 3.9e-11)?	SOLVED	s = 3.9e-06 mol/m^3, s_molL = 3.9e-09, log10_s = -8.40894
Find the molar solubility of AgCl in 0.010 M AgNO3 (Ksp = 1.8e-10).	SOLVED	s = 1.8e-05 mol/m^3, s_molL = 1.8e-08, log10_s = -7.74473
What is the molar solubility of PbI2 in 0.050 M Pb(NO3)2 (Ksp = 9.8e-9)?	SOLVED	s = 0.220872 mol/m^3, s_molL = 0.000220872, log10_s = -3.65586

pH from the hydronium concentration, `ph_from_hydronium`

pH, pOH and [OH-] from the hydronium (hydrogen ion) concentration at 25 °C.

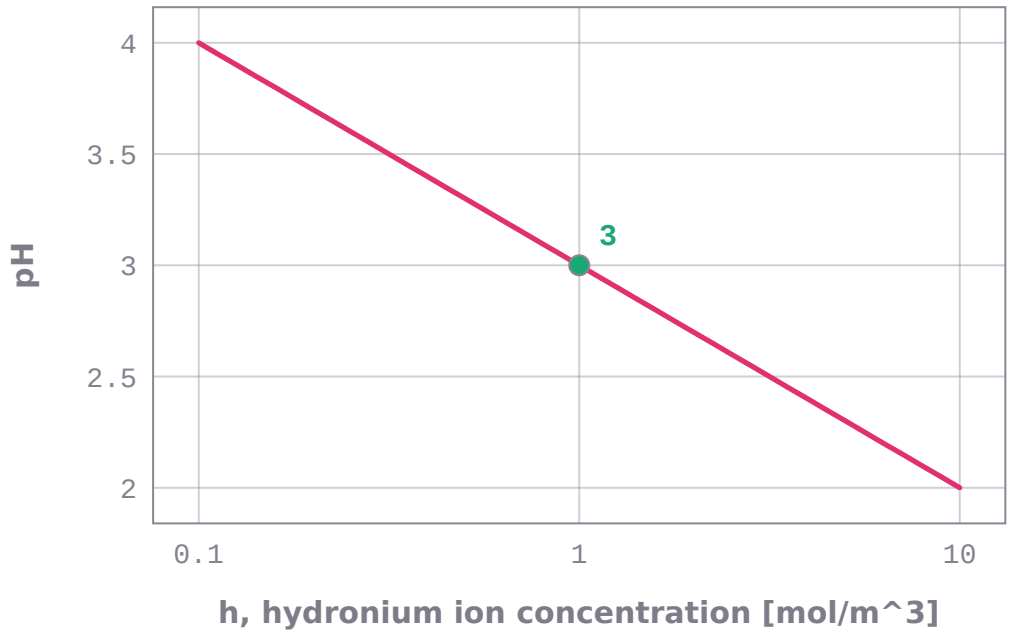
$$\mathrm{pH} = -\log_{10}[\mathrm{H_3O^+}]$$

input	meaning	condition
`h`	hydronium ion concentration	

output	meaning
`pH`	pH
`pOH`	pOH

output	meaning
`oh`	[OH-]

- **Method:** $pH = -log10$[H3O+]; pOH = 14 - pH (25 °C)
- **Assumes:** [H3O+] > 0 (mol/L); 25 °C $(Kw = 1.0e-14)$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH (pH) against h (hydronium ion concentration) from 0.1 mol/m^3 to 10 mol/m^3 on a logarithmic axis. 41 points computed by chemistry-ai999x; marked: the worked example, h = 1 mol/m^3, pH = 3.

```
chemistry_ai999x.solve('ph_from_hydronium', h=1.0)
# -> pH = 3, pOH = 11, oh = 1e-08 mol/m^3   (SOLVED)
```

```
chemistry_ai999x.solve('ph_from_hydronium', h=0.025)
# -> pH = 4.60206, pOH = 9.39794, oh = 4e-07 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of a solution with [H+] = 1.0e-3 M?	SOLVED	pH = 3, pOH = 11, oh = 1e-08 mol/m^3
Calculate the pH if the hydronium ion concentration is 2.5 × 10^-5 M.	SOLVED	pH = 4.60206, pOH = 9.39794, oh = 4e-07 mol/m^3

Hydronium concentration from pH, `hydronium_from_ph`

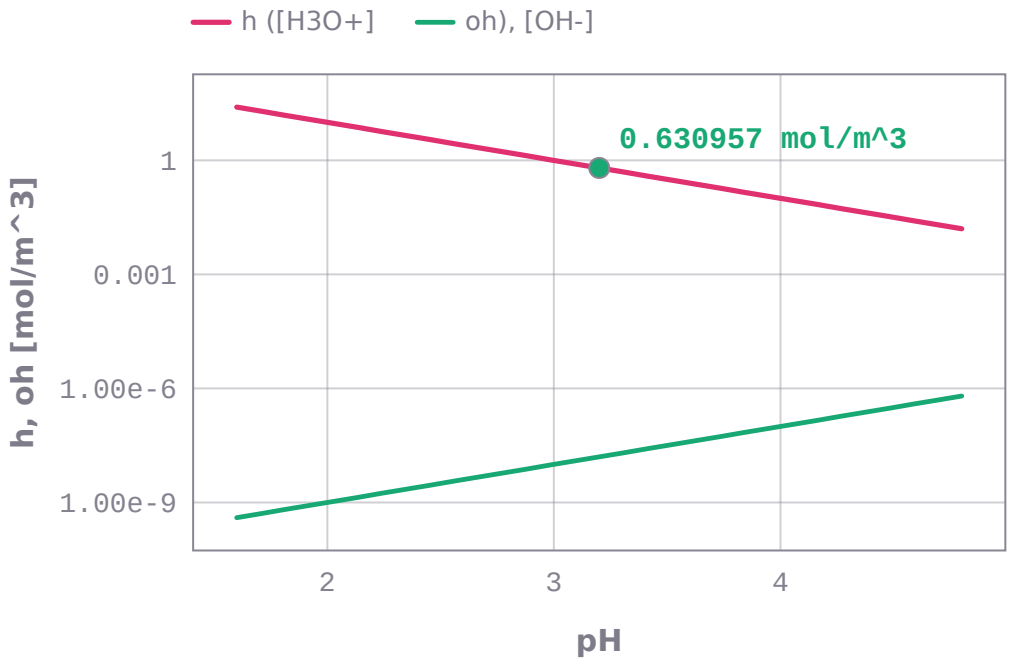
[H3O+], [OH-] and pOH from the pH at 25 °C.

$$[\mathrm{H_3O^+}] = 10^{-\mathrm{pH}}$$

input	meaning	condition
`pH`	pH	

output	meaning
`h`	[H3O+]
`h_molL`	[H3O+] in mol/L
`pOH`	pOH
`oh`	[OH-]

- **Method:** [H3O+] = 10^{-pH}
- **Assumes:** 25 °C $(Kw = 1.0e-14)$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



h ([H3O+]) against pH (pH) from 1.6 to 4.8, logarithmic output axis. Outputs with the same unit drawn together: oh ([OH-]). 31 points computed by chemistry-ai999x; marked: the worked example, pH = 3.2, h = 0.630957 mol/m^3.

```
chemistry_ai999x.solve('hydronium_from_ph', pH=3.2)
# -> h = 0.630957 mol/m^3, h_molL = 0.000630957, pOH = 10.8, oh = 1.58489e-08 mol/m^3
(SOLVED)
```

```
chemistry_ai999x.solve('hydronium_from_ph', pH=9.5)
# -> h = 3.16228e-07 mol/m^3, h_molL = 3.16228e-10, pOH = 4.5, oh = 0.0316228 mol/m^3
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is [H+] in a solution of pH 3.20?	SOLVED	h = 0.630957 mol/m^3, h_molL = 0.000630957, pOH = 10.8, oh = 1.58489e-08 mol/m^3
Find the hydronium ion concentration of a solution with pH = 9.5.	SOLVED	h = 3.16228e-07 mol/m^3, h_molL = 3.16228e-10, pOH = 4.5, oh = 0.0316228 mol/m^3

pH from the hydroxide concentration, `ph_from_hydroxide`

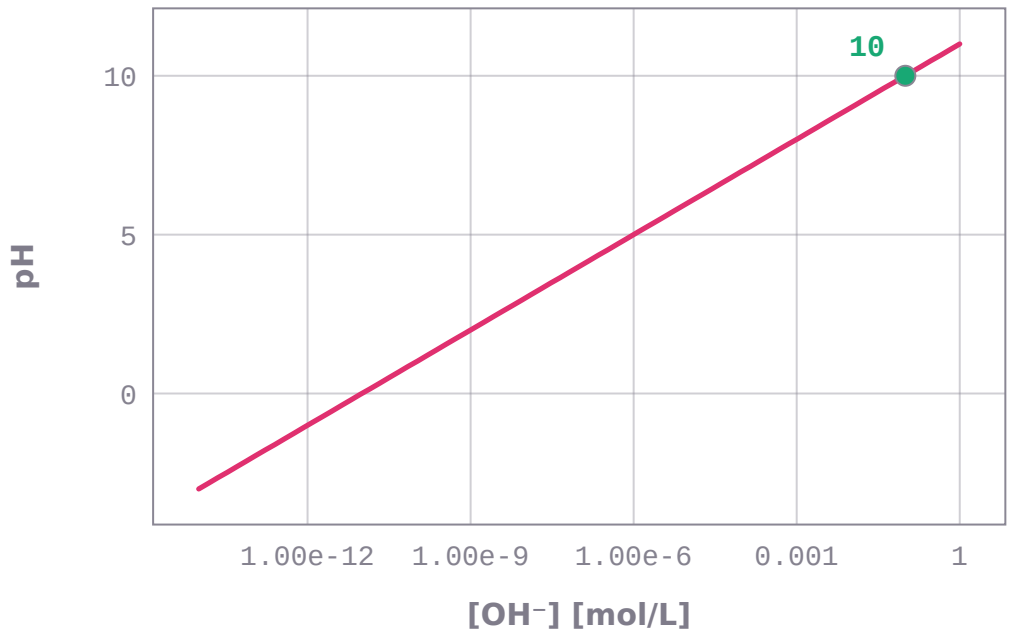
pOH and pH from the hydroxide ion concentration at 25 °C.

$$\mathrm{pH} = 14 + \log_{10}[\mathrm{OH}^-]$$

input	meaning	condition
`oh`	hydroxide ion concentration	

output	meaning
`pH`	pH
`pOH`	pOH

- **Method:** pOH = -log10[OH-]; pH = 14 - pOH (25 °C)
- **Assumes:** [OH-] > 0; 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH from the hydroxide concentration: pH against oh from 1.0000e-14 to 1 (logarithmic axis), 81 runs of the package; other inputs as in the worked example (). Marked: the worked example, oh = 0.1, pH = 10.

```
chemistry_ai999x.solve('ph_from_hydroxide', oh=0.1)
# -> pH = 10, pOH = 4   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of a solution with [OH-] = 1.0e-4 M?	SOLVED	pH = 10, pOH = 4
Find the pOH and pH if the hydroxide ion concentration is 3.2 × 10^-3 M.	SOLVED	pH = 11.5051, pOH = 2.49485

pH of a strong monoprotic acid, `ph_strong_acid`

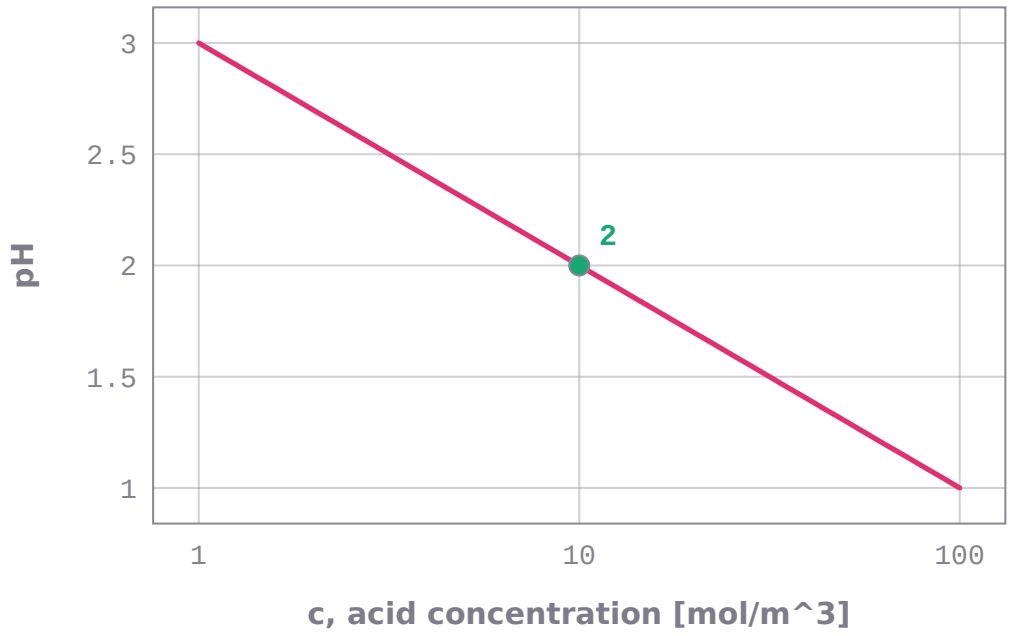
The pH of a strong monoprotic acid solution (water's own ionization included, so dilute solutions are right).

$$[\mathrm{H_3O^+}] = \frac{c}{2} + \sqrt{\frac{c^2}{4} + K_w}$$

input	meaning	condition
c	acid concentration	

output	meaning
pH	pH
h	[H3O+]

- **Method:** charge balance [H3O+] = c + Kw/[H3O+] solved exactly
- **Assumes:** monoprotic strong acid; 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH (pH) against c (acid concentration) from 1 mol/m^3 to 100 mol/m^3 on a logarithmic axis. 41 points computed by chemistry-ai999x; marked: the worked example, c = 10 mol/m^3, pH = 2.

```
chemistry_ai999x.solve('ph_strong_acid', c=10.0)
# -> pH = 2, h = 10 mol/m^3   (SOLVED)
```

```
chemistry_ai999x.solve('ph_strong_acid', c=1e-05)
# -> pH = 6.97829, h = 0.000105125 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of 0.010 M HCl?	SOLVED	pH = 2, h = 10 mol/m^3

question	outcome	result
Calculate the pH of a 1.0e-8 M HNO3 solution.	SOLVED	pH = 6.97829, h = 0.000105125 mol/m^3
What is the pH of 0.25 M hydrochloric acid?	SOLVED	pH = 0.60206, h = 250 mol/m^3

pH of a strong base, `ph_strong_base`

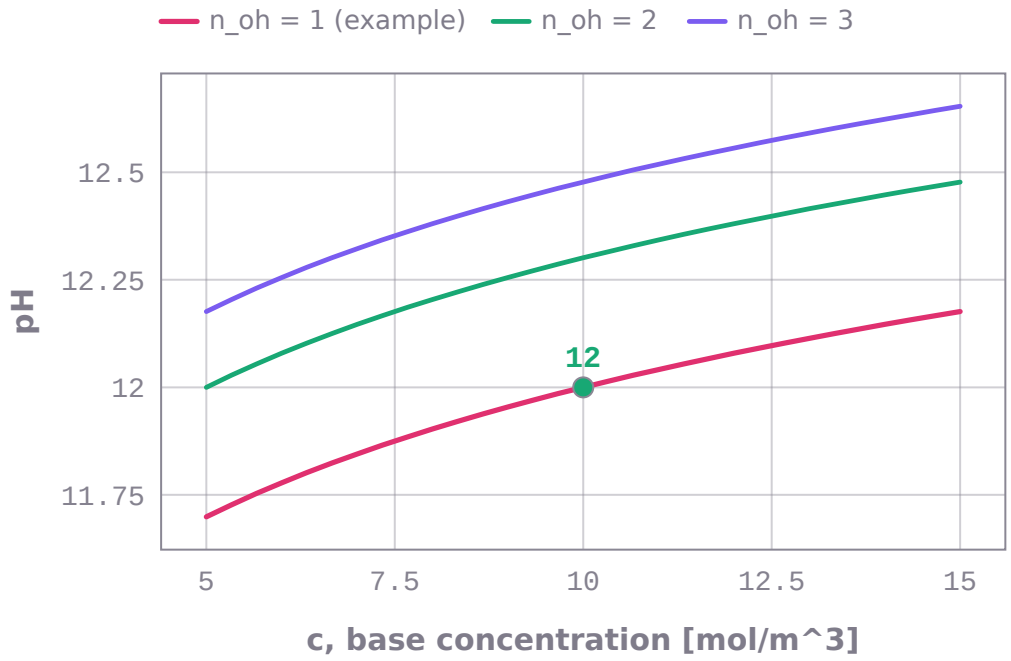
The pH and pOH of a strong (group 1 or 2) hydroxide solution.

$$[\mathrm{OH^-}] = nc, \quad \mathrm{pH} = 14 - \mathrm{pOH}$$

input	meaning	condition
`c`	base concentration	
`n_oh`	hydroxide ions per formula unit	

output	meaning
`pH`	pH
`pOH`	pOH

- **Method:** charge balance with water's ionization, [OH-] = n c + Kw/[OH-]
- **Assumes:** strong base fully dissociated; 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH (pH) against c (base concentration) from 5 mol/m^3 to 15 mol/m^3. One curve for each value of n_oh (hydroxide ions per formula unit): 1, 2, 3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, c = 10 mol/m^3, pH = 12.

```
chemistry_ai999x.solve('ph_strong_base', c=10.0, n_oh=1)
# -> pH = 12, pOH = 2   (SOLVED)
```

```
chemistry_ai999x.solve('ph_strong_base', c=5.0, n_oh=2)
# -> pH = 12, pOH = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of 0.010 M NaOH?	SOLVED	pH = 12, pOH = 2
Calculate the pH of 0.0050 M Ba(OH)2.	SOLVED	pH = 12, pOH = 2
What is the pH of a 0.020 M strong base?	NEEDS_INPUT	ph_strong_base needs n_oh (hydroxide ions per formula unit). Example: solve('ph_strong_base', c='10 mol/m^3', n_oh=1).

pH of a weak acid, `ph_weak_acid`

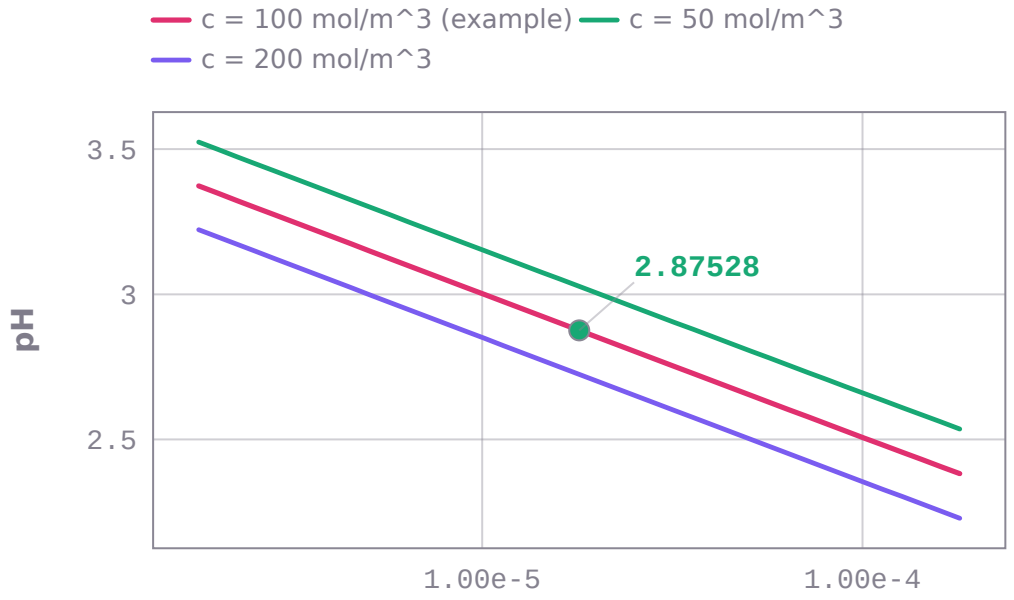
The pH and percent ionization of a weak monoprotic acid from Ka (or pKa) and its concentration.

$$[\mathrm{H_3O^+}] = \frac{K_a c}{K_a + [\mathrm{H_3O^+}]} + \frac{K_w}{[\mathrm{H_3O^+}]}$$

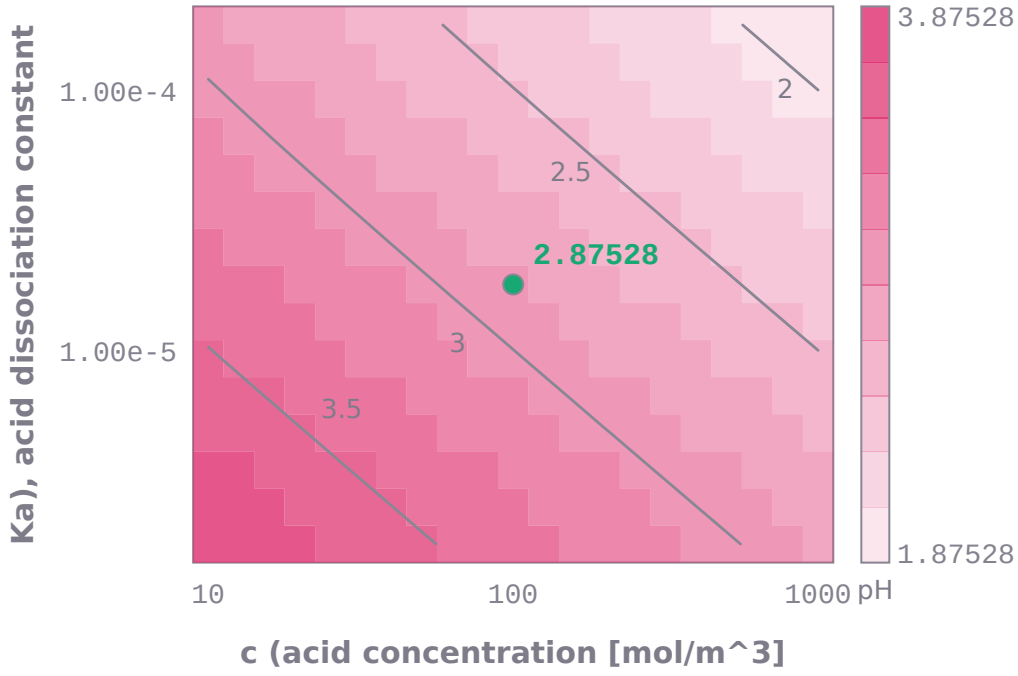
input	meaning	condition
`Ka`	acid dissociation constant	
`c`	acid concentration	

output	meaning
`pH`	pH
`h`	[H3O+]
`percent_ionized`	percent ionization

- **Method:** exact charge balance (no 5% approximation), solved by bisection
- **Assumes:** monoprotic acid; 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH (pH) against Ka (acid dissociation constant) from 1.8000e-06 to 0.00018 on a logarithmic axis. One curve for each value of c (acid concentration): 100 mol/m^3, 50 mol/m^3, 200 mol/m^3. 41 points per curve computed by chemistry-ai999x; marked: the worked example, Ka = 1.8000e-05, pH = 2.87528.



pH (pH) over c (acid concentration) from 10 mol/m^3 to 1000 mol/m^3 and Ka (acid dissociation constant) from 1.8000e-06 to 0.00018 (logarithmic axes): each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 1.87528 to dark 3.87528); iso-lines at pH = 2, 2.5, 3, 3.5. Marked: the worked example, c = 100 mol/m^3, Ka = 1.8000e-05, pH = 2.87528.

```
chemistry_ai999x.solve('ph_weak_acid', Ka=1.8e-05, c=100.0)
# -> pH = 2.87528, h = 1.33267 mol/m^3, percent_ionized = 1.33267   (SOLVED)
```

```
chemistry_ai999x.solve('ph_weak_acid', Ka=6.309573444801929e-05, c=50.0)
# -> pH = 2.75823, h = 1.7449 mol/m^3, percent_ionized = 3.48981   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of 0.10 M acetic acid (Ka = 1.8 × 10^-5)?	SOLVED	pH = 2.87528, h = 1.33267 mol/m^3, percent_ionized = 1.33267
Find the pH of a 0.050 M weak acid with pKa = 4.20.	SOLVED	pH = 2.75823, h = 1.7449 mol/m^3, percent_ionized = 3.48981
What is the percent ionization of 0.20 M HF (Ka = 6.8e-4)?	SOLVED	pH = 1.94589, h = 11.3269 mol/m^3, percent_ionized = 5.66343
What is the pH of HF (Ka = 6.8e-4)?	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: chemistry_ai999x.templates(); eve

pH of a weak base, `ph_weak_base`

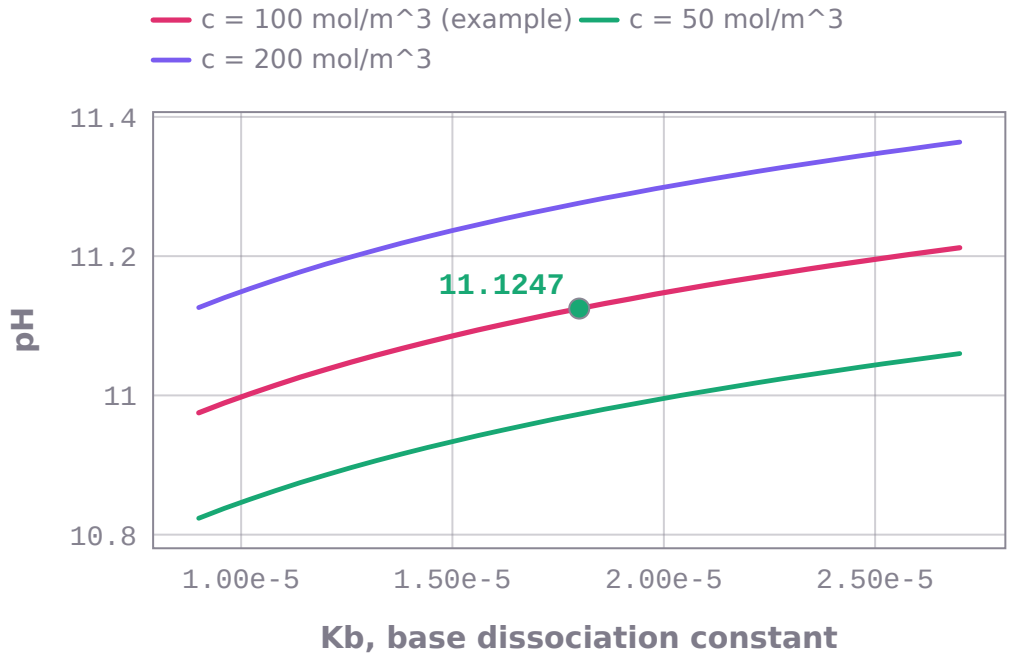
The pH, pOH and percent ionization of a weak base from Kb (or pKb) and its concentration.

$$[\mathrm{OH^-}] = \frac{K_b c}{K_b + [\mathrm{OH^-}]} + \frac{K_w}{[\mathrm{OH^-}]}$$

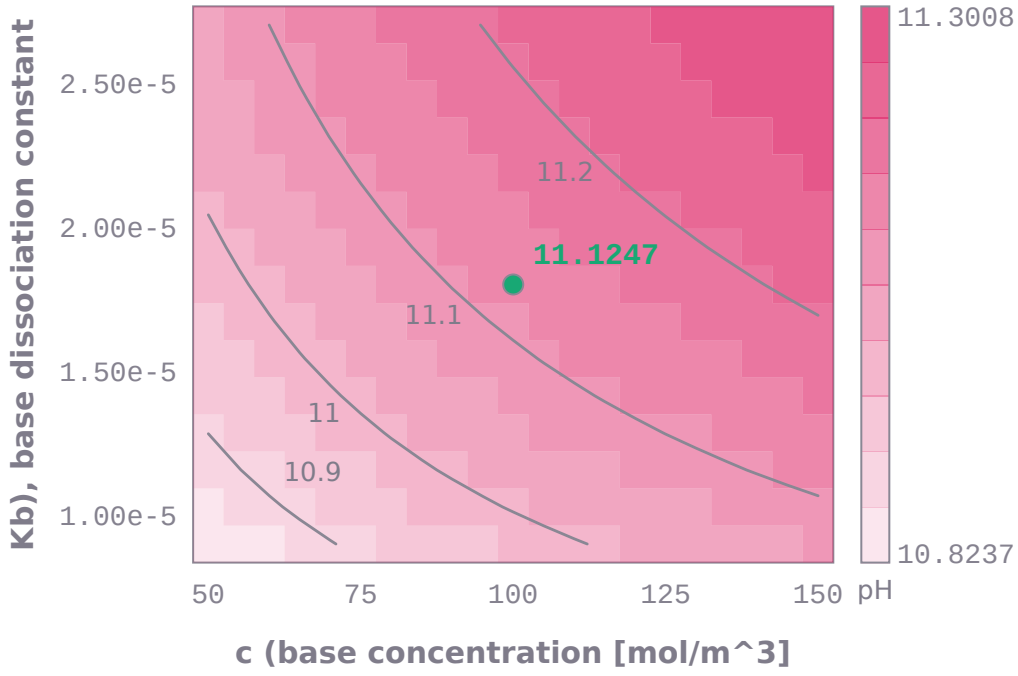
input	meaning	condition
`Kb`	base dissociation constant	
`c`	base concentration	

output	meaning
`pH`	pH
`pOH`	pOH
`percent_ionized`	percent ionization

- **Method:** exact charge balance for [OH-], solved by bisection; pH = 14 - pOH
- **Assumes:** 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH (pH) against Kb (base dissociation constant) from 9.0000e-06 to 2.7000e-05. One curve for each value of c (base concentration): 100 mol/m^3, 50 mol/m^3, 200 mol/m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, Kb = 1.8000e-05, pH = 11.1247.



pH (pH) over c (base concentration) from 50 mol/m^3 to 150 mol/m^3 and Kb (base dissociation constant) from 9.0000e-06 to 2.7000e-05: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 10.8237 to dark 11.3008); iso-lines at pH = 10.9, 11, 11.1, 11.2. Marked: the worked example, c = 100 mol/m^3, Kb = 1.8000e-05, pH = 11.1247.

```
chemistry_ai999x.solve('ph_weak_base', Kb=1.8e-05, c=100.0)
# -> pH = 11.1247, pOH = 2.87528, percent_ionized = 1.33267   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of 0.10 M NH3 (Kb = 1.8 × 10^-5)?	SOLVED	pH = 11.1247, pOH = 2.87528, percent_ionized = 1.33267

question	outcome	result
Find the pH of a 0.25 M weak base with pKb = 3.36.	SOLVED	pH = 12.0099, pOH = 1.9901, percent_ionized = 4.0922

Ka from the pH of a weak acid, `ka_from_ph`

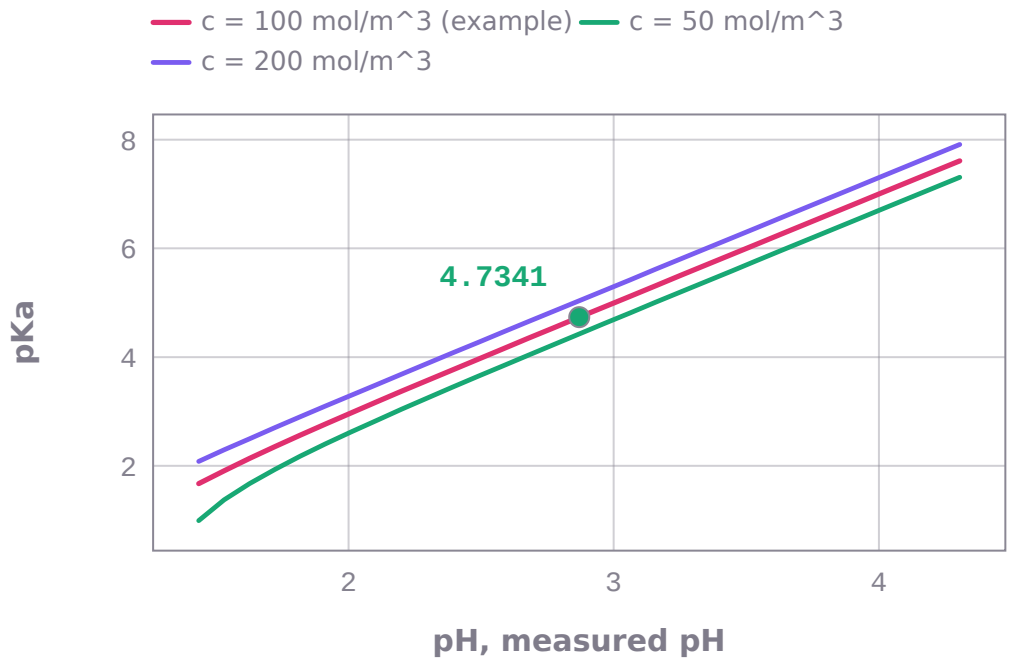
The acid dissociation constant of a weak monoprotic acid from the pH of a solution of known concentration.

$$K_a = \frac{[\mathrm{H_3O^+}][\mathrm{A^-}]}{c - [\mathrm{A^-}]},\ [\mathrm{A^-}] = [\mathrm{H_3O^+}] - K_w/[\mathrm{H_3O^+}]$$

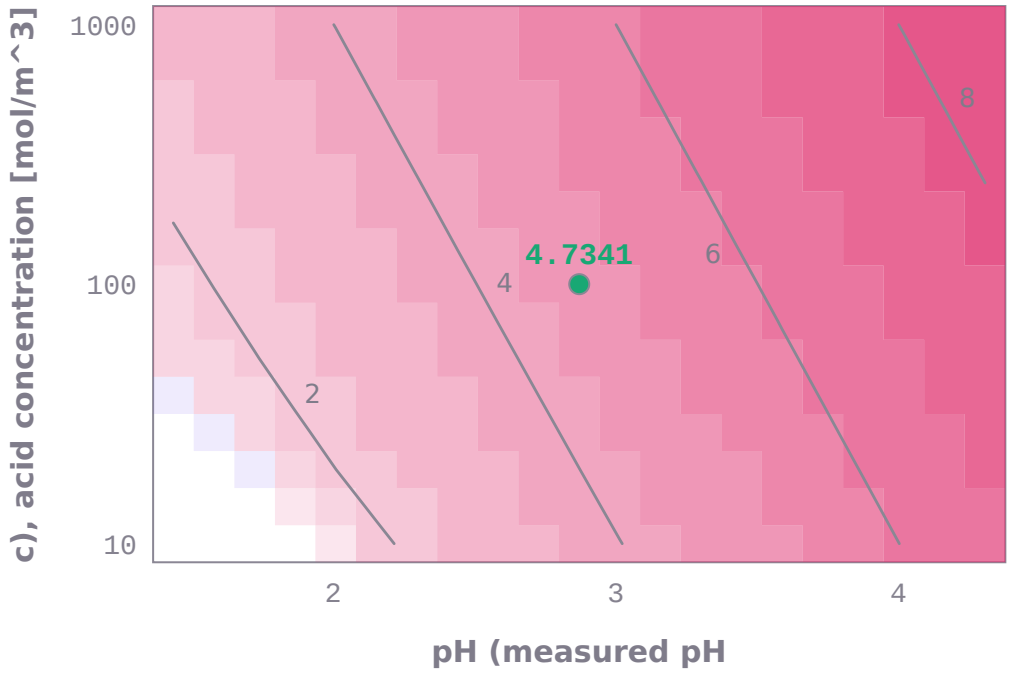
input	meaning	condition
`pH`	measured pH	
`c`	acid concentration	

output	meaning
`pKa`	pKa
`Ka`	Ka

- **Method:** [A-] from the charge balance at the measured pH, then the mass-action expression
- **Assumes:** monoprotic weak acid; 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pKa (pKa) against pH (measured pH) from 1.435 to 4.305. One curve for each value of c (acid concentration): 100 mol/m^3, 50 mol/m^3, 200 mol/m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, pH = 2.87, pKa = 4.7341.



pKa (pKa) over pH (measured pH) from 1.435 to 4.305 and c (acid concentration) from 10 mol/m^3 to 1000 mol/m^3 (logarithmic vertical axis): each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its sign and magnitude (second colour negative); iso-lines at pKa = 2, 4, 6, 8; blank cells: inputs the capability refuses. Marked: the worked example, pH = 2.87, c = 100 mol/m^3, pKa = 4.7341.

```
chemistry_ai999x.solve('ka_from_ph', pH=2.87, c=100.0)
# -> Ka = 1.84458e-05, pKa = 4.7341   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A 0.10 M solution of a weak acid has a pH of 2.87. What is its Ka?	SOLVED	Ka = 1.84458e-05, pKa = 4.7341
The pH of 0.050 M HNO2 is 2.30. Find the acid dissociation constant.	SOLVED	Ka = 0.000558344, pKa = 3.2531

Buffer pH (Henderson-Hasselbalch), `buffer_ph`

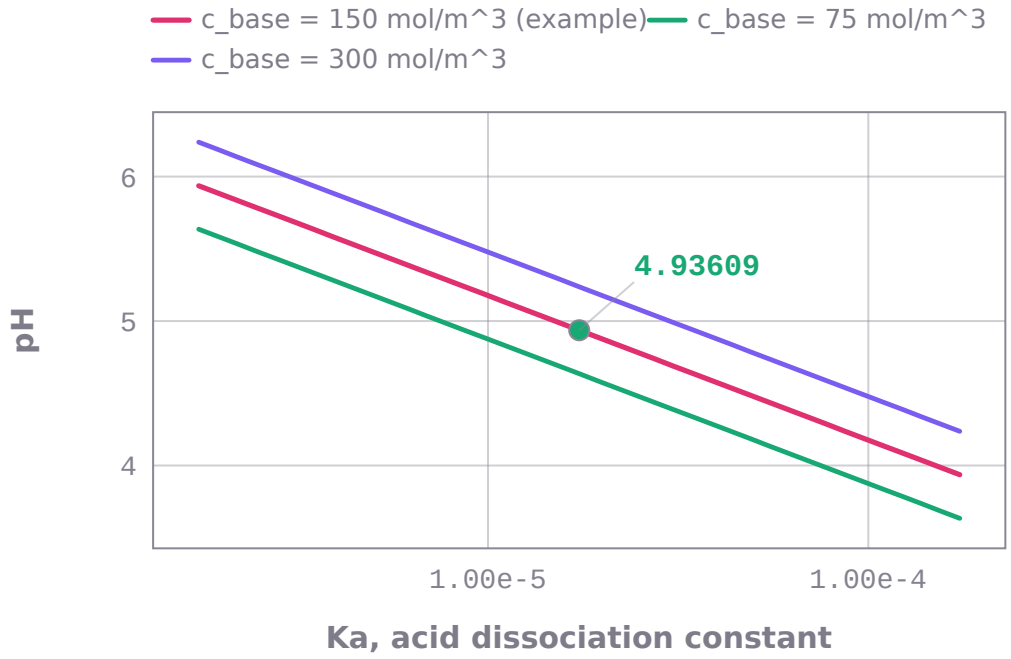
The pH of a buffer from the acid's Ka (or pKa) and the concentrations of the weak acid and its conjugate base.

$$\mathrm{pH} = \mathrm{p}K_a + \log_{10}\frac{[\mathrm{A}^-]}{[\mathrm{HA}]}$$

input	meaning	condition
`Ka`	acid dissociation constant	
`c_base`	conjugate base concentration	
`c_acid`	weak acid concentration	

output	meaning
`pH`	pH

- **Method:** Henderson-Hasselbalch equation
- **Assumes:** both concentrations well above Ka and [H3O+]
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH (pH) against Ka (acid dissociation constant) from 1.7378e-06 to 0.00017378 on a logarithmic axis. One curve for each value of c_base (conjugate base concentration): 150 mol/m^3, 75 mol/m^3, 300 mol/m^3. Other inputs as in the worked example: c_acid = 100.0 mol/m^3. 41 points per curve computed by chemistry-ai999x; marked: the worked example, Ka = 1.7378e-05, pH = 4.93609.

```
chemistry_ai999x.solve('buffer_ph', Ka=1.7378008287493764e-05, c_base=150.0, c_acid=100.0)
# -> pH = 4.93609   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of a buffer that is 0.10 M acetic acid and 0.15 M sodium acetate (pKa = 4.76)?	SOLVED	pH = 4.93609
Find the pH of a buffer with [HA] = 0.20 M, [A-] = 0.10 M and Ka = 1.8 × 10^-5.	SOLVED	pH = 4.4437

Titration: equivalence-point volume, `titration_equivalence_volume`

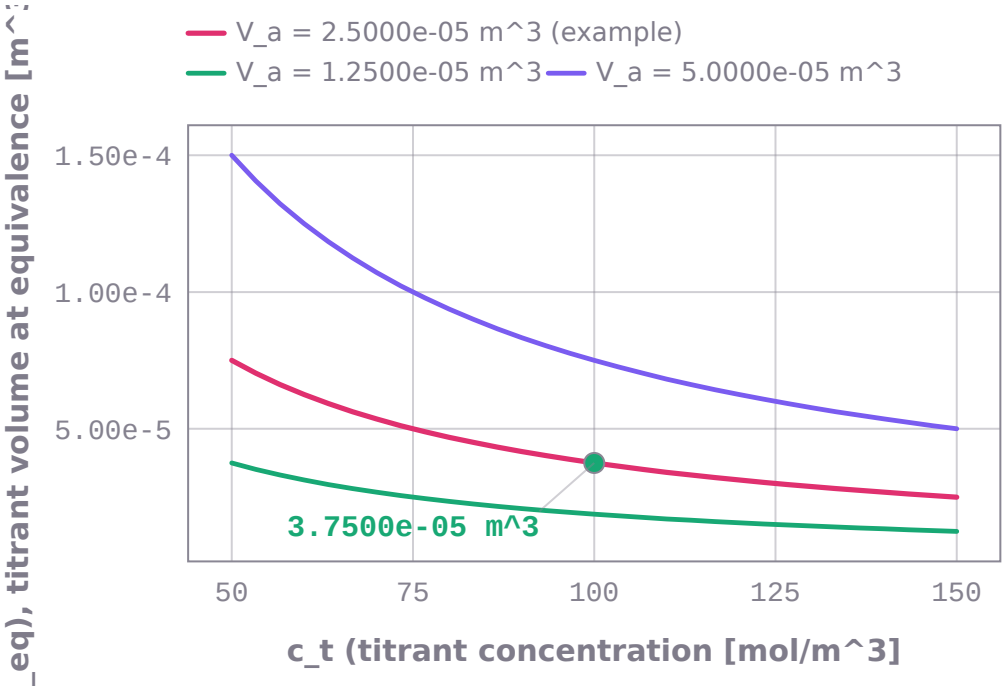
The volume of titrant needed to reach the equivalence point of a neutralization.

$$V_{eq} = \frac{\nu\, c_a V_a}{c_t}$$

input	meaning	condition
`c_t`	titrant concentration	
`V_a`	volume of the analyte	
`c_a`	analyte concentration	
`ratio`	moles of titrant per mole of analyte	

output	meaning
`V_eq`	titrant volume at equivalence
`V_eq_mL`	in mL

- **Method:** moles of titrant = ratio x moles of analyte
- **Assumes:** complete neutralization; the acid's protons and base's hydroxides from the formulas
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



V_eq (titrant volume at equivalence) against c_t (titrant concentration) from 50 mol/m^3 to 150 mol/m^3. One curve for each value of V_a (volume of the analyte): 2.5000e-05 m^3, 1.2500e-05 m^3, 5.0000e-05 m^3. Other inputs as in the worked example: c_a = 150.0 mol/m^3, ratio = 1.0. 31 points per curve computed by chemistry-ai999x; marked: the worked example, c_t = 100 mol/m^3, V_eq = 3.7500e-05 m^3.

```
chemistry_ai999x.solve('titration_equivalence_volume', c_t=100.0, V_a=2.5e-05, c_a=150.0,
ratio=1.0)
# -> V_eq = 3.75e-05 m^3, V_eq_mL = 37.5   (SOLVED)
```

```
chemistry_ai999x.solve('titration_equivalence_volume', c_t=100.0, V_a=2e-05, c_a=100.0,
ratio=2.0)
# -> V_eq = 4e-05 m^3, V_eq_mL = 40   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What volume of 0.100 M NaOH is needed to neutralize 25.0 mL of 0.150 M HCl?	SOLVED	V_eq = 3.75e-05 m^3, V_eq_mL = 37.5
What volume of 0.100 M NaOH is needed to neutralize 20.0 mL of 0.100 M H2SO4?	SOLVED	V_eq = 4e-05 m^3, V_eq_mL = 40
What volume of 0.200 M HCl neutralizes 30.0 mL of 0.100 M Ba(OH)2?	SOLVED	V_eq = 3e-05 m^3, V_eq_mL = 30

pH during a strong acid - strong base titration, `titration_ph_strong_acid_strong_base`

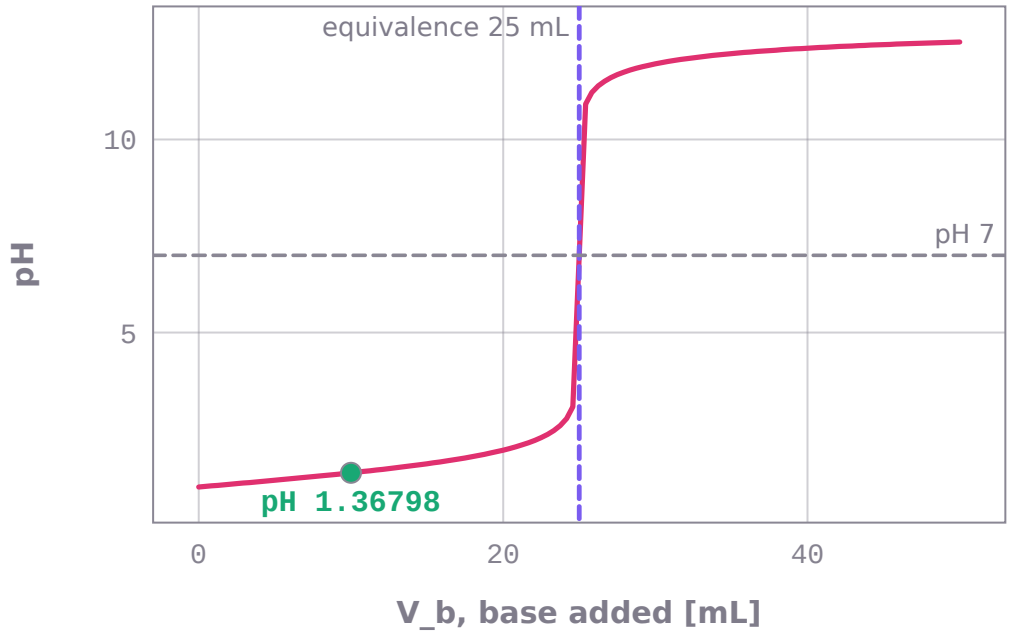
The pH after adding a volume of strong base to a strong monoprotic acid (exact, any point of the titration).

$$[\mathrm{H_3O^+}] - \frac{K_w}{[\mathrm{H_3O^+}]} = \frac{c_a V_a - c_b V_b}{V_a + V_b}$$

input	meaning	condition
`c_a`	acid concentration	
`V_a`	acid volume	
`c_b`	base concentration	
`V_b`	volume of base added	

output	meaning
`pH`	pH

- **Method:** excess strong acid or base after mixing, with water's ionization (exact at the equivalence point too)
- **Assumes:** strong monoprotic acid titrated with a strong monohydroxide base; 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Titration curve of a strong acid (100.0 mol/m³, 25 mL) with a strong base (100.0 mol/m³): the pH the package returned for 121 added volumes from 0 to twice the equivalence volume (equivalence at c_a·V_a/c_b = 25 mL, computed from the stated inputs). Marked: the worked example, V_b = 10 mL, pH = 1.36798.

```
chemistry_ai999x.solve('titration_ph_strong_acid_strong_base', c_a=100.0, V_a=2.5e-05,
c_b=100.0, V_b=1e-05)
# -> pH = 1.36798   (SOLVED)
```

```
chemistry_ai999x.solve('titration_ph_strong_acid_strong_base', c_a=100.0, V_a=2.5e-05,
c_b=100.0, V_b=2.5e-05)
# -> pH = 7   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
25.0 mL of 0.100 M HCl is titrated with 0.100 M NaOH. What is the pH after adding 10.0 mL of base?	SOLVED	pH = 1.36798
50.0 mL of 0.200 M HNO3 is titrated with 0.100 M KOH. Find the pH after adding 150 mL of base.	SOLVED	pH = 12.3979
25.0 mL of 0.100 M HCl is titrated with 0.100 M NaOH. What is the pH?	NEEDS_INPUT	titration_ph_strong_acid_strong_base needs V_b (volume of base added). Example: solve('titration_ph_strong_acid_strong_b

pH during a weak acid - strong base titration, `titration_ph_weak_acid_strong_base`

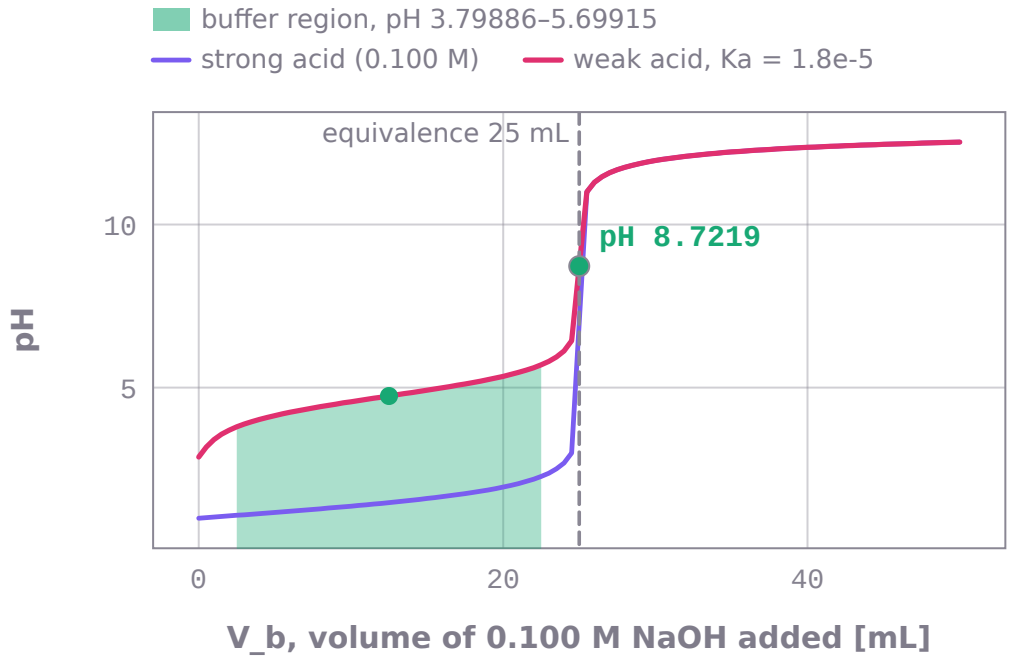
The pH after adding strong base to a weak monoprotic acid, at any point (exact charge balance).

$$[\mathrm{H_3O^+}] + [\mathrm{Na^+}] = \frac{K_w}{[\mathrm{H_3O^+}]} + \frac{K_a C}{K_a + [\mathrm{H_3O^+}]}$$

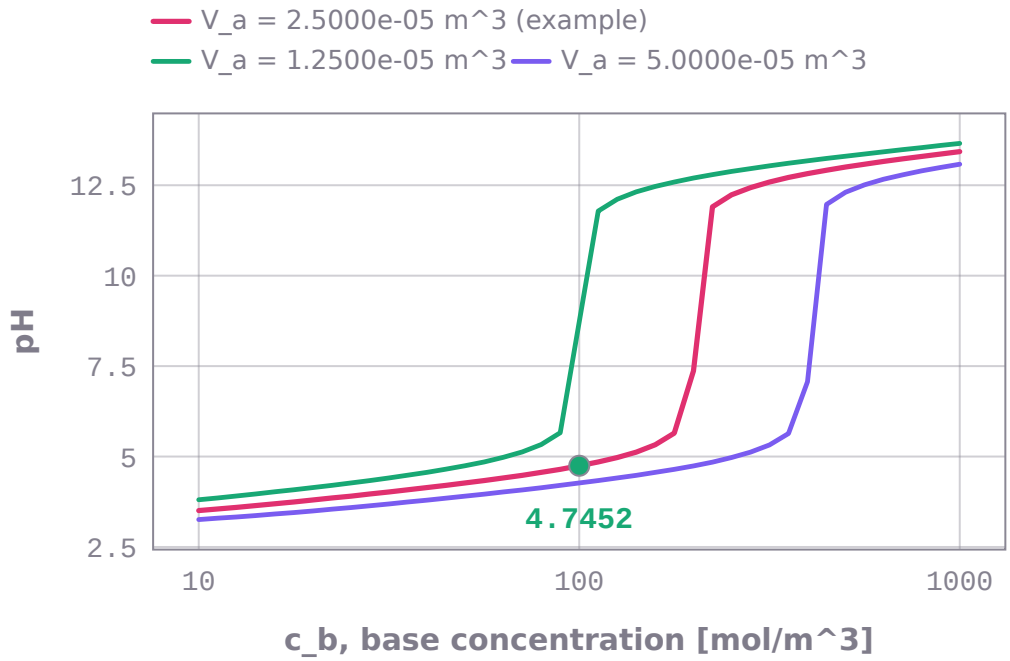
input	meaning	condition
`Ka`	acid dissociation constant	
`c_a`	acid concentration	
`V_a`	acid volume	
`c_b`	base concentration	
`V_b`	volume of base added	

output	meaning
`pH`	pH

- **Method:** exact charge balance after mixing, solved by bisection (before, at and after equivalence)
- **Assumes:** weak monoprotic acid, strong monohydroxide base; 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH against volume of 0.100 M NaOH added to 25.0 mL of 0.100 M acid, computed by the package at 101 volumes for a strong acid and a weak acid (Ka = 1.8e-5). Shaded: the weak acid's buffer region, 10 % to 90 % neutralised (pH 3.79886 to 5.69915); at half-equivalence pH = 4.7452 (= pKa). Marked: equivalence V_b = 25 mL, pH = 8.7219.



pH (pH) against c_b (base concentration) from 10 mol/m^3 to 1000 mol/m^3 on a logarithmic axis. One curve for each value of V_a (acid volume): 2.5000e-05 m^3, 1.2500e-05 m^3, 5.0000e-05 m^3. Other inputs as in the worked example: Ka = 1.8e-05, c_a = 100.0 mol/m^3, V_b = 1.25e-05 m^3. 41 points per curve computed by chemistry-ai999x; marked: the worked example, c_b = 100 mol/m^3, pH = 4.7452.

```
chemistry_ai999x.solve('titration_ph_weak_acid_strong_base', Ka=1.8e-05, c_a=100.0,
V_a=2.5e-05, c_b=100.0, V_b=1.25e-05)
# -> pH = 4.7452   (SOLVED)
```

```
chemistry_ai999x.solve('titration_ph_weak_acid_strong_base', Ka=1.8e-05, c_a=100.0,
V_a=2.5e-05, c_b=100.0, V_b=2.5e-05)
# -> pH = 8.7219   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
25.0 mL of 0.100 M CH3COOH (Ka = 1.8e-5) is titrated with 0.100 M NaOH. What is the pH after adding 12.5 mL of base?	SOLVED	pH = 4.7452
25.0 mL of 0.100 M CH3COOH (Ka = 1.8e-5) is titrated with 0.100 M NaOH. What is the pH after adding 25.0 mL of base?	SOLVED	pH = 8.7219

pH of a diprotic acid, `ph_diprotic_acid`

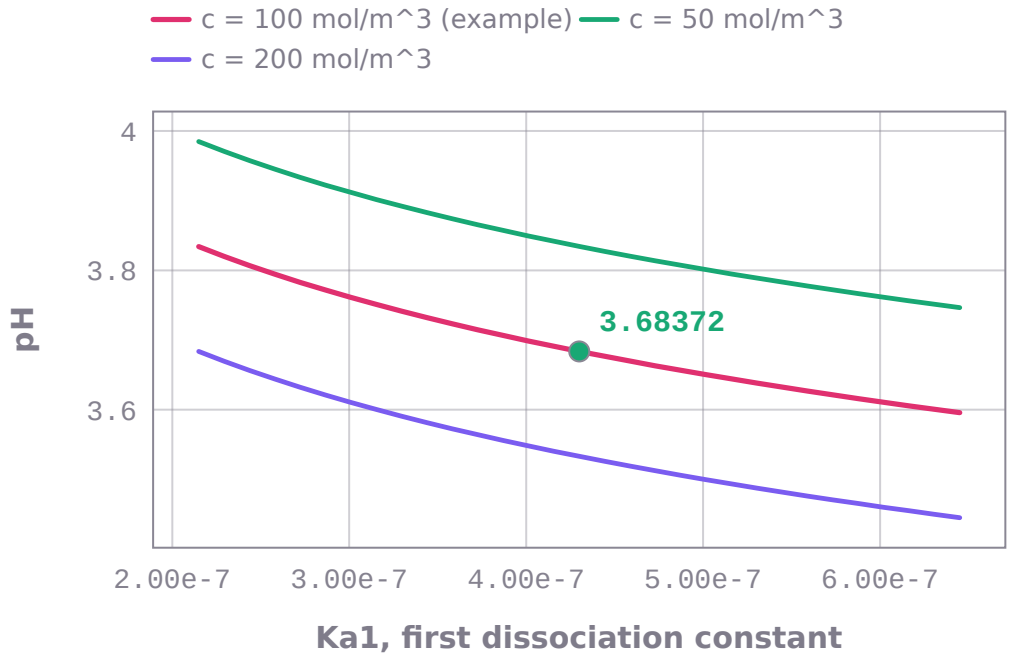
The pH of a diprotic acid H2A from both dissociation constants, solved exactly.

$$[\mathrm{H^+}] = C\frac{K_1[\mathrm{H^+}] + 2K_1K_2}{[\mathrm{H^+}]^2 + K_1[\mathrm{H^+}] + K_1K_2} + \frac{K_w}{[\mathrm{H^+}]}$$

input	meaning	condition
`Ka1`	first dissociation constant	
`Ka2`	second dissociation constant	
`c`	acid concentration	

output	meaning
`pH`	pH

- **Method:** exact charge balance of H2A, solved by bisection
- **Assumes:** 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH (pH) against Ka1 (first dissociation constant) from 2.1500e-07 to 6.4500e-07. One curve for each value of c (acid concentration): 100 mol/m^3, 50 mol/m^3, 200 mol/m^3. Other inputs as in the worked example: Ka2 = 4.8e-11. 31 points per curve computed by chemistry-ai999x; marked: the worked example, Ka1 = 4.3000e-07, pH = 3.68372.

```
chemistry_ai999x.solve('ph_diprotic_acid', Ka1=4.3e-07, Ka2=4.8e-11, c=100.0)
# -> pH = 3.68372   (SOLVED)
```

```
chemistry_ai999x.solve('ph_diprotic_acid', Ka1=0.059, Ka2=6.4e-05, c=100.0)
# -> pH = 1.27699   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of 0.10 M H2CO3 (Ka1 = 4.3e-7, Ka2 = 4.8e-11)?	SOLVED	pH = 3.68372
Find the pH of 0.10 M oxalic acid with Ka1 = 5.9e-2 and Ka2 = 6.4e-5.	SOLVED	pH = 1.27699

pH of a salt of a weak acid, `ph_salt_of_weak_acid`

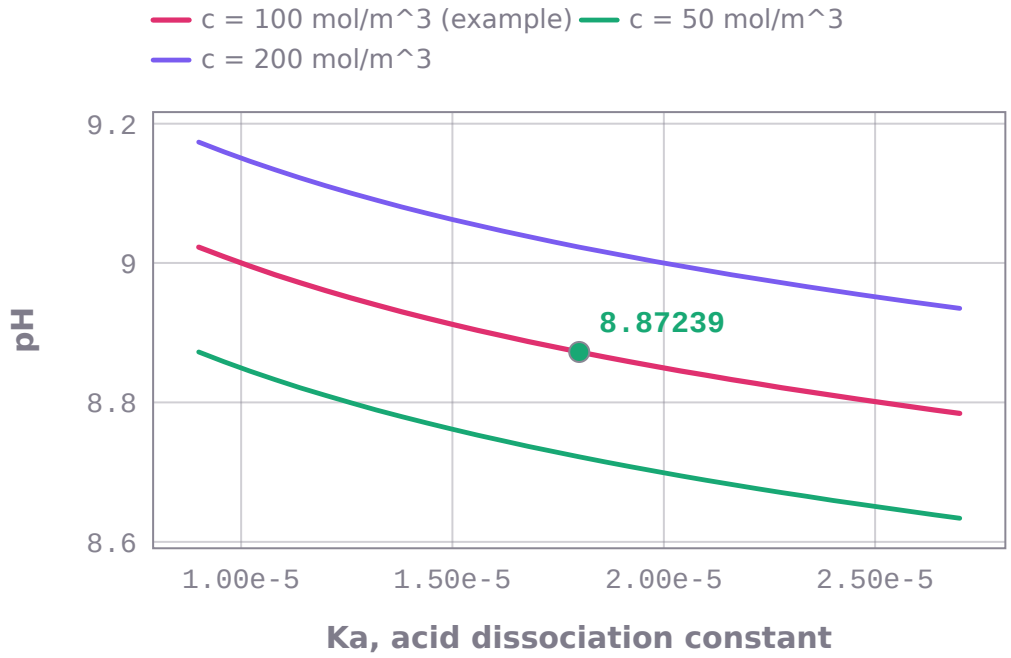
The pH of a solution of a salt of a weak acid and a strong base (e.g. sodium acetate) from the acid's Ka.

$$K_b = \frac{K_w}{K_a}$$

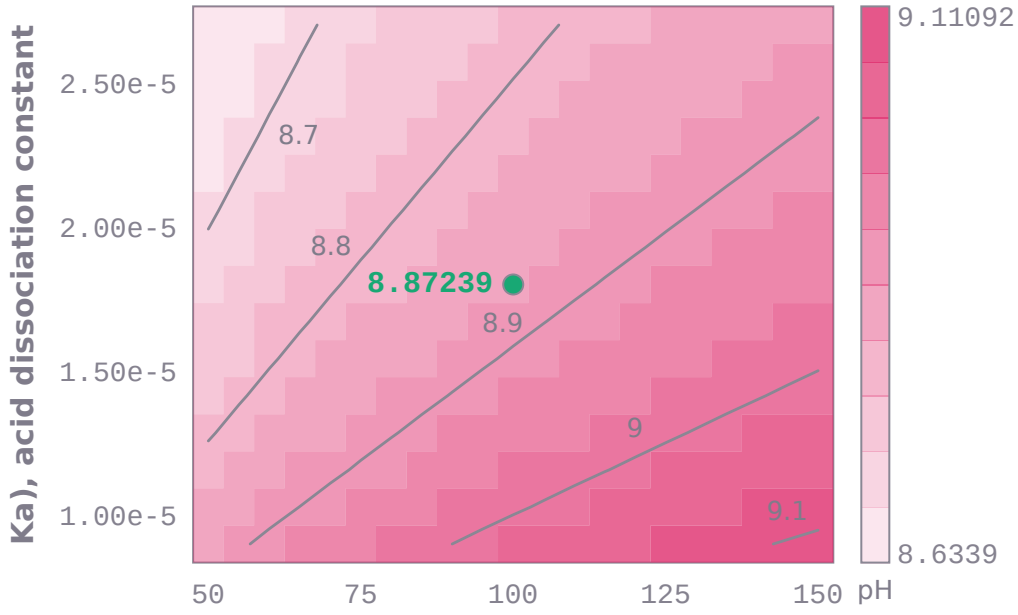
input	meaning	condition
`Ka`	acid dissociation constant	
`c`	salt concentration	

output	meaning
`pH`	pH
`Kb`	Kb of the anion (Kw/Ka)

- **Method:** the anion as a weak base with $Kb = Kw/Ka$, exact charge balance
- **Assumes:** salt of a weak monoprotic acid and a strong base; 25 °C
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



pH (pH) against Ka (acid dissociation constant) from 9.0000e-06 to 2.7000e-05. One curve for each value of c (salt concentration): 100 mol/m^3, 50 mol/m^3, 200 mol/m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, Ka = 1.8000e-05, pH = 8.87239.



pH (pH) over c (salt concentration) from 50 mol/m^3 to 150 mol/m^3 and Ka (acid dissociation constant) from 9.0000e-06 to 2.7000e-05: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 8.6339 to dark 9.11092); iso-lines at pH = 8.7, 8.8, 8.9, 9, 9.1. Marked: the worked example, c = 100 mol/m^3, Ka = 1.8000e-05, pH = 8.87239.

```
chemistry_ai999x.solve('ph_salt_of_weak_acid', Ka=1.8e-05, c=100.0)
# -> pH = 8.87239, Kb = 5.55556e-10   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the pH of 0.10 M sodium acetate (Ka of acetic acid = 1.8e-5)?	SOLVED	pH = 8.87239, Kb = 5.55556e-10

question	outcome	result
Find the pH of a 0.20 M solution of the salt NaF if Ka for HF is 6.8e-4.	SOLVED	pH = 8.235, Kb = 1.47059e-11

Zero-order integrated rate law, `zero_order_concentration`

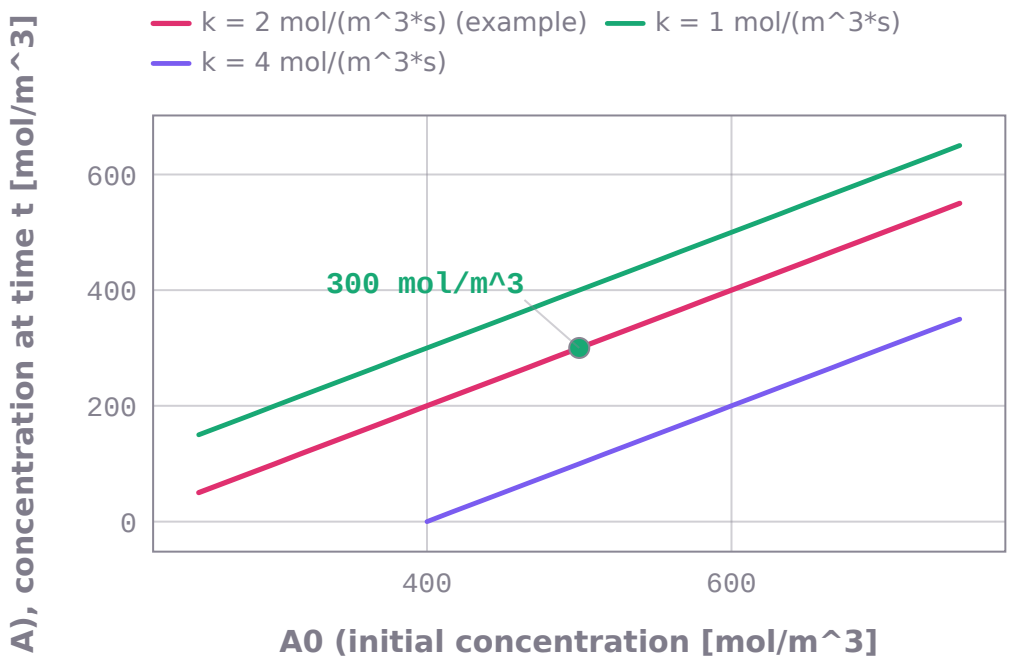
The concentration of a reactant after a time in a zero-order reaction.

$$[A] = [A]_0 - kt$$

input	meaning	condition
`A0`	initial concentration	
`k`	zero-order rate constant	
`t`	elapsed time	

output	meaning
`A`	concentration at time t
`A_molL`	in mol/L

- **Method:** [A] = [A]0 − k t
- **Assumes:** [A]0, k, $t > 0$; [A] stays >= 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



A (concentration at time t) against A0 (initial concentration) from 250 mol/m^3 to 750 mol/m^3. One curve for each value of k (zero-order rate constant): 2 mol/(m^3*s), 1 mol/(m^3*s), 4 mol/(m^3*s). Other inputs as in the worked example: t = 100.0 s. 31 points per curve computed by chemistry-ai999x; marked: the worked example, A0 = 500 mol/m^3, A = 300 mol/m^3.

```
chemistry_ai999x.solve('zero_order_concentration', A0=500.0, k=2.0, t=100.0)
# -> A = 300 mol/m^3, A_molL = 0.3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A zero-order reaction has k = 0.0020 M/s. If [A]0 = 0.50 M, what is the concentration after 100 s?	SOLVED	A = 300 mol/m^3, A_molL = 0.3
A zero-order reaction has k = 0.0020 M/s. If [A]0 = 0.50 M, what is the concentration after 1000 s?	INVALID_INPUT	Zero-order integrated rate law: the reactant is used up before this time (A0 - k t < 0).

First-order integrated rate law, `first_order_concentration`

The concentration of a reactant after a time in a first-order reaction.

$$[A] = [A]_0\, e^{-kt}$$

input	meaning	condition
`A0`	initial concentration	
`k`	first-order rate constant	
`t`	elapsed time	

output	meaning
`A`	concentration at time t
`A_molL`	in mol/L
`fraction_left`	fraction remaining

- **Method:** [A] = [A]0 exp(-k t)
- **Assumes:** [A]0, k, $t > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Reactant concentration against time from A0 = 500 mol/m³ for zero-, first- and second-order rate laws with the same initial rate (2 mol/(m³·s)), each computed by the package at up to 121 times. At t = 200 s: 100, 224.664 and 277.778 mol/m³; marked: t = 200 s, [A] = 224.664 mol/m³ (first order).



A (concentration at time t) against k (first-order rate constant) from 0.0015 1/s to 0.15 1/s on a logarithmic axis. One curve for each value of A0 (initial concentration): 800 mol/m^3, 400 mol/m^3, 1600 mol/m^3. Other inputs as in the worked example: t = 60.0 s. 41 points per curve computed by chemistry-ai999x; marked: the worked example, k = 0.015 1/s, A = 325.256 mol/m^3.

```
chemistry_ai999x.solve('first_order_concentration', A0=800.0, k=0.015, t=60.0)
# -> A = 325.256 mol/m^3, A_molL = 0.325256, fraction_left = 0.40657   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A first-order reaction has k = 0.0150 s^-1. If [A]0 = 0.800 M, what is the concentration after 60.0 s?	SOLVED	A = 325.256 mol/m^3, A_molL = 0.325256, fraction_left = 0.40657
For a first-order decomposition with k = 2.5e-3 min^-1 and an initial concentration of 0.20 M, find the concentration after 2.0 h.	SOLVED	A = 148.164 mol/m^3, A_molL = 0.148164, fraction_left = 0.740818

Second-order integrated rate law, `second_order_concentration`

The concentration of a reactant after a time in a second-order reaction (rate = k[A]^2).

$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

input	meaning	condition
`A0`	initial concentration	
`k`	second-order rate constant	
`t`	elapsed time	

output	meaning
`A`	concentration at time t
`A_molL`	in mol/L

- **Method:** 1/[A] = 1/[A]$0 + k$ t
- **Assumes:** [A]0, k, $t > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



A (concentration at time t) against k (second-order rate constant) from 5.0000e-05 m^3/(mol*s) to 0.005 m^3/(mol*s) on a logarithmic axis. One curve for each value of t (elapsed time): 30 s, 15 s, 60 s. Other inputs as in the worked example: A0 = 200.0 mol/m^3. 41 points per curve computed by chemistry-ai999x; marked: the worked example, k = 0.0005 m^3/(mol*s), A = 50 mol/m^3.

```
chemistry_ai999x.solve('second_order_concentration', A0=200.0, k=0.0005, t=30.0)
# -> A = 50 mol/m^3, A_molL = 0.05   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A second-order reaction has k = 0.50 M^-1 s^-1. If [A]0 = 0.20 M, what is the concentration after 30 s?	SOLVED	A = 50 mol/m^3, A_molL = 0.05

Half-life of a first-order reaction, `first_order_half_life`

The half-life of a first-order reaction from its rate constant.

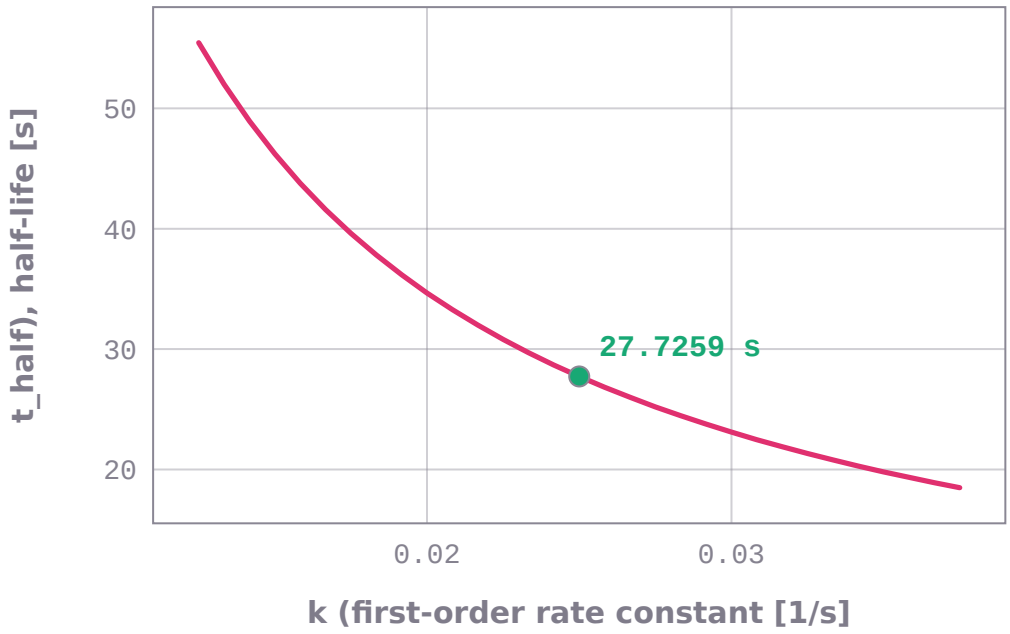
$$t_{1/2} = \frac{\ln 2}{k}$$

input	meaning	condition
k	first-order rate constant	

output	meaning
t_half	half-life

- **Method:** $t_1/2 = ln\ 2/k$

- **Assumes:** $k > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



t_half (half-life) against k (first-order rate constant) from 0.0125 1/s to 0.0375 1/s. 31 points computed by chemistry-ai999x; marked: the worked example, k = 0.025 1/s, t_half = 27.7259 s.

```
chemistry_ai999x.solve('first_order_half_life', k=0.025)
# -> t_half = 27.7259 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the half-life of a first-order reaction with k = 0.0250 s^-1?	SOLVED	t_half = 27.7259 s
Find the half-life of a first-order process whose rate constant is 1.2e-4 min^-1.	SOLVED	t_half = 346574 s

Time to reach a concentration (first order), `first_order_time`

How long a first-order reaction takes to bring a reactant from one concentration to another.

$$t = \frac{1}{k} \ln \frac{[A]_0}{[A]}$$

input	meaning	condition
`A0`	initial concentration	
`A`	final concentration	
`k`	first-order rate constant	

output	meaning
`t`	time needed

- **Method:** $t = ln$([A]0/[A]) / k
- **Assumes:** 0 < [A] < [A]0; $k > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



t (time needed) against k (first-order rate constant) from 0.015 1/s to 0.045 1/s. One curve for each value of A0 (initial concentration): 500 mol/m^3, 250 mol/m^3, 1000 mol/m^3. Other inputs as in the worked example: A = 100.0 mol/m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, k = 0.03 1/s, t = 53.6479 s.

```
chemistry_ai999x.solve('first_order_time', A0=500.0, A=100.0, k=0.03)
# -> t = 53.6479 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A first-order reaction has k = 0.0300 s^-1. How long does it take for the concentration to fall from 0.500 M to 0.100 M?	SOLVED	t = 53.6479 s

Arrhenius equation: rate constant at another temperature, `arrhenius_k2`

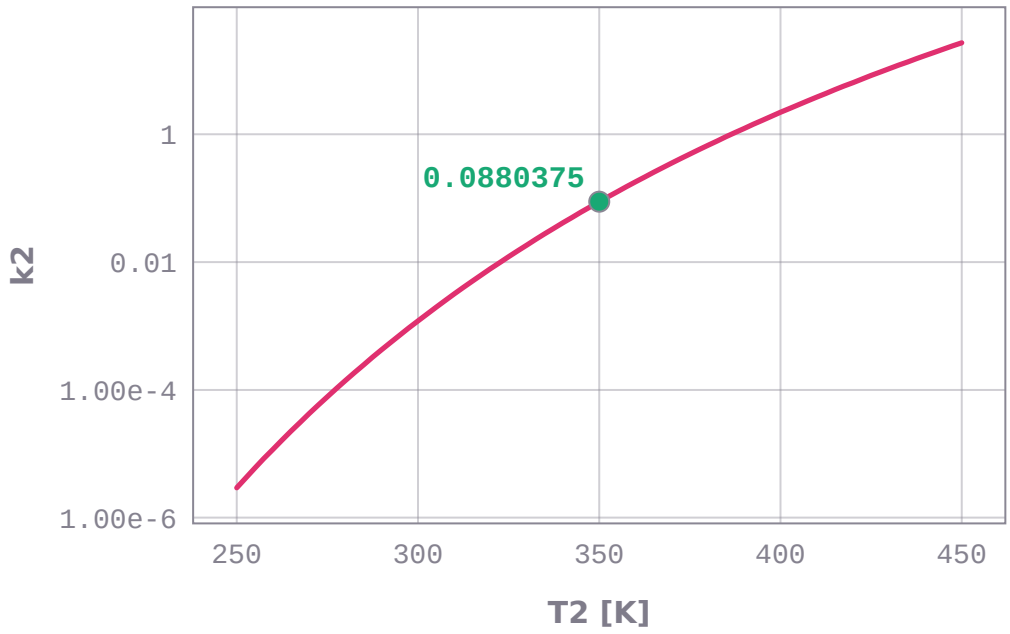
The rate constant at a second temperature from the rate constant at a first one and the activation energy.

$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

input	meaning	condition
`k1`	rate constant at T1 (any units)	
`T1`	first temperature	
`T2`	second temperature	
`Ea`	activation energy	

output	meaning
`k2`	rate constant at T2 (units of k1)
`ratio`	k_2/k_1

- **Method:** two-point Arrhenius equation
- **Assumes:** k1, T1, T2, $Ea > 0$; Ea independent of T
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Arrhenius equation: rate constant at another temperature: k2 against T2 from 250 to 450, 81 runs of the package; other inputs as in the worked example (k1 = 0.0012, T1 = 300.0, Ea = 75000.0). Marked: the worked example, T2 = 350, k2 = 0.0880375.

```
chemistry_ai999x.solve('arrhenius_k2', k1=0.0012, T1=300.0, T2=350.0, Ea=75000.0)
# -> k2 = 0.0880375, ratio = 73.3646   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A reaction has k = 1.2e-3 s^-1 at 300 K and Ea = 75.0 kJ/mol. What is k at 350 K?	SOLVED	k2 = 0.0880375, ratio = 73.3646
k1 = 0.050 M^-1 s^-1, T1 = 298 K, T2 = 318 K and the activation energy is 52 kJ/mol. Find the rate constant at T2.	SOLVED	k2 = 0.187161, ratio = 3.74321

Activation energy from two rate constants, `activation_energy`

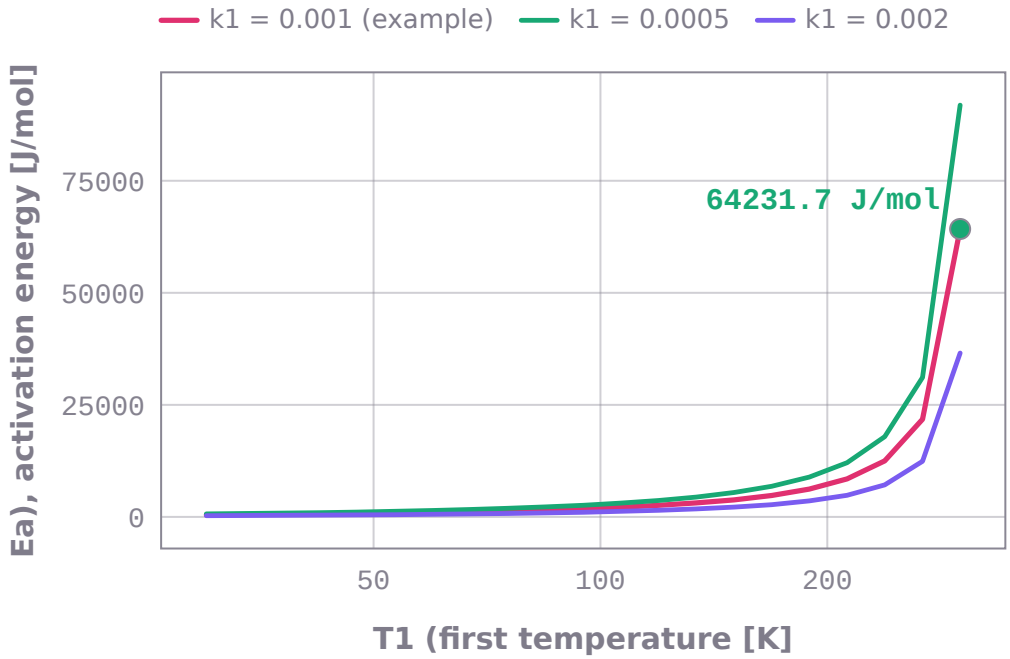
The Arrhenius activation energy from rate constants measured at two temperatures.

$$E_a = \frac{R \ln(k_2/k_1)}{1/T_1 - 1/T_2}$$

input	meaning	condition
`k1`	rate constant at T1	
`T1`	first temperature	
`k2`	rate constant at T2	
`T2`	second temperature	

output	meaning
`Ea`	activation energy
`Ea_kJ`	in kJ/mol

- **Method:** two-point Arrhenius equation solved for Ea
- **Assumes:** k1, $k_2 > 0$; $T_1 \neq T_2$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Ea (activation energy) against T1 (first temperature) from 30 K to 300 K on a logarithmic axis. One curve for each value of k1 (rate constant at T1): 0.001, 0.0005, 0.002. Other inputs as in the worked example: k2 = 0.005, T2 = 320.0 K. 21 points per curve computed by chemistry-ai999x; marked: the worked example, T1 = 300 K, Ea = 64231.7 J/mol.

```
chemistry_ai999x.solve('activation_energy', k1=0.001, T1=300.0, k2=0.005, T2=320.0)
# -> Ea = 64231.7 J/mol, Ea_kJ = 64.2317   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
k1 = 1.0e-3 s^-1 at T1 = 300 K and k2 = 5.0e-3 s^-1 at T2 = 320 K. What is the activation energy?	SOLVED	Ea = 64231.7 J/mol, Ea_kJ = 64.2317

Arrhenius equation: rate constant, `arrhenius_rate_constant`

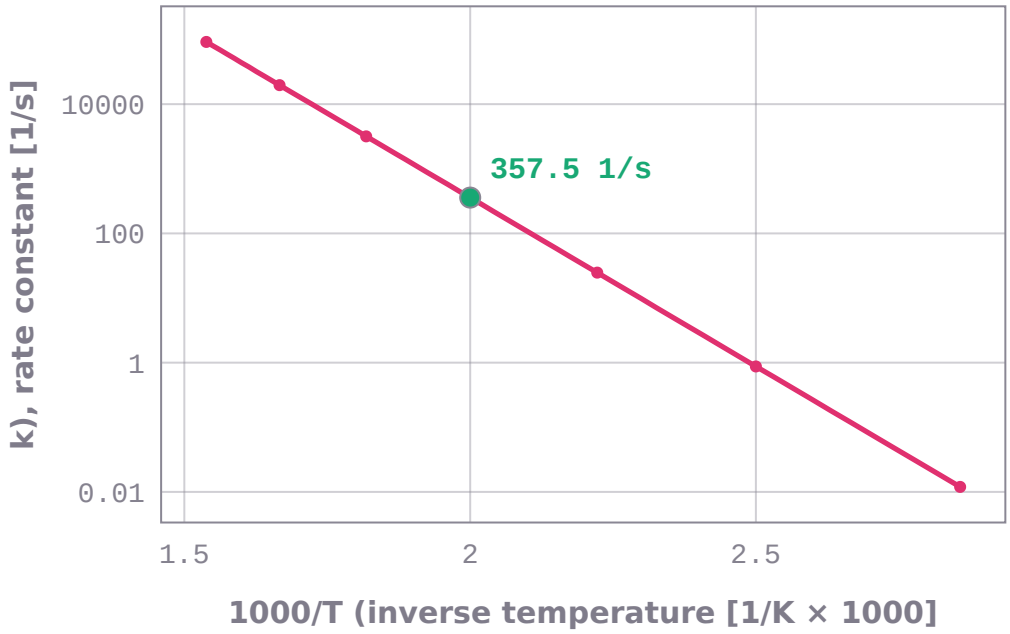
The rate constant from the pre-exponential factor, the activation energy and the temperature.

$$k = A\,e^{-E_a/RT}$$

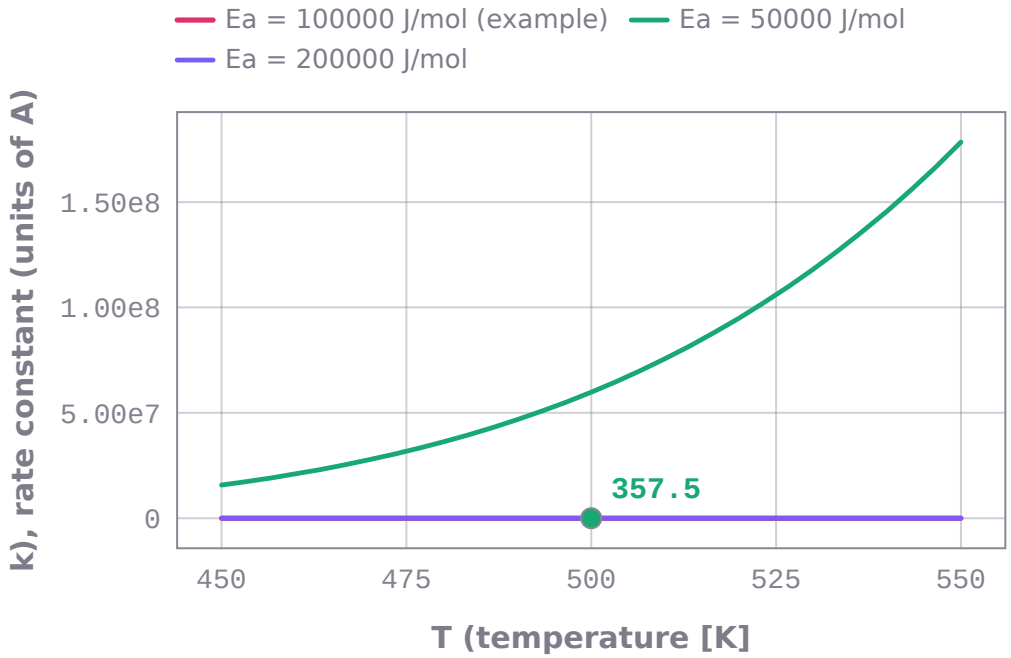
input	meaning	condition
A	pre-exponential factor (units of k)	
Ea	activation energy	
T	temperature	

output	meaning
k	rate constant (units of A)

- **Method:** $k = A$ exp(-Ea/RT)
- **Assumes:** A, Ea, $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Arrhenius plot: the rate constant (logarithmic axis) against 1000/T for A = 1.0000e+13 1/s and Ea = 100 kJ/mol, computed by the package at 61 temperatures from 350 K to 650 K, a straight line of slope −Ea/R. Marked: T = 500 K, k = 357.5 1/s.



k (rate constant (units of A)) against T (temperature) from 450 K to 550 K. One curve for each value of Ea (activation energy): 100000 J/mol, 50000 J/mol, 200000 J/mol. Other inputs as in the worked example: A = 10000000000000.0. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T = 500 K, k = 357.5.

```
chemistry_ai999x.solve('arrhenius_rate_constant', A=10000000000000.0, Ea=100000.0, T=500.0)
# -> k = 357.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A reaction has a pre-exponential factor of 1.0e13 s^-1 and Ea = 100 kJ/mol. What is the rate constant at 500 K?	SOLVED	k = 357.5

Reaction order from initial rates, `reaction_order_from_rates`

The order with respect to one reactant from two initial-rate experiments (other concentrations fixed).

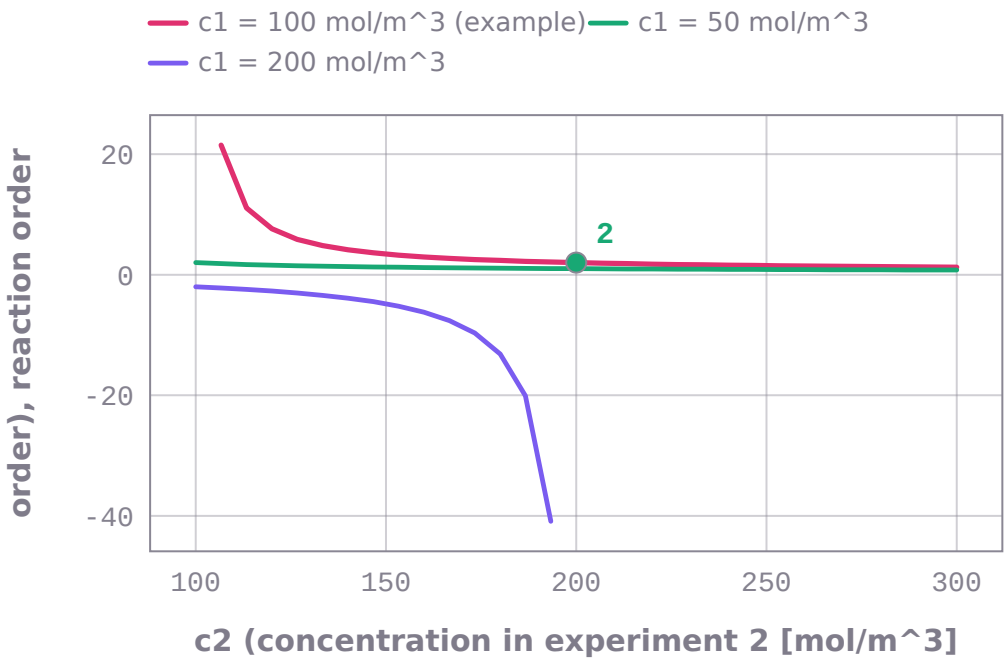
$$n = \frac{\ln(r_2/r_1)}{\ln(c_2/c_1)}$$

input	meaning	condition
`c1`	concentration in experiment 1	
`r1`	initial rate in experiment 1	
`c2`	concentration in experiment 2	
`r2`	initial rate in experiment 2	

output	meaning
`order`	reaction order

- **Method:** ratio of the two rate laws
- **Assumes:** $c_1 \neq c_2$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



order (reaction order) against c2 (concentration in experiment 2) from 100 mol/m^3 to 300 mol/m^3. One curve for each value of c1 (concentration in experiment 1): 100 mol/m^3, 50 mol/m^3, 200 mol/m^3. Other inputs as in the worked example: r1 = 2.0 mol/(m^3*s), r2 = 8.0 mol/(m^3*s). 30 points per curve computed by chemistry-ai999x; marked: the worked example, c2 = 200 mol/m^3, order = 2.

```
chemistry_ai999x.solve('reaction_order_from_rates', c1=100.0, r1=2.0, c2=200.0, r2=8.0)
# -> order = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
c1 = 0.10 M, r1 = 2.0e-3 M/s, c2 = 0.20 M, r2 = 8.0e-3 M/s. What is the order with respect to A?	SOLVED	order = 2

Consecutive first-order reactions (A -> B -> C), `consecutive_first_order`

Concentrations of the reactant and the intermediate of A -> B -> C (both steps first order) at a time.

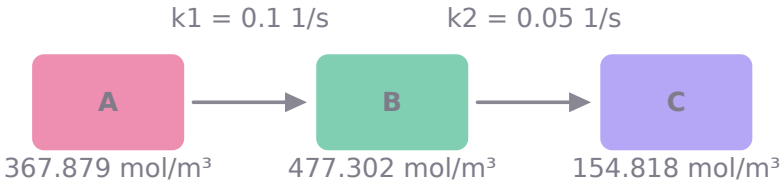
$$[B] = \frac{k_1 [A]_0}{k_2 - k_1} \left(e^{-k_1 t} - e^{-k_2 t} \right)$$

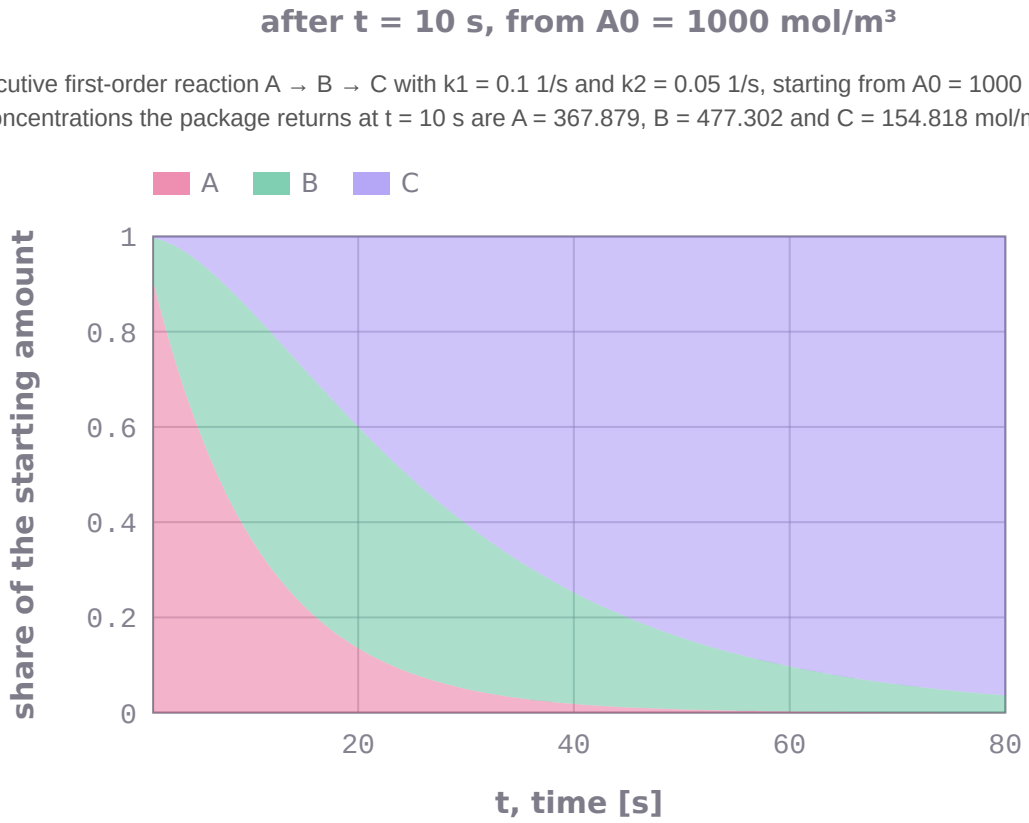
input	meaning	condition
`A0`	initial concentration of A (B and C start at 0)	
`k1`	rate constant of A -> B	
`k2`	rate constant of B -> C	
`t`	elapsed time	

output	meaning
`B`	intermediate concentration

output	meaning
`A`	reactant
`C`	product

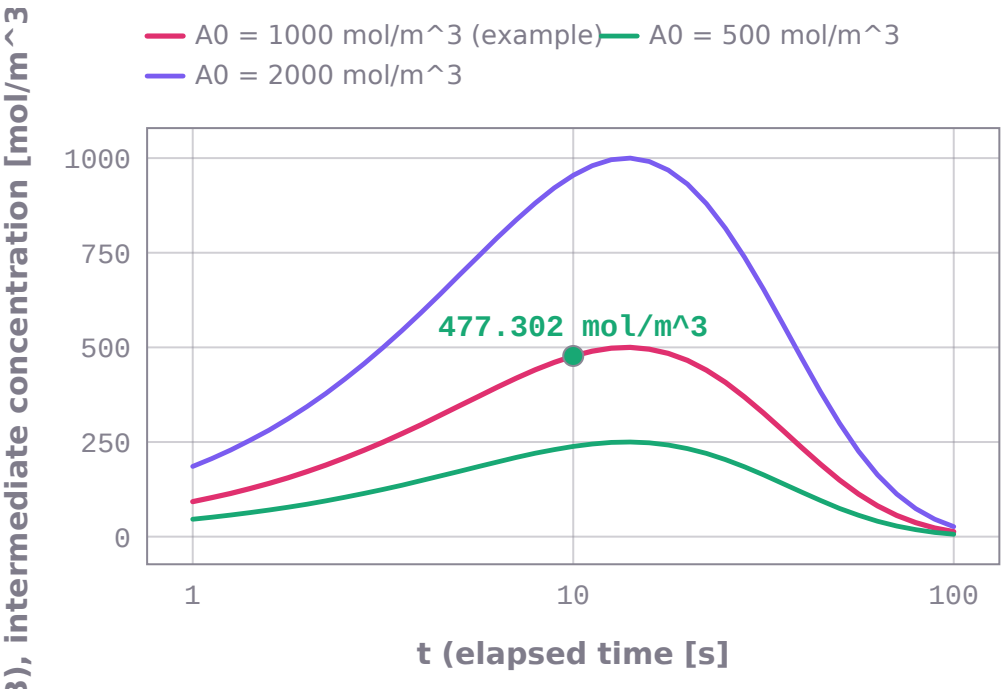
- **Method:** analytic solution of the linear rate equations (the $k_1 = k_2$ limit included)
- **Assumes:** both steps first order; B and C absent at $t = 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)





The consecutive first-order reaction A → B → C with k1 = 0.1 1/s and k2 = 0.05 1/s, starting from A0 = 1000 mol/m³: the concentrations the package returns at t = 10 s are A = 367.879, B = 477.302 and C = 154.818 mol/m³.

How the starting 1000 mol/m³ of A divides between A, B and C over time (stacked shares, k1 = 0.1 1/s, k2 = 0.05 1/s), each concentration computed by the package at 80 times. B is largest near t = 14 s (499.977 mol/m³, a share of 0.499977); at t = 80 s the shares are 0.000335463, 0.0359604 and 0.963704.



B (intermediate concentration) against t (elapsed time) from 1 s to 100 s on a logarithmic axis. One curve for each value of A0 (initial concentration of A (B and C start at 0)): 1000 mol/m^3, 500 mol/m^3, 2000 mol/m^3. Other inputs as in the worked example: k1 = 0.1 1/s, k2 = 0.05 1/s. 41 points per curve computed by chemistry-ai999x; marked: the worked example, t = 10 s, B = 477.302 mol/m^3.

```
chemistry_ai999x.solve('consecutive_first_order', A0=1000.0, k1=0.1, k2=0.05, t=10.0)
# -> B = 477.302 mol/m^3, A = 367.879 mol/m^3, C = 154.818 mol/m^3   (SOLVED)
```

```
chemistry_ai999x.solve('consecutive_first_order', A0=1000.0, k1=0.1, k2=0.1, t=10.0)
# -> B = 367.879 mol/m^3, A = 367.879 mol/m^3, C = 264.241 mol/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For A -> B -> C with k1 = 0.10 s^-1, k2 = 0.050 s^-1 and [A]0 = 1.0 M, what is the concentration of the intermediate B after 10 s?	SOLVED	B = 477.302 mol/m^3, A = 367.879 mol/m^3, C = 154.818 mol/m^3

Heat from a temperature change, `calorimetry_heat`

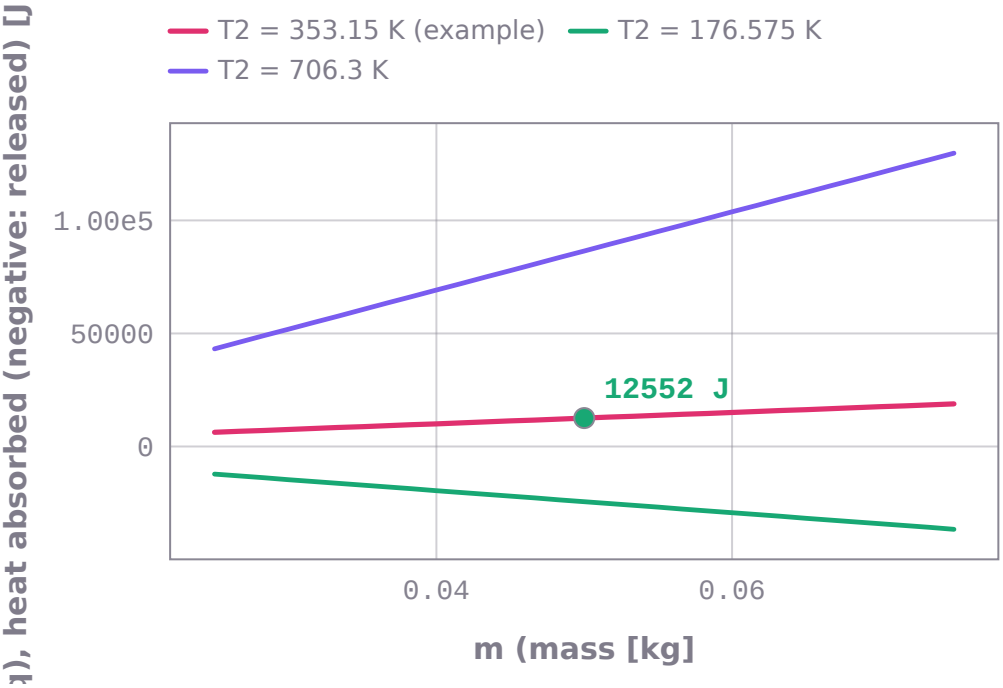
The heat absorbed (positive) or released (negative) by a mass of substance whose temperature changes.

$$q = mc(T_2 - T_1)$$

input	meaning	condition
`m`	mass	
`c`	specific heat capacity	
`T1`	initial temperature	
`T2`	final temperature	

output	meaning
`q`	heat absorbed (negative: released)

- **Method:** $q = m$ c dT
- **Assumes:** no phase change
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



q (heat absorbed (negative: released)) against m (mass) from 0.025 kg to 0.075 kg. One curve for each value of T2 (final temperature): 353.15 K, 176.575 K, 706.3 K. Other inputs as in the worked example: c = 4184.0 J/(kg*K), T1 = 293.15 K. 31 points per curve computed by chemistry-ai999x; marked: the worked example, m = 0.05 kg, q = 12552 J.

```
chemistry_ai999x.solve('calorimetry_heat', m=0.05, c=4184.0, T1=293.15, T2=353.15)
# -> q = 12552 J   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How much heat is needed to warm 50.0 g of water from 20.0 °C to 80.0 °C (c = 4.184 J/(g·°C))?	SOLVED	q = 12552 J
How much heat is released when 200 g of iron (c = 0.449 J/(g·K)) cools from 400 K to 300 K?	SOLVED	q = -8980 J

Reaction enthalpy from coffee-cup calorimetry, `enthalpy_from_calorimetry`

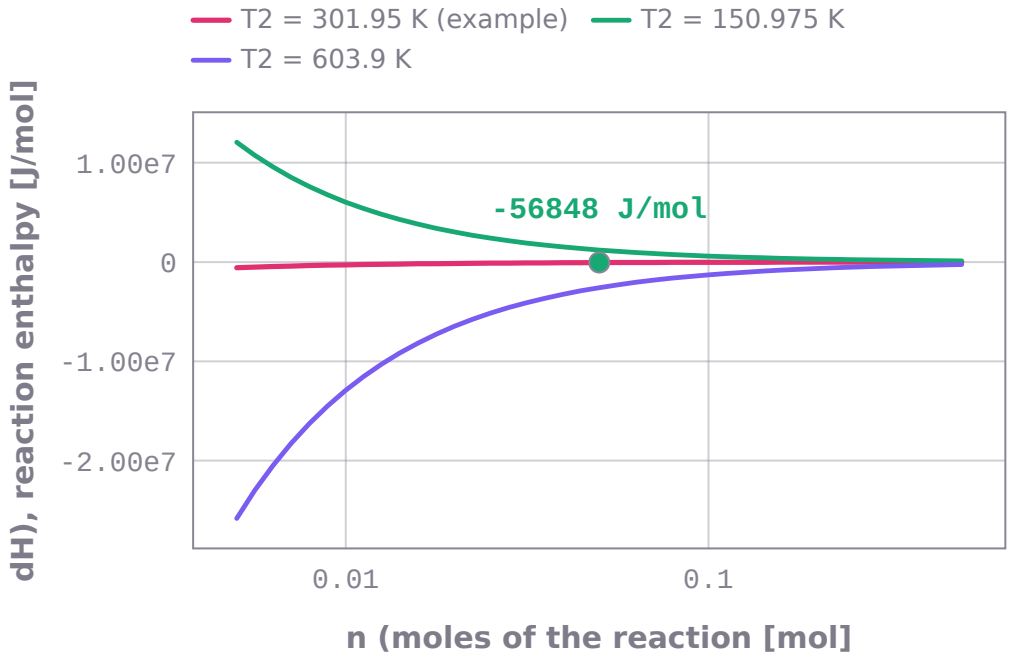
The molar enthalpy of a reaction from the temperature change of the solution it takes place in.

$$\Delta H = -\frac{mc(T_2 - T_1)}{n}$$

input	meaning	condition
`m`	mass of the solution	
`c`	specific heat of the solution	
`T1`	initial temperature	
`T2`	final temperature	
`n`	moles of the reaction (limiting reactant)	

output	meaning
`dH`	reaction enthalpy
`dH_kJ`	in kJ/mol
`q_solution`	heat taken up by the solution

- **Method:** heat gained by the solution, with the opposite sign, per mole
- **Assumes:** no heat lost to the calorimeter or surroundings; constant pressure
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dH (reaction enthalpy) against n (moles of the reaction (limiting reactant)) from 0.005 mol to 0.5 mol on a logarithmic axis. One curve for each value of T2 (final temperature): 301.95 K, 150.975 K, 603.9 K. Other inputs as in the worked example: m = 0.1 kg, c = 4180.0 J/(kg*K), T1 = 295.15 K. 41 points per curve computed by chemistry-ai999x; marked: the worked example, n = 0.05 mol, dH = -56848 J/mol.

```
chemistry_ai999x.solve('enthalpy_from_calorimetry', m=0.1, c=4180.0, T1=295.15, T2=301.95,
n=0.05)
# -> dH = -56848 J/mol, dH_kJ = -56.848, q_solution = 2842.4 J   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
When 0.0500 mol of HCl reacts in 100 g of solution (c = 4.18 J/(g·°C)), the temperature rises from 22.0 °C to 28.8 °C. What is the enthalpy of reaction?	SOLVED	dH = -56848 J/mol, dH_kJ = -56.848, q_solution = 2842.4 J

Reaction enthalpy from enthalpies of formation (Hess's law), `reaction_enthalpy_from_formation`

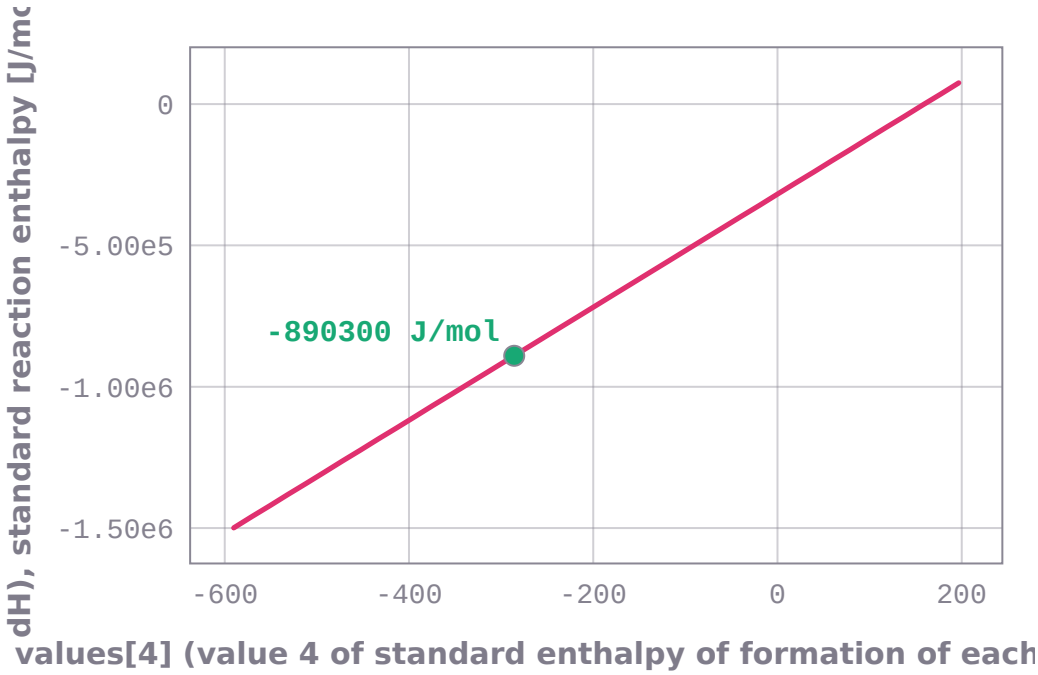
The standard reaction enthalpy from the standard enthalpies of formation of every species (balanced equation).

$$\Delta H^\circ_{rxn} = \sum \nu_p \Delta H^\circ_f(p) - \sum \nu_r \Delta H^\circ_f(r)$$

input	meaning	condition
`equation`	balanced equation	
`values`	standard enthalpy of formation of each species in the order written (kJ/mol)	

output	meaning
`dH`	standard reaction enthalpy
`dH_kJ`	in kJ/mol

- **Method:** coefficient-weighted sum of formation enthalpies, products minus reactants
- **Assumes:** the equation as written conserves every element; one value per species, in kJ/mol
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dH (standard reaction enthalpy) as value 4 of values (standard enthalpy of formation of each species in the order written (kJ/mol)) moves from -590.25 to 196.75, the other values as given (-74.8, 0, -393.5, -285.8); 41 runs of chemistry-ai999x. Other inputs: equation = CH4 + 2 O2 -> CO2 + 2 H2O. Marked: the worked example, values[4] = -285.8, dH = -890300 J/mol.

```
chemistry_ai999x.solve('reaction_enthalpy_from_formation', equation='CH4 + 2 O2 -> CO2 + 2 H2O', values='-74.8, 0, -393.5, -285.8')
# -> dH = -890300 J/mol, dH_kJ = -890.3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Using enthalpies of formation CH4 = -74.8, O2 = 0, CO2 = -393.5, H2O = -285.8 (kJ/mol), find ΔH° for CH4 + 2 O2 -> CO2 + 2 H2O.	SOLVED	dH = -890300 J/mol, dH_kJ = -890.3
Using enthalpies of formation CH4 = -74.8, O2 = 0, CO2 = -393.5, H2O = -285.8 (kJ/mol), find ΔH° for CH4 + O2 -> CO2 + H2O.	INVALID_INPUT	Reaction enthalpy from enthalpies of formation (Hess's law): the equation as written is not balanced; balance it first.

Gibbs energy change (dG = dH - T dS), `gibbs_free_energy`

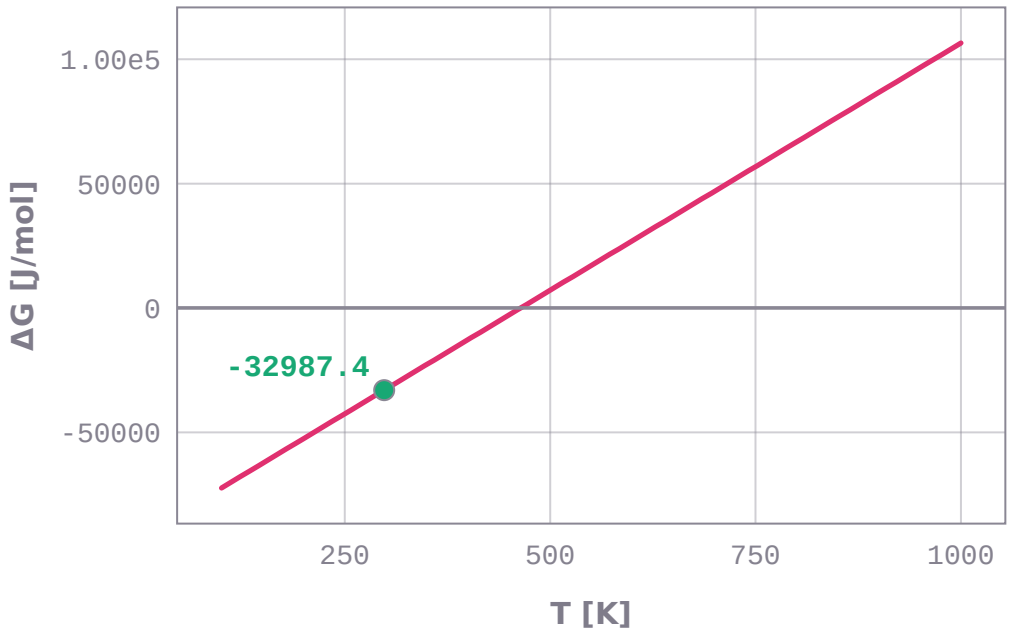
The Gibbs energy change of a process from its enthalpy and entropy changes, and whether it is spontaneous.

$$\Delta G = \Delta H - T\Delta S$$

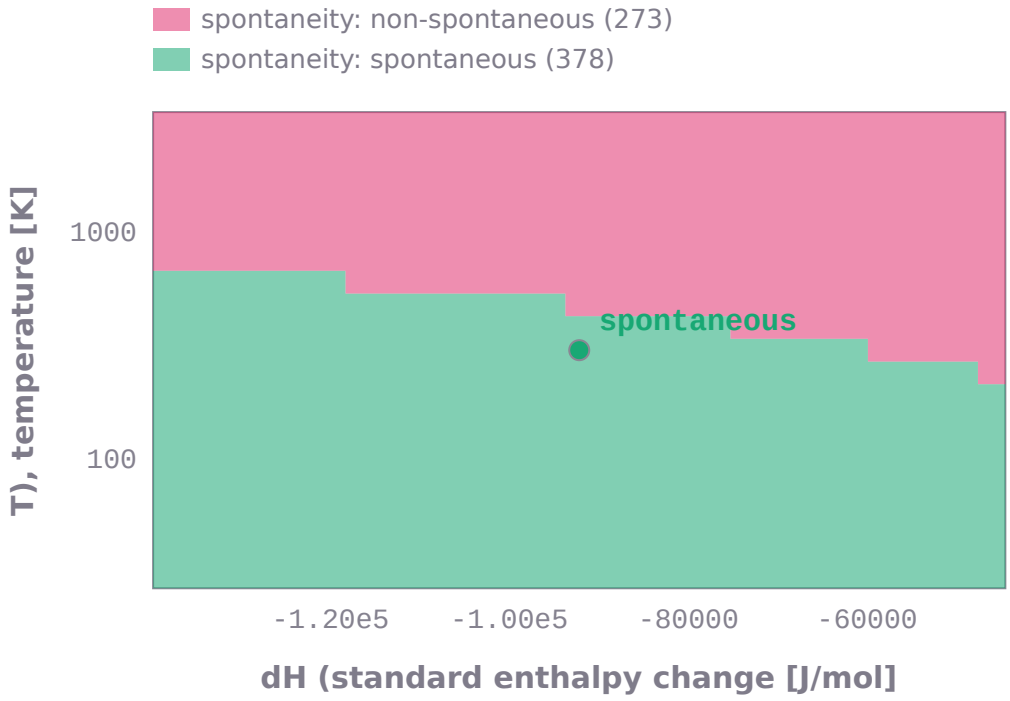
input	meaning	condition
`dH`	standard enthalpy change	
`dS`	entropy change	
`T`	temperature	

output	meaning
`dG`	Gibbs energy change
`dG_kJ`	in kJ/mol
`spontaneity`	spontaneous / non-spontaneous

- **Method:** $dG = dH - T$ dS
- **Assumes:** dH and dS taken as constant
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Gibbs energy change (dG = dH - T dS): dG against T from 100 to 1000, 81 runs of the package; other inputs as in the worked example (dH = -92200.0, dS = -198.7). Marked: the worked example, T = 298, dG = -32987.4.



spontaneous / non-spontaneous (spontaneity) over dH (standard enthalpy change) from -138300 J/mol to -46100 J/mol and T (temperature) from 29.8 K to 2980 K (logarithmic vertical axis): each of the 31 × 21 cells is one run of chemistry-ai999x, coloured by the verdict it returned, non-spontaneous: 273 cells, spontaneous: 378 cells. Other inputs as in the worked example: dS = -198.7 J/(mol*K). Marked: the worked example, dH = -92200 J/mol, T = 298 K → spontaneous.

```
chemistry_ai999x.solve('gibbs_free_energy', dH=-92200.0, dS=-198.7, T=298.0)
# -> dG = -32987.4 J/mol, dG_kJ = -32.9874, spontaneity = spontaneous   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For a reaction ΔH = -92.2 kJ/mol and ΔS = -198.7 J/(mol·K). What is ΔG at 298 K? Is it spontaneous?	SOLVED	dG = -32987.4 J/mol, dG_kJ = -32.9874, spontaneity = spontaneous
ΔH = 178 kJ/mol, ΔS = 161 J/(mol·K). Find ΔG at 1000 K.	SOLVED	dG = 17000 J/mol, dG_kJ = 17, spontaneity = non-spontaneous

Temperature at which dG = 0, `crossover_temperature`

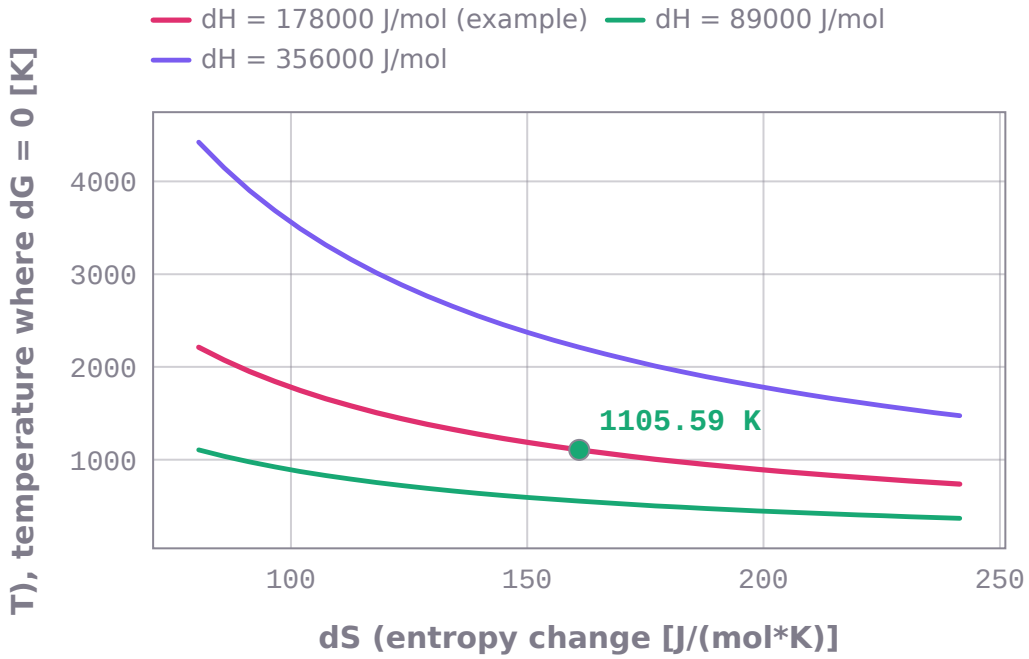
The temperature at which a reaction changes between spontaneous and non-spontaneous $(dG = 0)$.

$$T = \frac{\Delta H}{\Delta S}$$

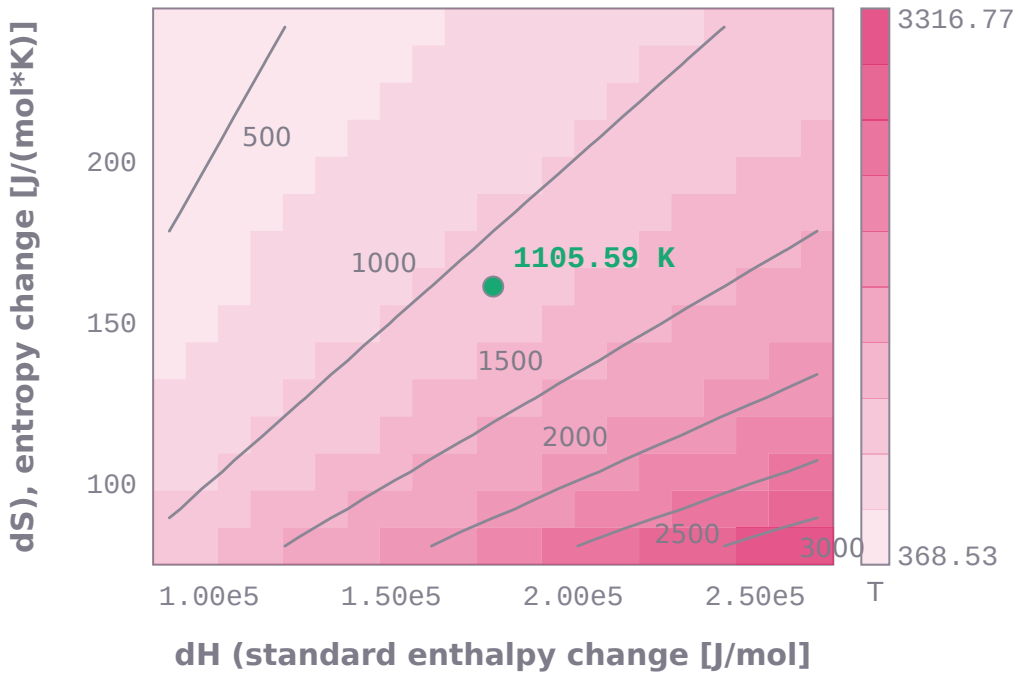
input	meaning	condition
`dH`	standard enthalpy change	
`dS`	entropy change	

output	meaning
`T`	temperature where $dG = 0$
`T_C`	in °C

- **Method:** $T = dH/dS$
- **Assumes:** dH and dS of the same sign
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



T (temperature where dG = 0) against dS (entropy change) from 80.5 J/(mol*K) to 241.5 J/(mol*K). One curve for each value of dH (standard enthalpy change): 178000 J/mol, 89000 J/mol, 356000 J/mol. 31 points per curve computed by chemistry-ai999x; marked: the worked example, dS = 161 J/(mol*K), T = 1105.59 K.



T (temperature where dG = 0) over dH (standard enthalpy change) from 89000 J/mol to 267000 J/mol and dS (entropy change) from 80.5 J/(mol*K) to 241.5 J/(mol*K): each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 368.53 K to dark 3316.77 K); iso-lines at T = 500, 1000, 1500, 2000, 2500, 3000 K. Marked: the worked example, dH = 178000 J/mol, dS = 161 J/(mol*K), T = 1105.59 K.

```
chemistry_ai999x.solve('crossover_temperature', dH=178000.0, dS=161.0)
# -> T = 1105.59 K, T_C = 832.44   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For ΔH = 178 kJ/mol and ΔS = 161 J/(mol·K), above what temperature is the reaction spontaneous?	SOLVED	T = 1105.59 K, T_C = 832.44
ΔH = -50 kJ/mol and ΔS = 100 J/(mol·K). At what temperature does the reaction become spontaneous?	INVALID_INPUT	Temperature at which dG = 0: dH and dS must have the same sign for a temperature where dG = 0 to exist.

Heat of reaction for a given mass, `heat_from_reaction_mass`

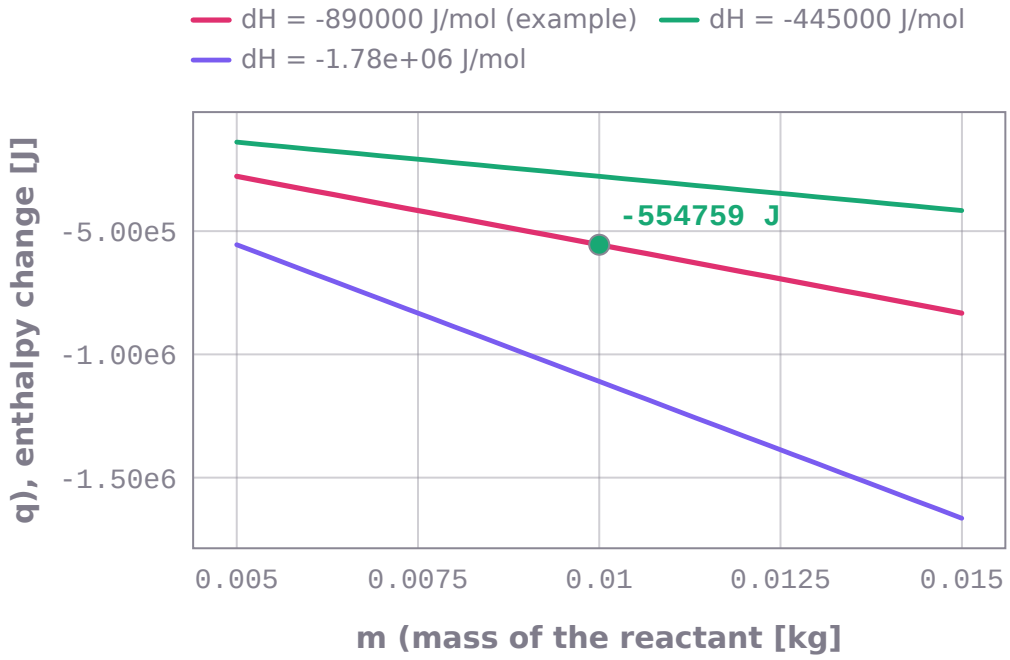
The heat of a reaction for a given mass of a reactant, from its molar enthalpy (per mole of that reactant).

$$q = \frac{m}{M}\Delta H$$

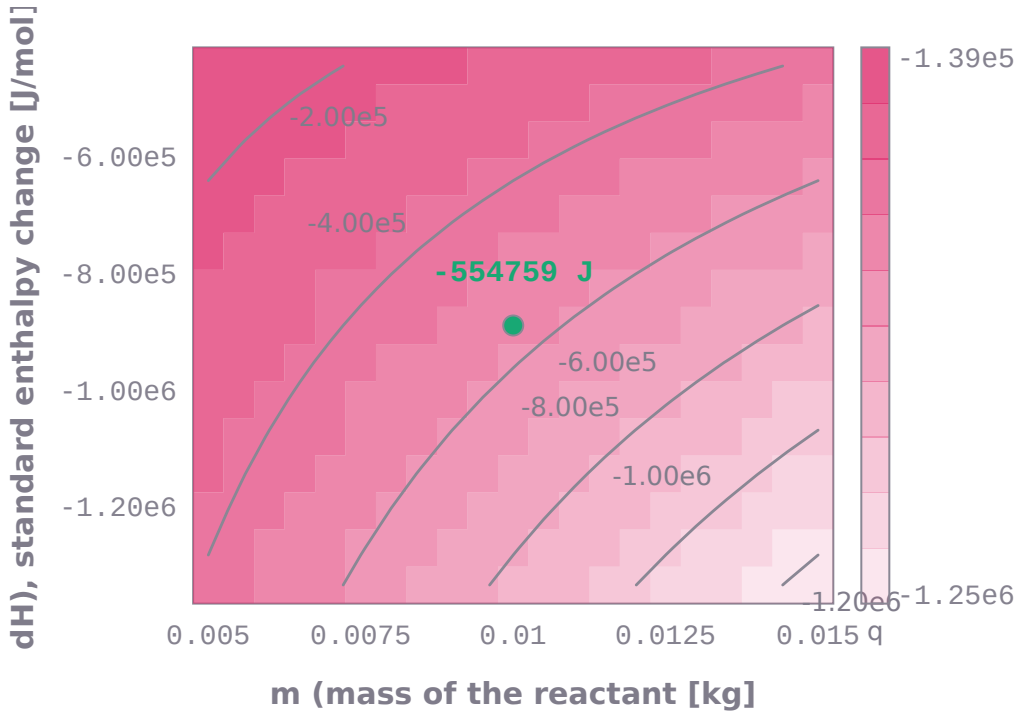
input	meaning	condition
`formula`	formula of the reactant	
`m`	mass of the reactant	
`dH`	standard enthalpy change	

output	meaning
`q`	enthalpy change (negative: heat released)
`heat_released`	heat released

- **Method:** moles of reactant times its molar enthalpy
- **Assumes:** dH is per mole of the given reactant
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



q (enthalpy change (negative: heat released)) against m (mass of the reactant) from 0.005 kg to 0.015 kg. One curve for each value of dH (standard enthalpy change): -890000 J/mol, -445000 J/mol, -1.78e+06 J/mol. Other inputs as in the worked example: formula = CH4. 31 points per curve computed by chemistry-ai999x; marked: the worked example, m = 0.01 kg, q = -554759 J.



q (enthalpy change (negative: heat released)) over m (mass of the reactant) from 0.005 kg to 0.015 kg and dH (standard enthalpy change) from -1.335e+06 J/mol to -445000 J/mol: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light -1.24821e+06 J to dark -138690 J); iso-lines at q = -1.20e6, -1.00e6, -8.00e5, -6.00e5, -4.00e5, -2.00e5 J. Other inputs as in the worked example: formula = CH4. Marked: the worked example, m = 0.01 kg, dH = -890000 J/mol, q = -554759 J.

```
chemistry_ai999x.solve('heat_from_reaction_mass', formula='CH4', m=0.01, dH=-890000.0)
# -> q = -554759 J, heat_released = 554759 J   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How much heat is released when 10.0 g of CH4 burns? ΔH = -890 kJ/mol.	SOLVED	q = -554759 J, heat_released = 554759 J

Clausius-Clapeyron: vapor pressure at another temperature, `clausius_clapeyron_pressure`

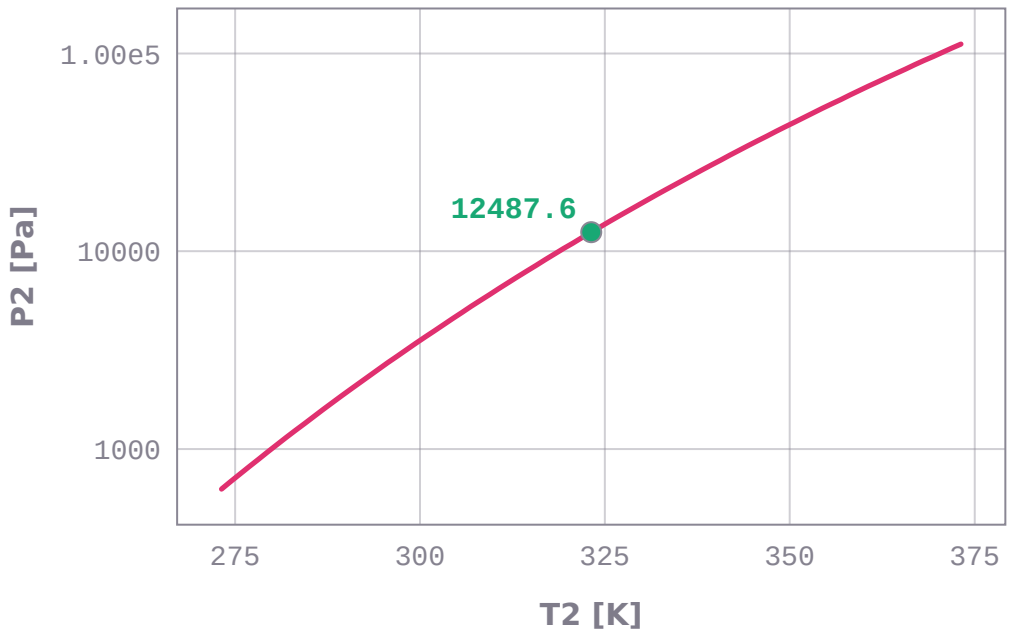
The vapor pressure of a liquid at a second temperature from one known point and the enthalpy of vaporization.

$$\ln \frac{P_2}{P_1} = -\frac{\Delta H_{vap}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

input	meaning	condition
`P1`	vapor pressure at T1	
`T1`	first temperature	
`T2`	second temperature	
`dH`	enthalpy of vaporization	

output	meaning
`P2`	vapor pressure at T2

- **Method:** integrated Clausius-Clapeyron equation
- **Assumes:** dHvap constant; ideal vapor
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Clausius-Clapeyron: vapor pressure at another temperature: P2 against T2 from 273.15 to 373.15, 81 runs of the package; other inputs as in the worked example (P1 = 3173.0723684210525, T1 = 298.15, dH = 43900.0). Marked: the worked example, T2 = 323.15, P2 = 12487.6.

```
chemistry_ai999x.solve('clausius_clapeyron_pressure', P1=3173.0723684210525, T1=298.15,
T2=323.15, dH=43900.0)
# -> P2 = 12487.6 Pa   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Water has a vapor pressure of 23.8 torr at 25 °C and ΔHvap = 43.9 kJ/mol. What is its vapor pressure at 50 °C?	SOLVED	P2 = 12487.6 Pa

Clausius-Clapeyron: enthalpy of vaporization, `clausius_clapeyron_enthalpy`

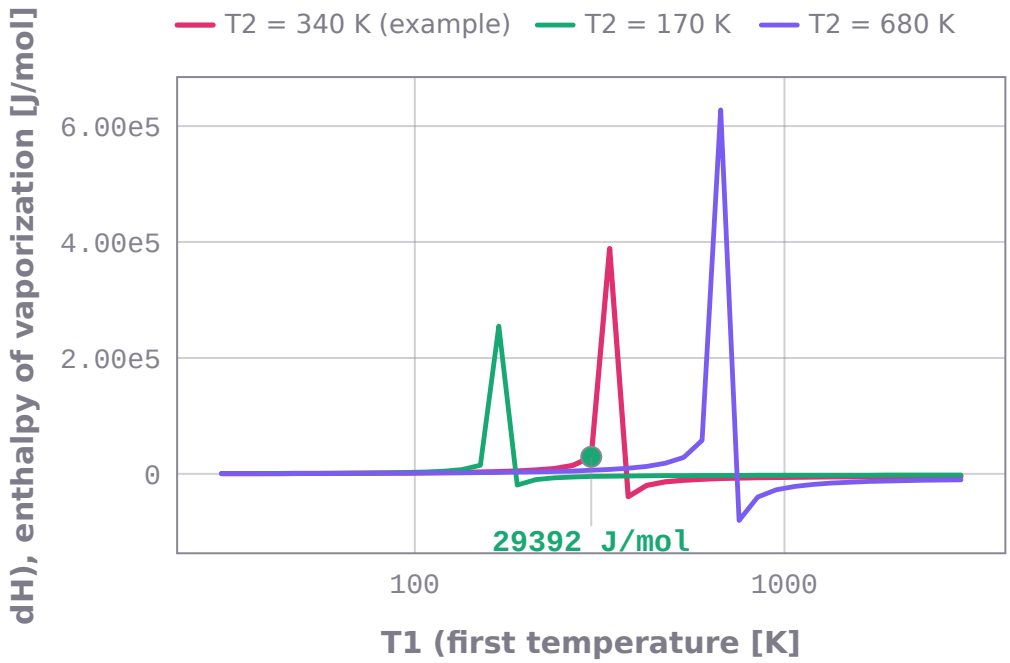
The enthalpy of vaporization from vapor pressures at two temperatures.

$$\Delta H_{vap} = \frac{R \ln(P_2/P_1)}{1/T_1 - 1/T_2}$$

input	meaning	condition
`P1`	vapor pressure at T1	
`T1`	first temperature	
`P2`	vapor pressure at T2	
`T2`	second temperature	

output	meaning
`dH`	enthalpy of vaporization

- **Method:** two-point Clausius-Clapeyron equation
- **Assumes:** $T_1 \neq T_2$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dH (enthalpy of vaporization) against T1 (first temperature) from 30 K to 3000 K on a logarithmic axis. One curve for each value of T2 (second temperature): 340 K, 170 K, 680 K. Other inputs as in the worked example: P1 = 13332.236842105263 Pa, P2 = 53328.94736842105 Pa. 41 points per curve computed by chemistry-ai999x; marked: the worked example, T1 = 300 K, dH = 29392 J/mol.

```
chemistry_ai999x.solve('clausius_clapeyron_enthalpy', P1=13332.236842105263, T1=300.0,
P2=53328.94736842105, T2=340.0)
# -> dH = 29392 J/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
P1 = 100 torr at T1 = 300 K and P2 = 400 torr at T2 = 340 K. What is the enthalpy of vaporization?	SOLVED	dH = 29392 J/mol

Gibbs energy from a cell potential, `delta_g_from_cell_potential`

The (standard) Gibbs energy change of a cell reaction from its potential (dG = -nFE).

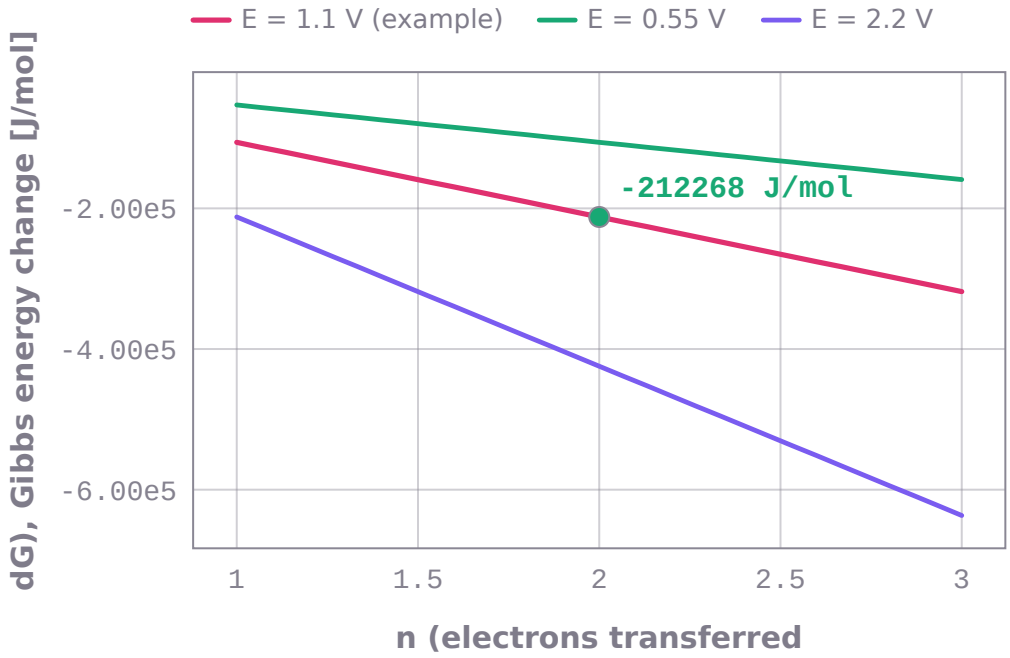
$$\Delta G = -nFE$$

input	meaning	condition
`n`	electrons transferred	
`E`	cell potential	

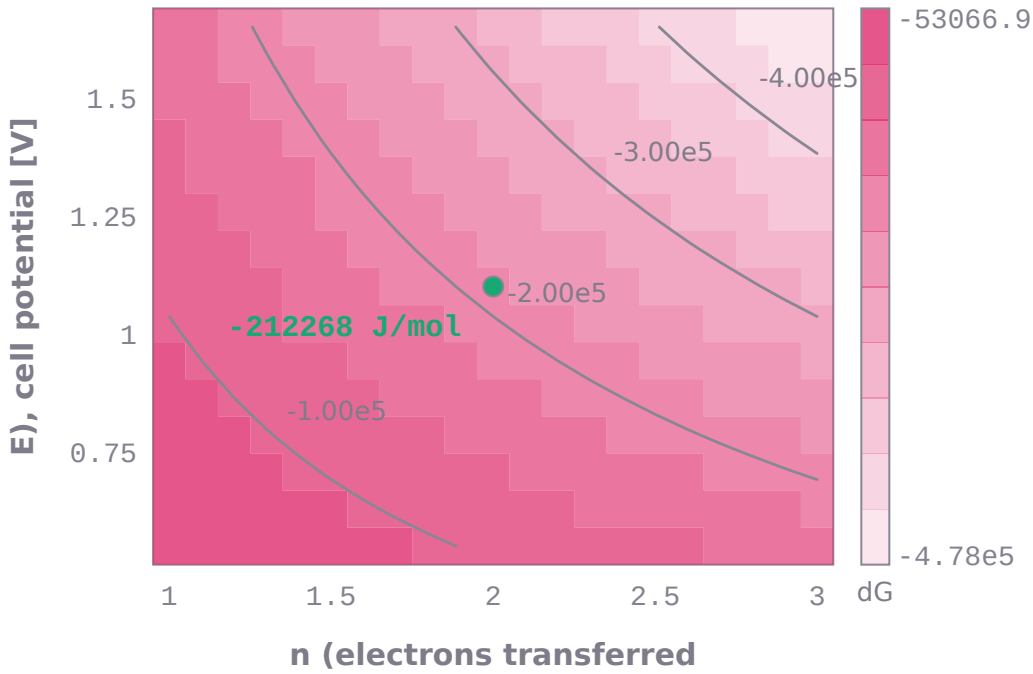
output	meaning
`dG`	Gibbs energy change
`dG_kJ`	in kJ/mol

- **Method:** $dG = -n$ F E
- **Assumes:** $n > 0$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dG (Gibbs energy change) against n (electrons transferred) from 1 to 3. One curve for each value of E (cell potential): 1.1 V, 0.55 V, 2.2 V. 31 points per curve computed by chemistry-ai999x; marked: the worked example, n = 2, dG = -212268 J/mol.



dG (Gibbs energy change) over n (electrons transferred) from 1 to 3 and E (cell potential) from 0.55 V to 1.65 V: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light -477602 J/mol to dark -53066.9 J/mol); iso-lines at dG = -4.00e5, -3.00e5, -2.00e5, -1.00e5 J/mol. Marked: the worked example, n = 2, E = 1.1 V, dG = -212268 J/mol.

```
chemistry_ai999x.solve('delta_g_from_cell_potential', n=2.0, E=1.1)
# -> dG = -212268 J/mol, dG_kJ = -212.268   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A cell has E° = 1.10 V and n = 2. What is ΔG°?	SOLVED	dG = -212268 J/mol, dG_kJ = -212.268
A cell has E° = 1.10 V. What is ΔG°?	NEEDS_INPUT	delta_g_from_cell_potential needs n (electrons transferred). Example: solve('delta_g_from_cell_potential', n=2, E='1.1 V

Equilibrium constant from a standard cell potential, `k_from_cell_potential`

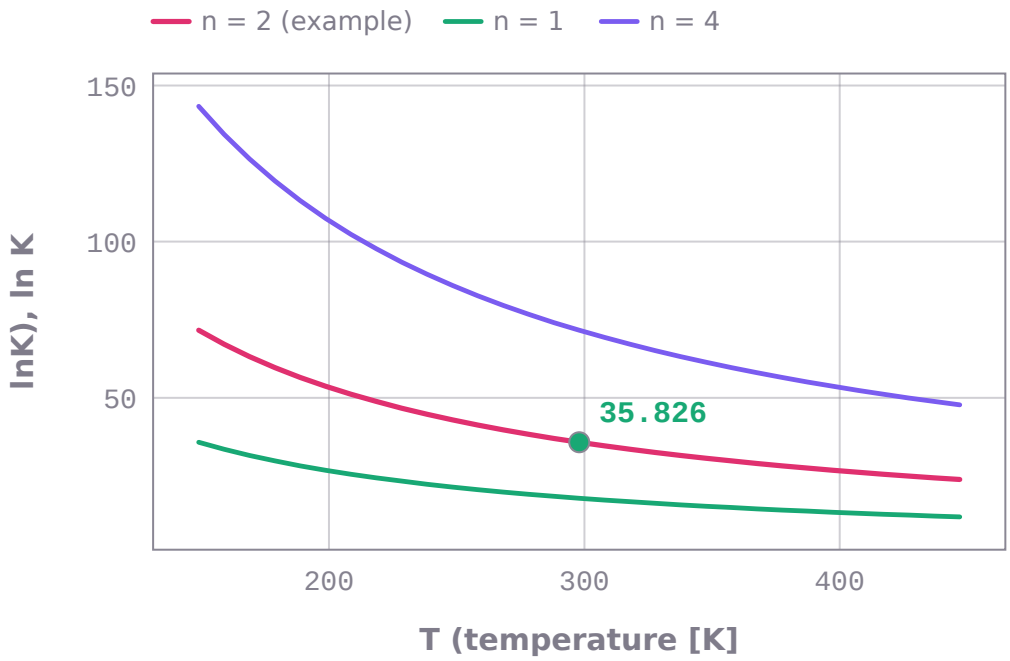
The equilibrium constant of a cell reaction from its standard potential (ln K = nFE°/RT).

$$\ln K = \frac{nFE^\circ}{RT}$$

input	meaning	condition
`n`	electrons transferred	
`E`	standard cell potential	
`T`	temperature	

output	meaning
`lnK`	ln K
`K`	K

- **Method:** ln $K = n$ F E° / (R T)
- **Assumes:** n, $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



lnK (ln K) against T (temperature) from 149 K to 447 K. One curve for each value of n (electrons transferred): 2, 1, 4. Other inputs as in the worked example: E = 0.46 V. 31 points per curve computed by chemistry-ai999x; marked: the worked example, T = 298 K, lnK = 35.826.

```
chemistry_ai999x.solve('k_from_cell_potential', n=2.0, E=0.46, T=298.0)
# -> lnK = 35.826, K = 3.62282e+15   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A cell has E° = 0.46 V with n = 2 at 298 K. What is the equilibrium constant?	SOLVED	lnK = 35.826, K = 3.62282e+15

Nernst equation, `nernst_potential`

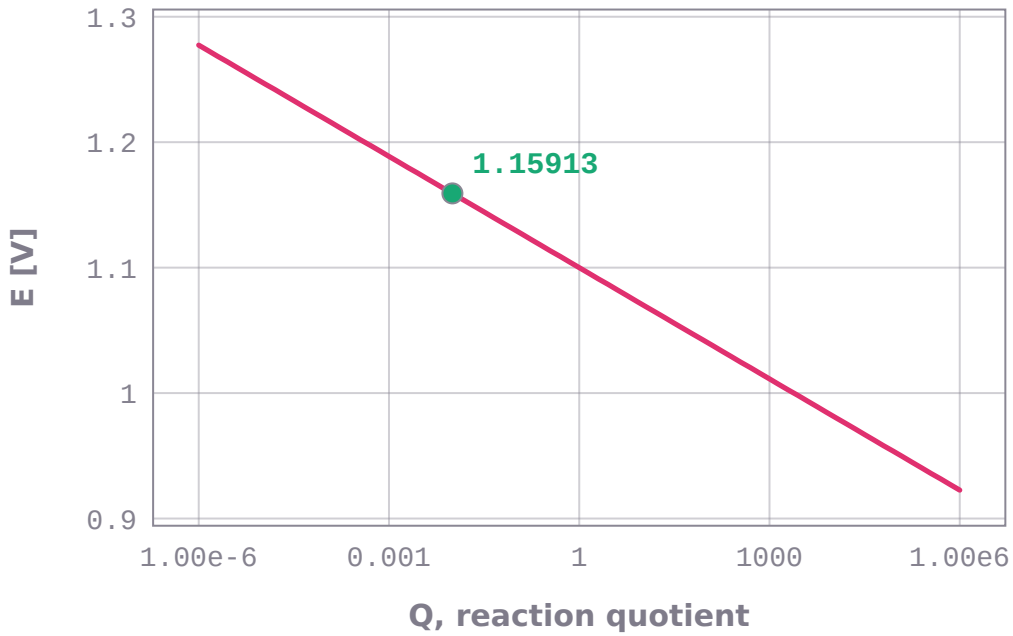
The cell potential at non-standard conditions from E°, the electrons transferred and the reaction quotient.

$$E = E^\circ - \frac{RT}{nF}\ln Q$$

input	meaning	condition
`E0`	standard cell potential	
n	electrons transferred	
`Q`	reaction quotient	
`T`	temperature	

output	meaning
`E`	cell potential

- **Method:** $E = E^\circ$ - (RT/nF) ln Q
- **Assumes:** n, Q, $T > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Nernst equation: E against Q from 1.0000e-06 to 1e+06 (logarithmic axis), 81 runs of the package; other inputs as in the worked example (E0 = 1.1, n = 2.0, T = 298.0). Marked: the worked example, Q = 0.01, E = 1.15913.

```
chemistry_ai999x.solve('nernst_potential', E0=1.1, n=2.0, Q=0.01, T=298.0)
# -> E = 1.15913 V   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
E° = 1.10 V, n = 2, Q = 0.010 at 298 K. What is the cell potential?	SOLVED	E = 1.15913 V
Use the Nernst equation: E° = 0.80 V, n = 1, reaction quotient 50, T = 310 K. Find E.	SOLVED	E = 0.695495 V

Mass deposited in electrolysis (Faraday's law), `electrolysis_mass`

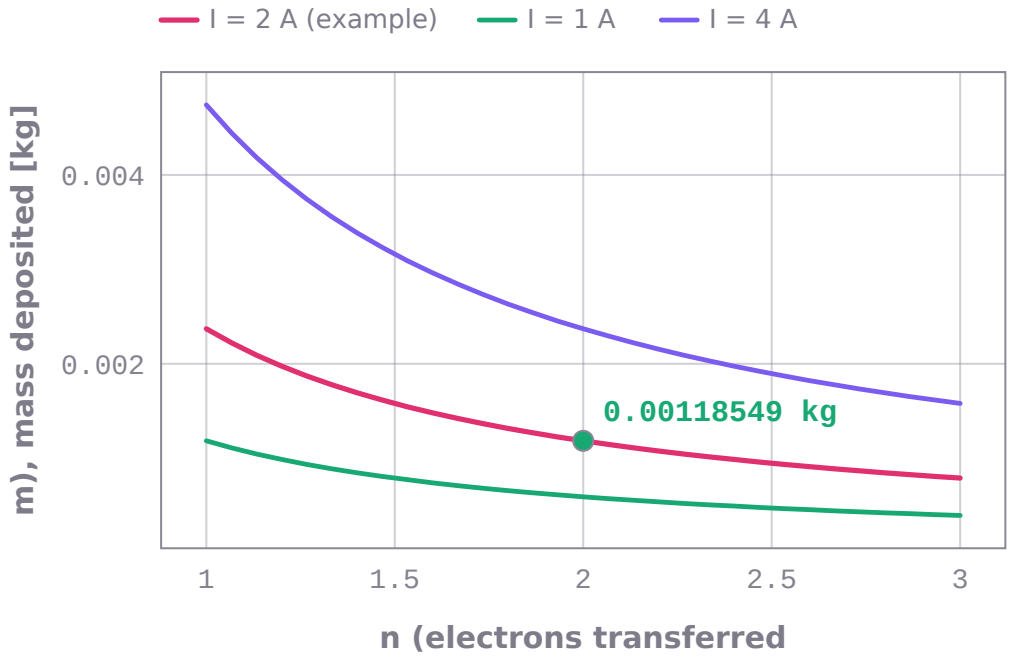
The mass of a metal deposited by a current flowing for a time.

$$m = \frac{It}{nF} M$$

input	meaning	condition
`metal`	the element deposited (symbol or name)	
`I`	current	
`t`	time	
`n`	electrons transferred	

output	meaning
`m`	mass deposited
`m_g`	in grams

- **Method:** charge / (n F) gives moles of metal; times its molar mass
- **Assumes:** 100% current efficiency
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



m (mass deposited) against n (electrons transferred) from 1 to 3. One curve for each value of I (current): 2 A, 1 A, 4 A. Other inputs as in the worked example: metal = Cu, t = 1800.0 s. 31 points per curve computed by chemistry-ai999x; marked: the worked example, n = 2, m = 0.00118549 kg.

```
chemistry_ai999x.solve('electrolysis_mass', metal='Cu', I=2.0, t=1800.0, n=2.0)
# -> m = 0.00118549 kg, m_g = 1.18549   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What mass of copper is deposited from a Cu2+ solution by a current of 2.00 A flowing for 30.0 min?	SOLVED	m = 0.00118549 kg, m_g = 1.18549
What mass of silver is plated from Ag+ by 0.500 A in 2.00 h?	SOLVED	m = 0.0040247 kg, m_g = 4.0247

Time for electrolysis (Faraday's law), `electrolysis_time`

How long a current must flow to deposit a given mass of a metal.

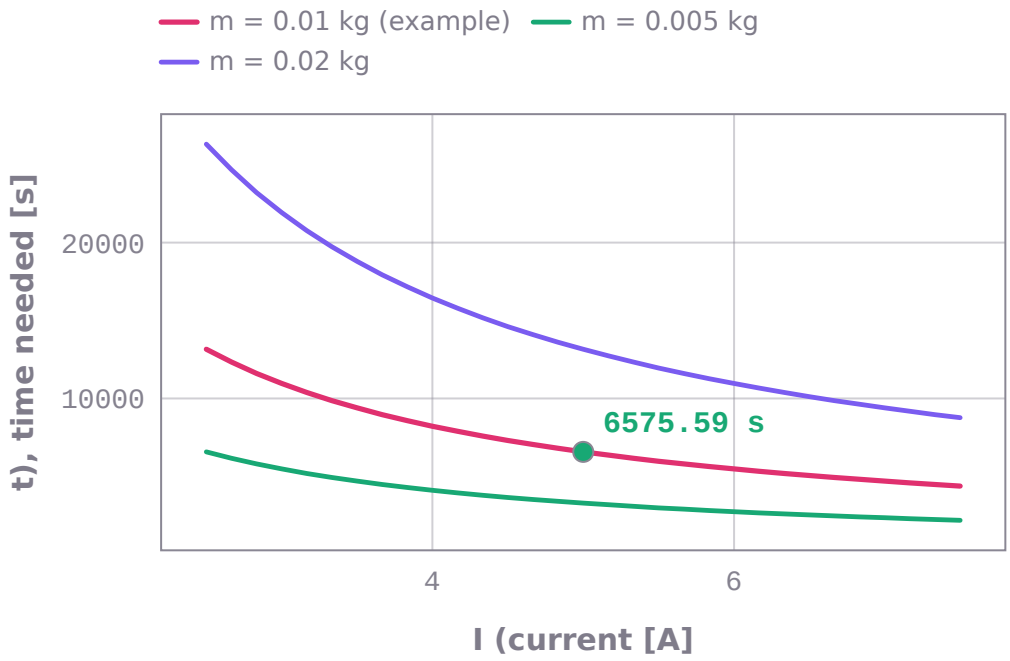
$$t = \frac{mnF}{MI}$$

input	meaning	condition
`metal`	the element deposited (symbol or name)	
`m`	mass to deposit	
`I`	current	
`n`	electrons transferred	

output	meaning
`t`	time needed

- **Method:** $t = m$ n F / (M I)

- **Assumes:** 100% current efficiency
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



t (time needed) against I (current) from 2.5 A to 7.5 A. One curve for each value of m (mass to deposit): 0.01 kg, 0.005 kg, 0.02 kg. Other inputs as in the worked example: metal = Ni, n = 2.0. 31 points per curve computed by chemistry-ai999x; marked: the worked example, I = 5 A, t = 6575.59 s.

```
chemistry_ai999x.solve('electrolysis_time', metal='Ni', m=0.01, I=5.0, n=2.0)
# -> t = 6575.59 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
How long must a current of 5.00 A flow to deposit 10.0 g of Ni from a Ni2+ solution?	SOLVED	t = 6575.59 s

Energy of a photon from its wavelength, `photon_energy`

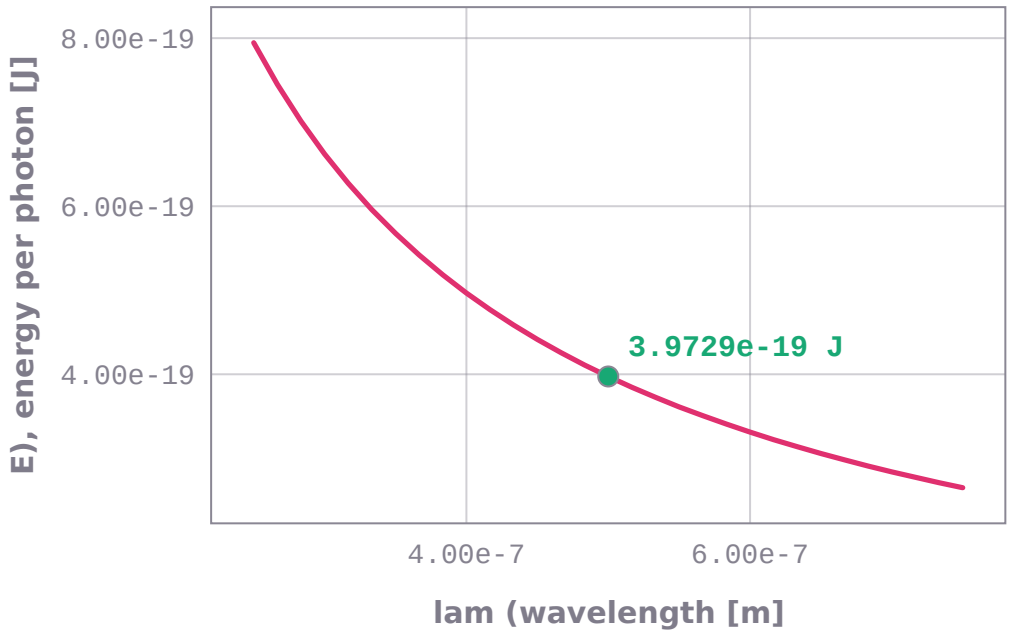
The energy (per photon and per mole of photons) and frequency of light of a given wavelength.

$$E = \frac{hc}{\lambda}$$

input	meaning	condition
`lam`	wavelength	

output	meaning
`E`	energy per photon
`E_eV`	in eV
`E_molar`	energy per mole of photons
`nu`	frequency

- **Method:** $E = h$ c / lambda
- **Assumes:** $\lambda > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



E (energy per photon) against lam (wavelength) from 2.5000e-07 m to 7.5000e-07 m. 31 points computed by chemistry-ai999x; marked: the worked example, lam = 5.0000e-07 m, E = 3.9729e-19 J.

```
chemistry_ai999x.solve('photon_energy', lam=5e-07)
# -> E = 3.97289e-19 J, E_eV = 2.47968, E_molar = 239253 J/mol, nu = 5.99585e+14 Hz
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the energy of a photon of wavelength 500 nm?	SOLVED	E = 3.97289e-19 J, E_eV = 2.47968, E_molar = 239253 J/mol, nu = 5.99585e+14 Hz
Calculate the energy per mole of photons of light with a wavelength of 254 nm.	SOLVED	E = 7.82065e-19 J, E_eV = 4.88127, E_molar = 470971 J/mol, nu = 1.18029e+15 Hz

Wavelength of a photon from its energy, `photon_wavelength`

The wavelength of a photon of a given energy.

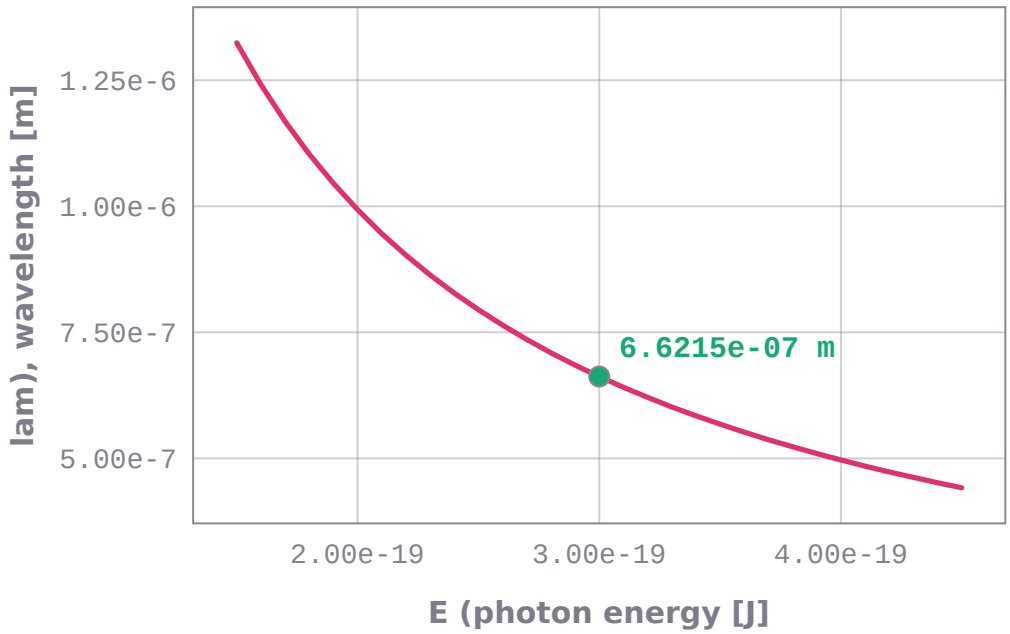
$$\lambda = \frac{hc}{E}$$

input	meaning	condition
`E`	photon energy	

output	meaning
`lam`	wavelength

output	meaning
`lam_nm`	in nm

- **Method:** $\lambda = h$ c / E
- **Assumes:** $E > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



lam (wavelength) against E (photon energy) from 1.5000e-19 J to 4.5000e-19 J. 31 points computed by chemistry-ai999x; marked: the worked example, E = 3.0000e-19 J, lam = 6.6215e-07 m.

```
chemistry_ai999x.solve('photon_wavelength', E=3e-19)
# -> lam = 6.62149e-07 m, lam_nm = 662.149   (SOLVED)
```

```
chemistry_ai999x.solve('photon_wavelength', E=4.005441585e-19)
# -> lam = 4.95937e-07 m, lam_nm = 495.937   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the wavelength of a photon with an energy of 3.0e-19 J?	SOLVED	lam = 6.62149e-07 m, lam_nm = 662.149
Find the wavelength of a 2.5 eV photon.	SOLVED	lam = 4.95937e-07 m, lam_nm = 495.937

Beer-Lambert law: absorbance, `beer_lambert_absorbance`

The absorbance of a solution from its molar absorptivity, path length and concentration.

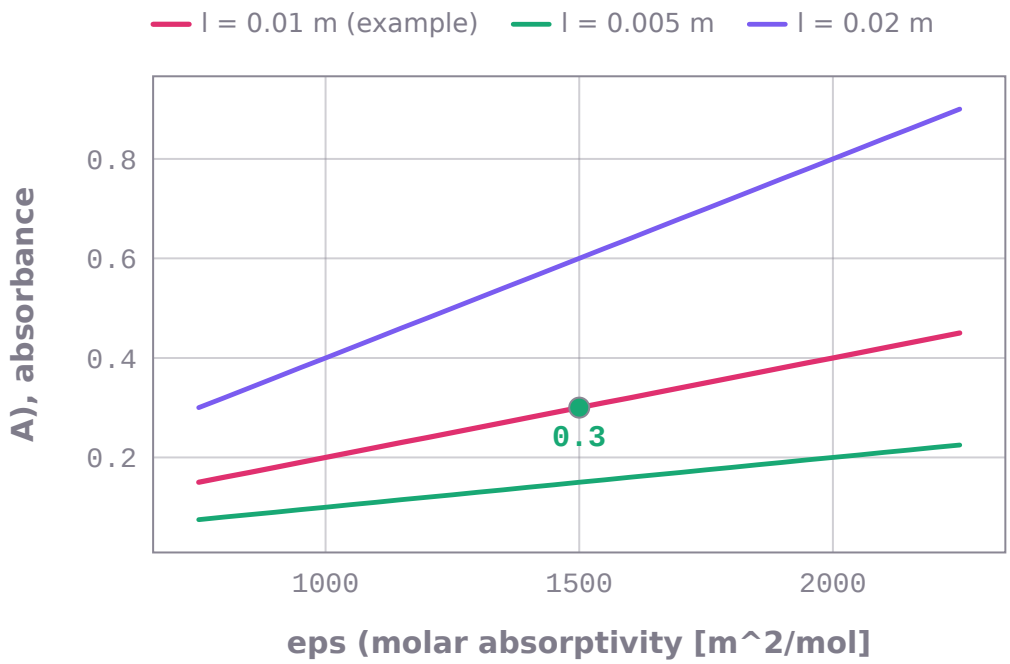
$$A = \varepsilon l c$$

input	meaning	condition
`eps`	molar absorptivity	

input	meaning	condition
`l`	path length	
`c`	concentration	

output	meaning
`A`	absorbance
`T_percent`	percent transmittance

- **Method:** A = epsilon l c
- **Assumes:** dilute solution, monochromatic light
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



A (absorbance) against eps (molar absorptivity) from 750 m^2/mol to 2250 m^2/mol. One curve for each value of l (path length): 0.01 m, 0.005 m, 0.02 m. Other inputs as in the worked example: c = 0.02 mol/m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, eps = 1500 m^2/mol, A = 0.3.

```
chemistry_ai999x.solve('beer_lambert_absorbance', eps=1500.0, l=0.01, c=0.02)
# -> A = 0.3, T_percent = 50.1187   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A 1.00 cm cell holds a 2.0e-5 M solution with molar absorptivity 15000 L/(mol·cm). What is the absorbance?	SOLVED	A = 0.3, T_percent = 50.1187

Beer-Lambert law: concentration, `beer_lambert_concentration`

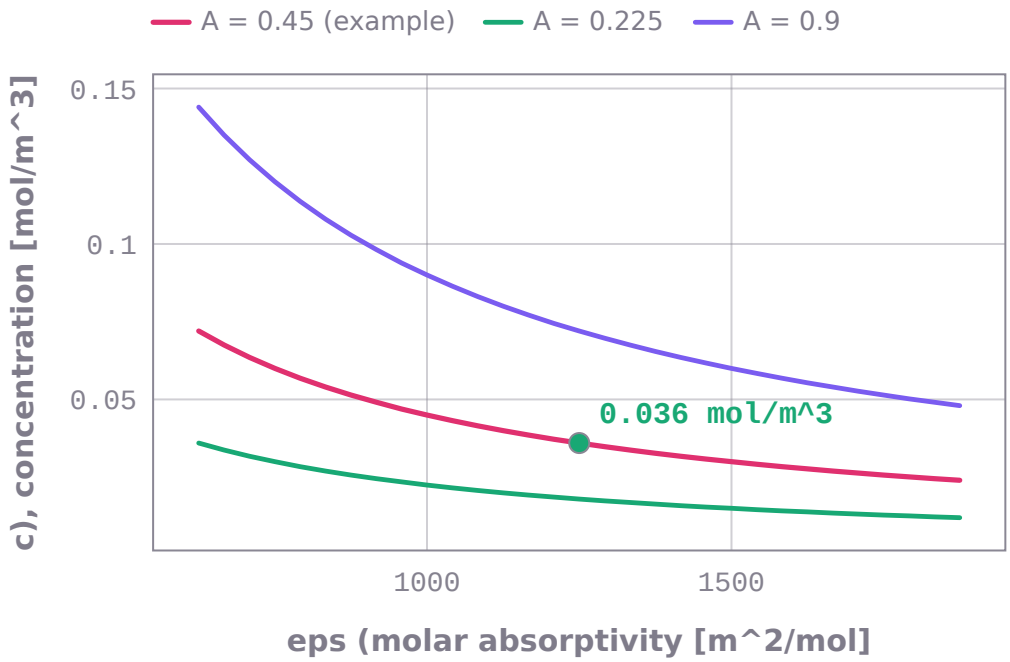
The concentration of a solution from its absorbance, molar absorptivity and path length.

$$c = \frac{A}{\varepsilon l}$$

input	meaning	condition
`A`	absorbance	
`eps`	molar absorptivity	
`l`	path length	

output	meaning
`c`	concentration
`c_molL`	in mol/L

- **Method:** c = A / (epsilon l)
- **Assumes:** dilute solution
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



c (concentration) against eps (molar absorptivity) from 625 m^2/mol to 1875 m^2/mol. One curve for each value of A (absorbance): 0.45, 0.225, 0.9. Other inputs as in the worked example: l = 0.01 m. 31 points per curve computed by chemistry-ai999x; marked: the worked example, eps = 1250 m^2/mol, c = 0.036 mol/m^3.

```
chemistry_ai999x.solve('beer_lambert_concentration', A=0.45, eps=1250.0, l=0.01)
# -> c = 0.036 mol/m^3, c_molL = 3.6e-05   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A solution has an absorbance of 0.450 in a 1.00 cm cell; ε = 1.25e4 L/(mol·cm). What is its concentration?	SOLVED	c = 0.036 mol/m^3, c_molL = 3.6e-05

Hydrogen atom transition wavelength (Rydberg), `hydrogen_transition_wavelength`

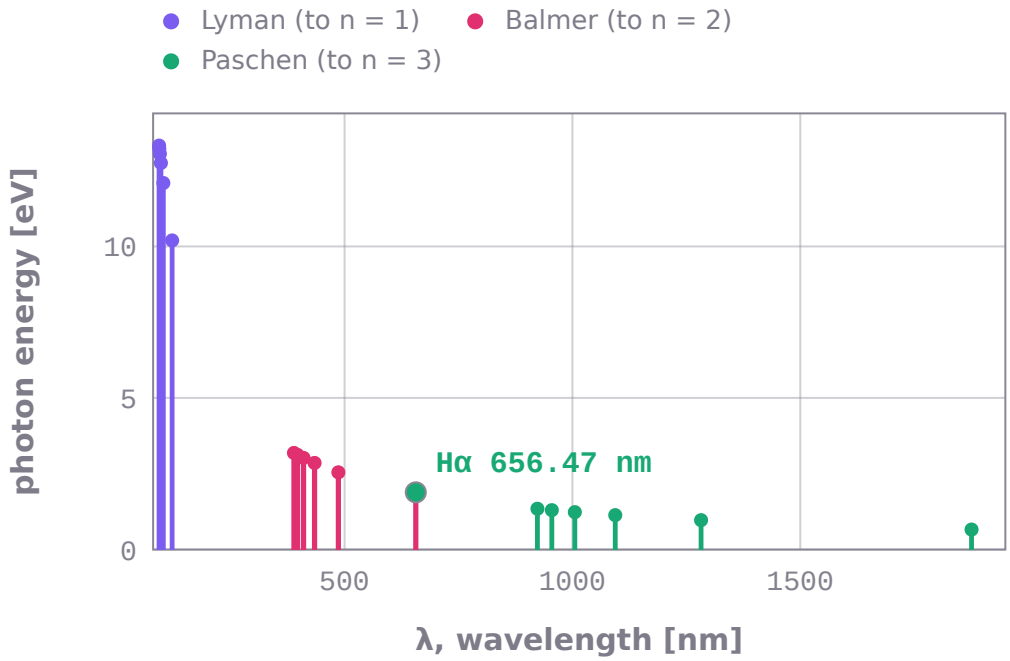
The wavelength of the photon emitted or absorbed in a transition between two levels of hydrogen.

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

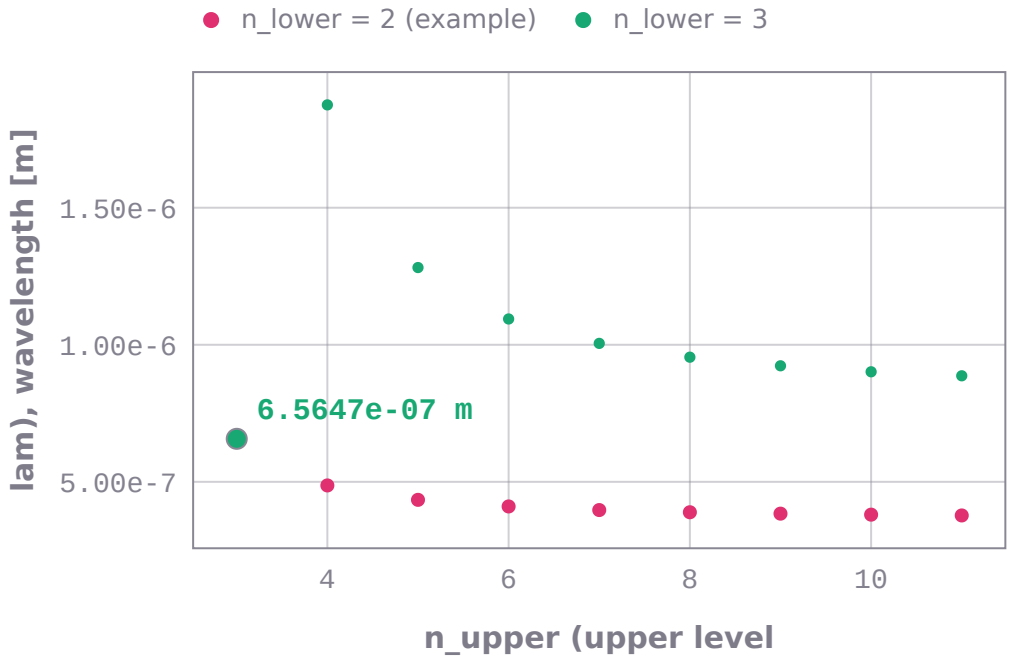
input	meaning	condition
`n_upper`	upper level	
`n_lower`	lower level	

output	meaning
`lam`	wavelength
`lam_nm`	in nm
`E`	photon energy

- **Method:** Rydberg formula with the hydrogen Rydberg constant (reduced-mass corrected)
- **Assumes:** hydrogen atom
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The hydrogen line spectrum: wavelength of the first six lines of three series, computed by the package (Rydberg formula), each drawn at its photon energy (also computed by the package). Ranges: Lyman 121.568–93.0758 nm; Balmer 656.47–389.019 nm; Paschen 1875.63–923.16 nm. Marked: Hα (n = 3 → 2), λ = 656.47 nm, E = 1.88865 eV.



lam (wavelength) against n_upper (upper level) from 1 to 11. One curve for each value of n_lower (lower level): 2, 3. 9 points per curve computed by chemistry-ai999x; marked: the worked example, n_upper = 3, lam = 6.5647e-07 m.

```
chemistry_ai999x.solve('hydrogen_transition_wavelength', n_upper=3, n_lower=2)
# -> lam = 6.5647e-07 m, lam_nm = 656.47, E = 3.02595e-19 J   (SOLVED)
```

```
chemistry_ai999x.solve('hydrogen_transition_wavelength', n_upper=2, n_lower=1)
# -> lam = 1.21568e-07 m, lam_nm = 121.568, E = 1.63401e-18 J   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the wavelength of light emitted when an electron in hydrogen falls from n = 3 to n = 2?	SOLVED	lam = 6.5647e-07 m, lam_nm = 656.47, E = 3.02595e-19 J
Find the wavelength of the hydrogen transition from n = 1 to n = 4.	SOLVED	lam = 9.72548e-08 m, lam_nm = 97.2548, E = 2.04252e-18 J

Degree of unsaturation, `degree_of_unsaturation`

The number of rings plus pi bonds implied by a molecular formula.

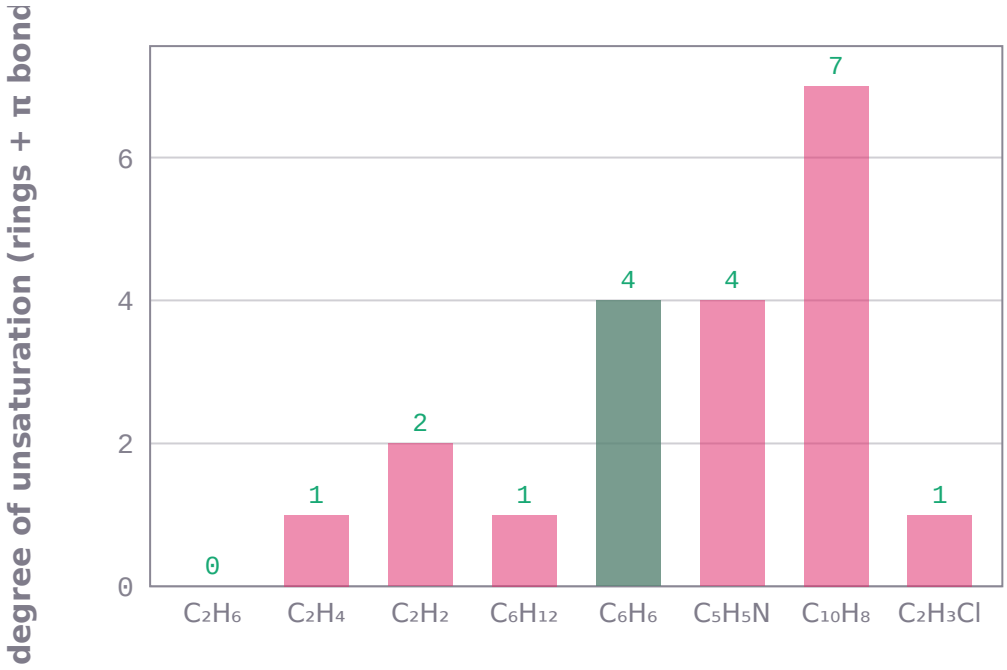
$$\mathrm{DoU} = \frac{2C + 2 + N - H - X}{2}$$

input	meaning	condition
`formula`	molecular formula	

output	meaning
`dou`	degree of unsaturation (rings + pi bonds)

- **Method:** $(2C + 2 + N - H - X)/2$ (Si as C, P and B as N, O and S ignored)
- **Assumes:** a neutral organic formula

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Degree of unsaturation (rings plus π bonds) of eight molecules, each returned by the package from its formula: ethane 0, ethene 1, ethyne 2, cyclohexane 1, benzene 4, pyridine 4, naphthalene 7, vinyl chloride 1. The worked example, C_6H_6 (benzene), gives 4: one ring and three double bonds.

```
chemistry_ai999x.solve('degree_of_unsaturation', formula='C6H6')
# -> dou = 4   (SOLVED)
```

```
chemistry_ai999x.solve('degree_of_unsaturation', formula='C5H9NO')
# -> dou = 2   (SOLVED)
```

```
chemistry_ai999x.solve('degree_of_unsaturation', formula='C4H8Cl2')
# -> dou = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the degree of unsaturation of C6H6?	SOLVED	dou = 4
Find the degree of unsaturation for C5H9NO.	SOLVED	dou = 2
What is the degree of unsaturation of C2H7?	INVALID_INPUT	Degree of unsaturation: this formula has no valid structure (negative or half-integer unsaturation).

Vibrational wavenumber of a diatomic molecule, `vibrational_wavenumber`

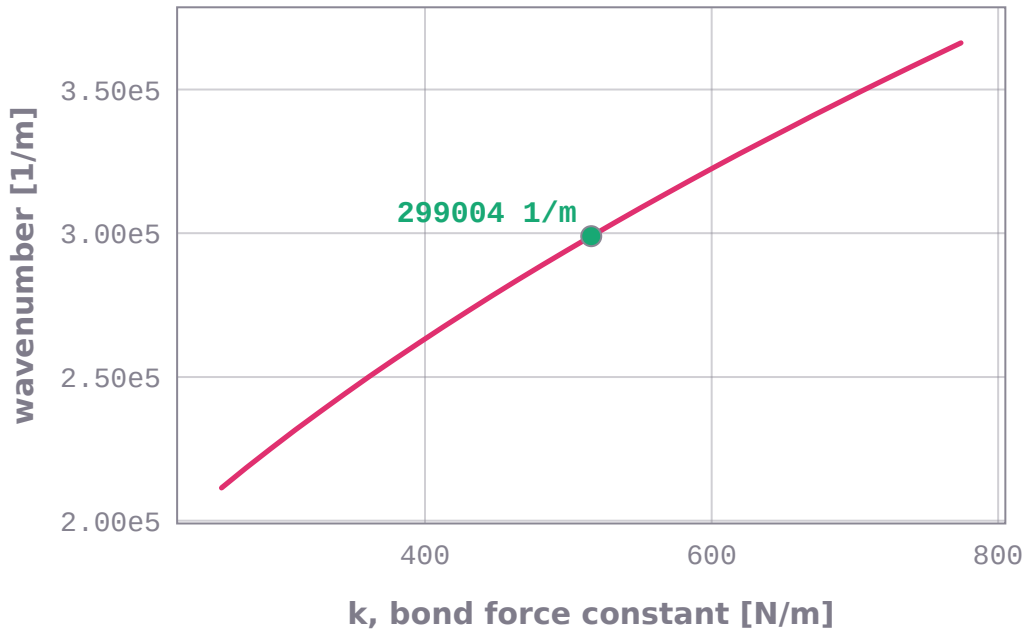
The harmonic vibrational wavenumber of a diatomic molecule from its bond force constant.

$$\tilde{\nu} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}$$

input	meaning	condition
`molecule`	diatomic molecule, e.g. HCl or CO	
`k`	bond force constant	

output	meaning
`wavenumber`	wavenumber
`wavenumber_cm`	in cm^{-1}
`nu`	vibrational frequency

- **Method:** harmonic oscillator with the reduced mass of the most abundant isotopes
- **Assumes:** harmonic approximation; most abundant isotopes (1H, 12C, 16O, 35Cl ...)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



wavenumber (wavenumber) against k (bond force constant) from 258 N/m to 774 N/m. Other inputs as in the worked example: molecule = HCl. 31 points computed by chemistry-ai999x; marked: the worked example, k = 516 N/m, wavenumber = 299004 1/m.

```
chemistry_ai999x.solve('vibrational_wavenumber', molecule='HCl', k=516.0)
# -> wavenumber = 299004 1/m, wavenumber_cm = 2990.04, nu = 8.96391e+13 Hz   (SOLVED)
```

```
chemistry_ai999x.solve('vibrational_wavenumber', molecule='CO', k=1902.0)
# -> wavenumber = 216989 1/m, wavenumber_cm = 2169.89, nu = 6.50517e+13 Hz   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
What is the vibrational wavenumber of HCl if its force constant is 516 N/m?	SOLVED	wavenumber = 299004 1/m, wavenumber_cm = 2990.04, nu = 8.96391e+13 Hz
Find the vibrational frequency of CO with a force constant of 1902 N/m.	SOLVED	wavenumber = 216989 1/m, wavenumber_cm = 2169.89, nu = 6.50517e+13 Hz

Average atomic mass from isotopes, `average_atomic_mass`

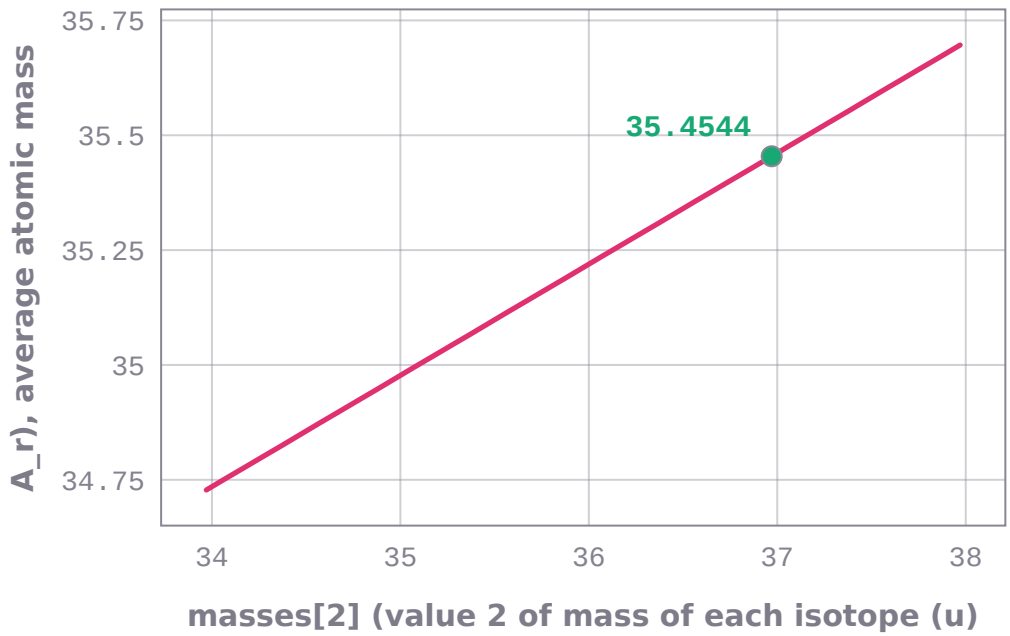
The average (relative) atomic mass of an element from the masses and natural abundances of its isotopes.

$$\bar{A} = \sum_i f_i \, m_i$$

input	meaning	condition
`masses`	mass of each isotope (u)	
`abundances`	abundance of each isotope, in the same order (fractions adding up to 1, or percentages adding up to 100)	

output	meaning
`A_r`	average atomic mass (u, numerically g/mol)
`M`	molar mass of the element

- **Method:** abundance-weighted sum of the isotope masses (abundances used as written)
- **Assumes:** one abundance per isotope; abundances add up to 1 (within 0.002) or to 100 (within 0.2); masses > 0, abundances >= 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



A_r (average atomic mass (u, numerically g/mol)) as value 2 of masses (mass of each isotope (u)) moves from 33.97 to 37.97, the other values as given (34.97, 36.97); 41 runs of chemistry-ai999x. Other inputs: abundances = 75.78, 24.22. Marked: the worked example, masses[2] = 36.97, A_r = 35.4544.

```
chemistry_ai999x.solve('average_atomic_mass', masses='34.97, 36.97', abundances='75.78, 24.22')
# -> A_r = 35.4544, M = 0.0354544 kg/mol   (SOLVED)
```

```
chemistry_ai999x.solve('average_atomic_mass', masses='23.985, 24.986, 25.983',
abundances='0.7899, 0.1000, 0.1101')
# -> A_r = 24.3051, M = 0.0243051 kg/mol   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Isotope masses are 34.97 u and 36.97 u with abundances 75.78% and 24.22%. What is the average atomic mass?	SOLVED	A_r = 35.4544, M = 0.0354544 kg/mol
Find the average atomic mass of magnesium from isotope masses 23.985, 24.986 and 25.983 u and abundances 0.7899, 0.1000 and 0.1101.	SOLVED	A_r = 24.3051, M = 0.0243051 kg/mol
Isotope masses are 34.97 u and 36.97 u with abundances 75.78% and 30.00%. What is the average atomic mass?	INVALID_INPUT	Average atomic mass from isotopes: the abundances must add up to 1 (fractions) or to 100 (percent).

Abundances of two isotopes from the average atomic mass, `isotope_abundance`

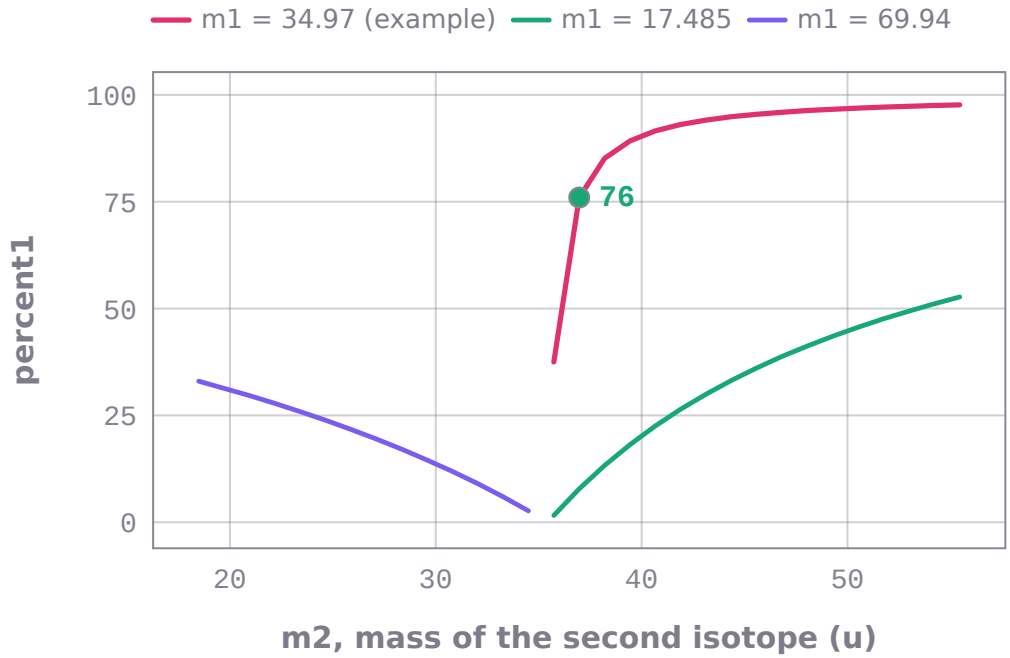
The natural abundances of an element's two isotopes from their masses and the average atomic mass.

$$x_1 = \frac{\bar{A} - m_2}{m_1 - m_2}, \quad x_2 = 1 - x_1$$

input	meaning	condition
`m1`	mass of the first isotope (u)	
`m2`	mass of the second isotope (u)	
`A_r`	average atomic mass (u)	

output	meaning
`x1`	fractional abundance of the first isotope
`x2`	fractional abundance of the second isotope
`percent1`	percent abundance of the first isotope
`percent2`	percent abundance of the second isotope

- **Method:** the average mass is x1 $m_1 + (1 - x_1)$ m2, solved for x1
- **Assumes:** two isotopes of different mass; the average mass lies strictly between them
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



percent1 (percent abundance of the first isotope) against m2 (mass of the second isotope (u)) from 18.485 to 55.455. One curve for each value of m1 (mass of the first isotope (u)): 34.97, 17.485, 69.94. Other inputs as in the worked example: A_r = 35.45. 17 points per curve computed by chemistry-ai999x; marked: the worked example, m2 = 36.97, percent1 = 76.

```
chemistry_ai999x.solve('isotope_abundance', m1=34.97, m2=36.97, A_r=35.45)
# -> x1 = 0.76, x2 = 0.24, percent1 = 76, percent2 = 24   (SOLVED)
```

```
chemistry_ai999x.solve('isotope_abundance', m1=62.93, m2=64.93, A_r=63.55)
# -> x1 = 0.69, x2 = 0.31, percent1 = 69, percent2 = 31   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The average atomic mass of chlorine is 35.45 u. Its isotope masses are 34.97 u and 36.97 u. What are the percent abundances of the two isotopes?	SOLVED	x1 = 0.76, x2 = 0.24, percent1 = 76, percent2 = 24
The average atomic mass of copper is 63.55 u and the isotope masses are 62.93 u and 64.93 u. Find the abundance of each isotope.	SOLVED	x1 = 0.69, x2 = 0.31, percent1 = 69, percent2 = 31
The average atomic mass of copper is 66.00 u and the isotope masses are 62.93 u and 64.93 u. Find the abundance of each isotope.	INVALID_INPUT	Abundances of two isotopes from the average atomic mass: the average atomic mass must lie between the two isotope masses

Standard cell potential from two half-reactions, `standard_cell_potential`

The standard potential of a galvanic cell from the standard reduction potentials of its cathode and anode half-reactions, and whether the cell reaction is spontaneous.

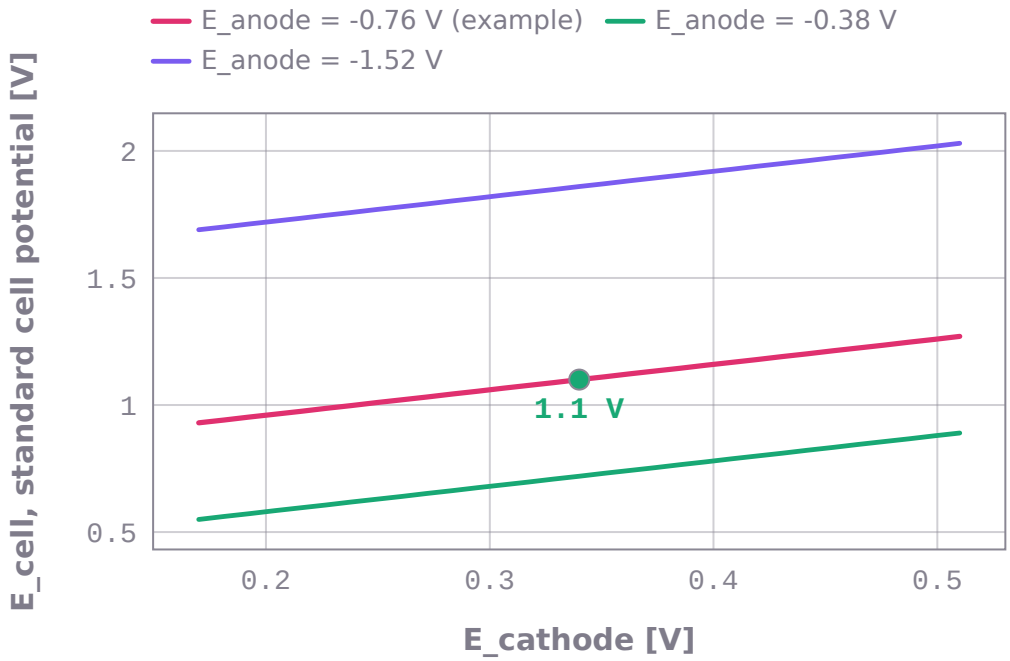
$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

input	meaning	condition
`E_cathode`	standard reduction potential of the cathode half-reaction	

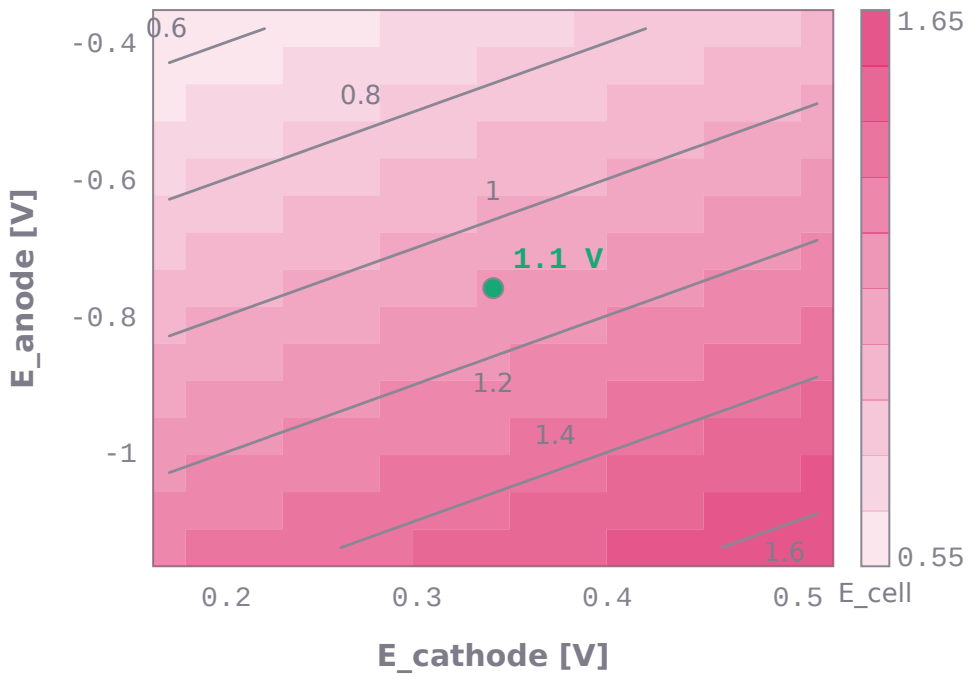
input	meaning	condition
`E_anode`	standard reduction potential of the anode half-reaction	

output	meaning
`E_cell`	standard cell potential
`spontaneity`	spontaneous (galvanic) or not (electrolytic)

- **Method:** both potentials are reduction potentials; the anode runs as an oxidation, so its potential is subtracted
- **Assumes:** both values are standard REDUCTION potentials; |E°| <= 3.5 V
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



E_cell (standard cell potential) against E_cathode (standard reduction potential of the cathode half-reaction) from 0.17 V to 0.51 V. One curve for each value of E_anode (standard reduction potential of the anode half-reaction): -0.76 V, -0.38 V, -1.52 V. 31 points per curve computed by chemistry-ai999x; marked: the worked example, E_cathode = 0.34 V, E_cell = 1.1 V.



E_cell (standard cell potential) over E_cathode (standard reduction potential of the cathode half-reaction) from 0.17 V to 0.51 V and E_anode (standard reduction potential of the anode half-reaction) from -1.14 V to -0.38 V: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 0.55 V to dark 1.65 V); iso-lines at E_cell = 0.6, 0.8, 1, 1.2, 1.4, 1.6 V. Marked: the worked example, E_cathode = 0.34 V, E_anode = -0.76 V, E_cell = 1.1 V.

```
chemistry_ai999x.solve('standard_cell_potential', E_cathode=0.34, E_anode=-0.76)
# -> E_cell = 1.1 V, spontaneity = spontaneous (galvanic)   (SOLVED)
```

```
chemistry_ai999x.solve('standard_cell_potential', E_cathode=0.34, E_anode=0.8)
# -> E_cell = -0.46 V, spontaneity = non-spontaneous (electrolytic)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
E°(cathode) = 0.34 V and E°(anode) = -0.76 V. What is the standard cell potential?	SOLVED	E_cell = 1.1 V, spontaneity = spontaneous (galvanic)
The cathode half-reaction has E° = 0.80 V and the anode half-reaction has E° = 0.34 V. Find the standard cell potential.	SOLVED	E_cell = 0.46 V, spontaneity = spontaneous (galvanic)
E°(cathode) = 4.20 V and E°(anode) = -0.76 V. What is the standard cell potential?	INVALID_INPUT	Standard cell potential from two half-reactions: standard reduction potentials lie between about -3.1 V and +3.0 V (|E°|

Reaction enthalpy from bond enthalpies, `bond_enthalpy_reaction`

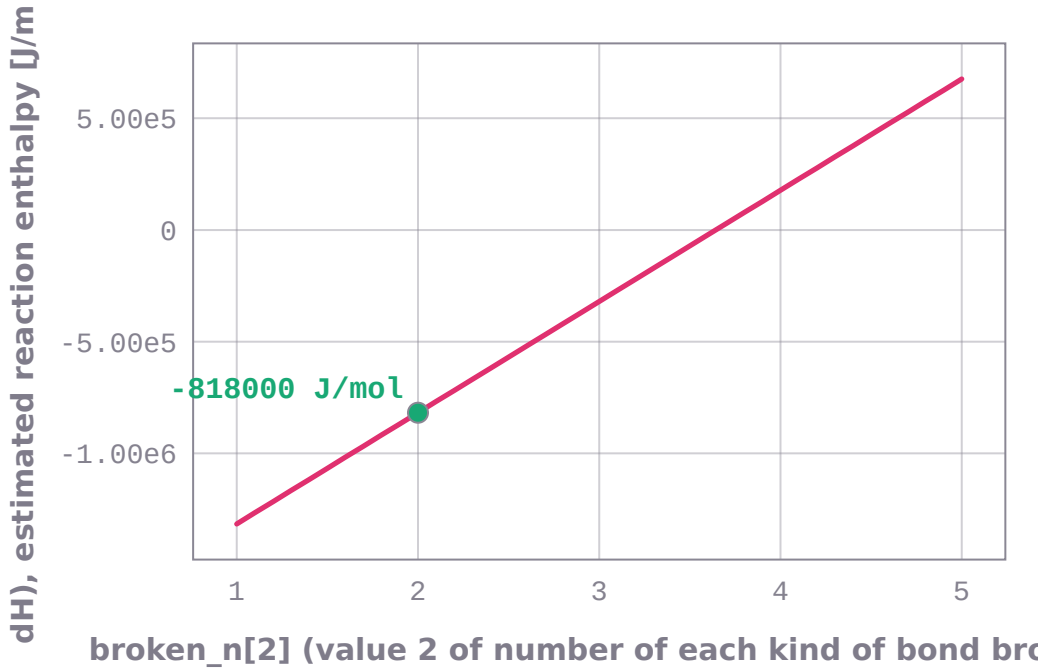
An estimate of a gas-phase reaction enthalpy from the average bond enthalpies of the bonds broken and formed.

$$\Delta H \approx \sum_{\text{broken}} n\,D - \sum_{\text{formed}} n\,D$$

input	meaning	condition
`broken_n`	number of each kind of bond broken	
`broken_D`	bond enthalpy of each kind of bond broken (kJ/mol), same order	
`formed_n`	number of each kind of bond formed	
`formed_D`	bond enthalpy of each kind of bond formed (kJ/mol), same order	

output	meaning
`dH`	estimated reaction enthalpy
`dH_kJ`	in kJ/mol
`broken_kJ`	energy to break the bonds (kJ/mol)
`formed_kJ`	energy released forming the bonds (kJ/mol)

- **Method:** energy to break the bonds minus energy released forming the new ones (average bond enthalpies)
- **Assumes:** bond counts and bond enthalpies > 0; one count per bond enthalpy; an estimate: average bond enthalpies, gas phase
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dH (estimated reaction enthalpy) as value 2 of broken_n (number of each kind of bond broken) moves from 1 to 5, the other values as given (4, 2); 41 runs of chemistry-ai999x. Other inputs: broken_D = 413, 498, formed_n = 2, 4, formed_D = 799, 467. Marked: the worked example, broken_n[2] = 2, dH = -818000 J/mol.

```
chemistry_ai999x.solve('bond_enthalpy_reaction', broken_n='4, 2', broken_D='413, 498',
formed_n='2, 4', formed_D='799, 467')
# -> dH = -818000 J/mol, dH_kJ = -818, broken_kJ = 2648, formed_kJ = 3466   (SOLVED)
```

```
chemistry_ai999x.solve('bond_enthalpy_reaction', broken_n='1, 1', broken_D='436, 243',
formed_n='2', formed_D='432')
# -> dH = -185000 J/mol, dH_kJ = -185, broken_kJ = 679, formed_kJ = 864   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bonds broken: 4 × 413, 2 × 498; bonds formed: 2 × 799, 4 × 467 (all in kJ/mol). Estimate the reaction enthalpy from these bond enthalpies.	SOLVED	dH = -818000 J/mol, dH_kJ = -818, broken_kJ = 2648, formed_kJ = 3466
Bonds broken: 436, 243; bonds formed: 2 × 432. Estimate the reaction enthalpy from bond enthalpies in kJ/mol.	SOLVED	dH = -185000 J/mol, dH_kJ = -185, broken_kJ = 679, formed_kJ = 864
Bonds broken: 436, -243; bonds formed: 2 × 432. Estimate the reaction enthalpy from bond enthalpies in kJ/mol.	INVALID_INPUT	Reaction enthalpy from bond enthalpies: bond enthalpies are positive (the energy needed to break a bond).

Henry's law: solubility of a gas, `henry_law_solubility`

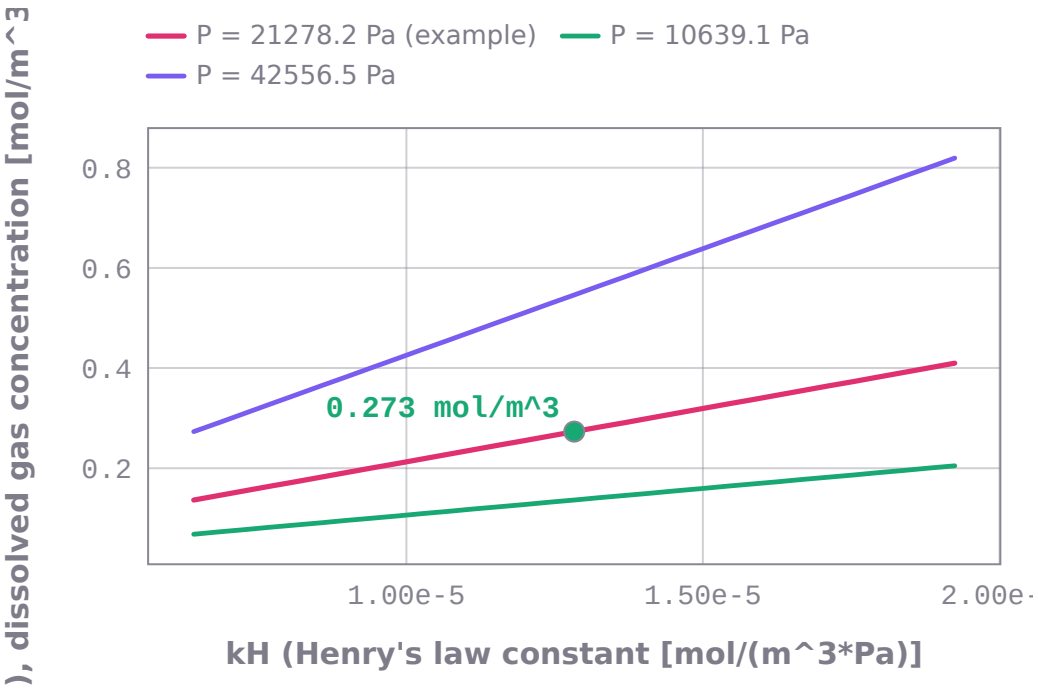
The concentration of a dissolved gas from its Henry's law constant and its partial pressure.

$$c = k_H\, P$$

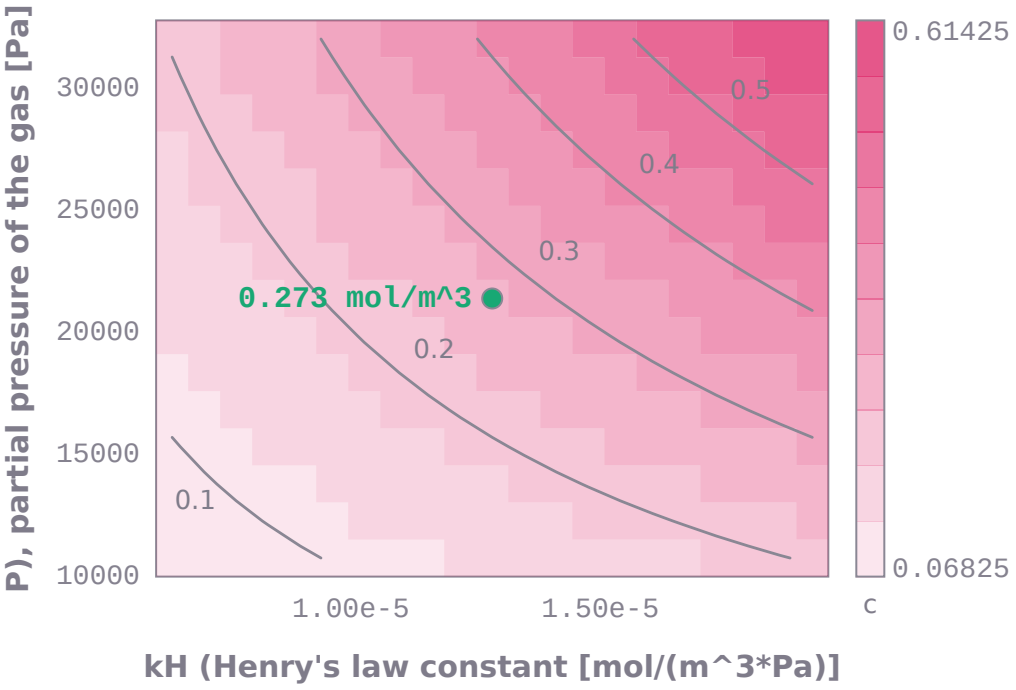
input	meaning	condition
`kH`	Henry's law constant (solubility form, $c = kH$ P)	
`P`	partial pressure of the gas	

output	meaning
`c`	dissolved gas concentration
`c_molL`	in mol/L

- **Method:** $c = kH$ P
- **Assumes:** kH, $P > 0$; dilute solution, no reaction with the solvent
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



c (dissolved gas concentration) against kH (Henry's law constant (solubility form, c = kH P)) from 6.4150e-06 mol/(m^3*Pa) to 1.9245e-05 mol/(m^3*Pa). One curve for each value of P (partial pressure of the gas): 21278.2 Pa, 10639.1 Pa, 42556.5 Pa. 31 points per curve computed by chemistry-ai999x; marked: the worked example, kH = 1.2830e-05 mol/(m^3*Pa), c = 0.273 mol/m^3.



c (dissolved gas concentration) over kH (Henry's law constant (solubility form, c = kH P)) from 6.4150e-06 mol/(m^3*Pa) to 1.9245e-05 mol/(m^3*Pa) and P (partial pressure of the gas) from 10639.1 Pa to 31917.4 Pa: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 0.06825 mol/m^3 to dark 0.61425 mol/m^3); iso-lines at c = 0.1, 0.2, 0.3, 0.4, 0.5 mol/m^3. Marked: the worked example, kH = 1.2830e-05 mol/(m^3*Pa), P = 21278.2 Pa, c = 0.273 mol/m^3.

```
chemistry_ai999x.solve('henry_law_solubility', kH=1.2830002467308166e-05, P=21278.25)
# -> c = 0.273 mol/m^3, c_molL = 0.000273   (SOLVED)
```

```
chemistry_ai999x.solve('henry_law_solubility', kH=0.00033555391068344434, P=405300.0)
# -> c = 136 mol/m^3, c_molL = 0.136   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The Henry's law constant for O2 in water is 1.3e-3 M/atm. What is the solubility of O2 at a partial pressure of 0.21 atm?	SOLVED	c = 0.273 mol/m^3, c_molL = 0.000273
Henry's law constant of CO2 = 3.4e-2 M/atm. Find the solubility of CO2 when its partial pressure is 4.0 atm.	SOLVED	c = 136 mol/m^3, c_molL = 0.136
The Henry's law constant for N2 in water is 6.1e-4 M/atm. What is the solubility of N2 at a partial pressure of -0.78 atm?	INVALID_INPUT	Henry's law: solubility of a gas: P must be a positive number.

Reaction quotient and the direction of shift, `reaction_quotient_direction`

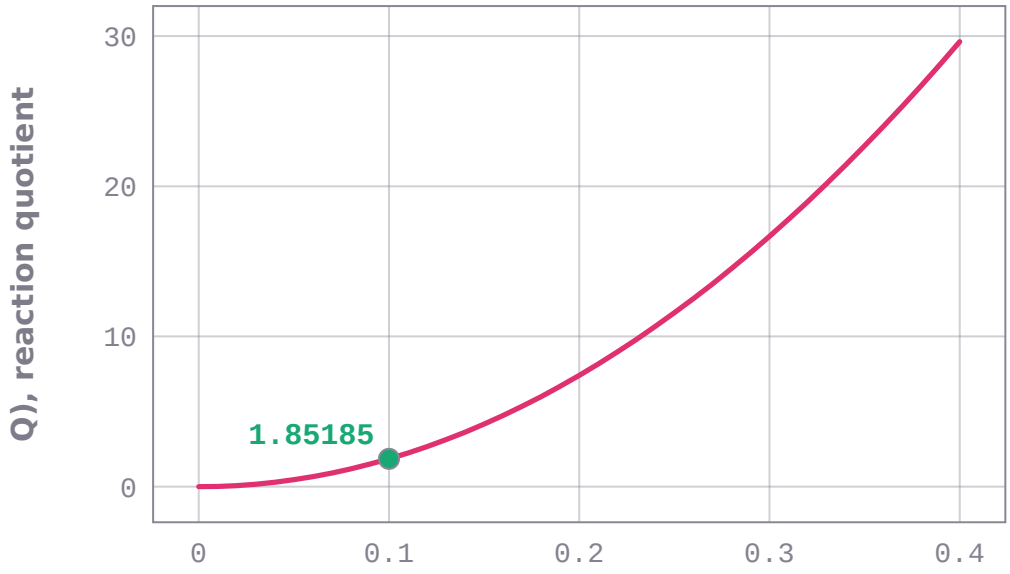
The reaction quotient Q from the current concentrations (or partial pressures) and, compared with K, the direction in which the reaction proceeds to reach equilibrium.

$$Q = \prod_i c_i^{\nu_i};\ Q < K \Rightarrow \text{forward},\ Q > K \Rightarrow \text{reverse}$$

input	meaning	condition
`equation`	the reaction as written (its coefficients define Q and K)	
`conc`	current concentration of each species in the order written (mol/L, or atm for Kp)	
`K`	equilibrium constant	

output	meaning
`Q`	reaction quotient
`log10_Q`	log10 Q
`direction`	direction of the shift

- **Method:** mass-action product of the current concentrations, compared with K
- **Assumes:** one reaction, coefficients as written; no pure solids or liquids in the equation; every concentration > 0; $K > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Q (reaction quotient) as value 3 of conc (current concentration of each species in the order written (mol/L, or atm for Kp)) moves from 1.3878e-17 to 0.4, the other values as given (0.20, 0.30, 0.10); 41 runs of chemistry-ai999x. Other inputs: equation = N2 + 3 H2 ⇌ 2 NH3, K = 0.105. Marked: the worked example, conc[3] = 0.1, Q = 1.85185.

```
chemistry_ai999x.solve('reaction_quotient_direction', equation='N2 + 3 H2 ⇌ 2 NH3',
conc='0.20, 0.30, 0.10', K=0.105)
# -> Q = 1.85185, log10_Q = 0.267606, direction = reverse (toward reactants)   (SOLVED)
```

```
chemistry_ai999x.solve('reaction_quotient_direction', equation='H2 + I2 ⇌ 2 HI',
conc='0.10, 0.20, 0.40', K=50.0)
# -> Q = 8, log10_Q = 0.90309, direction = forward (toward products)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
For N2 + 3 H2 ⇌ 2 NH3, Kc = 0.105. The concentrations are [N2] = 0.20 M, [H2] = 0.30 M and [NH3] = 0.10 M. Calculate the reaction quotient Q and predict the direction of the shift.	SOLVED	Q = 1.85185, log10_Q = 0.267606, direction = reverse (toward reactants)
For H2 + I2 ⇌ 2 HI, K = 50.0. The concentrations are [H2] = 0.10 M, [I2] = 0.20 M and [HI] = 0.40 M. Find the reaction quotient and the direction of the shift.	SOLVED	Q = 8, log10_Q = 0.90309, direction = forward (toward products)
For N2O4 ⇌ 2 NO2, Kc = 4.6e-3. The concentrations are [N2O4] = 0.10 M and [NO2] = 0 M. Calculate the reaction quotient and the direction of the shift.	INVALID_INPUT	Reaction quotient and the direction of shift: every concentration must be above zero.

Precipitation test (Q versus Ksp), `precipitation_test`

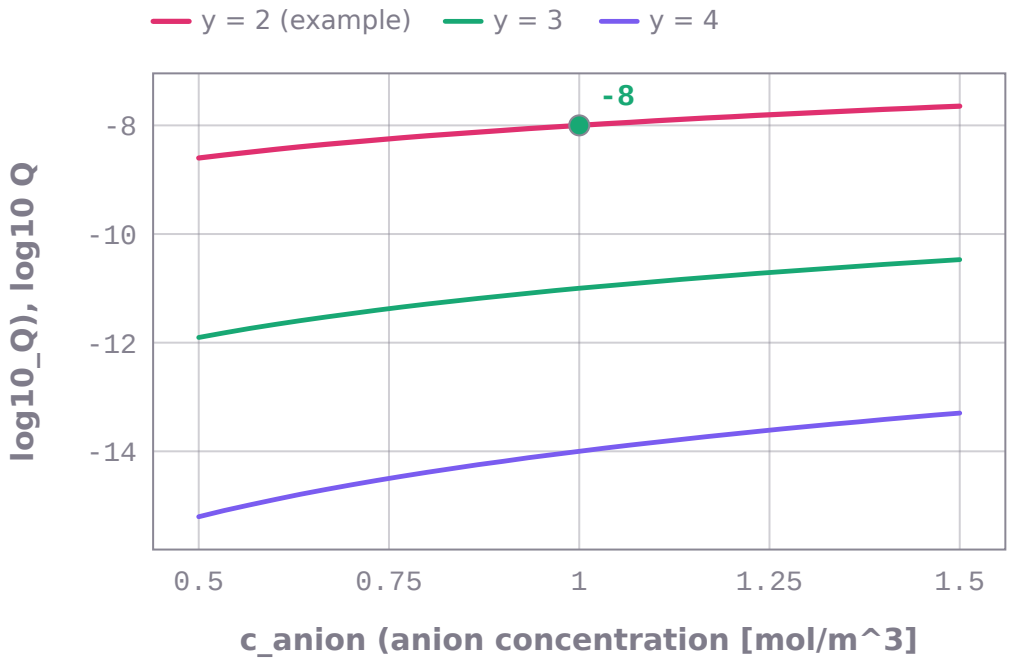
Whether a salt MxAy precipitates from a solution with given ion concentrations: the ion product Q compared with Ksp.

$$Q = [M]^x[A]^y;\ Q > K_{sp} \Rightarrow \text{precipitate}$$

input	meaning	condition
`Ksp`	solubility product	
`x`	cations per formula unit	
`y`	anions per formula unit	
`c_cation`	cation concentration (after any mixing)	
`c_anion`	anion concentration (after any mixing)	

output	meaning
`Q`	ion product Q
`log10_Q`	log10 Q
`result`	precipitate forms or not

- **Method:** ion product from the stated ion concentrations, compared with Ksp (in logarithms)
- **Assumes:** Ksp > 0; ion concentrations > 0 (mol/L, after mixing); the bracketed ions are the salt's own
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



log10_Q (log10 Q) against c_anion (anion concentration (after any mixing)) from 0.5 mol/m^3 to 1.5 mol/m^3. One curve for each value of y (anions per formula unit): 2, 3, 4. Other inputs as in the worked example: Ksp = 9.8e-09, x = 1, c_cation = 10.0 mol/m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, c_anion = 1 mol/m^3, log10_Q = -8.

```
chemistry_ai999x.solve('precipitation_test', Ksp=9.8e-09, x=1, y=2, c_cation=10.0,
c_anion=1.0)
# -> Q = 1e-08, log10_Q = -8, result = precipitate forms (Q > Ksp)   (SOLVED)
```

```
chemistry_ai999x.solve('precipitation_test', Ksp=1.8e-10, x=1, y=1, c_cation=0.01,
c_anion=0.001)
# -> Q = 1e-11, log10_Q = -11, result = no precipitate (Q < Ksp)   (SOLVED)
```

```
chemistry_ai999x.solve('precipitation_test', Ksp=3.9e-11, x=1, y=2, c_cation=0.2,
c_anion=1.0)
# -> Q = 2e-10, log10_Q = -9.69897, result = precipitate forms (Q > Ksp)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The Ksp of PbI2 is 9.8e-9. If [Pb2+] = 0.010 M and [I-] = 0.0010 M after mixing, will a precipitate form?	SOLVED	Q = 1e-08, log10_Q = -8, result = precipitate forms (Q > Ksp)
The Ksp of AgCl is 1.8e-10. If [Ag+] = 1.0e-5 M and [Cl-] = 1.0e-6 M, will a precipitate form?	SOLVED	Q = 1e-11, log10_Q = -11, result = no precipitate (Q < Ksp)
The Ksp of CaF2 is 3.9e-11. If [Ca2+] = 2.0e-4 M and [F-] = 1.0e-3 M, will a precipitate form?	SOLVED	Q = 2e-10, log10_Q = -9.69897, result = precipitate forms (Q > Ksp)
The Ksp of PbI2 is 9.8e-9. If [Pb2+] = -0.010 M and [I-] = 0.0010 M after mixing, will a precipitate form?	INVALID_INPUT	Precipitation test (Q versus Ksp): c_cation must be a positive number.

Standard reaction entropy from standard molar entropies,

`reaction_entropy_from_standard_entropies`

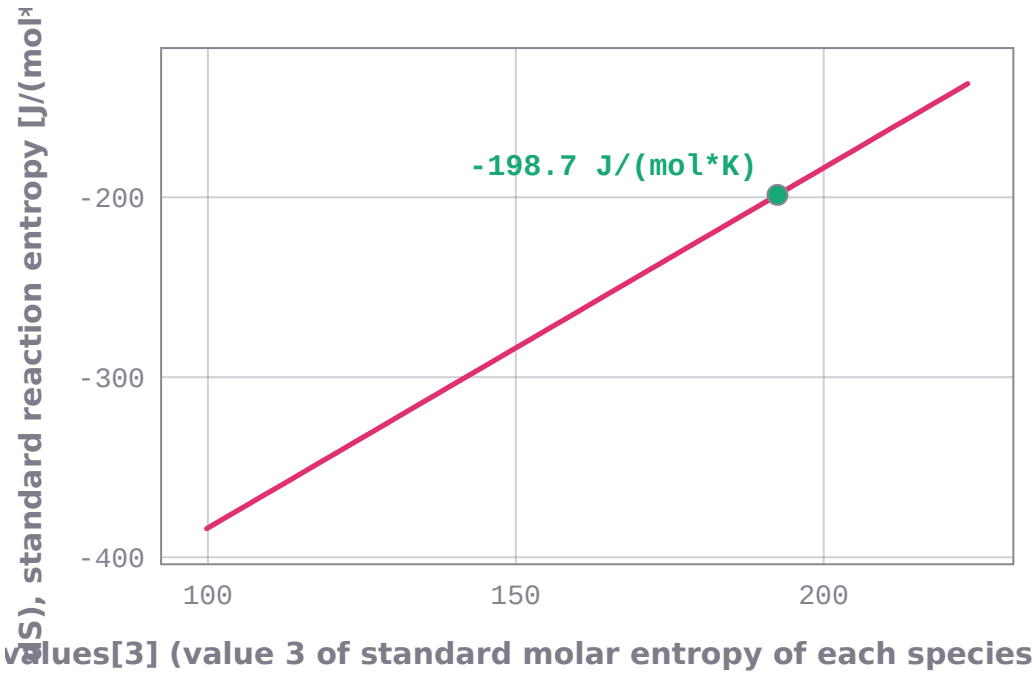
The standard entropy change of a reaction from the standard molar entropies of every species (balanced equation).

$$\Delta S^\circ_{rxn} = \sum \nu_p S^\circ(p) - \sum \nu_r S^\circ(r)$$

input	meaning	condition
`equation`	balanced equation	
`values`	standard molar entropy of each species in the order written (J/(mol·K))	

output	meaning
`dS`	standard reaction entropy

- **Method:** coefficient-weighted sum of standard molar entropies, products minus reactants
- **Assumes:** the equation as written conserves every element; one value per species, in J/(mol·K)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dS (standard reaction entropy) as value 3 of values (standard molar entropy of each species in the order written (J/(mol·K))) moves from 99.8 to 223.4, the other values as given (191.6, 130.7, 192.5); 41 runs of chemistry-ai999x. Other inputs: equation = N2 + 3 H2 -> 2 NH3. Marked: the worked example, values[3] = 192.5, dS = -198.7 J/(mol*K).

```
chemistry_ai999x.solve('reaction_entropy_from_standard_entropies', equation='N2 + 3 H2 -> 2 NH3', values='191.6, 130.7, 192.5')
# -> dS = -198.7 J/(mol*K)   (SOLVED)
```

```
chemistry_ai999x.solve('reaction_entropy_from_standard_entropies', equation='CaCO3 -> CaO + CO2', values='92.9, 39.8, 213.7')
# -> dS = 160.6 J/(mol*K)   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Using standard molar entropies N2 = 191.6, H2 = 130.7, NH3 = 192.5 (J/(mol·K)), find ΔS° for N2 + 3 H2 -> 2 NH3.	SOLVED	dS = -198.7 J/(mol*K)
Standard molar entropies: CaCO3 = 92.9, CaO = 39.8, CO2 = 213.7 J/(mol·K). Calculate the entropy change ΔS° for CaCO3 -> CaO + CO2.	SOLVED	dS = 160.6 J/(mol*K)
Using standard molar entropies N2 = 191.6, H2 = 130.7, NH3 = 192.5 (J/(mol·K)), find ΔS° for N2 + H2 -> NH3.	INVALID_INPUT	Standard reaction entropy from standard molar entropies: the equation as written is not balanced; balance it first.

Standard reaction Gibbs energy from Gibbs energies of formation,

`reaction_gibbs_from_formation`

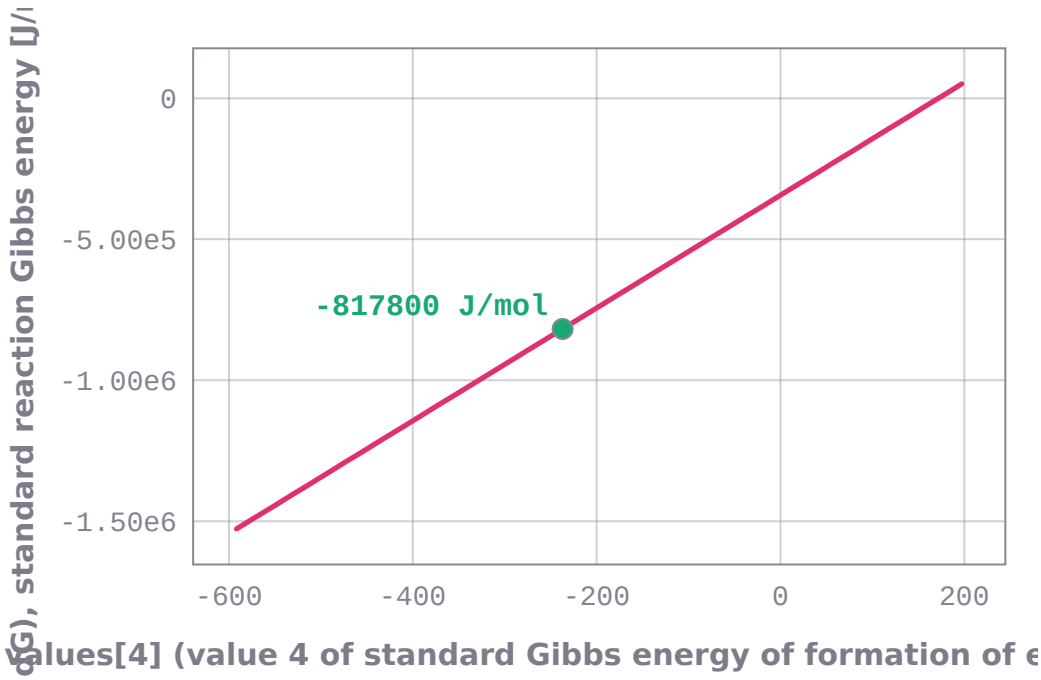
The standard Gibbs energy change of a reaction from the standard Gibbs energies of formation of every species (balanced equation), and whether it is spontaneous under standard conditions.

$$\Delta G^\circ_{rxn} = \sum \nu_p \Delta G^\circ_f(p) - \sum \nu_r \Delta G^\circ_f(r)$$

input	meaning	condition
`equation`	balanced equation	
`values`	standard Gibbs energy of formation of each species in the order written (kJ/mol)	

output	meaning
`dG`	standard reaction Gibbs energy
`dG_kJ`	in kJ/mol
`spontaneity`	spontaneous / non-spontaneous under standard conditions

- **Method:** coefficient-weighted sum of Gibbs energies of formation, products minus reactants
- **Assumes:** the equation as written conserves every element; one value per species, in kJ/mol
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dG (standard reaction Gibbs energy) as value 4 of values (standard Gibbs energy of formation of each species in the order written (kJ/mol)) moves from -591.6 to 197.2, the other values as given (-50.8, 0, -394.4, -237.1); 41 runs of chemistry-ai999x. Other inputs: equation = CH4 + 2 O2 -> CO2 + 2 H2O. Marked: the worked example, values[4] = -237.1, dG = -817800 J/mol.

```
chemistry_ai999x.solve('reaction_gibbs_from_formation', equation='CH4 + 2 O2 -> CO2 + 2 H2O', values='-50.8, 0, -394.4, -237.1')
# -> dG = -817800 J/mol, dG_kJ = -817.8, spontaneity = spontaneous   (SOLVED)
```

```
chemistry_ai999x.solve('reaction_gibbs_from_formation', equation='N2 + 3 H2 -> 2 NH3', values='0, 0, -16.4')
# -> dG = -32800 J/mol, dG_kJ = -32.8, spontaneity = spontaneous   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Using standard Gibbs energies of formation CH4 = -50.8, O2 = 0, CO2 = -394.4, H2O = -237.1 (kJ/mol), find ΔG° for CH4 + 2 O2 -> CO2 + 2 H2O.	SOLVED	dG = -817800 J/mol, dG_kJ = -817.8, spontaneity = spontaneous

question	outcome	result
Standard Gibbs energies of formation: NH3 = -16.4, N2 = 0, H2 = 0 kJ/mol. Find ΔG° for N2 + 3 H2 -> 2 NH3.	SOLVED	dG = -32800 J/mol, dG_kJ = -32.8, spontaneity = spontaneous
Using standard Gibbs energies of formation CH4 = -50.8, O2 = 0, CO2 = -394.4, H2O = -237.1 (kJ/mol), find ΔG° for CH4 + O2 -> CO2 + H2O.	INVALID_INPUT	Standard reaction Gibbs energy from Gibbs energies of formation: the equation as written is not balanced; balance it fir

Mass defect and nuclear binding energy, `nuclear_binding_energy`

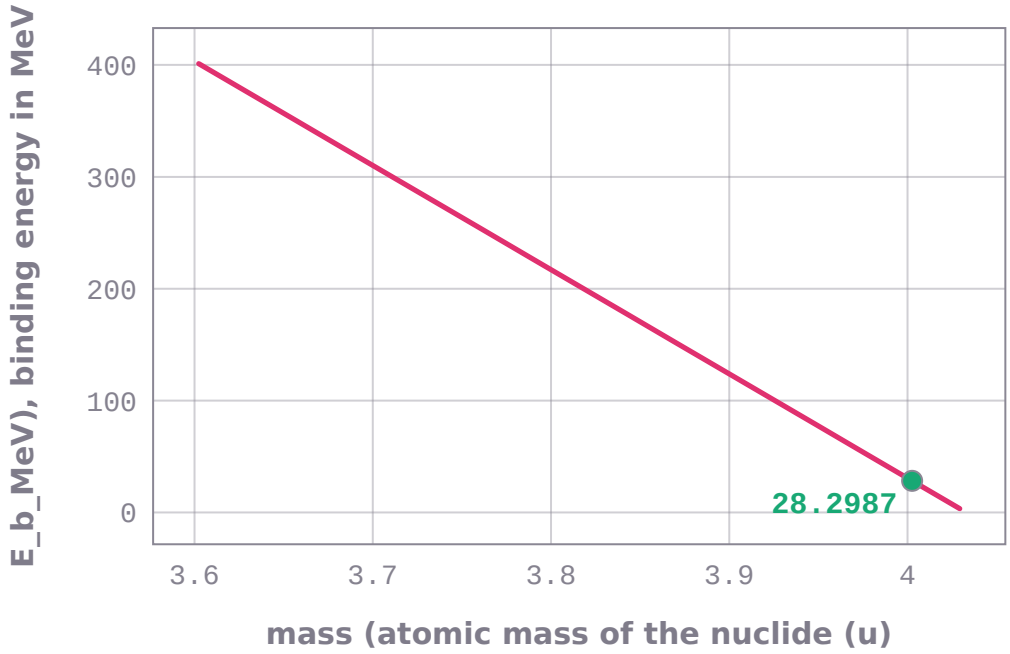
The mass defect, binding energy and binding energy per nucleon of a nuclide from its atomic mass.

$$\Delta m = Z\,m(^1\mathrm{H}) + (A - Z)\,m_n - m_{atom}, \quad E_b = \Delta m c^2$$

input	meaning	condition
`Z`	atomic number (protons)	
`A`	mass number (nucleons)	
`mass`	atomic mass of the nuclide (u)	

output	meaning
`dm`	mass defect (u)
`E_b`	binding energy
`E_b_MeV`	binding energy in MeV
`E_per_nucleon_MeV`	binding energy per nucleon in MeV

- **Method:** mass defect with atomic masses (Z hydrogen atoms carry the electrons), times 931.494 MeV per u
- **Assumes:** $1 \le Z \le A$, $A \ge 2$; atomic mass in u, within 0.5 u of A; mass defect > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



E_b_MeV (binding energy in MeV) against mass (atomic mass of the nuclide (u)) from 3.60234 to 4.40286. Other inputs as in the worked example: Z = 2, A = 4. 17 points computed by chemistry-ai999x; marked: the worked example, mass = 4.0026, E_b_MeV = 28.2987.

```
chemistry_ai999x.solve('nuclear_binding_energy', Z=2, A=4, mass=4.0026)
# -> dm = 0.0303799, E_b = 4.53395e-12 J, E_b_MeV = 28.2987, E_per_nucleon_MeV = 7.07467
(SOLVED)
```

```
chemistry_ai999x.solve('nuclear_binding_energy', Z=26, A=56, mass=55.9349)
# -> dm = 0.528498, E_b = 7.8874e-11 J, E_b_MeV = 492.293, E_per_nucleon_MeV = 8.79095
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the nuclear binding energy of a nuclide with Z = 2, A = 4 and atomic mass 4.0026 u.	SOLVED	dm = 0.0303799, E_b = 4.53395e-12 J, E_b_MeV = 28.2987, E_per_nucleon_MeV = 7.07467
What is the mass defect and binding energy per nucleon of a nuclide with Z = 26, A = 56 and atomic mass 55.9349 u?	SOLVED	dm = 0.528498, E_b = 7.8874e-11 J, E_b_MeV = 492.293, E_per_nucleon_MeV = 8.79095
Find the nuclear binding energy of a nuclide with Z = 26, A = 56 and atomic mass 65.9349 u.	INVALID_INPUT	Mass defect and nuclear binding energy: the atomic mass (in u) must lie within 0.5 u of the mass number A.

Amount remaining after radioactive decay (half-life), `radioactive_decay_remaining`

The mass of a radioactive nuclide that remains after a time, from its half-life.

$$m = m_0\left(\tfrac{1}{2}\right)^{t/t_{1/2}}$$

input	meaning	condition
`m0`	initial mass	

input	meaning	condition
`t_half`	half-life	
`t`	elapsed time	

output	meaning
`m`	mass remaining
`m_g`	in grams
`fraction`	fraction remaining
`half_lives`	number of half-lives

- **Method:** $m = m_0\ (1/2)^{t/t_1/2}$
- **Assumes:** m0, $t_1/2 > 0$; $t \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



m (mass remaining) against t_half (half-life) from 69292.8 s to 6.92928e+06 s on a logarithmic axis, logarithmic output axis. One curve for each value of m0 (initial mass): 0.0001 kg, 5.0000e-05 kg, 0.0002 kg. Other inputs as in the worked example: t = 2078784.0 s. 41 points per curve computed by chemistry-ai999x; marked: the worked example, t_half = 692928 s, m = 1.2500e-05 kg.

```
chemistry_ai999x.solve('radioactive_decay_remaining', m0=0.0001, t_half=692928.0,
t=2078784.0)
# -> m = 1.25e-05 kg, m_g = 0.0125, fraction = 0.125, half_lives = 3   (SOLVED)
```

```
chemistry_ai999x.solve('radioactive_decay_remaining', m0=0.005, t_half=180825048000.0,
t=361650096000.0)
# -> m = 0.00125 kg, m_g = 1.25, fraction = 0.25, half_lives = 2   (SOLVED)
```

```
chemistry_ai999x.solve('radioactive_decay_remaining', m0=0.001, t_half=30.0, t=15.0)
# -> m = 0.000707107 kg, m_g = 0.707107, fraction = 0.707107, half_lives = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The half-life of iodine-131 is 8.02 days. How much of a 100 mg sample remains after 24.06 days?	SOLVED	m = 1.25e-05 kg, m_g = 0.0125, fraction = 0.125, half_lives = 3
Carbon-14 has a half-life of 5730 years. What mass of a 5.00 g sample is left after 11460 years?	SOLVED	m = 0.00125 kg, m_g = 1.25, fraction = 0.25, half_lives = 2
The half-life of iodine-131 is 8.02 days. How much of a 100 mg sample remains after -5.0 days?	INVALID_INPUT	Amount remaining after radioactive decay (half-life): the elapsed time cannot be negative.

Age from the fraction remaining (radioactive dating), `radioactive_dating_age`

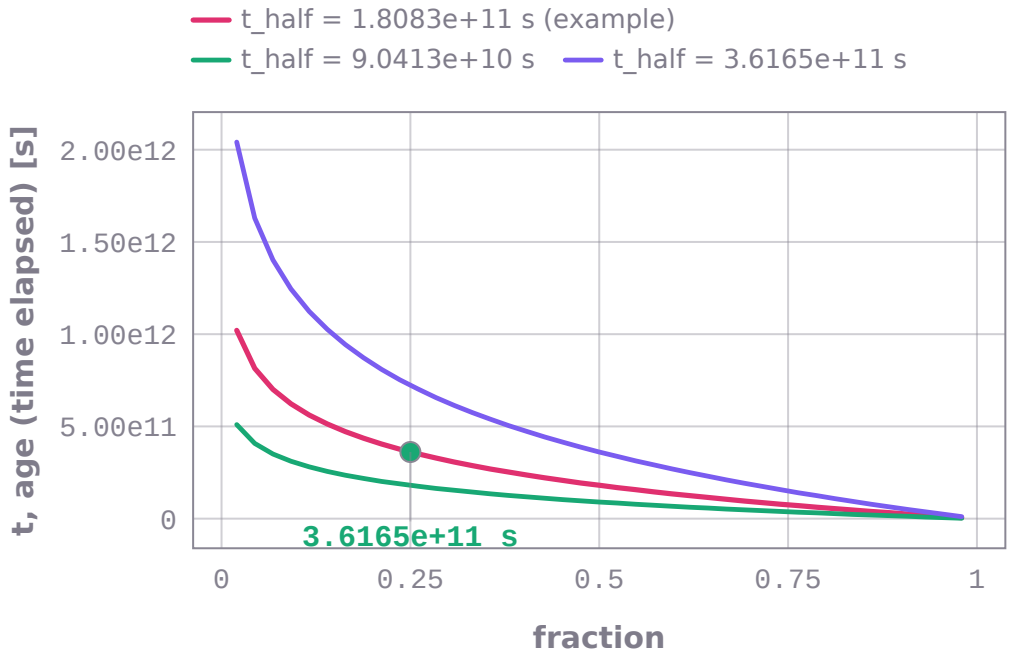
The age of a sample (the time elapsed) from the fraction of a radioactive nuclide remaining and its half-life.

$$t = t_{1/2} \log_2 \frac{1}{f}$$

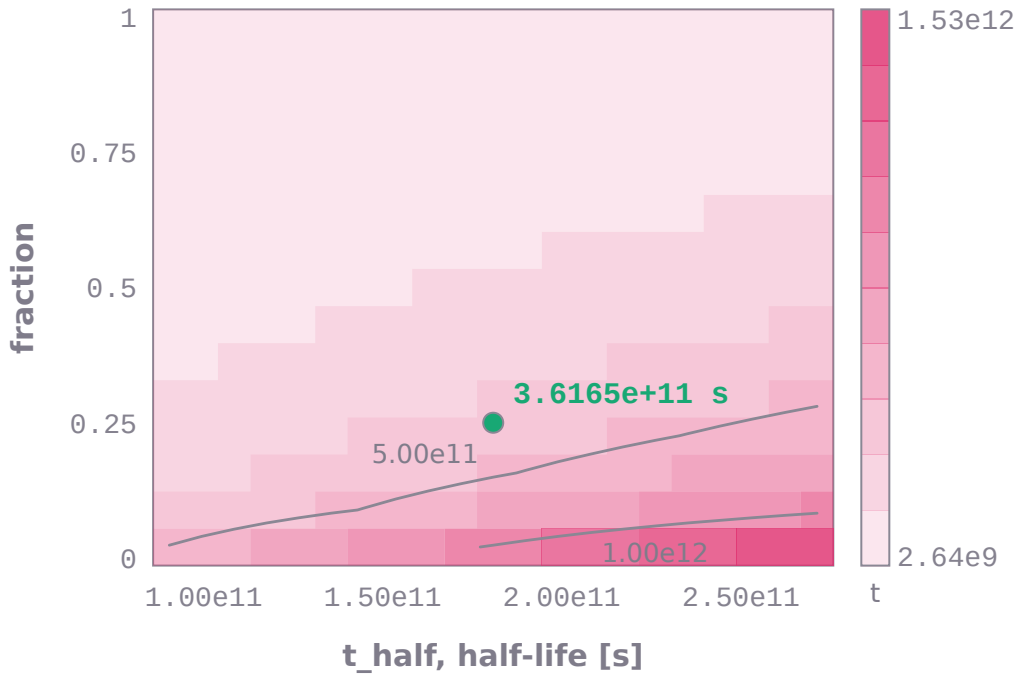
input	meaning	condition
`fraction`	fraction of the original nuclide remaining $(0 < f \leq 1)$	
`t_half`	half-life	

output	meaning
`t`	age (time elapsed)
`t_years`	in years
`half_lives`	number of half-lives

- **Method:** $t = t_1/2$ ln(1/f) / ln 2
- **Assumes:** $0 < f \leq 1$; $t_1/2 > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



t (age (time elapsed)) against fraction (fraction of the original nuclide remaining (0 < f <= 1)) from 0.02 to 0.98. One curve for each value of t_half (half-life): 1.8083e+11 s, 9.0413e+10 s, 3.6165e+11 s. 41 points per curve computed by chemistry-ai999x; marked: the worked example, fraction = 0.25, t = 3.6165e+11 s.



t (age (time elapsed)) over t_half (half-life) from 9.0413e+10 s to 2.7124e+11 s and fraction (fraction of the original nuclide remaining (0 < f <= 1)) from 0.02 to 0.98: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 2.6352e+09 s to dark 1.5308e+12 s); iso-lines at t = 5.00e11, 1.00e12 s. Marked: the worked example, t_half = 1.8083e+11 s, fraction = 0.25, t = 3.6165e+11 s.

```
chemistry_ai999x.solve('radioactive_dating_age', fraction=0.25, t_half=180825048000.0)
# -> t = 3.6165e+11 s, t_years = 11460, half_lives = 2   (SOLVED)
```

```
chemistry_ai999x.solve('radioactive_dating_age', fraction=0.1, t_half=180825048000.0)
# -> t = 6.00688e+11 s, t_years = 19034.6, half_lives = 3.32193   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A wood sample has 25% of the original carbon-14 remaining. The half-life of carbon-14 is 5730 years. How old is the sample?	SOLVED	t = 3.6165e+11 s, t_years = 11460, half_lives = 2
The fraction remaining is 0.10 and the half-life is 5730 years. What is the age of the sample?	SOLVED	t = 6.00688e+11 s, t_years = 19034.6, half_lives = 3.32193
The fraction remaining is 1.5 and the half-life is 5730 years. What is the age of the sample?	INVALID_INPUT	Age from the fraction remaining (radioactive dating): the fraction remaining lies between 0 and 1.

Conjugate acid-base constants (Ka Kb = Kw), `conjugate_dissociation_constant`

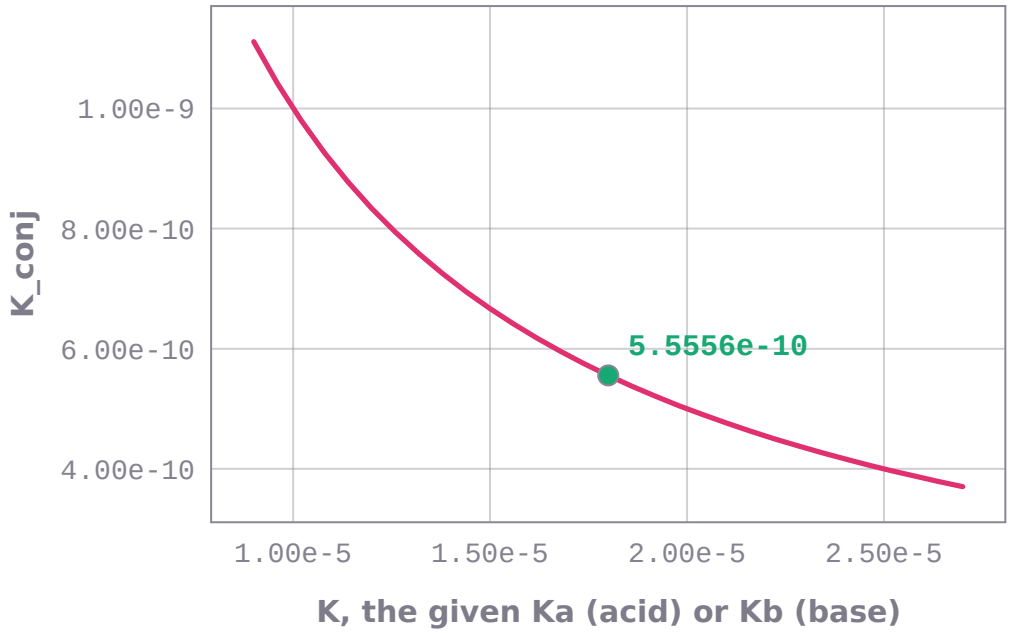
The dissociation constant of the conjugate base of an acid (or of the conjugate acid of a base), with pKa and pKb, at 25 °C.

$$K_a K_b = K_w, \quad \mathrm{p}K_a + \mathrm{p}K_b = 14$$

input	meaning	condition
`K`	the given Ka (acid) or Kb (base)	
`known`	which constant is given	

output	meaning
`K_conj`	dissociation constant of the conjugate
`pK`	pK of the given constant
`pK_conj`	pK of the conjugate
`conjugate`	which constant the conjugate's is

- **Method:** K_conj = Kw / K; $pK = -log10$ K
- **Assumes:** $K > 0$; 25 °C $(Kw = 1.0e-14)$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



K_conj (dissociation constant of the conjugate) against K (the given Ka (acid) or Kb (base)) from 9.0000e-06 to 2.7000e-05. Other inputs as in the worked example: known = Ka. 31 points computed by chemistry-ai999x; marked: the worked example, K = 1.8000e-05, K_conj = 5.5556e-10.

```
chemistry_ai999x.solve('conjugate_dissociation_constant', K=1.8e-05, known='Ka')
# -> K_conj = 5.55556e-10, pK = 4.74473, pK_conj = 9.25527, conjugate = Kb of the conjugate
base   (SOLVED)
```

```
chemistry_ai999x.solve('conjugate_dissociation_constant', K=1e-05, known='Kb')
# -> K_conj = 1e-09, pK = 5, pK_conj = 9, conjugate = Ka of the conjugate acid   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Ka of acetic acid = 1.8e-5. What is Kb of its conjugate base?	SOLVED	K_conj = 5.55556e-10, pK = 4.74473, pK_conj = 9.25527, conjugate = Kb of the conjugate base
The pKa of an acid HA is 3.00. Find Kb of its conjugate base.	SOLVED	K_conj = 1e-11, pK = 3, pK_conj = 11, conjugate = Kb of the conjugate base
The pKb of a base B is 5.00. What is Ka of its conjugate acid?	SOLVED	K_conj = 1e-09, pK = 5, pK_conj = 9, conjugate = Ka of the conjugate acid
Ka of an acid HA = -1.0e-5. What is Kb of its conjugate base?	INVALID_INPUT	Conjugate acid-base constants (Ka Kb = Kw): K must be a positive number.

Specific rotation from an observed rotation, `specific_rotation`

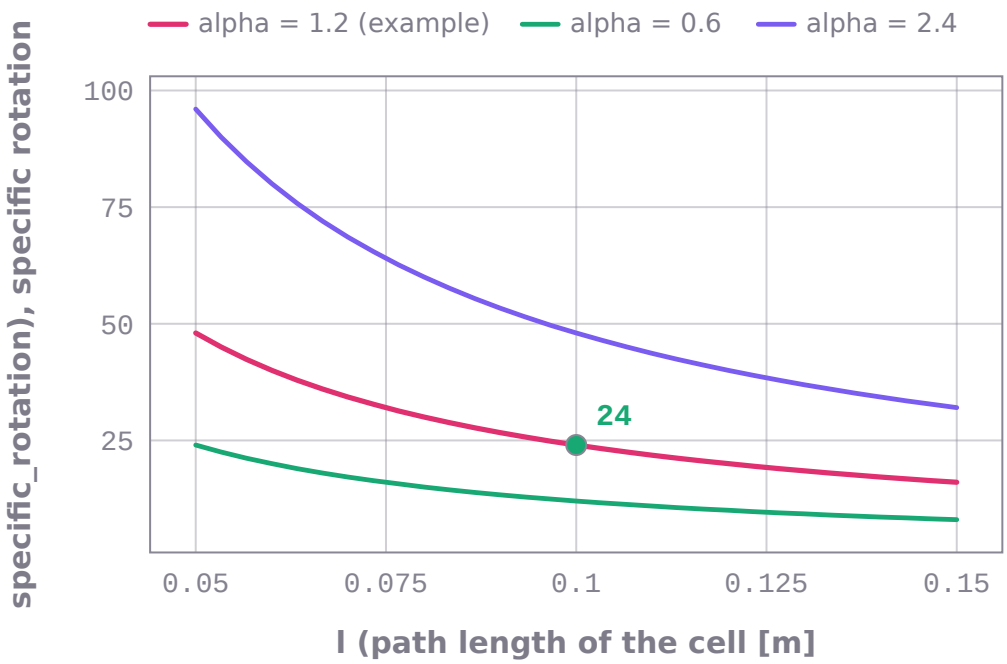
The specific rotation of an optically active compound from the observed rotation, the path length and the concentration of its solution.

$$[\alpha] = \frac{\alpha}{l\,c} \quad (l \text{ in dm},\ c \text{ in g/mL})$$

input	meaning	condition
`alpha`	observed rotation (degrees)	
`l`	path length of the cell	
`c`	concentration in mass per volume (g/mL)	

output	meaning
`specific_rotation`	specific rotation (degrees mL g^{-1} dm^-1)

- **Method:** observed rotation over path length (dm) times concentration (g/mL)
- **Assumes:** l, $c > 0$; rotation in degrees, written with °
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



specific_rotation (specific rotation (degrees mL g^-1 dm^-1)) against l (path length of the cell) from 0.05 m to 0.15 m. One curve for each value of alpha (observed rotation (degrees)): 1.2, 0.6, 2.4. Other inputs as in the worked example: c = 50.0 kg/m^3. 31 points per curve computed by chemistry-ai999x; marked: the worked example, l = 0.1 m, specific_rotation = 24.

```
chemistry_ai999x.solve('specific_rotation', alpha=1.2, l=0.1, c=50.0)
# -> specific_rotation = 24   (SOLVED)
```

```
chemistry_ai999x.solve('specific_rotation', alpha=-4.32, l=0.2, c=120.0)
# -> specific_rotation = -18   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A solution of 0.0500 g/mL of a compound in a 1.00 dm tube shows an observed rotation of +1.20°. What is its specific rotation?	SOLVED	specific_rotation = 24
The observed rotation is -4.32° for a 0.120 g/mL solution in a 2.00 dm cell. Find the specific rotation.	SOLVED	specific_rotation = -18

question	outcome	result
The observed rotation is -4.32° for a 0.120 g/mL solution in a 0 dm cell. Find the specific rotation.	INVALID_INPUT	Specific rotation from an observed rotation: l must be a positive number.

Enantiomeric excess from optical rotation, `enantiomeric_excess_from_rotation`

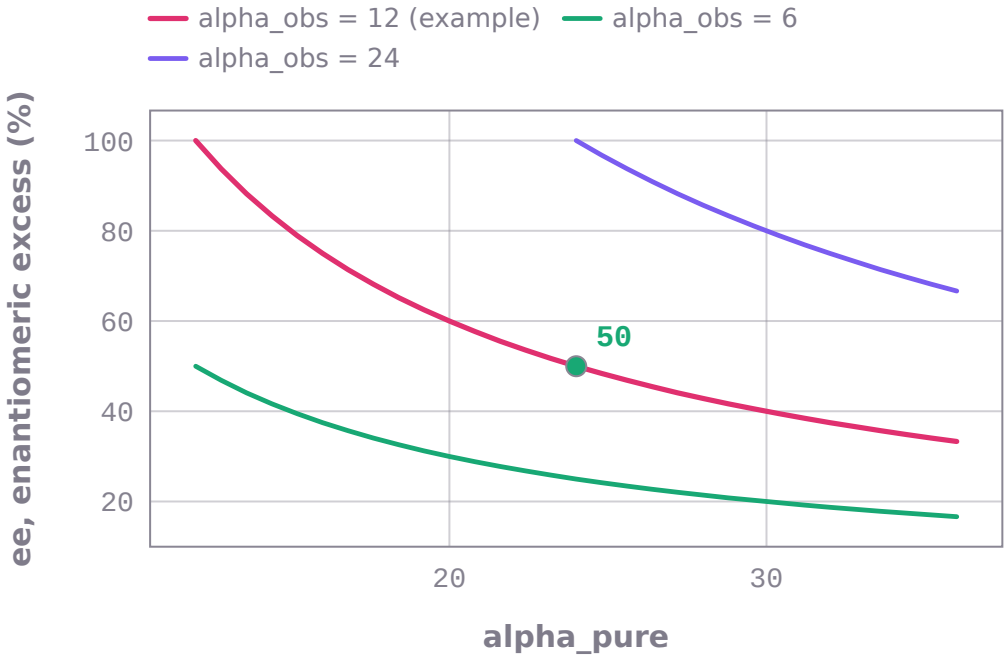
The enantiomeric excess (optical purity) of a mixture and the percent of each enantiomer, from its specific rotation and that of the pure enantiomer.

$$ee = \frac{[\alpha]_{mixture}}{[\alpha]_{pure}} \times 100\%, \quad \%_{major} = 50 + \frac{ee}{2}$$

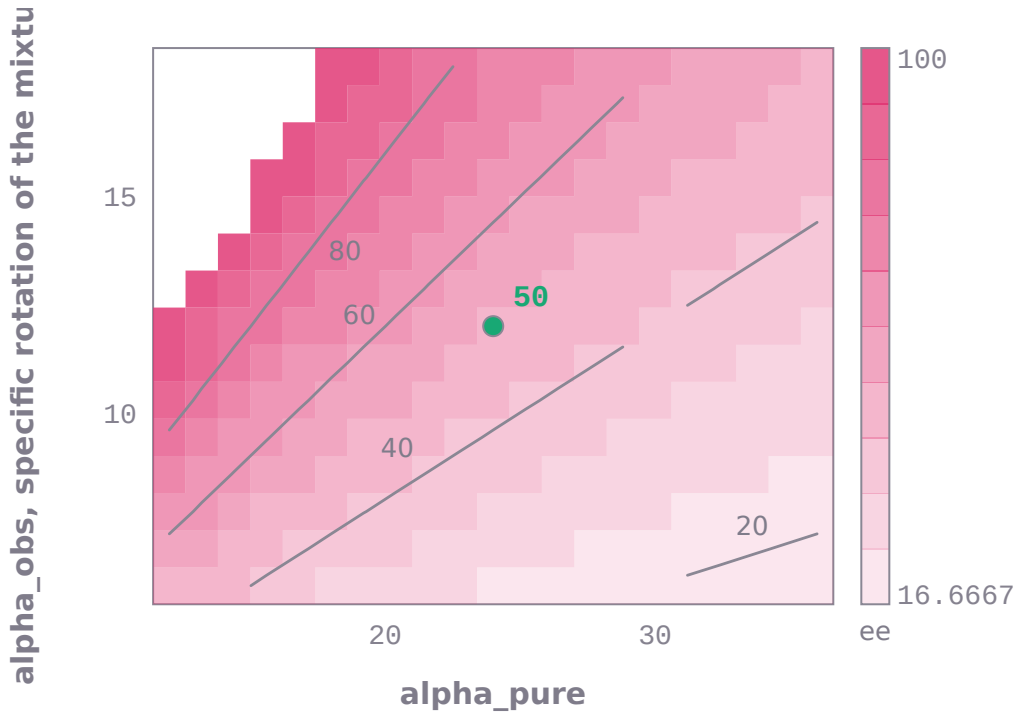
input	meaning	condition
`alpha_obs`	specific rotation of the mixture (degrees)	
`alpha_pure`	specific rotation of the pure enantiomer (degrees)	

output	meaning
`ee`	enantiomeric excess (%)
`major_percent`	percent of the major enantiomer
`minor_percent`	percent of the minor enantiomer
`predominant`	which enantiomer predominates

- **Method:** optical purity = ratio of the specific rotations; the major enantiomer is 50% + $ee/2$
- **Assumes:** the pure enantiomer's rotation is not zero; |observed| <= |pure|
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



ee (enantiomeric excess (%)) against alpha_pure (specific rotation of the pure enantiomer (degrees)) from 12 to 36. One curve for each value of alpha_obs (specific rotation of the mixture (degrees)): 12, 6, 24. 31 points per curve computed by chemistry-ai999x; marked: the worked example, alpha_pure = 24, ee = 50.



ee (enantiomeric excess (%)) over alpha_pure (specific rotation of the pure enantiomer (degrees)) from 12 to 36 and alpha_obs (specific rotation of the mixture (degrees)) from 6 to 18: each of the 21 × 15 cells is one run of chemistry-ai999x, shaded by its value (light 16.6667 to dark 100); iso-lines at ee = 20, 40, 60, 80; blank cells: inputs the capability refuses. Marked: the worked example, alpha_pure = 24, alpha_obs = 12, ee = 50.

```
chemistry_ai999x.solve('enantiomeric_excess_from_rotation', alpha_obs=12.0,
alpha_pure=24.0)
# -> ee = 50, major_percent = 75, minor_percent = 25, predominant = the same enantiomer as
the pure sample   (SOLVED)
```

```
chemistry_ai999x.solve('enantiomeric_excess_from_rotation', alpha_obs=-6.0,
alpha_pure=24.0)
# -> ee = 25, major_percent = 62.5, minor_percent = 37.5, predominant = the opposite
enantiomer   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The observed specific rotation of a mixture is +12.0° and the pure enantiomer has a specific rotation of +24.0°. What is the enantiomeric excess?	SOLVED	ee = 50, major_percent = 75, minor_percent = 25, predominant = the same enantiomer as the pure sample
The observed rotation of a sample is -6.0° and the pure enantiomer has a specific rotation of +24.0°. What is the optical purity (enantiomeric excess)?	SOLVED	ee = 25, major_percent = 62.5, minor_percent = 37.5, predominant = the opposite enantiomer
The observed specific rotation of a mixture is +30.0° and the pure enantiomer has a specific rotation of +24.0°. What is the enantiomeric excess?	INVALID_INPUT	Enantiomeric excess from optical rotation: the observed rotation cannot be larger in size than the pure enantiomer's.

Enantiomeric excess from the composition of a mixture,

`enantiomeric_excess_from_composition`

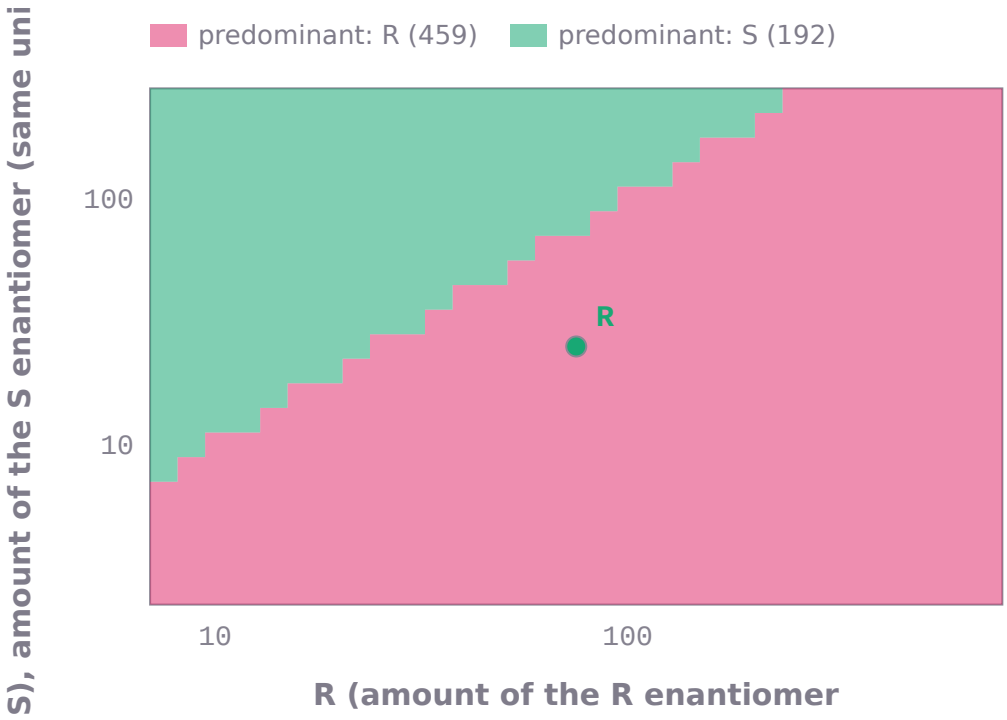
The enantiomeric excess and the percent of each enantiomer from the amounts of the R and S enantiomers.

$$ee = \frac{|n_R - n_S|}{n_R + n_S} \times 100\%$$

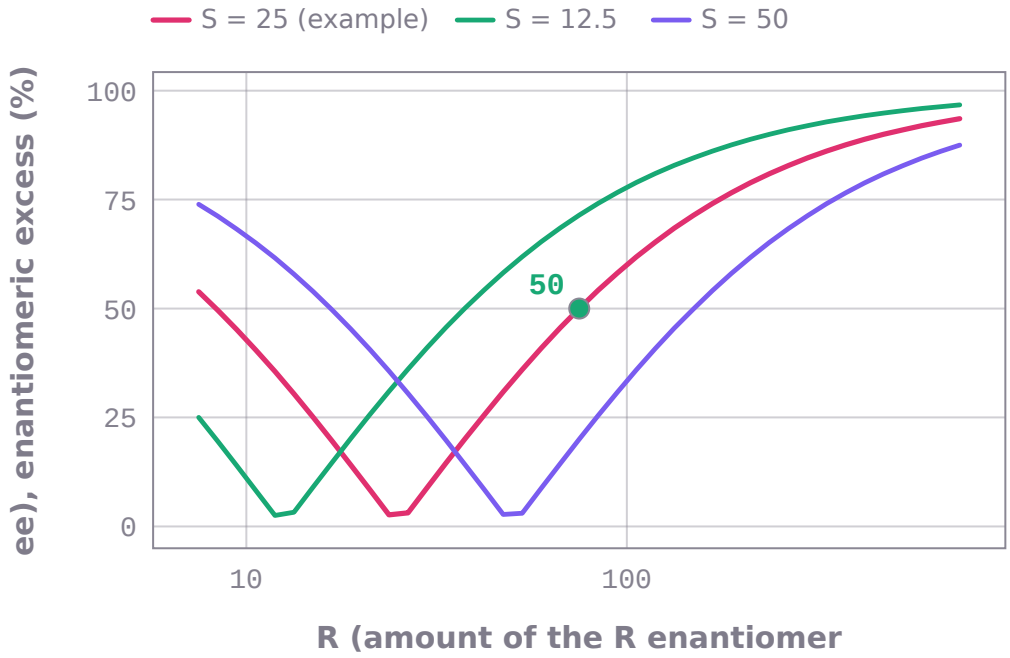
input	meaning	condition
`R`	amount of the R enantiomer (any unit, the same for both)	
`S`	amount of the S enantiomer (same unit)	

output	meaning
`ee`	enantiomeric excess (%)
`percent_R`	percent R
`percent_S`	percent S
`predominant`	which enantiomer predominates

- **Method:** difference of the two amounts over their sum
- **Assumes:** amounts >= 0, not both zero; one unit for both
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



which enantiomer predominates (predominant) over R (amount of the R enantiomer (any unit, the same for both)) from 7.5 to 750 and S (amount of the S enantiomer (same unit)) from 2.5 to 250 (logarithmic axes): each of the 31 × 21 cells is one run of chemistry-ai999x, coloured by the verdict it returned, R: 459 cells, S: 192 cells. Marked: the worked example, R = 75, S = 25 → R.



ee (enantiomeric excess (%)) against R (amount of the R enantiomer (any unit, the same for both)) from 7.5 to 750 on a logarithmic axis. One curve for each value of S (amount of the S enantiomer (same unit)): 25, 12.5, 50. 41 points per curve computed by chemistry-ai999x; marked: the worked example, R = 75, ee = 50.

```
chemistry_ai999x.solve('enantiomeric_excess_from_composition', R=75.0, S=25.0)
# -> ee = 50, percent_R = 75, percent_S = 25, predominant = R   (SOLVED)
```

```
chemistry_ai999x.solve('enantiomeric_excess_from_composition', R=1.0, S=3.0)
# -> ee = 50, percent_R = 25, percent_S = 75, predominant = S   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A mixture contains 75% of the R enantiomer and 25% of the S enantiomer. What is the enantiomeric excess?	SOLVED	ee = 50, percent_R = 75, percent_S = 25, predominant = R
A sample contains 1.0 g of the (R)-enantiomer and 3.0 g of the (S)-enantiomer. Find the enantiomeric excess.	SOLVED	ee = 50, percent_R = 25, percent_S = 75, predominant = S
A sample contains -1.0 g of the (R)-enantiomer and 3.0 g of the (S)-enantiomer. Find the enantiomeric excess.	INVALID_INPUT	Enantiomeric excess from the composition of a mixture: amounts cannot be negative.